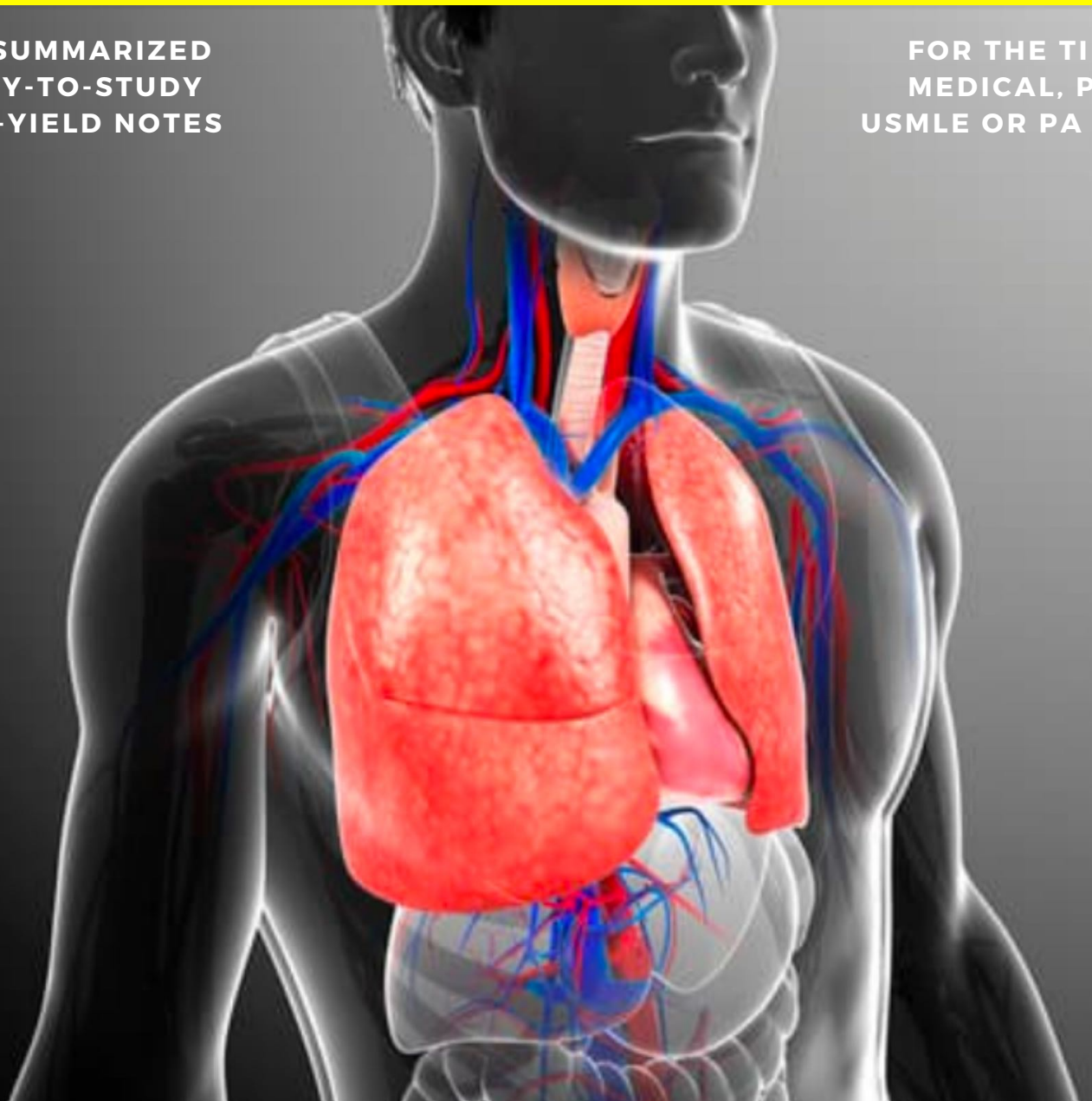


ANATOMY, PHYSIOLOGY & PATHOLOGY NOTES OF THE RESPIRATORY SYSTEM

FOURTH EDITION

**PRE-SUMMARIZED
READY-TO-STUDY
HIGH-YIELD NOTES**

**FOR THE TIME-POOR
MEDICAL, PRE-MED,
USMLE OR PA STUDENT**



**MED
STUDENT
NOTES
.COM**

192 PAGES





A Message From Our Team

Studying medicine or any health-related degree can be stressful; believe us, we know from experience! The human body is an incredibly complex organism, and finding a way to streamline your learning is crucial to succeeding in your exams and future profession. Our goal from the outset has been to create the greatest educational resource for the next generation of medical students, and to make them as affordable as possible.

In this fourth edition of our notes we have made a number of text corrections, formatting updates, and figure updates which we feel will enhance your study experience. We have also endeavoured to use only open-source images and/or provide attribution where possible.

If you are new to us, here are a few things to help get the most out of your notes:

1. **Once saved, the notes are yours for life!** However, we strongly advise that you download and save the files immediately upon purchasing for permanent offline access.
2. **Sharing notes is prohibited.** All files are share-protected and our system will automatically revoke access to and lock files if it detects a customer attempting to share or distribute our notes.
3. **Your license permits you to do the following:**
 - a. You may download/save/view files on up to 2 simultaneous devices.
 - b. You may save the files to an external hard drive for backup purposes only.
 - c. You may print your notes to hard copy on any home printer/photocopier.
4. **Once saved, you do not need to download the notes again.** You can simply transfer your file/s to your second personal device (eg: your iPad/tablet) without the need to re-download your files. If you wish to retire an old device, simply transfer your files to your new device, then delete the files from your old device.
5. **Each order has an allocated download allowance.** Exceeding your download quota will require you to request an extension by emailing us at admin@medstudentnotes.com and quoting your order number.
6. **If you have any further queries,** feel free to check out our FAQ page: <https://www.medstudentnotes.com/pages/faqs> or email us at admin@medstudentnotes.com
7. **Please ensure you use the SAME Email address for any future purchases.** That way, all of your files will be visible in the one place on your account page.
8. **Feel free to use the discount code "TAKE20OFF"** for 20% off any future purchases. (Valid for a limited time)
9. **If you find these notes valuable but you haven't yet got our [Bundle Deal](#), (which includes ALL of our subjects at an 80% discount), then NOW is the time! To upgrade, simply visit this link for instructions: <https://www.medstudentnotes.com/pages/bundle-upgrade-process> .**



Table Of Contents:

What's included: Ready-to-study anatomy, physiology and pathology notes of the respiratory system presented in succinct, intuitive and richly illustrated downloadable PDF documents. Once downloaded, you may choose to either print and bind them, or make annotations digitally on your iPad or tablet PC.

Anatomy & Physiology Notes:

- ANATOMY OF THE THORAX
- ANATOMY OF THE AIRWAYS
- AIRWAY MUCOSAL LININGS
- ALVEOLAR GAS EXCHANGE & GAS TRANSPORT
- THE PHYSIOLOGY OF BREATHING
- HEMOGLOBIN / HAEMOGLOBIN
- BODY ACID-BASE BALANCE
- CONTROL OF BREATHING
- FOETAL LUNG DEVELOPMENT & TRANSITION TO EXTRAUTERINE LIFE

Pathology Notes:

- HYPERBARIC & HYPOBARIC CONDITIONS
- AIRWAY HYPERSENSITIVITY & ASTHMA
- UPPER RESPIRATORY TRACT INFECTIONS (URTI'S)
 - COMMON COLD (ACUTE RHINITIS)
 - PHARYNGITIS (SORE THROAT)
 - (EPSTEIN BARR VIRUS) – INFECTIOUS MONONUCLEOSIS (GLANDULAR FEVER)
 - DIPHTHERIA
 - SCARLET FEVER ("STRAWBERRY TONGUE")
 - ACUTE LARYNGITIS
 - CHRONIC LARYNGITIS
 - ACUTE TONSILLITIS
 - ACUTE LARYNGOTRACHEOBRONCHITIS (CROUP)
 - ACUTE EPIGLOTTITIS
 - PERTUSSIS - WHOOPING COUGH:
 - MEASLES VIRUS
 - MUMPS VIRUS
 - RUBELLA VIRUS
 - ACUTE OTITIS MEDIA (AOM)
 - ACUTE SUPPURATIVE SINUSITIS (<4wks)
 - CHRONIC SINUSITIS (>3MTHS)
 - Q-FEVER
- LOWER RESPIRATORY TRACT INFECTIONS
 - BRONCHITIS (ACUTE)
 - BRONCHIECTASIS:
 - PNEUMONIAS
 - BRONCHIOLITIS
- GLOBAL RESPIRATORY VIRUSES
 - SEASONAL FLU (INFLUENZA A & B)
 - BIRD FLU (H5N1)
 - SWINE FLU (H1N1)
 - SARS & COVID – SEVERE ACUTE RESPIRATORY SYNDROME

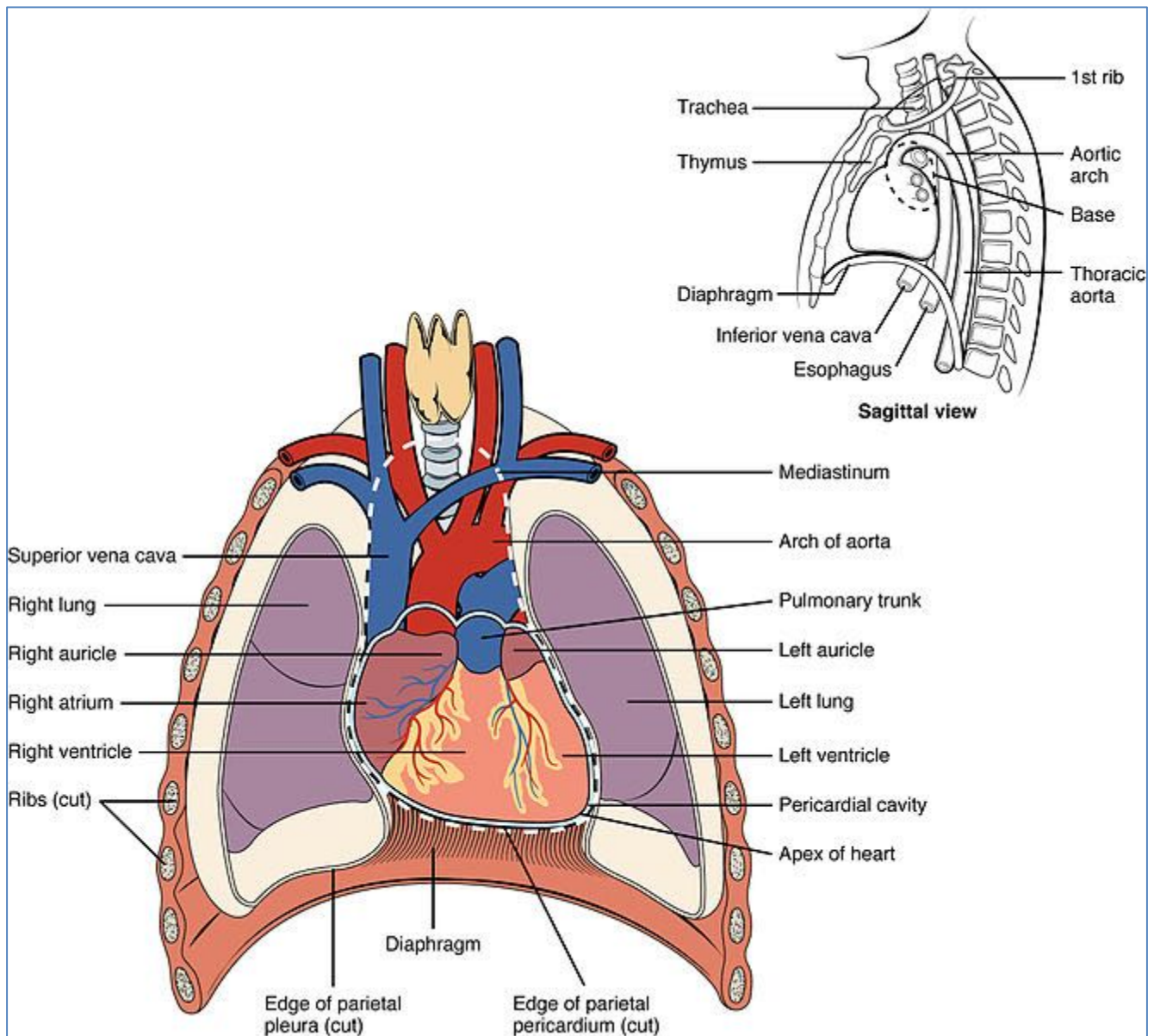
- OBSTRUCTIVE & RESTRICTIVE LUNG DISORDERS
 - CHRONIC OBSTRUCTIVE PULMONARY DISEASE
 - CHRONIC BRONCHITIS
 - EMPHYSEMA
 - IDIOPATHIC PULMONARY FIBROSIS (IPF)
 - SARCOIDOSIS
 - PNEUMOCONIOSES
 - CYSTIC FIBROSIS (CF)
 - PNEUMOTHORAX
- HEAD AND NECK PATHOLOGIES
 - SIALADENITIS (PAROTITIS):
 - SIALOLITHIASIS (SALIVARY GLAND CALCULI)
 - PLEOMORPHIC ADENOMA
 - WARTHIN'S TUMOUR
 - ORAL SQUAMOUS CELL CARCINOMA
 - NASAL POLYPS (Inflammatory)
 - NASOPHARYNGEAL CARCINOMA
 - LARYNGEAL TUMOURS
- LUNG CANCERS
 - BRONCHOGENIC LUNG CANCERS
 - MESOTHELIOMA
- HYPOXIA AND HYPERCAPNIA

ANATOMY OF THE THORAX

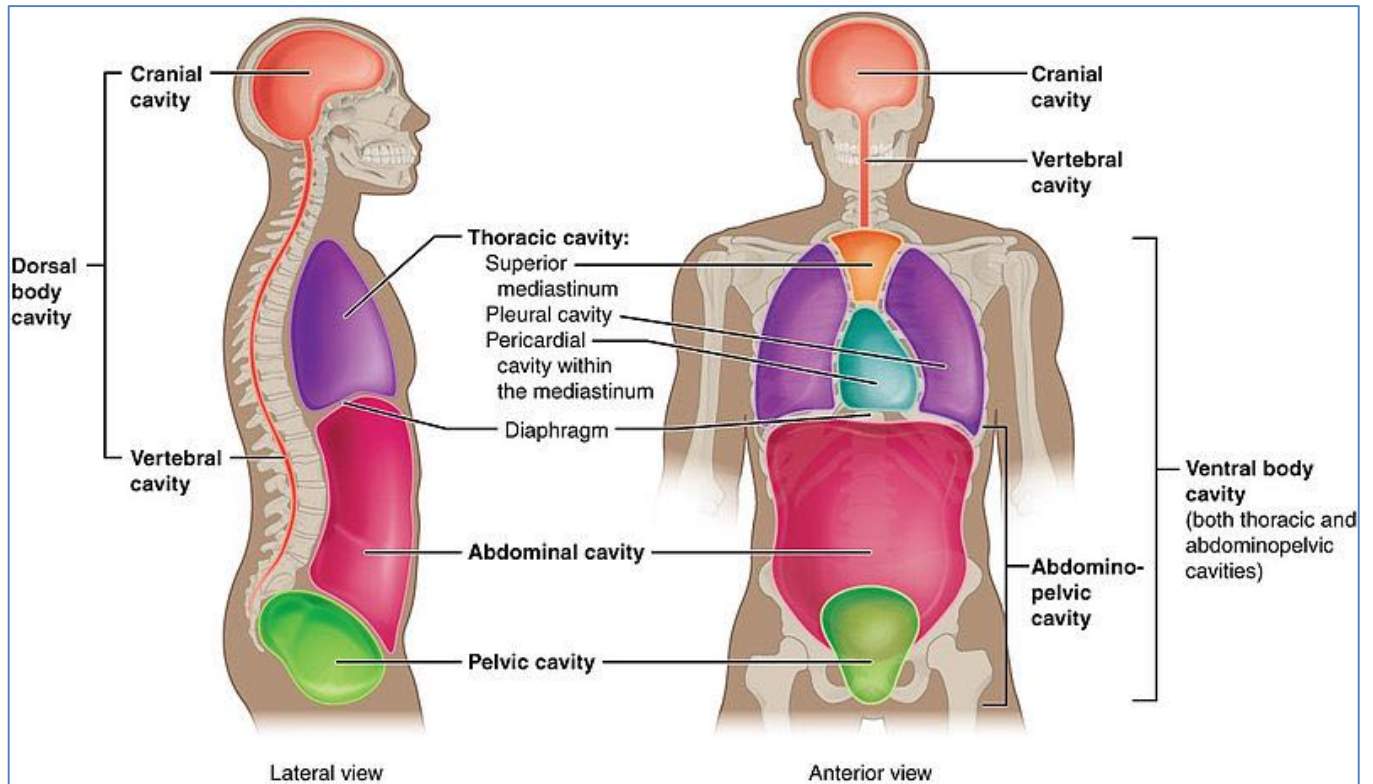
ANATOMY OF THE THORAX

Thoracic Overview:

- **3 Parts:**
 - Thoracic Cage (skeletal components)
 - Thoracic Wall (muscular components)
 - Thoracic Cavity (internal area)
- **3 Internal Compartments:**
 - **Central Mediastinum**
 - Containing the Heart/oesophagus/trachea/nerves/vessels
 - **Left Pleural Cavity**
 - Containing the L-Lung
 - **Right Pleural Cavity**
 - Containing the R-Lung



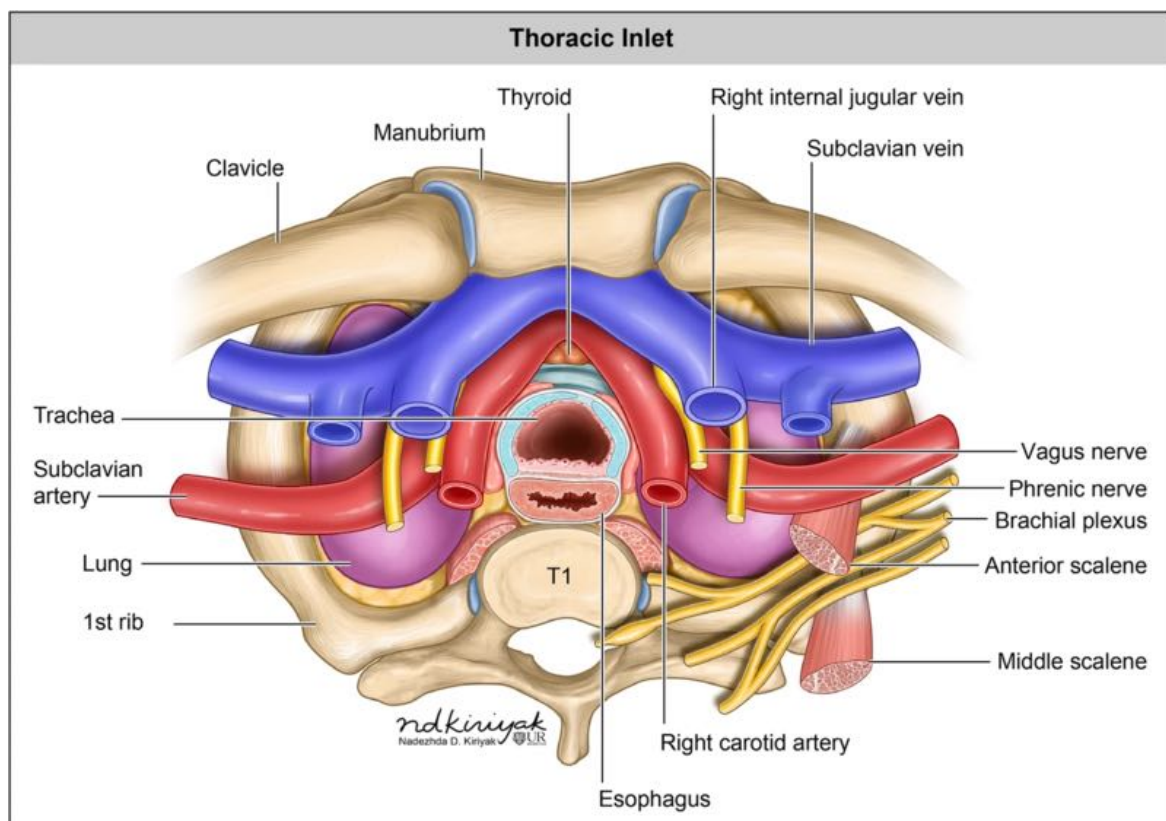
OpenStax College, CC BY 3.0 <<https://creativecommons.org/licenses/by/3.0>>, via Wikimedia Commons



Connexions, CC BY 3.0 <<https://creativecommons.org/licenses/by/3.0>>, via Wikimedia Commons

Relationship of thorax to Other Regions:

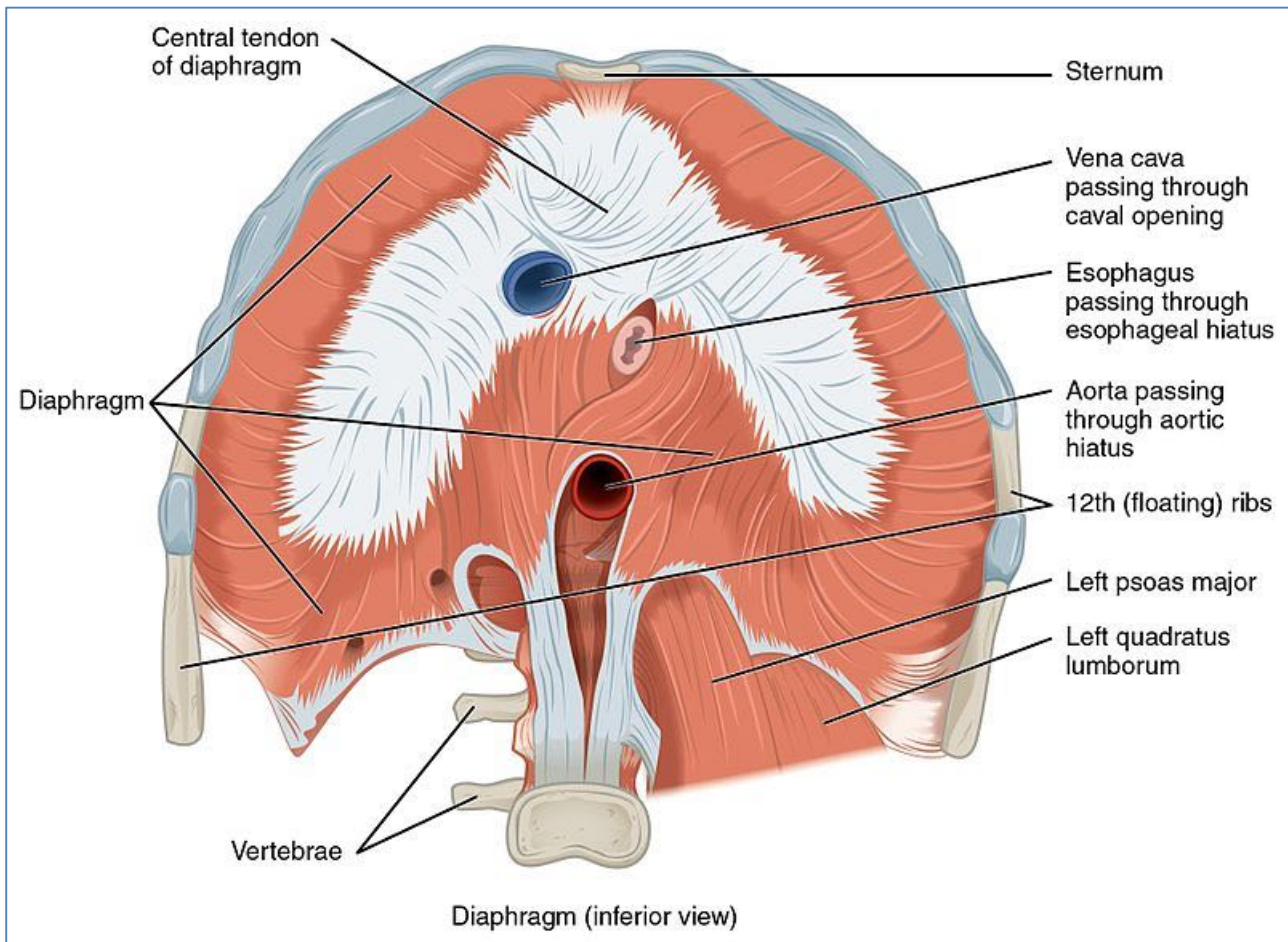
- **Neck:**
 - Trachea
 - Oesophagus
 - Major Nerves & Vessels



Nadezdha D. Kiriya, CC BY 4.0 <<https://creativecommons.org/licenses/by/4.0>>, via Wikimedia Commons

- **Abdomen:**

- Inferior Vena Cava
- Oesophagus
- Aorta



OpenStax, CC BY 4.0 <<https://creativecommons.org/licenses/by/4.0>>, via Wikimedia Commons

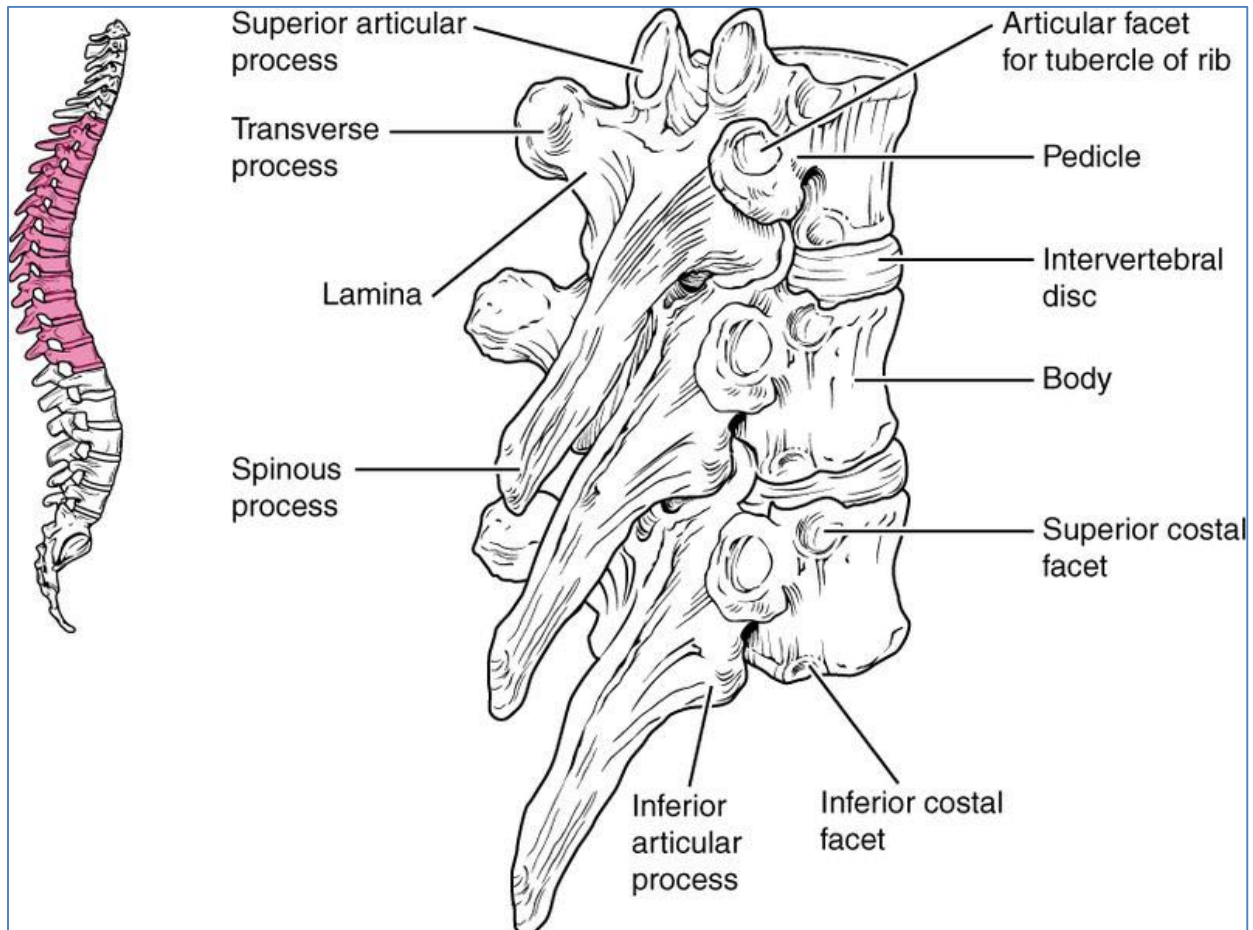
Thoracic Skeleton:

- **12 Thoracic Vertebrae:**

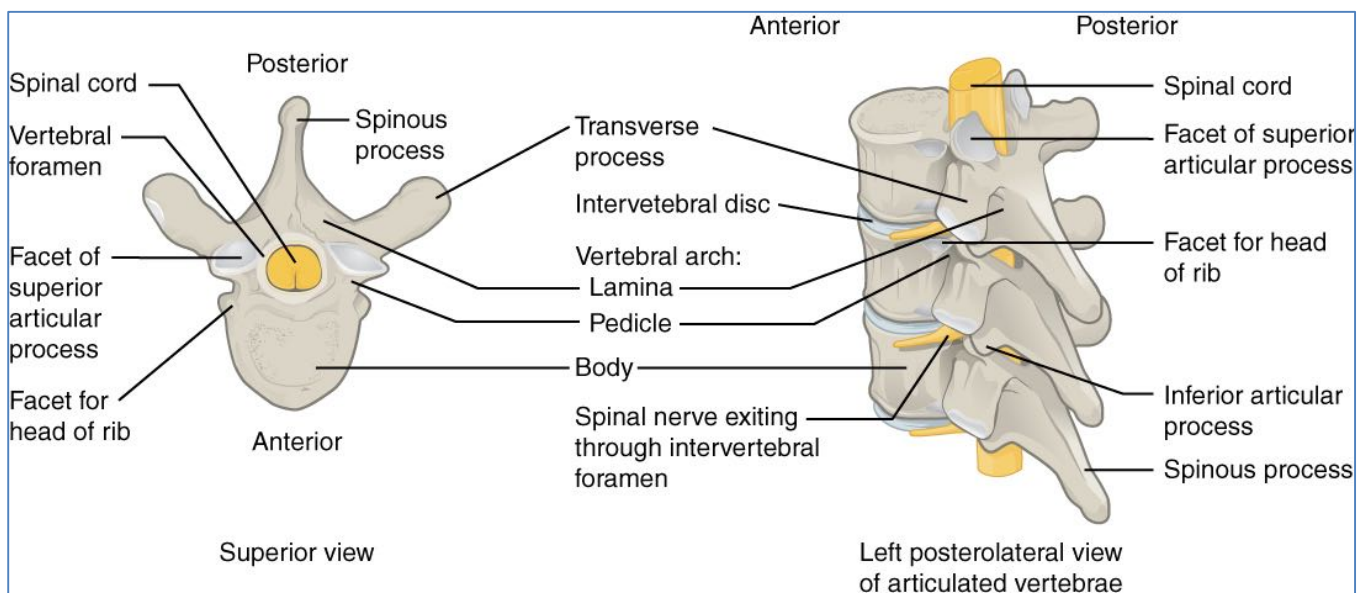
- T1 - T12

- **Distinguishing Features:**

- **Heart-Shaped Body** (for extra weight-bearing)
 - **Inferiorly Projecting Spine** (Allows a degree of mobility that would otherwise not be possible due to ribs)
 - **Large Transverse Processes** (To support articulations with ribs)
 - **Costal Demifacets** (small articulation points) for articulation with the ribs.

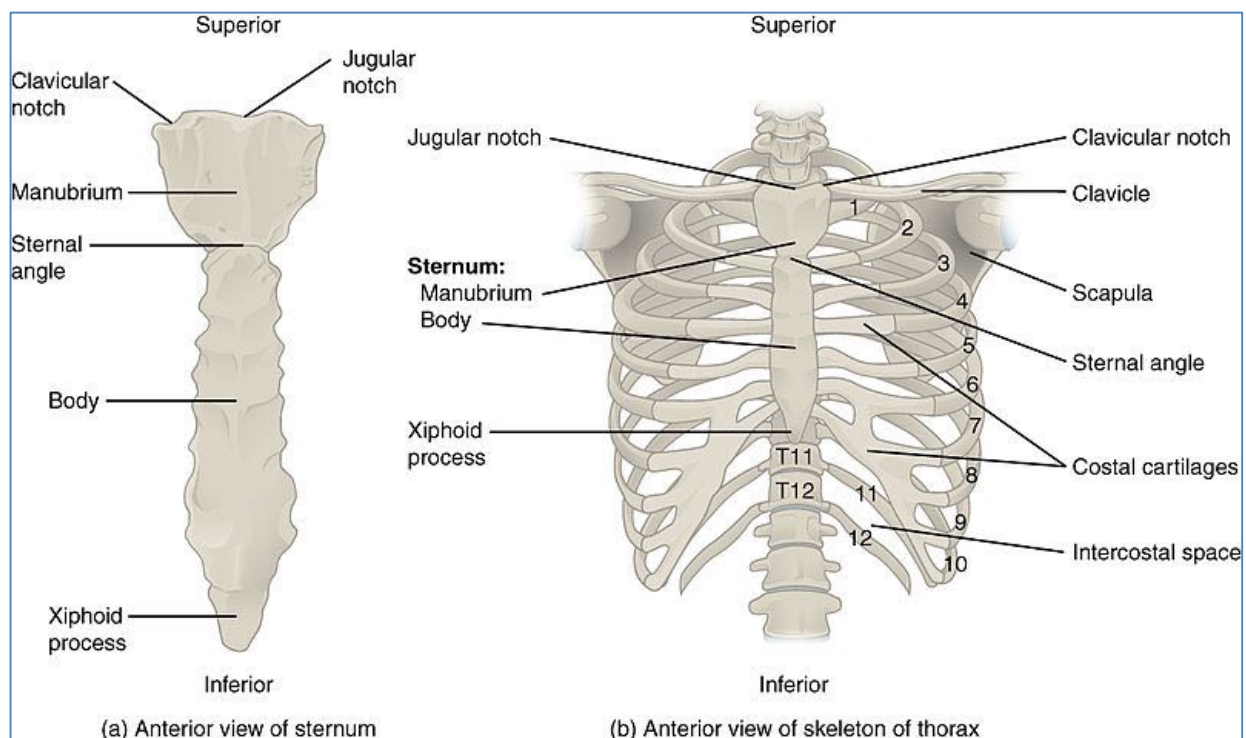


OpenStax College, CC BY 3.0 <<https://creativecommons.org/licenses/by/3.0>>, via Wikimedia Commons



OpenStax College, CC BY 3.0 <<https://creativecommons.org/licenses/by/3.0>>, via Wikimedia Commons

- **Sternum:**
 - **3 Parts:**
 - Manubrium
 - Body of Sternum
 - Xiphoid Process
 - **Sternal Angle:**
 - Between the Manubrium & the Body
 - Important **Landmark for:**
 - Bifurcation of Trachea
 - Aortic Crest
 - T-4 Vertebrae
 - 2nd Rib
 - **Articulations:**
 - **Ribs 1-7:** Via **Sternocostal Joints** (Synovial Joints)
 - **Ribs 8-10:** Have **Interchondral Joints** between the costal cartilages (i.e. Indirect articulation with sternum.)



OpenStax College, CC BY 3.0 <<https://creativecommons.org/licenses/by/3.0/>>, via Wikimedia Commons

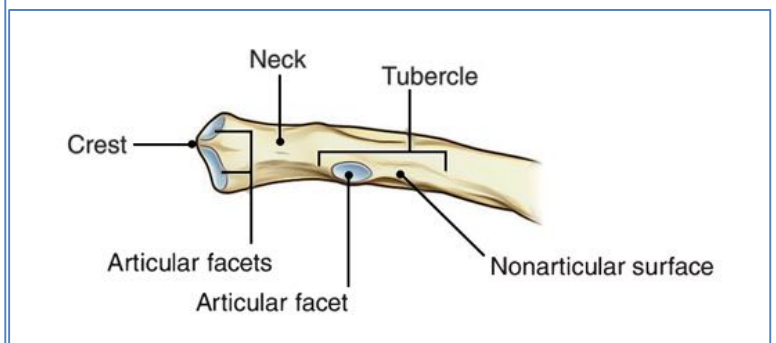
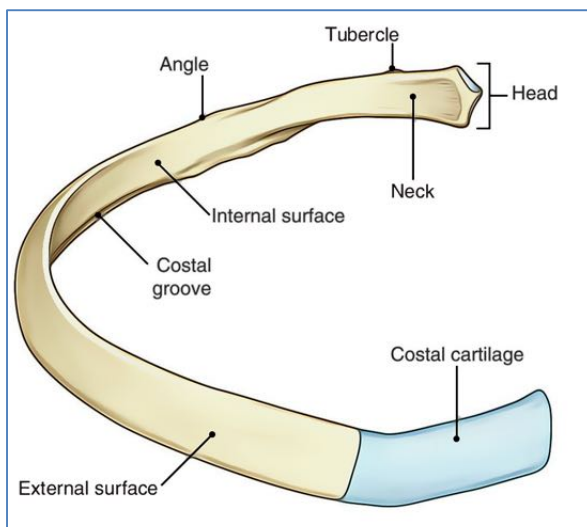
- **Note:** all bones of the Thoracic Cage are interconnected by articulations (cartilaginous & synovial), each offering a small amount of movement. However, despite limited movement of individual joints, their combined movements make the Thoracic Cage remarkably mobile.

- **12 Pairs of Ribs:**

- 1-7 = 'True' Ribs (attach directly to sternum)
- 8-12 = 'False' Ribs (don't attach directly to the sternum)
 - Ribs 11 & 12 are 'Floating' Ribs (insert into abdominal muscles & connective tissue.)

- **Distinguishing Features:**

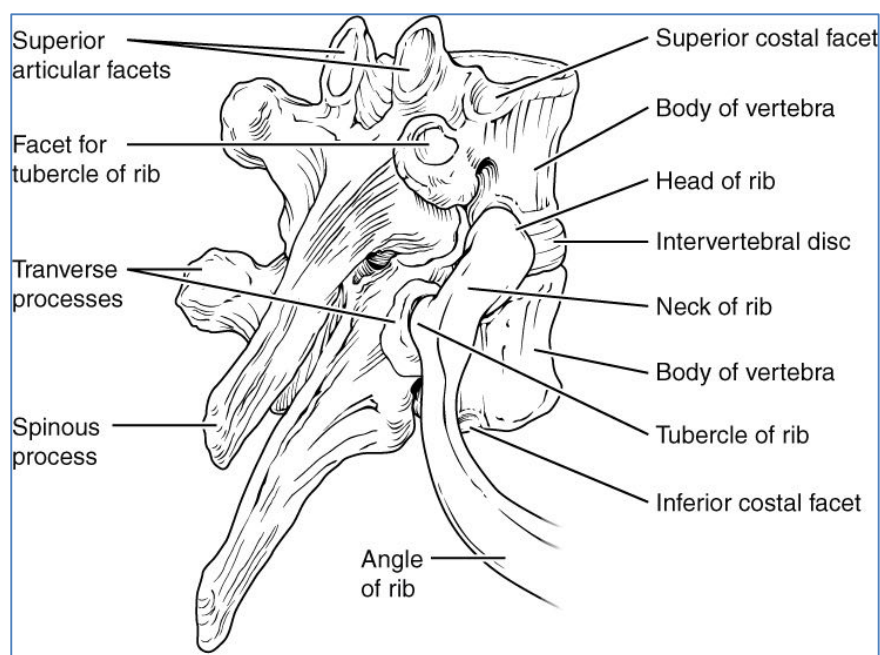
- **Posterior End:**
 - has a **head, neck & tubercle** (for attachment of ligaments & to vertebrae)
- **Curved shaft:**
 - Generally thin & flat
 - Oriented Vertically
 - Has a subcostal groove running on the inside of its inferior aspect.
- **SubCostal Groove:**
 - Houses the **Intercostals Nerve, Artery & Vein**
- **Anterior End:**
 - Sits more inferior than the posterior end.
 - Attach to sternum via **Costal Cartilage** – forms a *cartilaginous joint*



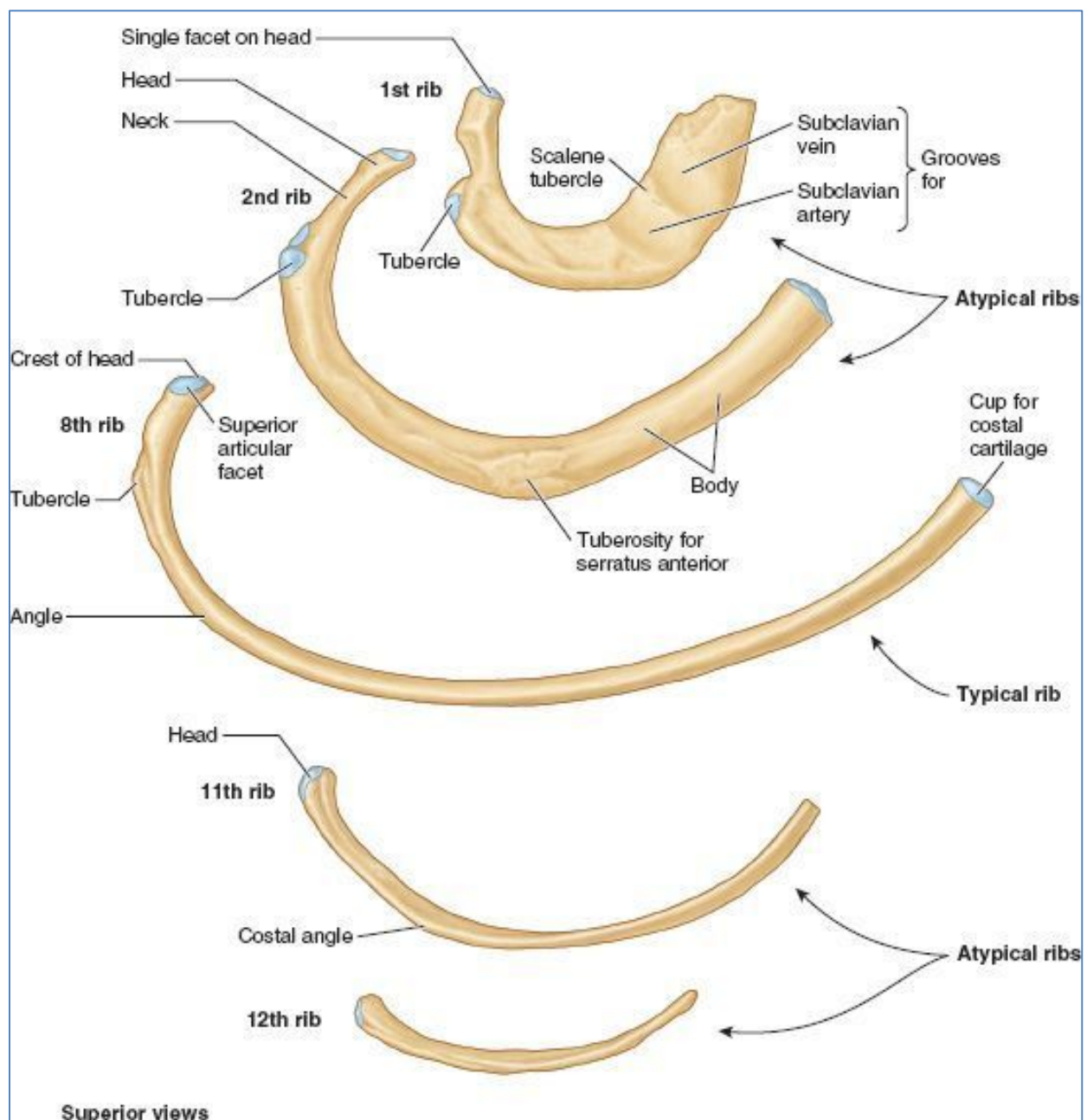
Source: Unattributable.

- **Typical Articulations:**

- Between **Head & Vertebrae** of the *same number*
- Between **Head & Vertebrae above**
- Between the **Tubercle & the Transverse Process** of the **Vertebrae** of the *same number*



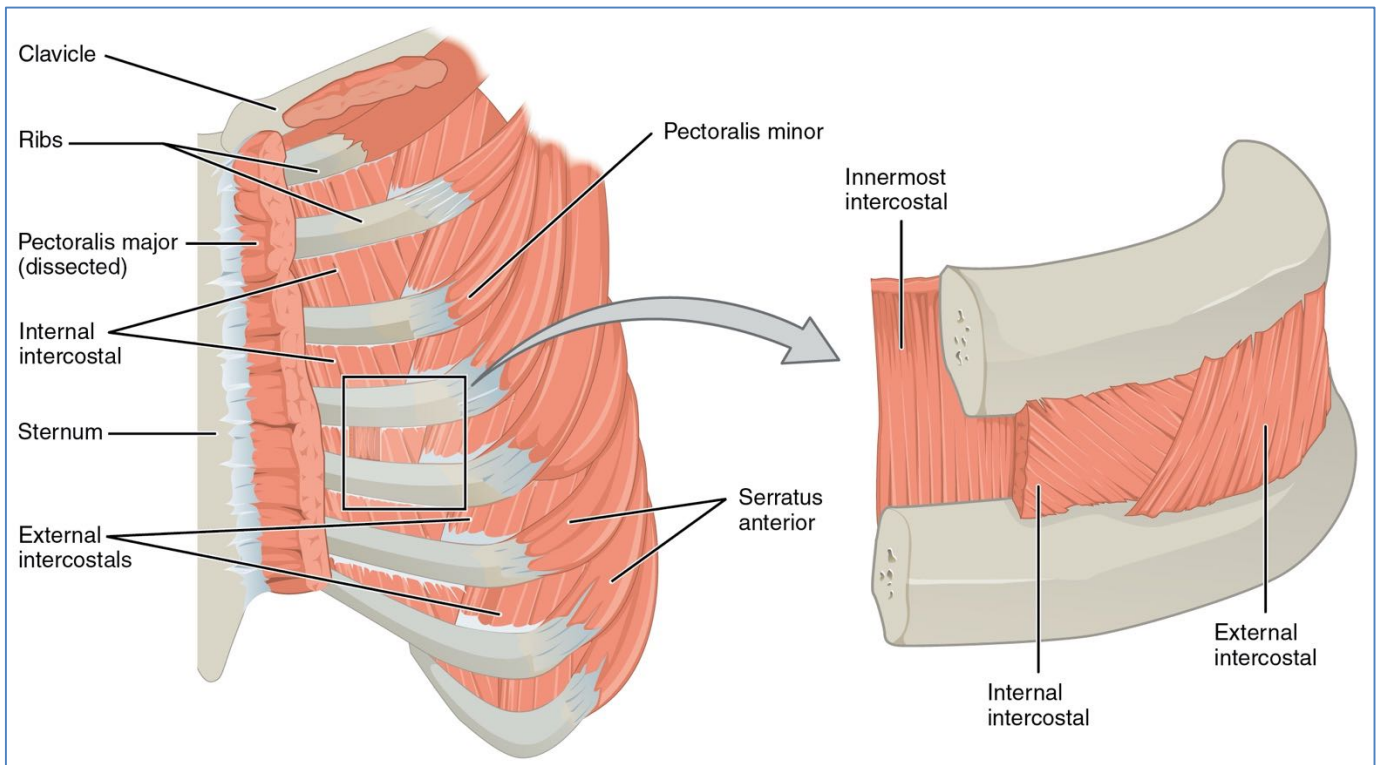
- **Atypical Ribs:**
 - Ribs 1, 2, 10, 11 & 12.
 - Why atypical?:
 - **Rib 1:**
 - Oriented horizontally (rather than vertically)
 - Much shorter
 - Articulates *only* with the *body* of T1.
 - Scalene Tubercle – attachment point of Anterior Scalene Muscle
 - Grooves for Subclavian Veins & Arteries
 - **Rib 2:**
 - Oriented horizontally (rather than vertically)
 - Otherwise typical
 - **Rib 8:**
 - Articulates *only with its own Vertebra* – only has 1 Facet on its head.
 - **Rib 11 & 12:**
 - Articulate *only with their own Vertebra*
 - No Tubercles / Necks
 - No Anterior Articulation



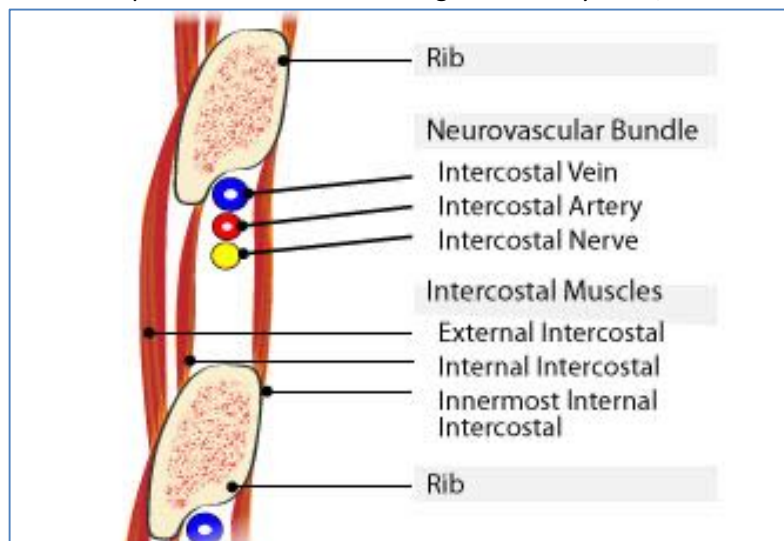
Source: Unattributable.

Thoracic Wall (Muscular Component):

- **3 Layers:**
 - **External Intercostal Muscle:**
 - Oriented Diagonally Inferio-Anteriorly
 - Incomplete Anteriorly → Transitions into the Anterior Intercostal *Membrane*
 - **Internal Intercostal Muscle:**
 - → Transitions into the Posterior Intercostal *Membrane*
 - **Innermost Intercostal Muscle:**
 - Oriented Diagonally Inferio-Posteriorly
 - Incomplete Posteriorly
- **Blood Supply:** (segmental)
 - **Posterior Intercostal Arteries** (Branches of Descending Aorta)
 - **Anterior Intercostal Arteries** (Branches of Internal Thoracic Arteries- From Subclavian Arteries)
- **Nerve Supply:**
 - Anterior Rami of Thoracic Spinal Nerves directly supply intercostals muscles.



OpenStax, CC BY 4.0 <<https://creativecommons.org/licenses/by/4.0>>, via Wikimedia Commons



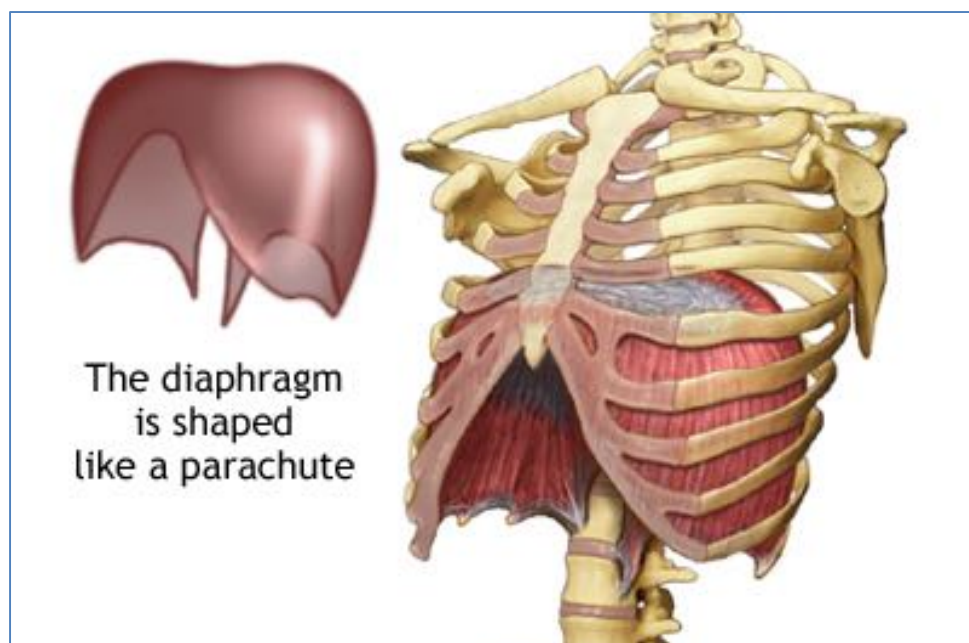
<https://csds.qld.edu.au/>

Accessory Muscles Of:

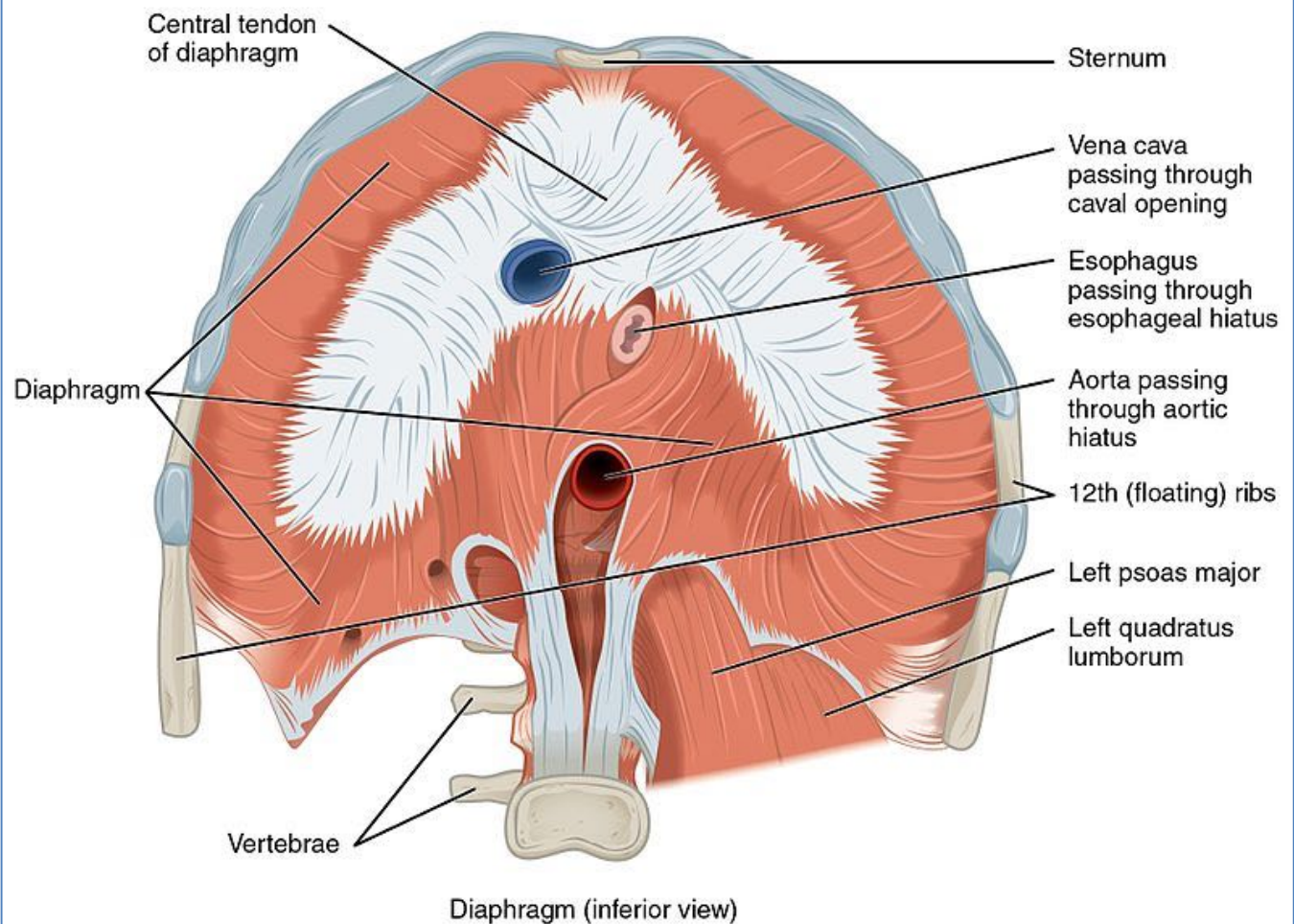
- **Inspiration:**
 - **Scalene Muscles**
 - **Sternocleidomastoid**
 - **External Intercostals**
 - **How:**
 - Pull the Ribs & Sternum *Superiorly* (i.e. Pump & Bucket-handle Movements)
- **Expiration:**
 - **Abdominal Wall Muscles**
 - By increasing intra-abdominal pressure (forces diaphragm up)
 - **Internal Intercostals**
 - Pull the Ribs & Sternum *Inferiorly* (i.e. Reverse of Pump & Bucket-Handle Movements)

Primary Muscle: The Diaphragm:

- A MusculoTendinous Structure
- Divides thorax from abdomen
- Primary muscle of respiration
- Contraction = Flattening (i.e. Downward movement) → Inspiration
- Relaxation = Doming into thoracic cavity (upward movement) → Expiration
- **Origins:**
 - Xiphoid Process
 - **Costal Margin** (approx 7th rib)
 - Lateral Lower Ribs (11 & 12)
 - Body of T12 Vertebra.
- **Inserts Onto:**
 - A central tendon
- **Blood Supply:**
 - Phrenic Arteries (superior & Inferior)
- **Venous Drainage:**
 - Brachiocephalic Veins
 - Azygous Veins
 - Inferior Vena Cava
- **Nerve Supply:**
 - Phrenic Nerve (C3, 4 & 5)
 - Receives sympathetic fibres from Cervical Ganglia → Voluntary & Autonomic Nerve Supply



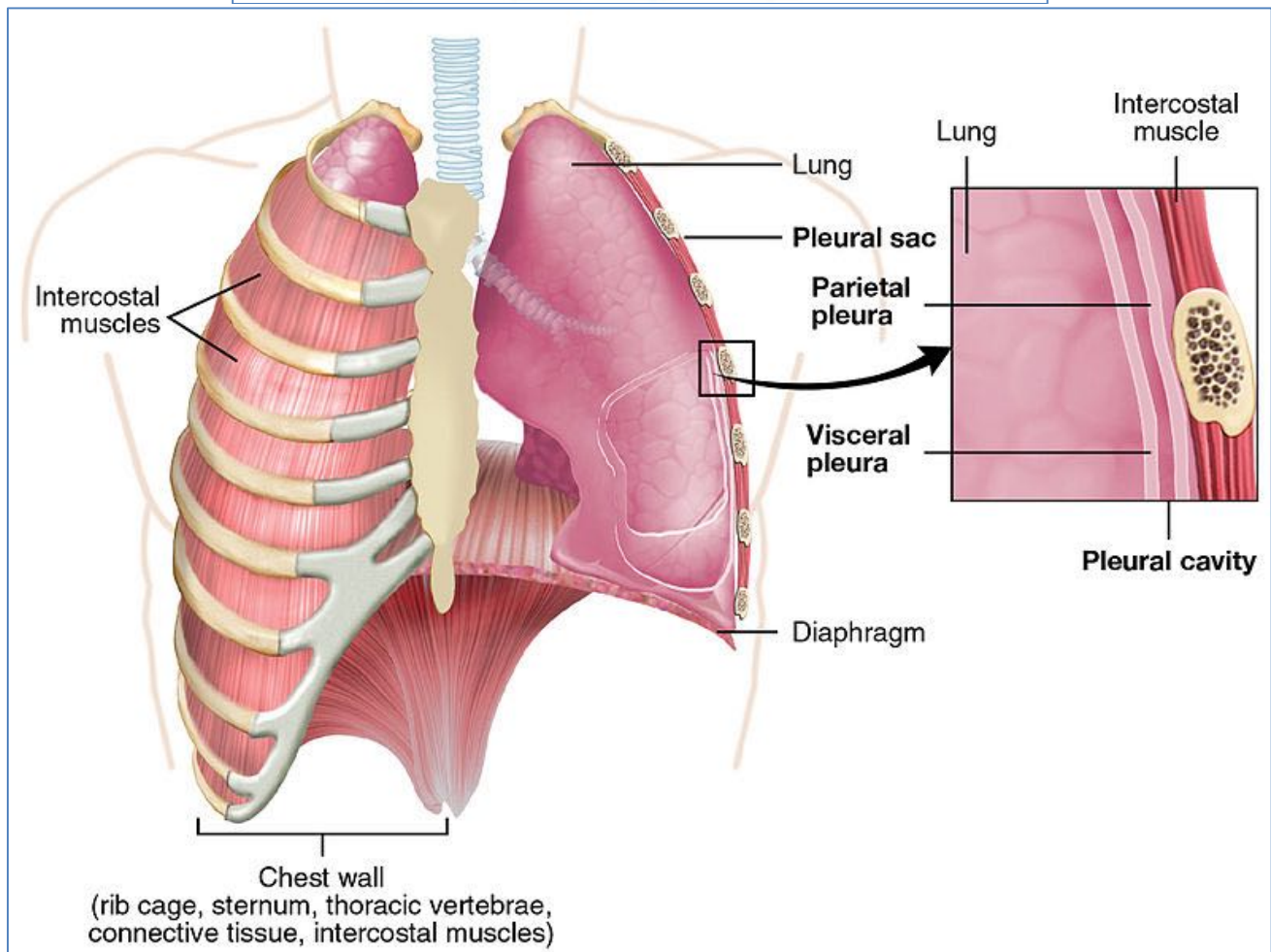
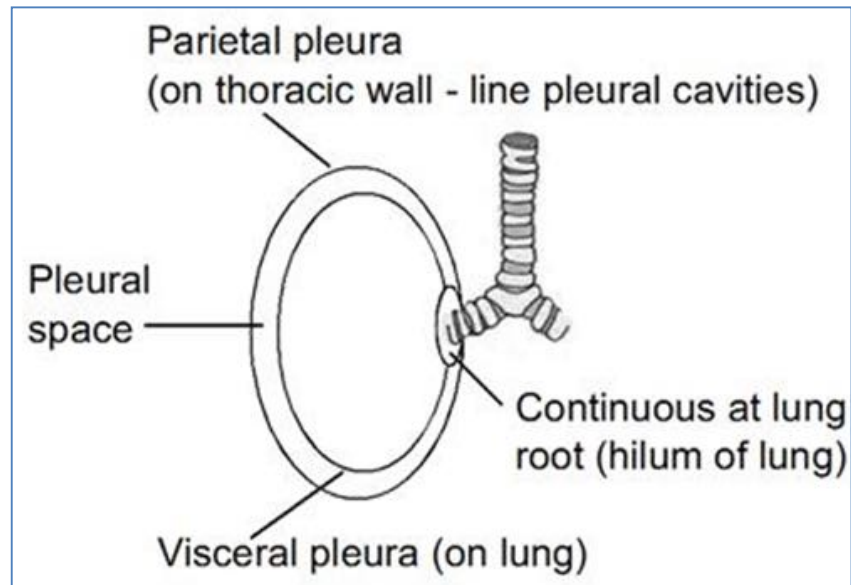
<https://medlineplus.gov/ency/imagepages/19072.htm>



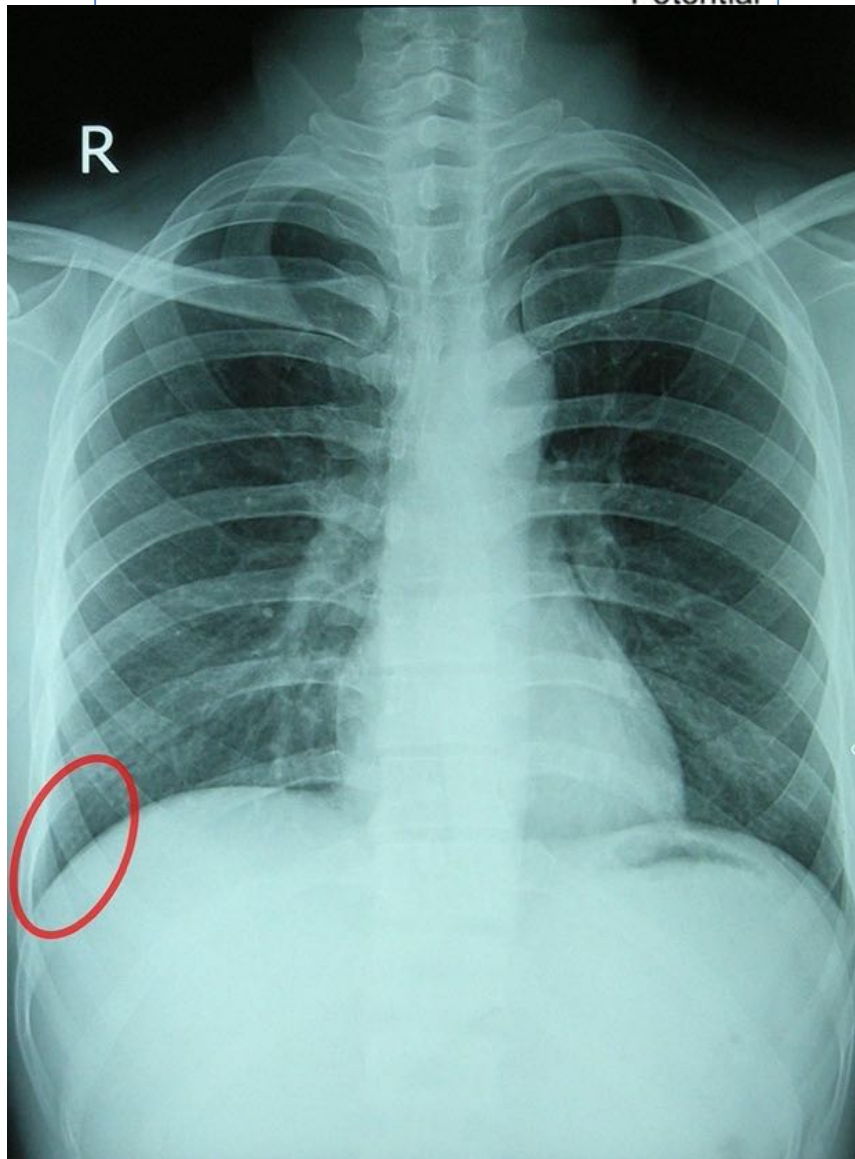
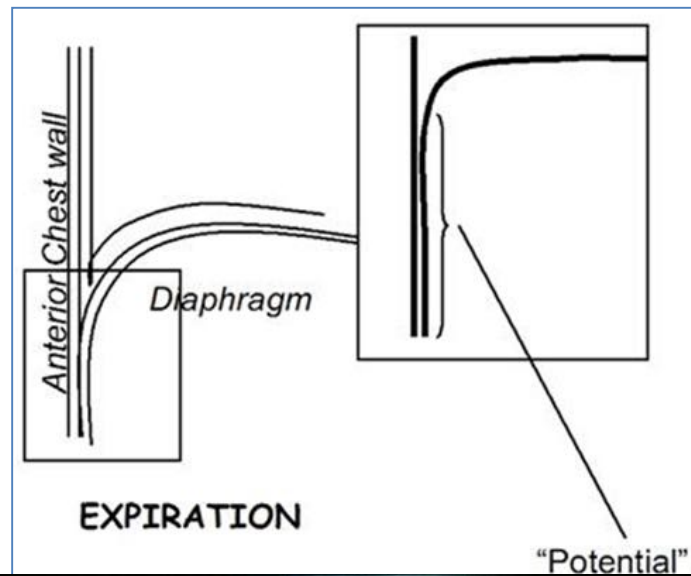
OpenStax, CC BY 4.0 <<https://creativecommons.org/licenses/by/4.0>>, via Wikimedia Commons

Pleura:

- Each are continuous Serous Sacs
 - Each has a Visceral 'pleura' & A Parietal 'pleura'
 - Between these layers is a 'potential' space aka. The "Pleural Space"
 - This Pleural Space is contains lubricating Serous Fluid
 - Fluid creates surface tension
 - Keeps the lung inflated even during expiration.
 - Keeps the pleurae together.
- Pleurae line the lung & Pulmonary Cavities



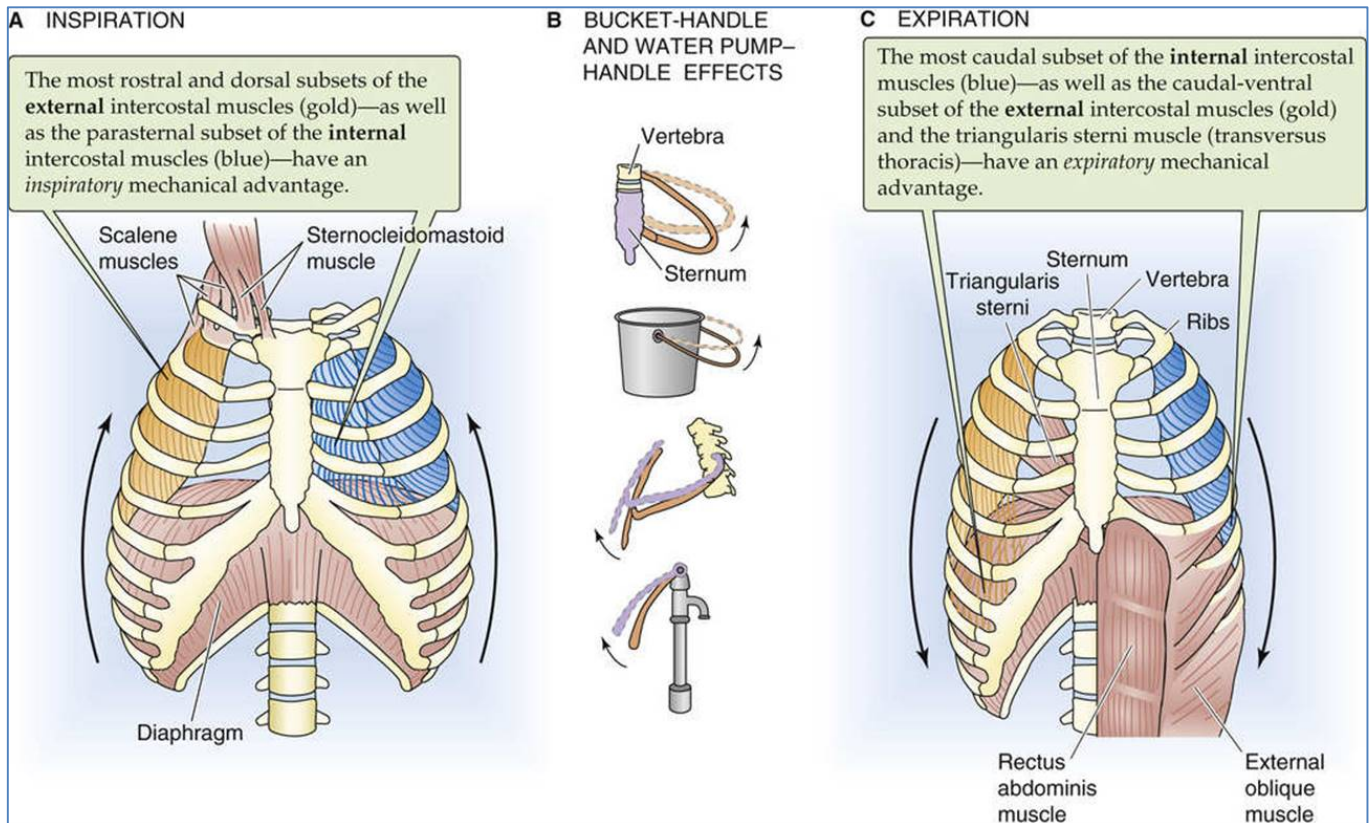
- **Costodiaphragmatic Recess:** (or just Diaphragmatic Recess)
 - 'Extra' space allocated to the lungs for use during forced inspiration
 - Allow extra expansion of the lungs



Mikael Häggström, CC0, via Wikimedia Commons

Thoracic Movements of Breathing:

- Brought about by muscles of the Thoracic Wall & Accessory Muscles
- Breathing is **not** just movement of the diaphragm
- Due to articulations, 2 groups of ribs create different movements:
 - **Upper 6 Ribs:**
 - *Pump Handle Action*
 - Increases **Anterio-Posterior Diameter** of Thoracic Cavity
 - **Lower 6 Ribs:**
 - *Bucket Handle Action*
 - Increases the **Transverse Diameter** of Thoracic Cavity



<https://doctorlib.info/physiology/medical/146.html>

ANATOMY OF THE AIRWAYS

ANATOMY OF THE AIRWAYS

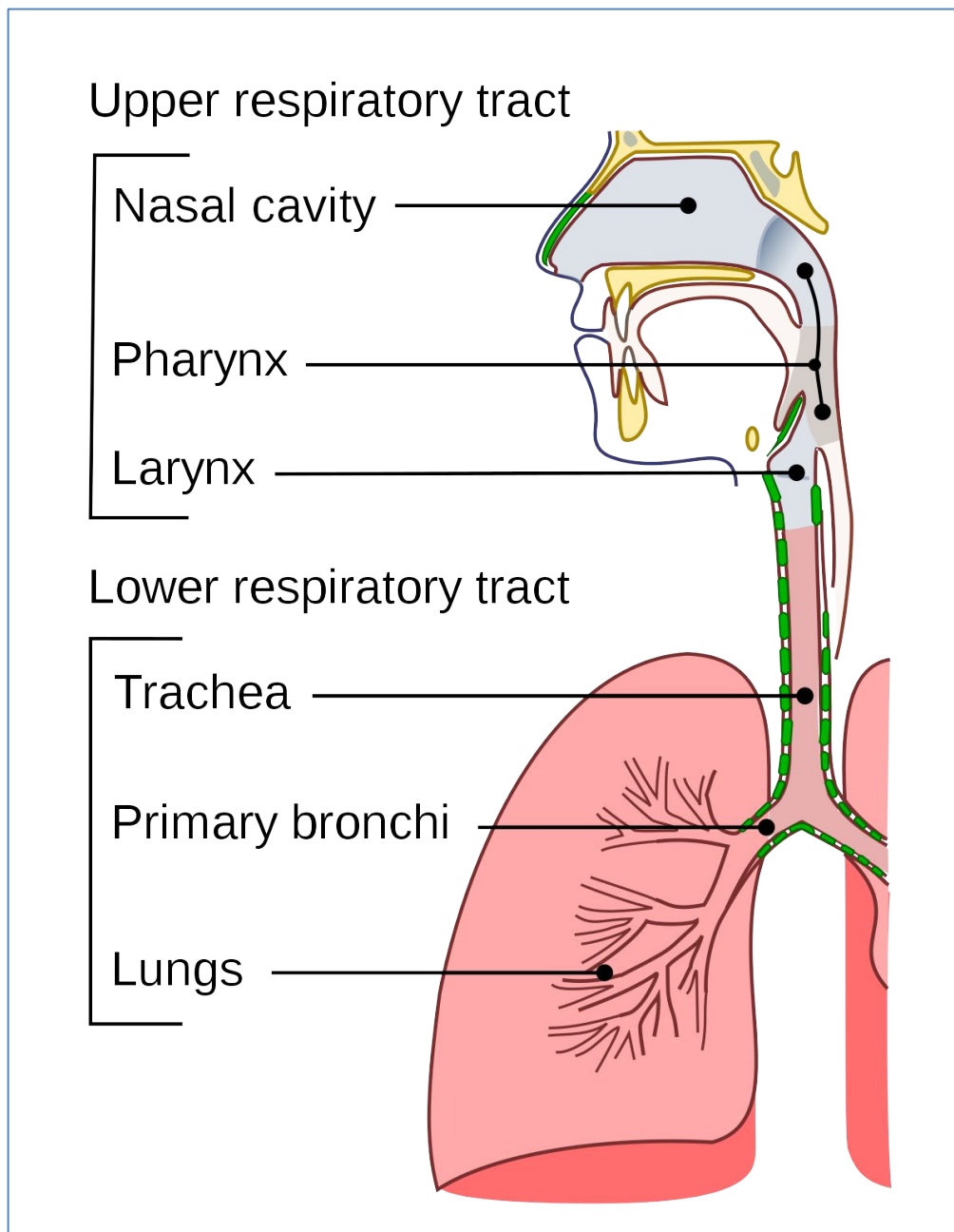
Structural Divisions:

- **Upper Airways:**

- Aka. 'Conducting' zones: Due to its conduit-like structure
- Functions:
 - Filter particulate matter from air (debris & dust)
 - Mucosal Epithelium:
 - Warm incoming air
 - Moisten incoming air
- Nose → Trachea

- **Lower Airways:**

- Aka. 'Respiratory' zones: Due to site of gas exchange
- Functions:
 - Facilitate Gas Exchange
 - O₂ in CO₂ out.
- Bronchi → Lung

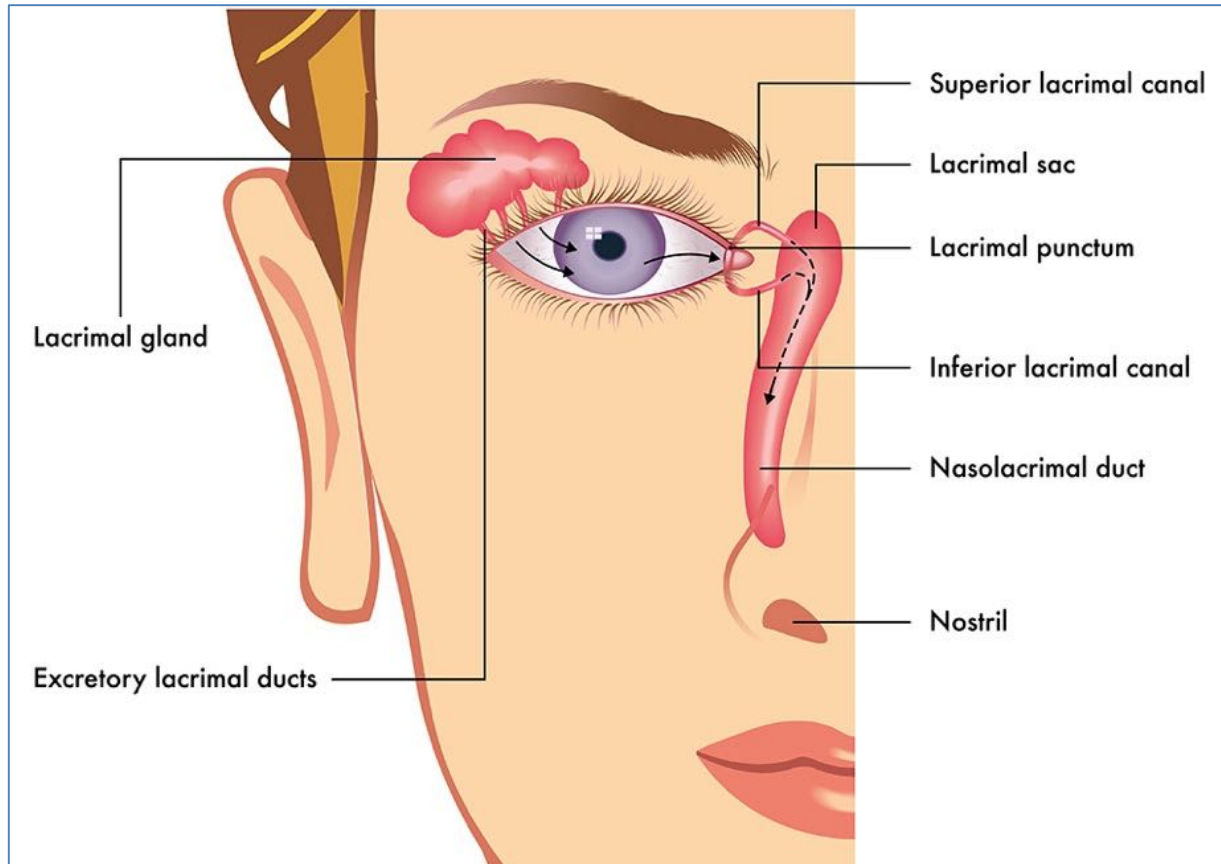


Lord Akryl, Jmarchn, Public domain, via Wikimedia Commons

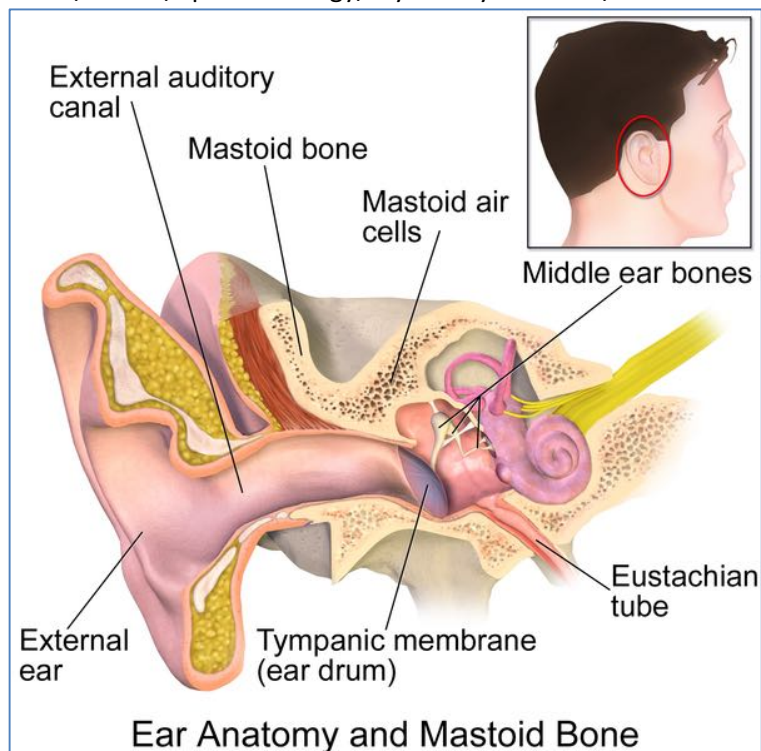
The Facial Skeleton:

- **Important Communication Routes Exist Between:**

- Eye Orbits & Nasal Cavities (**Nasolacrimal Duct (Tear Duct)**)
- Nasal Cavities & Paranasal Sinuses
- Nasal Cavities & Oral Cavities
- Ears & Pharynx (**Eustachian Tube** – equalises pressure within mid ear)
- Pharynx & Larynx

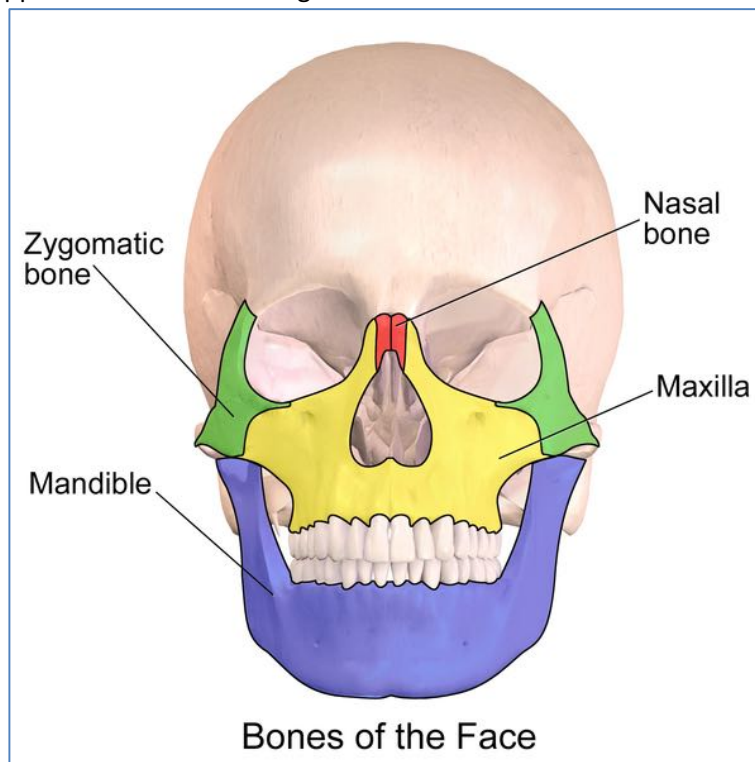


<https://healthcare.utah.edu/moran/ophthalmology/thyroid-eye-disease/nasolacrimal-duct-obstruction.php>



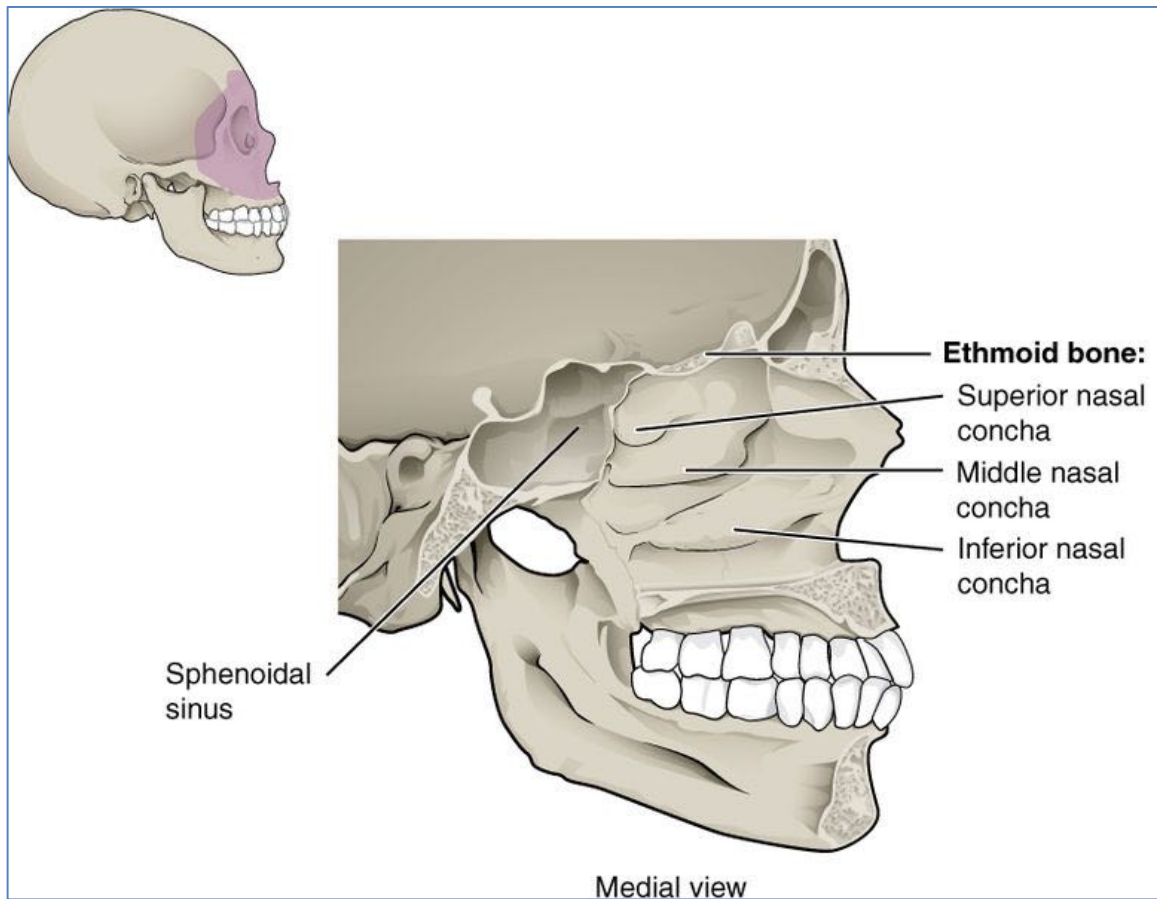
BruceBlaus, CC BY-SA 4.0 <<https://creativecommons.org/licenses/by-sa/4.0>>, via Wikimedia Commons

- **2 Maxillae:**
 - Fused Medially
 - Carry the upper teeth
 - Forms front 2/3 of hard palate.
 - 'Keystone' of the facial skeleton (Articulates with all facial bones except mandible)
- **Frontal Bone:**
 - Anterior Cranium
 - Contains the (frontal) sinuses
 - Connects to Ethmoid bone
- **Nasal Bones:**
 - Form the 'bridge' of the nose
 - Provide support for external cartilage -->nose structure

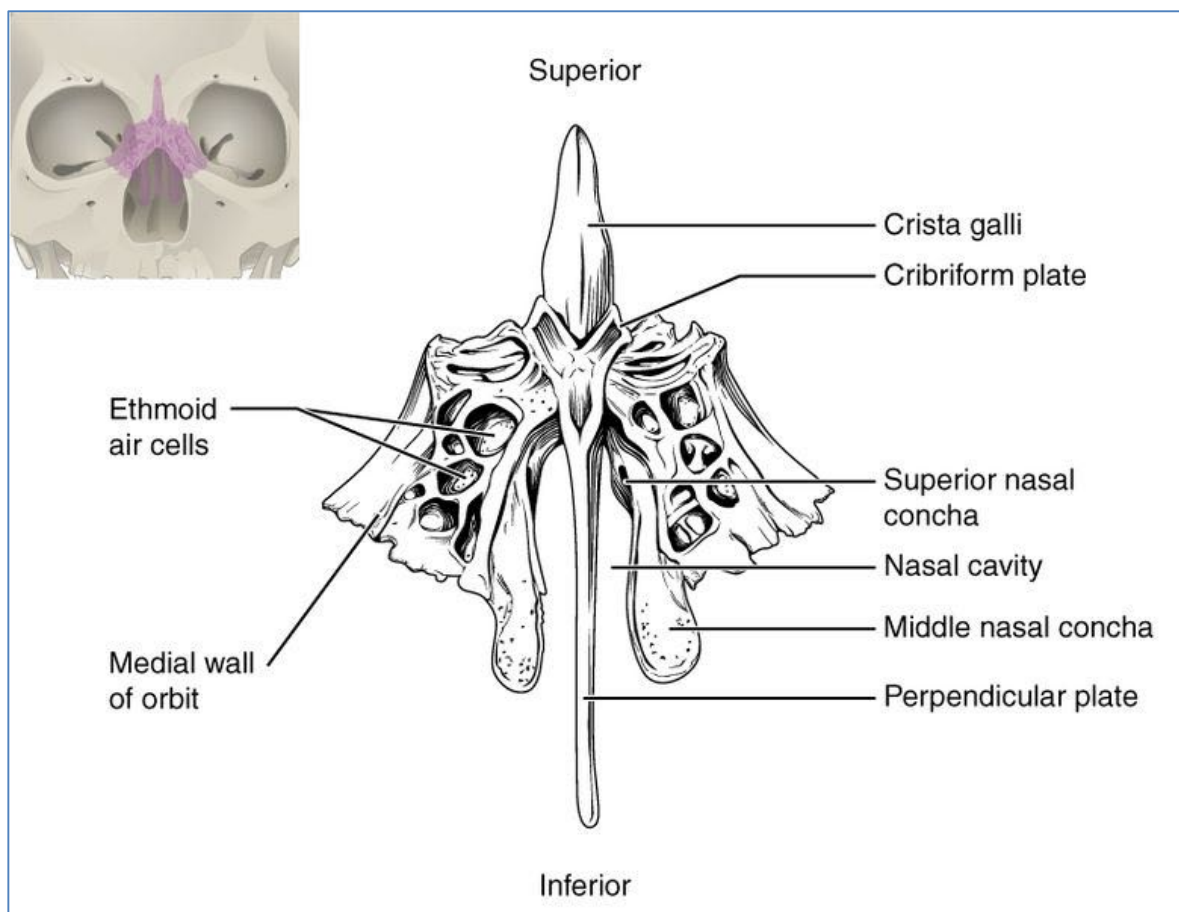


Blausen.com staff (2014). "Medical gallery of Blausen Medical 2014". WikiJournal of Medicine 1 (2). via Wikimedia Commons

- **Ethmoid:**
 - Forms majority of nasal cavity
 - Anchors the cartilage of the nose
 - Turbulates the air - moisten + warm + filters
 - **Important Components:**
 - **2 Cribriform Plates** - Punctured by *olfactory foramina* – for olfactory nerves
 - **Crista Gali** (inside cranium) - Triangular process between the Cribriform Plates – Anchors the brain.
 - **Perpendicular Plate** (superior part of nasal septum) - Separates R&L Nasal Cavities
 - **L & R lateral Masses** – riddled with *ethmoid sinuses*
 - **Superior Nasal Conchae**
 - Turbulates the air - moisten + warm + filters
 - **Middle Nasal Conchae**
 - Turbulates the air - moisten + warm + filters
 - **Inferior Nasal Conchae:**
 - Small scroll of bone
 - Sit in inferior portion of nasal cavity
 - Paired
 - Attach to part of maxilla
 - Turbulates the air - moisten + warm + filters



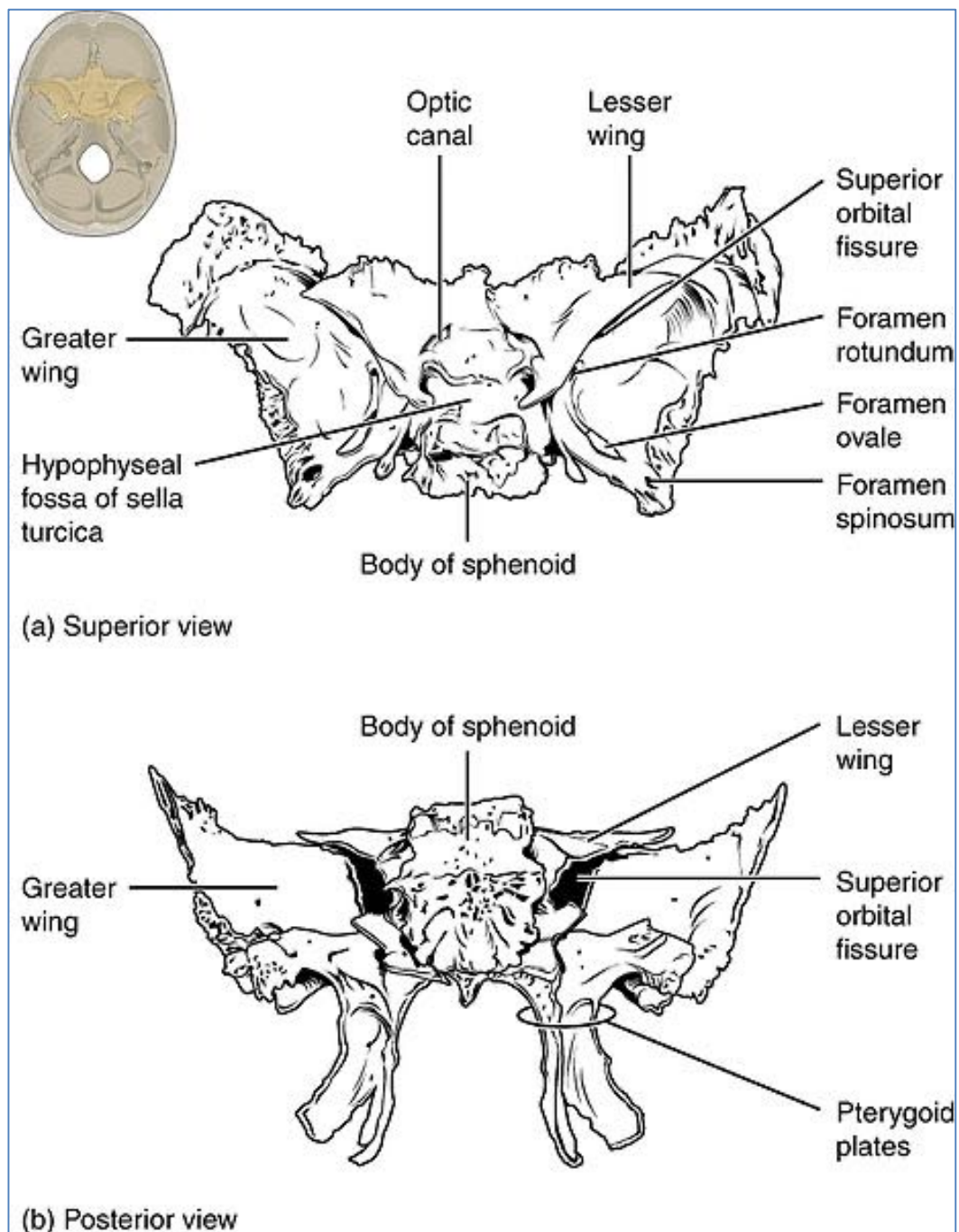
OpenStax College, CC BY 3.0 <<https://creativecommons.org/licenses/by/3.0>>, via Wikimedia Commons



OpenStax College, CC BY 3.0 <<https://creativecommons.org/licenses/by/3.0>>, via Wikimedia Commons

- **Sphenoid:**

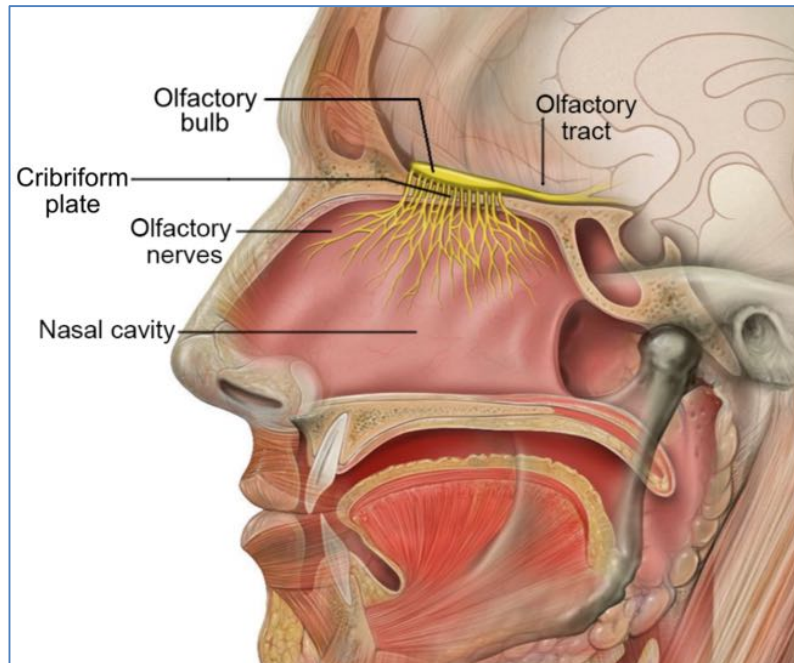
- 'keystone' of the cranium (articulates with all bones of cranium)
- Butterfly-shaped
- Contain paired Sphenoid sinuses



OpenStax College, CC BY 3.0 <<https://creativecommons.org/licenses/by/3.0/>>, via Wikimedia Commons

The Nose:

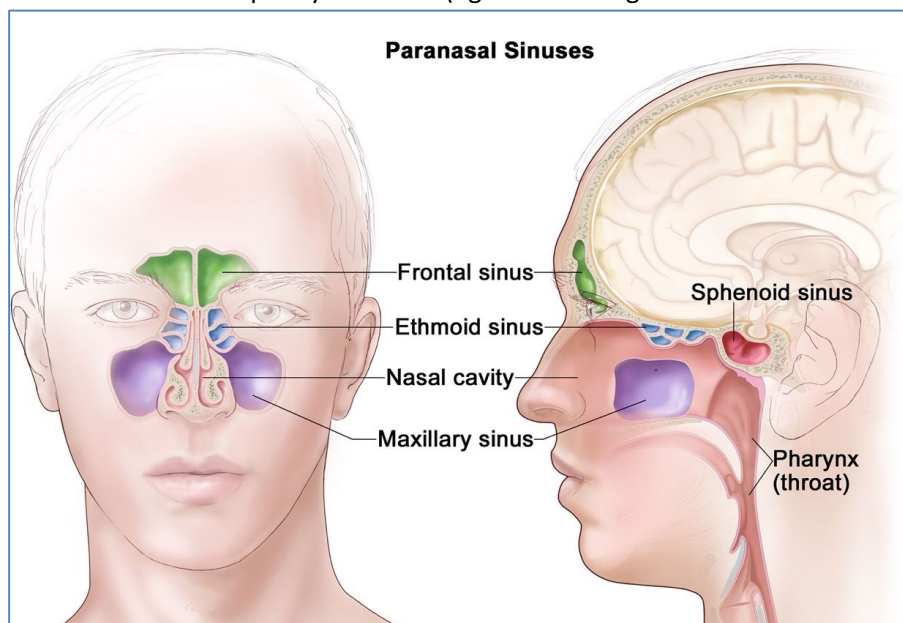
- Provides an airway for respiration
- Moistens & warms entering air
- Filters inspired air
- Doubles as a resonance chamber during speech
- Houses Olfactory (Smell) Receptors
- **External Nose:**
 - Skeletal framework consists of:
 - Nasal & Frontal bones Superiorly
 - Maxillary bones Laterally
 - Flexible Plates of Cartilage Inferiorly
- **Nasal Cavity:**
 - **Air Enters Through Nostrils:**
 - Lined with skin
 - Sebaceous & sweat glands
 - Numerous ***Vibrissae*** (Hair Follicles) – filter coarse particles from air.
 - **Epithelial Linings:**
 - **Olfactory Mucosa:**
 - Specialized epithelium involved with smell
 - Surface littered with olfactory neurons (receptors)
 - Olfactory neurons synapse with #1 CN (the Olfactory Nerve)
 - **Respiratory Mucosa:**
 - Pseudostratified Columnar Epithelium
 - Ciliated
 - Scattered Goblet Cells
 - Base of Lamina Propria rich in mucous & serous glands.
 - **Divided in the middle by the nasal septum:**
 - Perpendicular Plate of Ethmoid (upper 2/3)
 - Vomer of the Sphenoid (lower 1/3)
 - **Roof Formed by:**
 - Ethmoid Bone
 - Sphenoid Bone
 - **Floor Formed by the *Palate*:**
 - Hard Palate – Palatine Bone
 - Soft Palate (Uvula) – Musculo-Tendinous Structure
 - **Lateral Walls: *Conchae***
 - 3 on each wall
 - Superior/Middle/Inferior
 - Increase Mucosal Surface Area
 - Heat & Moistens the air during Inspiration.
 - Reclaims Heat & Precipitates Moisture during Expiration.
 - Enhance Turbulence – heavier, nongaseous particles are flung onto & stick to the mucosa



Patrick J. Lynch, medical illustrator, CC BY 2.5 <<https://creativecommons.org/licenses/by/2.5>>, via Wikimedia Commons

- **Paranasal Sinuses:**

- **Exist In 4 Bones:**
 - **Maxillae**
 - **Frontal**
 - **Sphenoid**
 - **Ethmoid**
- All continuous with nasal cavity
- Increase surface area
- Create turbulence
- Help to humidify & warm inflowing air.
- Lighten the skull
- Provides resonance for 'quality' of voice. (eg. Voice changes with blocked sinuses)



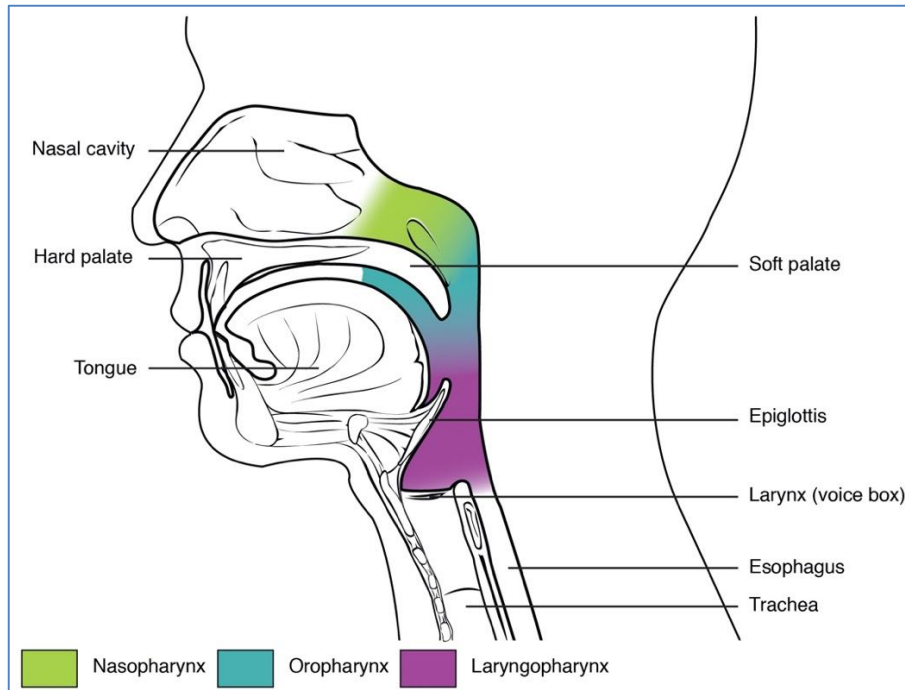
Public Domain: <https://www.cancer.gov/publications/dictionaries/cancer-terms/def/paranasal-sinus>

Note: During infection, the openings of the sinuses may become blocked by excessive mucus production.

- This can lead to excessive pressure inside the sinuses → pain

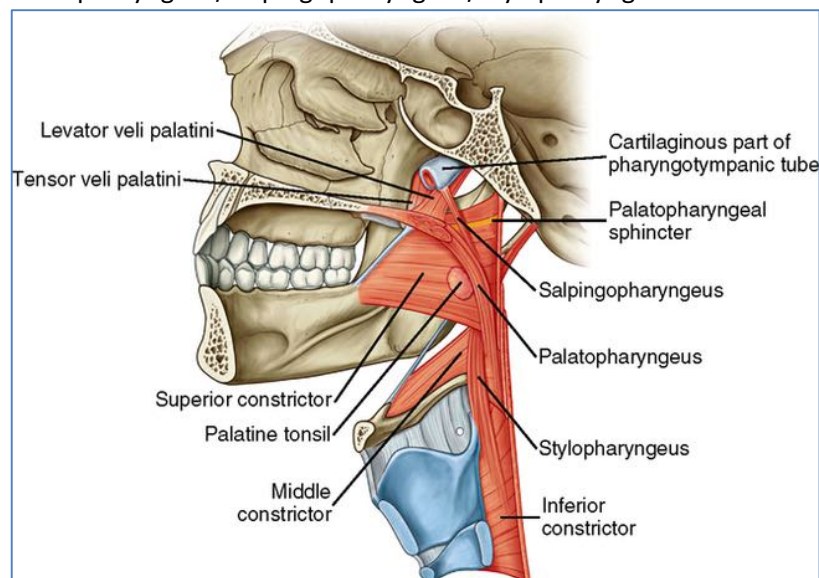
The Pharynx:

- Connects Nasal Cavities, Oral Cavity & Oesophagus
- **Epithelium of Each Pharyngeal Region:**
 - **Nasopharynx:**
 - Air passageway ONLY.
 - *Pseudostratified Ciliated Epithelium*
 - **Oropharynx:**
 - Both Food & Air Pass Through it. → More protection is needed.
 - *Stratified Squamous Epithelium*
 - **Laryngopharynx:**
 - Both Food & Air Pass Through it. → More protection is needed.
 - *Stratified Squamous Epithelium*
 - During swallowing, food has 'right-of-way' (breathing is halted temporarily)



OpenStax College, CC BY 3.0 <<https://creativecommons.org/licenses/by/3.0>>, via Wikimedia Commons

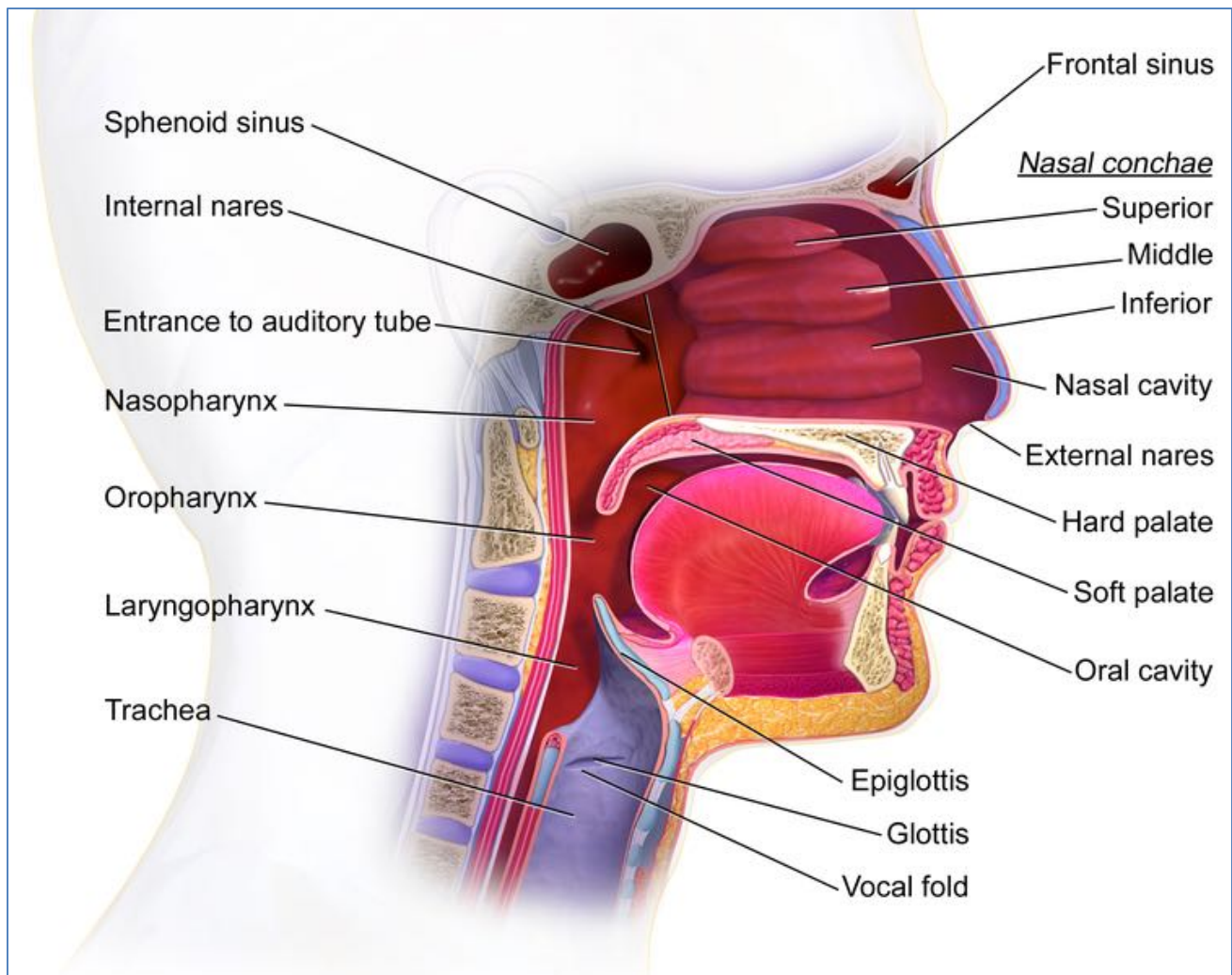
- **2 Muscle Groups of Pharynx:**
 - **3x Constrictor Muscles:** (move food down to the *laryngopharynx*)
 - Superior/Middle/Inferior
 - **3x Longitudinal Muscles:** (Elevate the Pharynx – prevent food in trachea)
 - Palatopharyngeus/Salpingopharyngeus/Stylopharyngeus



<https://www.tekportal.net/salpingopharyngeal-muscle/>

- **Soft Palate:**

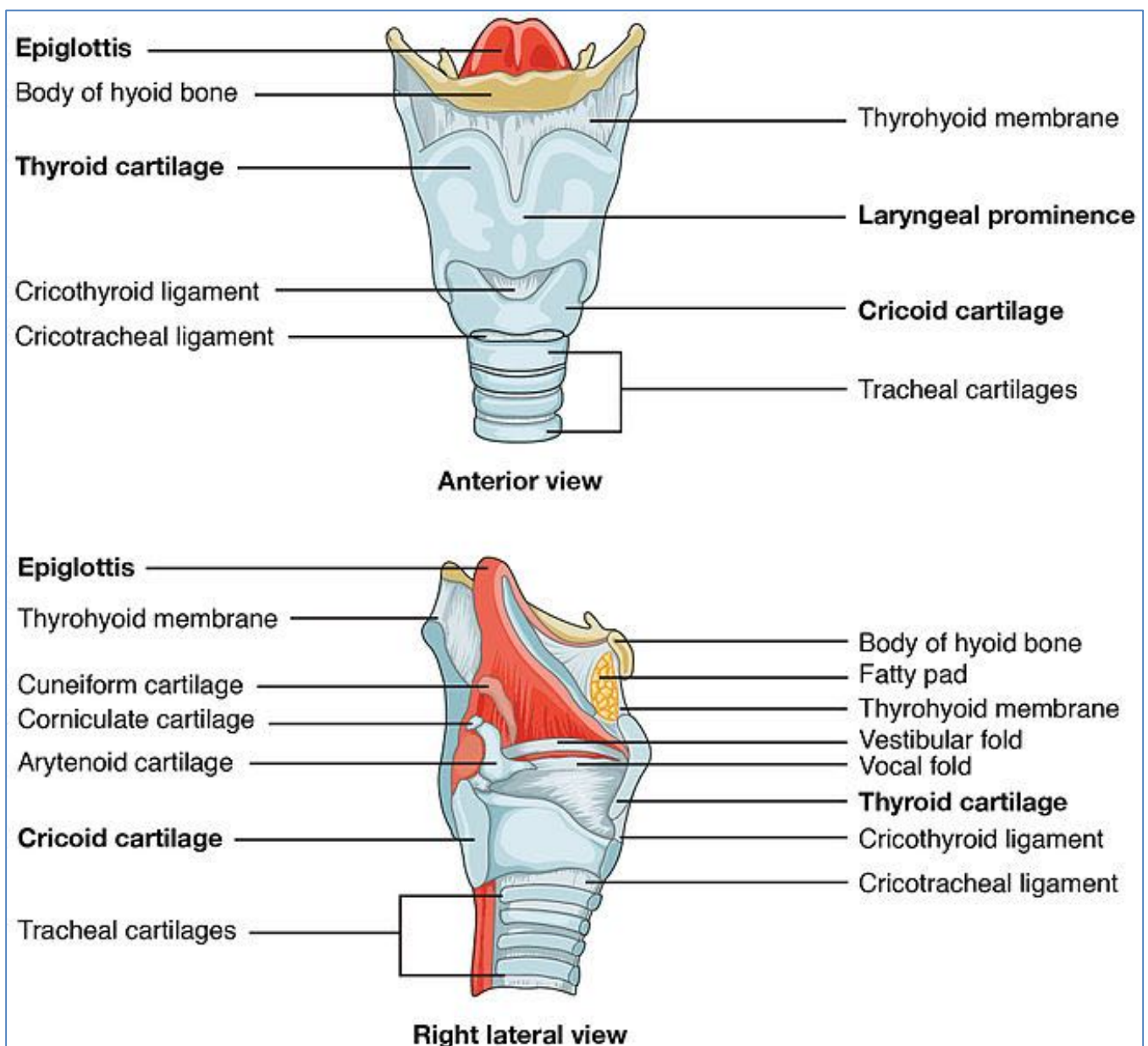
- Posterior aspect of oral cavity
- Separates Oral Cavity & Nasopharynx
- Involved during *Deglutition*
- Made of & Operated By 5 Muscles:
 - Levator Veli Palatini
 - Tensor Veli Palatini
 - Palatoglossus
 - Palatopharyngeus
 - Musculus Uvulae
- Supplied by *Vagus Nerve*



Blaesen.com staff (2014). "Medical gallery of Blaesen Medical 2014". WikiJournal of Medicine 1 (2). via Wikimedia Commons

The Larynx: ("Voicebox")

- Superiorly, it attaches to the Hyoid Bone
- Inferiorly, it merges with the Trachea
- **3 Functions:**
 - Provide an open airway (breathing)
 - Direct Air & Food into proper channels
 - Voice production. (Phonation)
- **Made of 9 Cartilages:**
 - **3 Unpaired Cartilages:**
 - Form the Tube-Like Skeletal Framework of Larynx
 - **Thyroid Cartilage**
 - **Cricoid Cartilage**
 - **Epiglottis**
 - **3 Paired Cartilages (6 total):**
 - Involved in moving the Vocal Ligaments (Adduction & Abduction)
 - **Arytenoid Cartilage**
 - **Cuneiform Cartilage**
 - **Corniculate Cartilage**

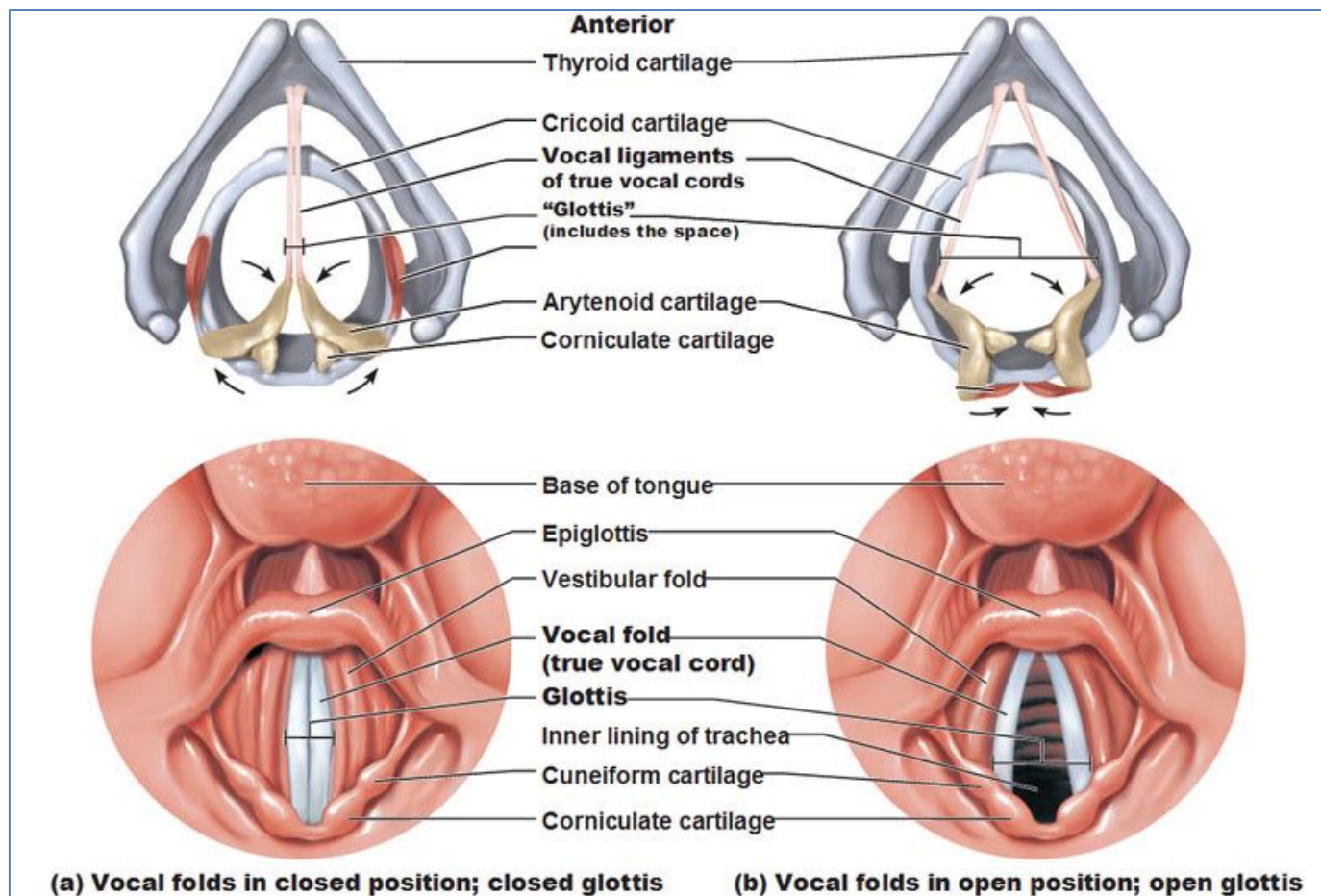


- **Vocal Ligaments: 'True Vocal Cords' ("Cricothyroid Ligament/Membrane")**

- Covered in mucosa
- Made of Elastic Fibres
- Fibres vibrate as air rushes up from lungs. (tighter = higher pitch)
- Appear white – no blood vessels
- Attach the *Arytenoid Cartilages* to the *Thyroid Cartilage*
- Form the 'Vocal Folds' or 'True Vocal Cords'.

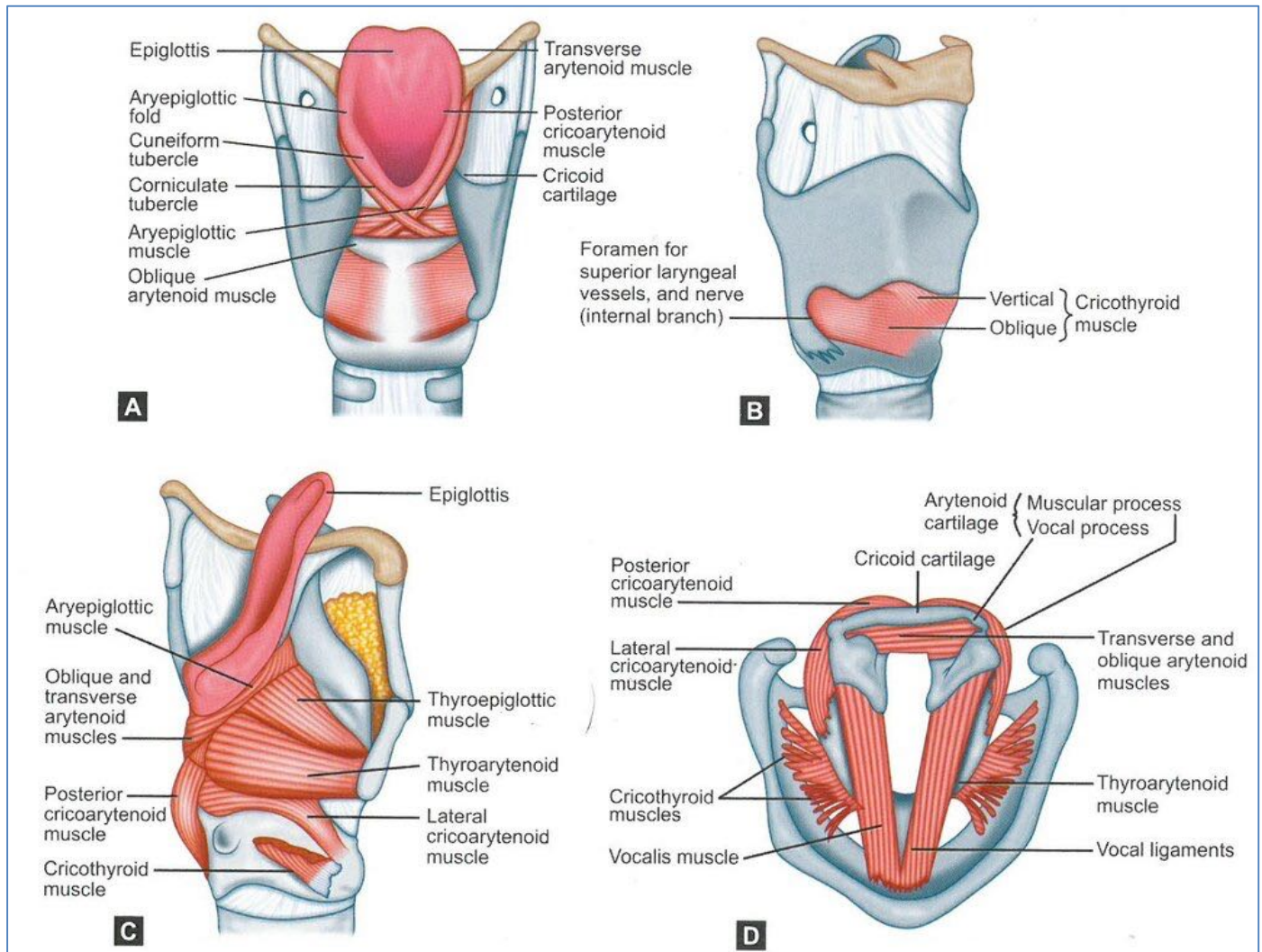
- **Vestibular Folds: 'False Vocal Cords' ("Quadrangular Ligament/Membrane")**

- Play no part in sound production
- Help to close the 'glottis' when swallowing.



<https://antranik.org/the-respiratory-system/>

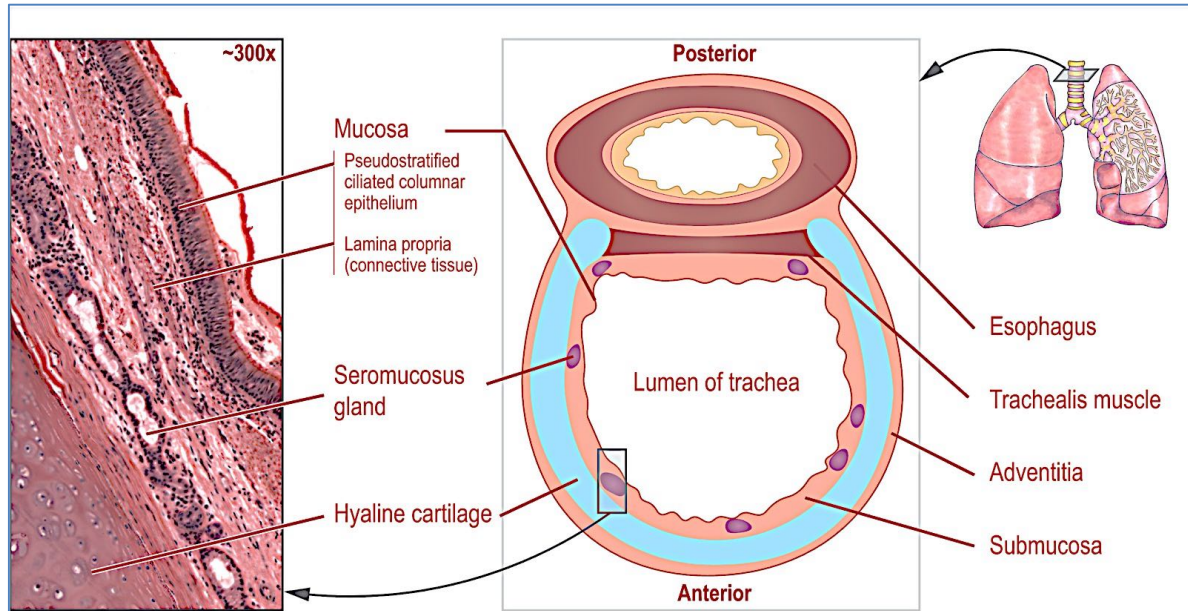
- **Muscles:** (aka. Intrinsic laryngeal muscles)
 - Work to affect tension/length/position of vocal cords.
 - All Controlled By the Vagus Nerve
 - **2x Cricothyroid Muscle**
 - **2x Vocalis**
 - **2x Transverse Arytenoids**
 - **2x Oblique Arytenoids**
 - **2x Posterior Crico-Arytenoids**
 - **2x Lateral Crico-Arytenoids**
 - **2x Thyromuscularis**



From Sataloff, R.T., Chowdhury, F., Portnoy, J., Hawkshaw, M.J., Joglekar S. Surgical Techniques in Otolaryngology – Head and Neck Surgery: Laryngeal Surgery.

Trachea:

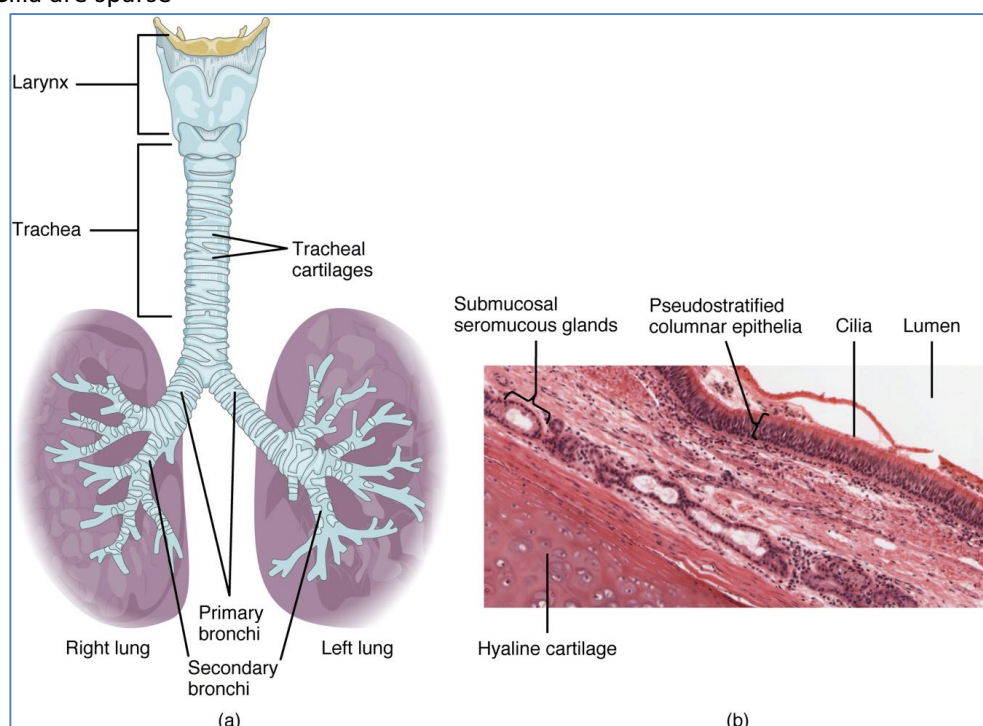
- The continuation of the pharynx
- A membranous tube of Conn. Tissue
 - + smooth muscle
 - Reinforced by 15-20 C-Shaped Cartilage Rings (incomplete posteriorly)
- Begins at C6
- Terminates at Bifurcation → Bronchi @ T4
 - Note: Right Bronchus is more vertical than the Left – hence inhaled objects tend to go down here.



PBrieux, CC BY-SA 4.0 <<https://creativecommons.org/licenses/by-sa/4.0/>>, via Wikimedia Commons

The Bronchial Tree:

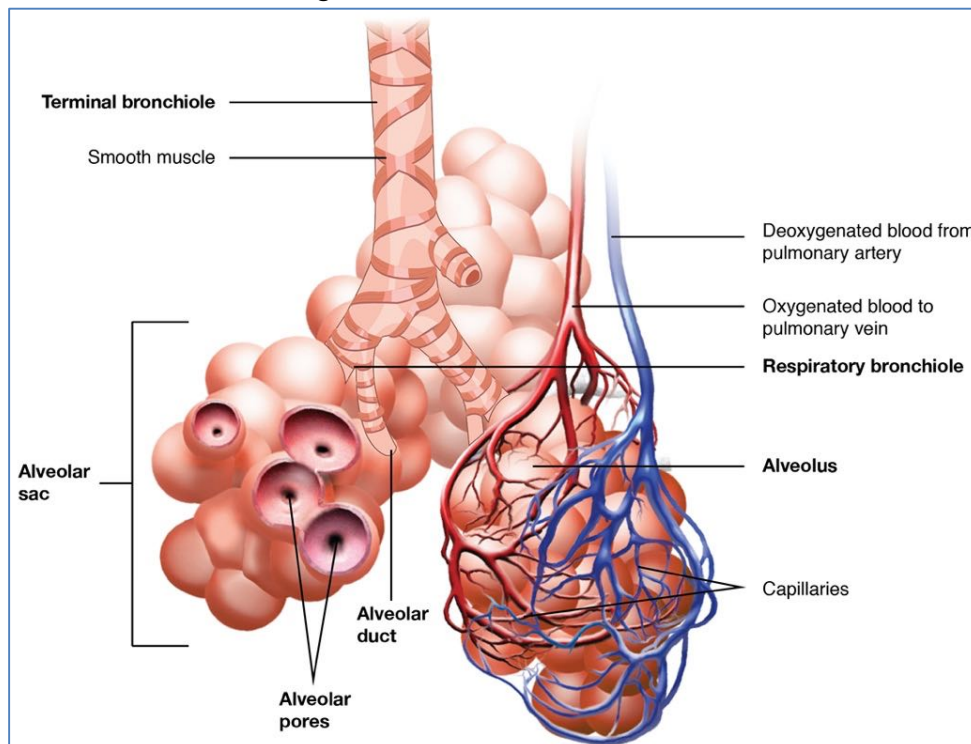
- Where *conducting structures* merge with *respiratory structures*.
- Once inside the lungs, the bronchi branch profusely until the *bronchioles* ("little bronchi") are <0.5mm thick.
- **Gradual Structural Changes:**
 - Cartilage rings replaced by irregular *plates* of cartilage.
 - No cartilage at all in *bronchioles*
 - Mucosal Epithelium thins from Pseudostratified → Columnar → Cuboidal in the bronchioles.
 - Cilia are sparse



Creative Commons: <https://guides.hostos.cuny.edu/bio140/7-29>

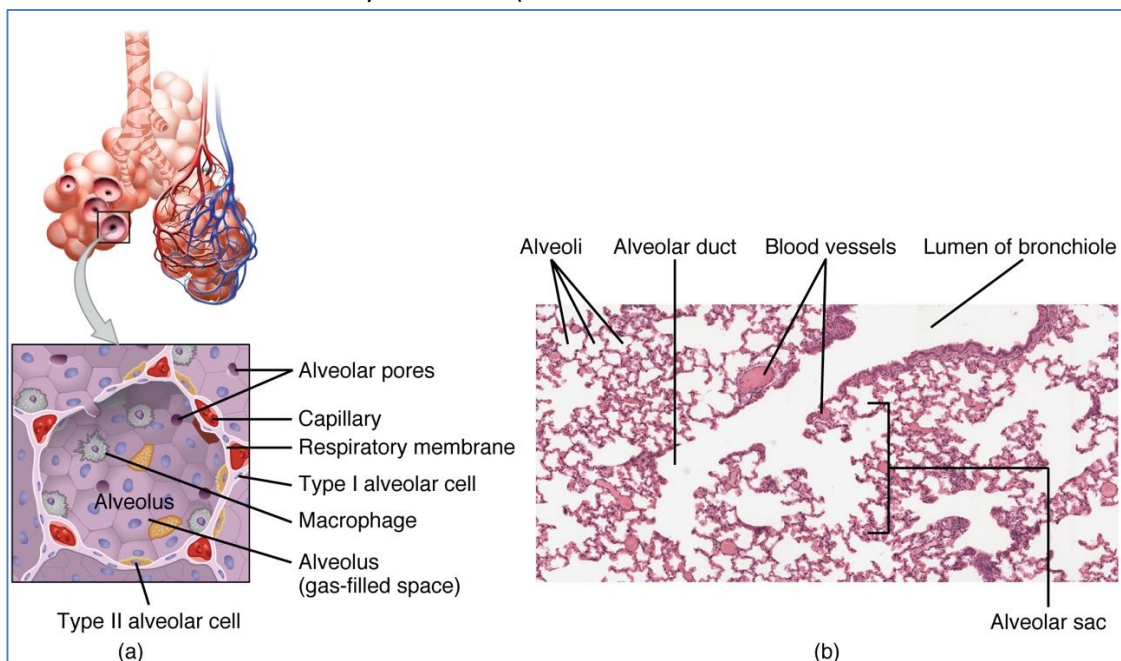
The Respiratory Zone:

- Formed by alveoli
- Gas Exchange happens in 2 Places:
 - Tube-Like Ducts
 - Balloon-Like Sacs
- Large Surface Area – for Gas Exchange



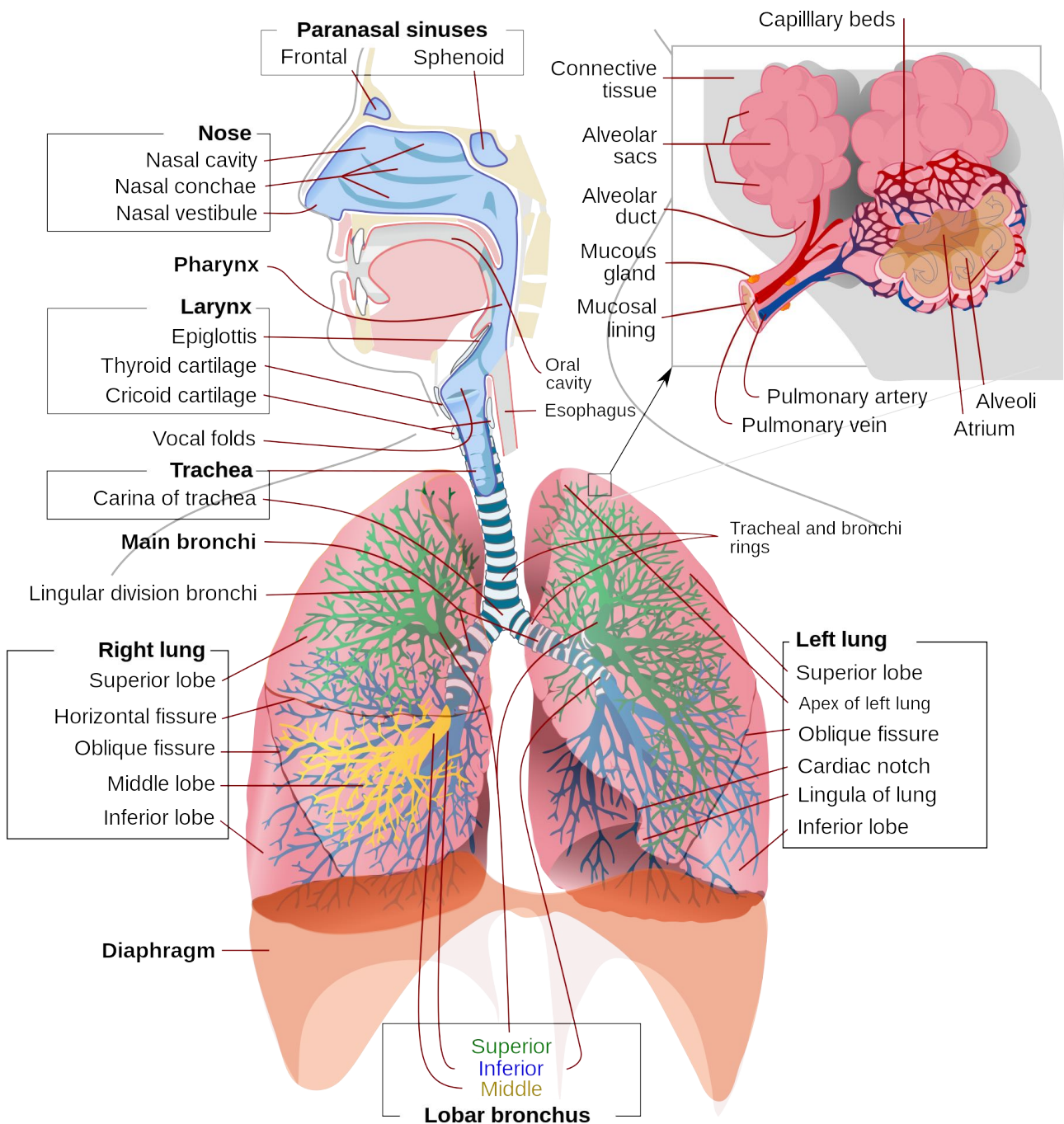
Creative Commons: <https://guides.hostos.cuny.edu/bio140/7-29>

- **2 Types of Alveolar Cells:**
 - **Type I Alveolar Cells:**
 - Aka. Squamous Alveolar Cells
 - Gas Exchange Alveolar
 - Make up the Alveoli Walls
 - **Type II Alveolar Cells:**
 - Aka. Great Alveolar Cells
 - Secrete Pulmonary Surfactant (lower the surface tension of water → easier breathing).



Creative Commons: <https://guides.hostos.cuny.edu/bio140/7-29>

Review of Entire Respiratory Anatomy:



https://en.wikipedia.org/wiki/File:Respiratory_system_complete_en.svg#filelinks

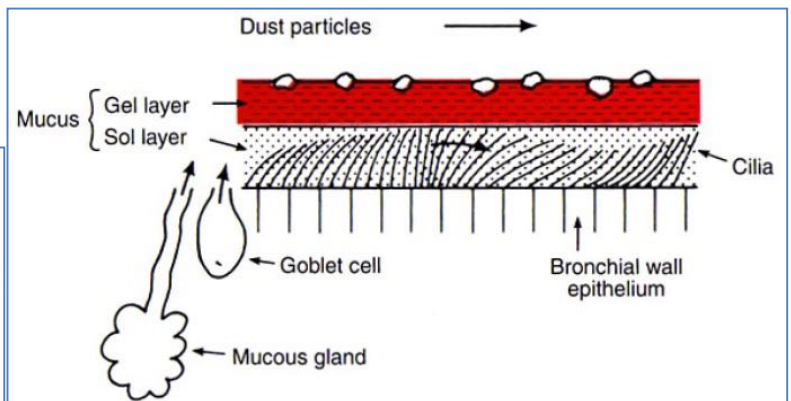
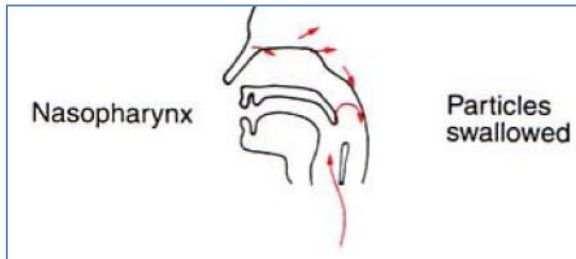
AIRWAY MUCOSAL LININGS

AIRWAY MUCOSAL LININGS

Clearance of Inhaled Particles:

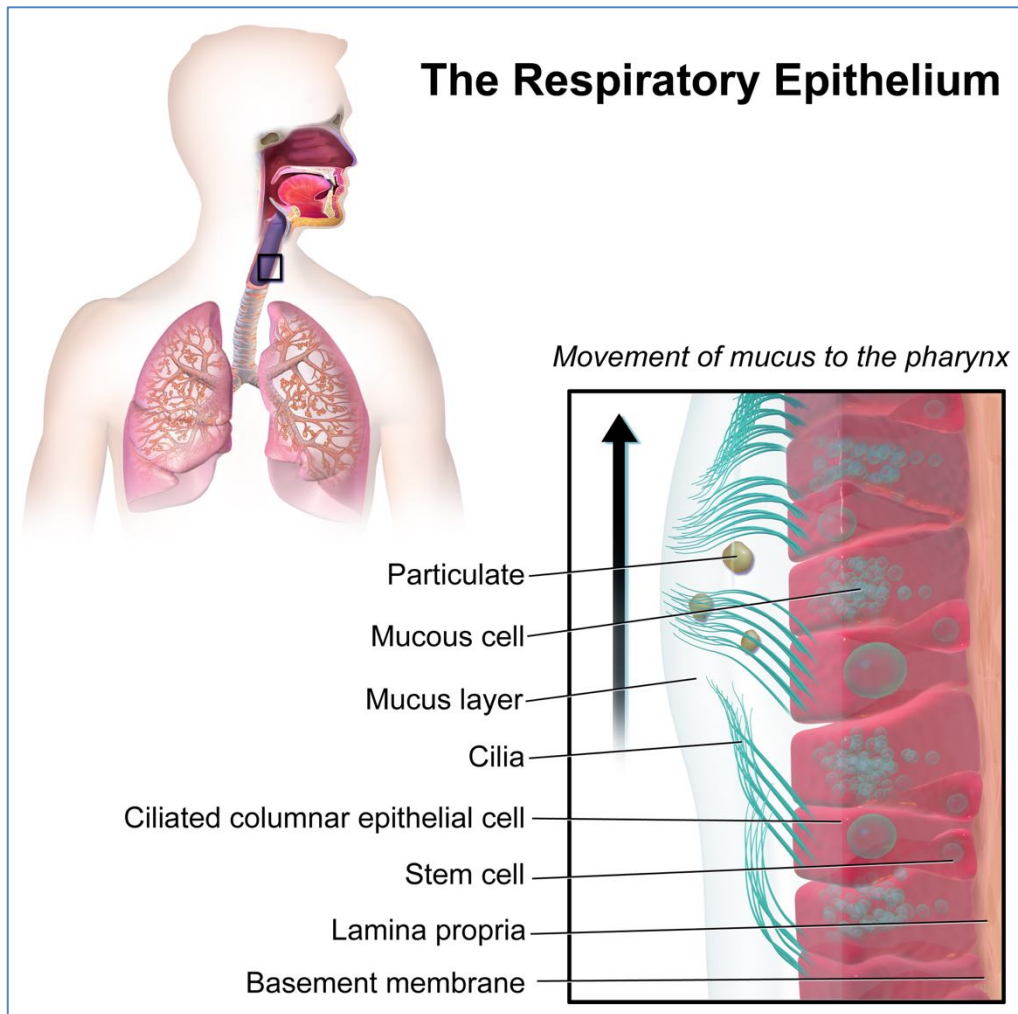
- Cilia in Nasopharynx:

- Cilia Beat Backwards towards Pharynx → Swallowed



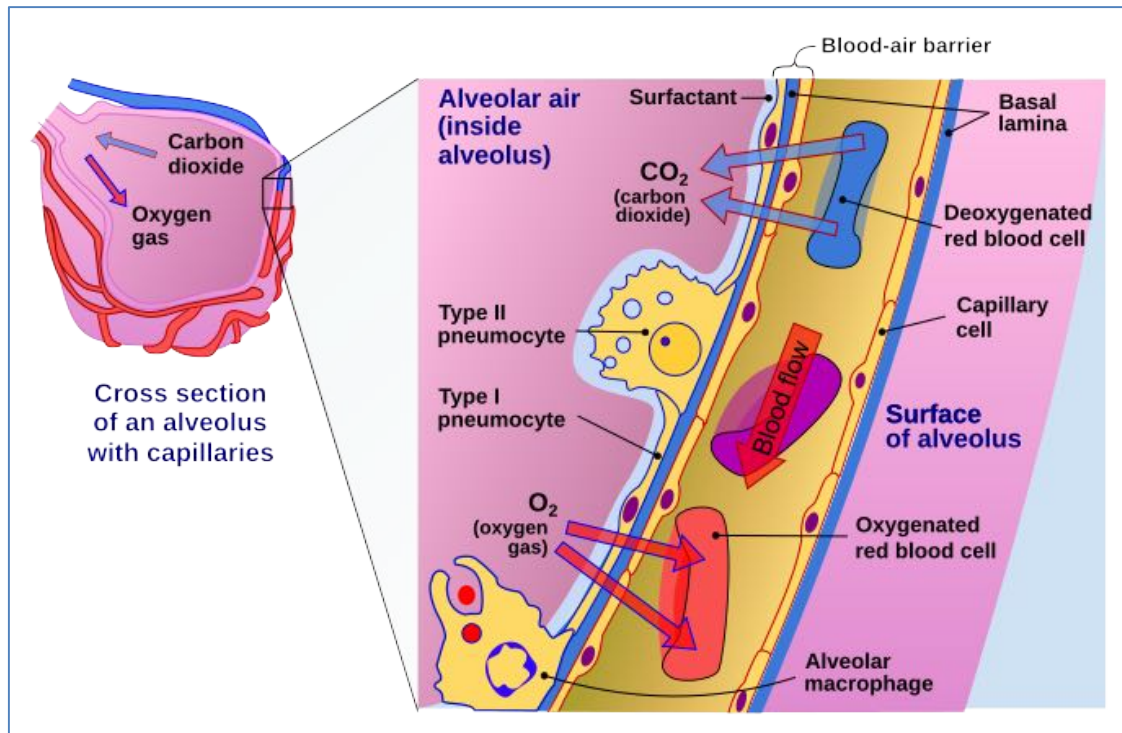
- Mucociliary Escalator:

- Ciliated Epithelium in *Conducting Zone*
- Mucous from Goblet Cells & Submucosal Glands → Traps Particles
- Cilia Beat Upwards towards Pharynx → Swallowed/Coughed Up.
 - 2 Strokes = Power & Recovery
 - Mucous Movement $\approx 1\text{-}2\text{mm/min}$
- Airway Surface Liquid Layer (Sol) Critical for Cilia Function.



- Alveoli Macrophages:

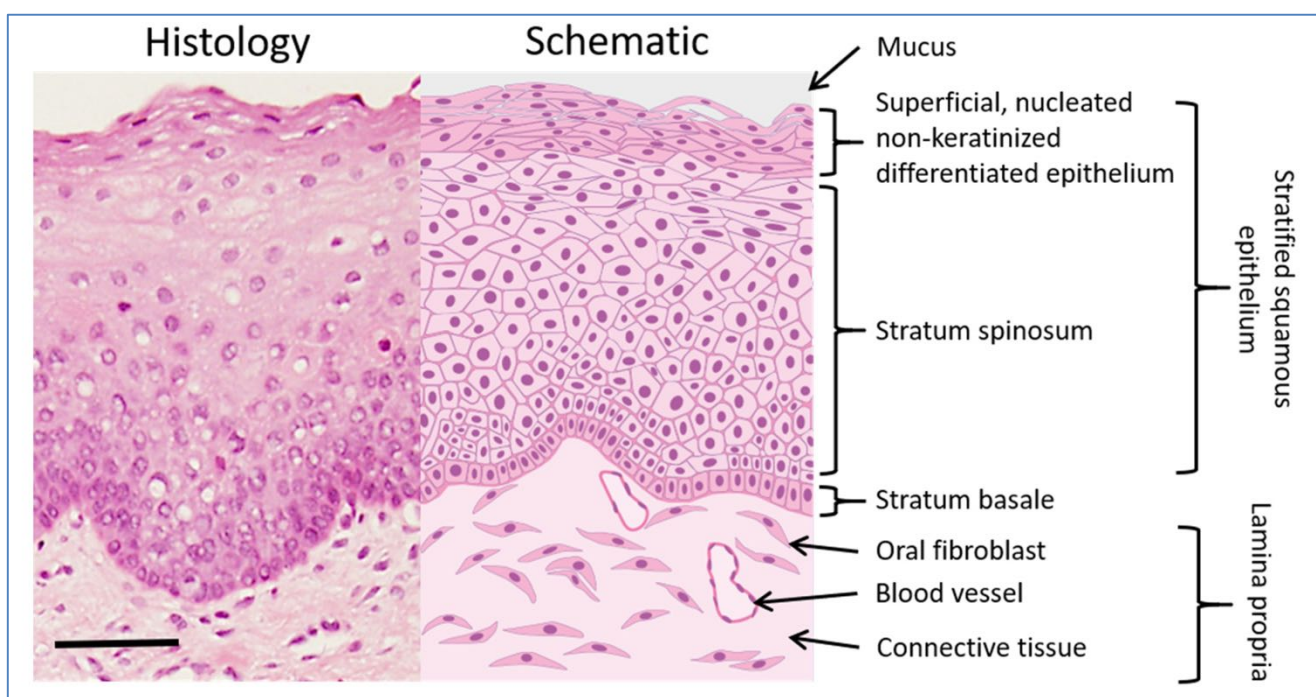
- Particles Phagocytosed by Alveolar Macrophages
- Able to Migrate *Into* Alveoli
- Use Destructive Enzymes to Destroy Foreign Particles + Bacteria.
 - Note: Dead Macrophages *Release* these enzymes → Lung Damage.
- Debris Are Either:
 - Dumped into Lymphatics
 - Delivered to Mucociliary Escalator



Delmalani18, CC BY-SA 4.0 <<https://creativecommons.org/licenses/by-sa/4.0/>>, via Wikimedia Commons

Mucosal Histology:

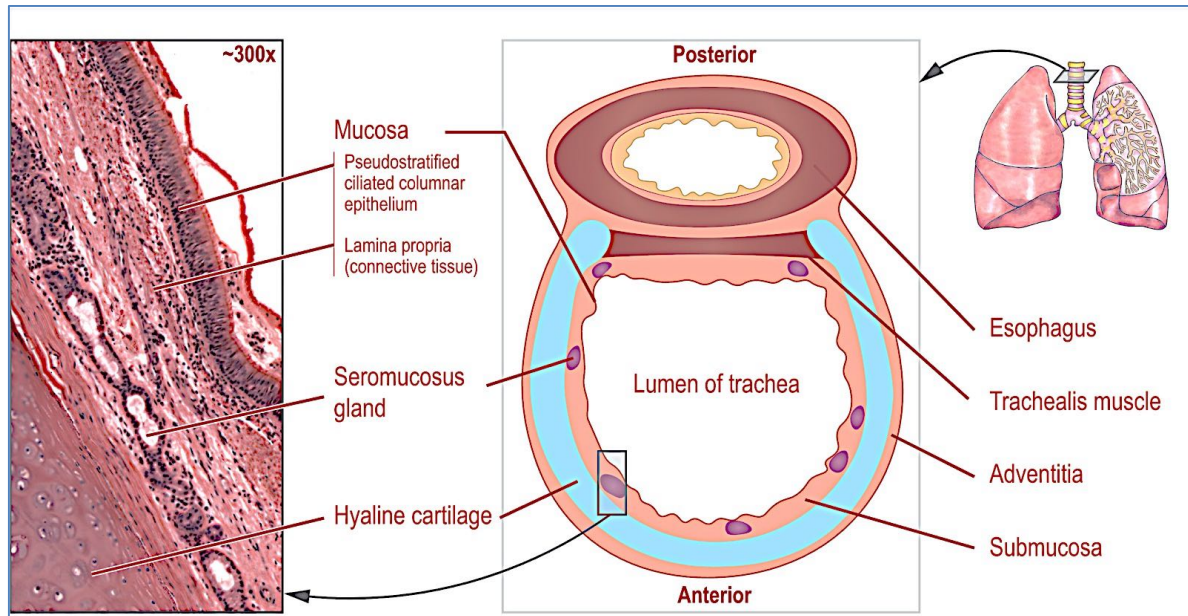
- Oropharynx + Laryngopharynx:



Courtesy of Prof. Hunter, Unit of Oral Pathology, University of Sheffield

- **Trachea:**

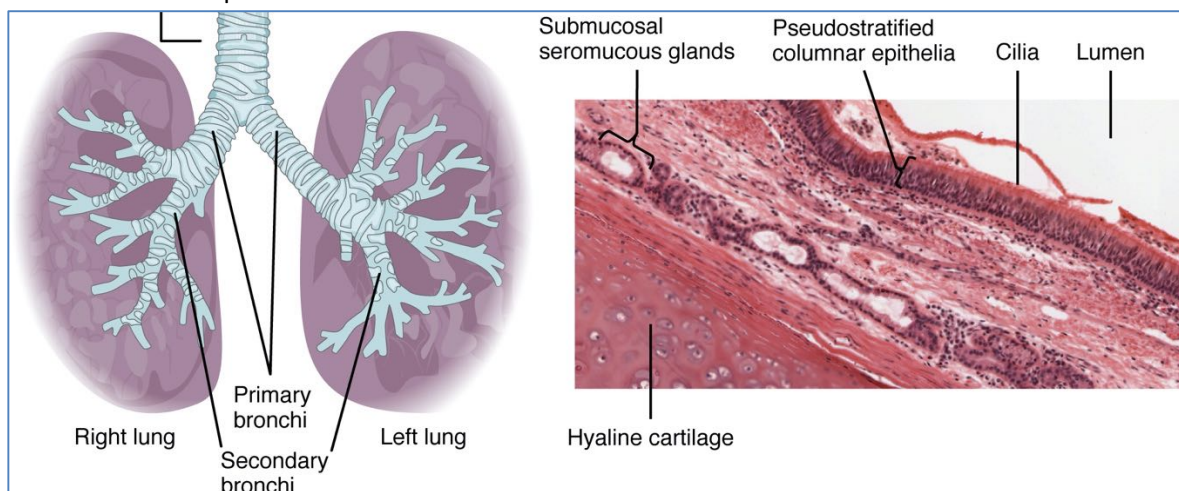
- **Mucosa:**
 - Ciliated Pseudostratified Epithelium
 - Goblet Cells → Mucous
 - Lamina Propria (of dense elastic fibres → high elasticity)
- **Submucosa:**
 - Submucosal Mucous Glands (Seromucous Glands)
- **Adventitia (Outer Covering):**
 - Connective Tissue
 - Cartilage Rings
 - Trachealis Muscle (- Constricts trachea during coughing)
- Note: Place at Highest Risk of Infection – Common Infection best @ 33-35°C (Lower Airways Hotter)



PBrieux, CC BY-SA 4.0 <<https://creativecommons.org/licenses/by-sa/4.0/>>, via Wikimedia Commons

- **Bronchi:**

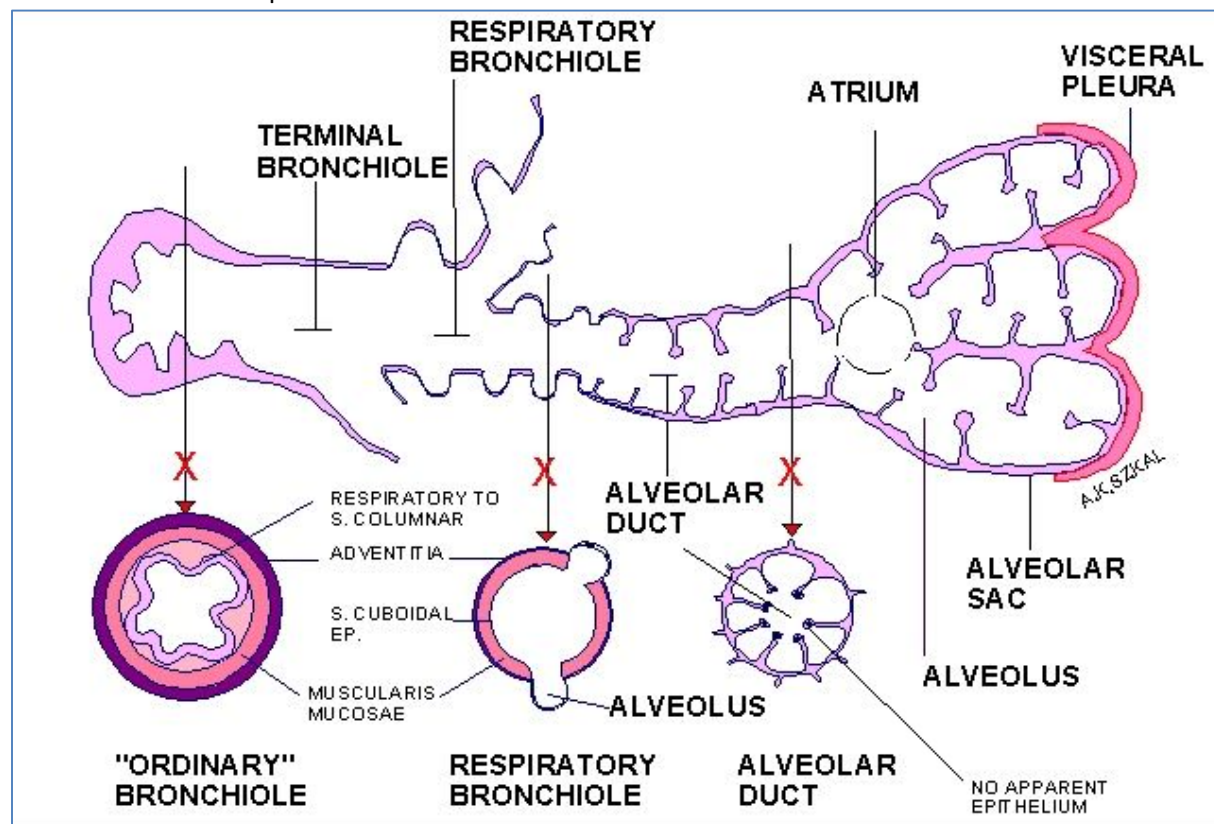
- **Mucosa:**
 - Ciliated Pseudostratified → Ciliated Columnar Epithelium (lower bronchi)
 - Goblet Cells → Mucous
 - Cilia Decreases
- **Submucosa:**
 - Submucosal Mucous Glands (Seromucous Glands)
- **Adventitia:**
 - Irregular Cartilage Rings → Cartilage Plates (lower bronchi)
 - Proportion of Smooth Muscle Increases



Creative Commons: <https://guides.hostos.cuny.edu/bio140/7-29>

Bronchioles:

- Sparsely-Ciliated Simple Cuboidal Epithelium
- No Goblet Cells
- No Submucosal Mucous Glands
- No Cartilage Support (Elastic Fibres Instead → Radial Traction)
- Increased Proportion of Smooth Muscle



Source: Unattributable

ALVEOLAR GAS EXCHANGE & GAS TRANSPORT

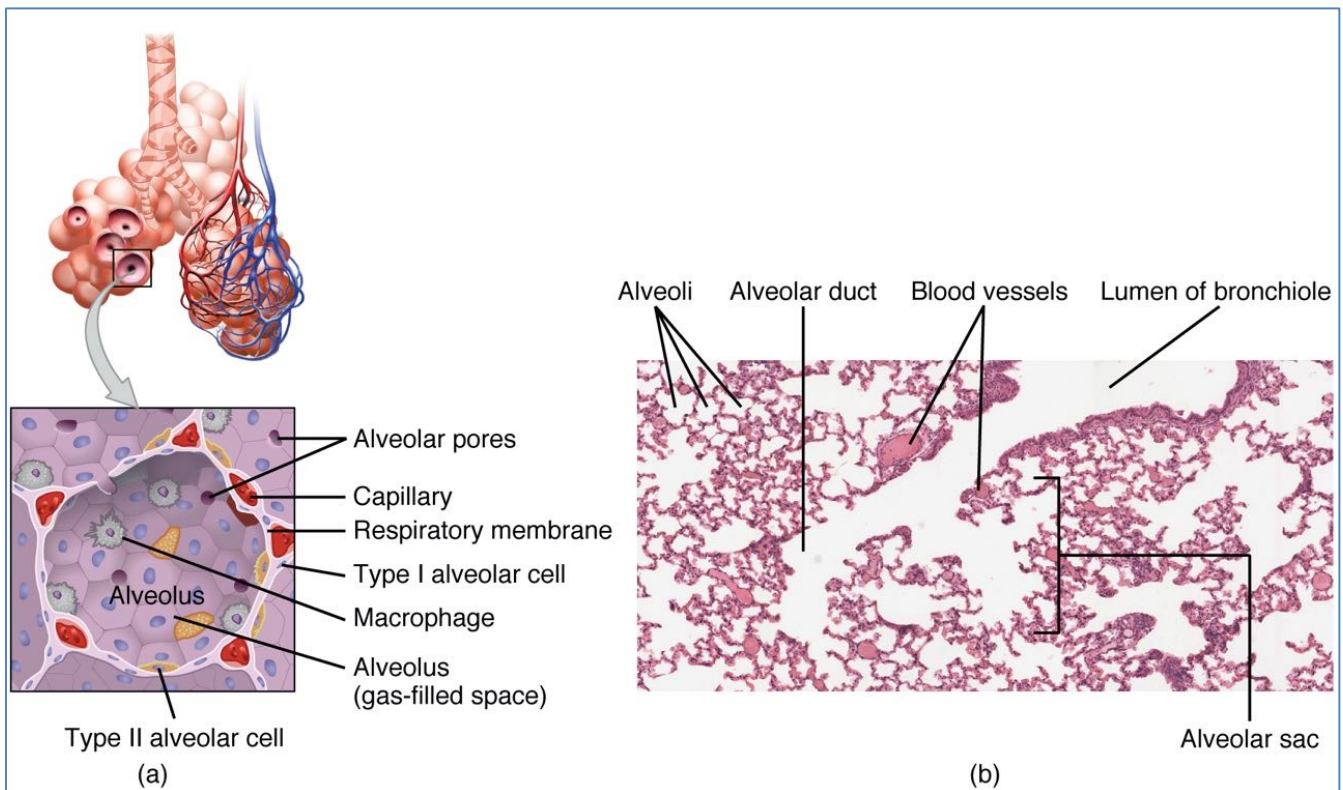
ALVEOLAR GAS EXCHANGE & GAS TRANSPORT

Body's Aim:

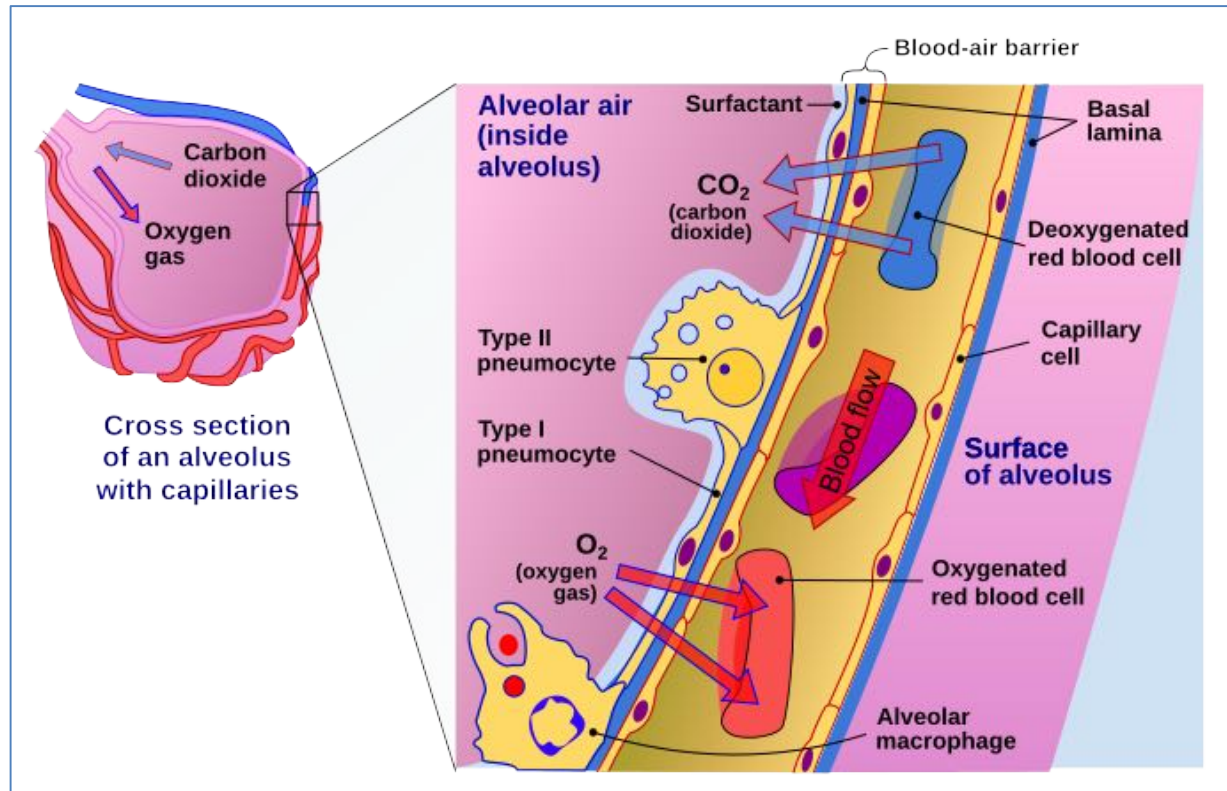
- Get O₂ from Lungs → Blood
- Get CO₂ from Blood → Lungs

Path of Oxygen Molecules From Air → Blood:

- **Convection: Air into Lungs (Active)**
 1. Atmosphere
 2. Alveoli
- **Diffusion: Air Diffuses into Blood (Passive)**
 3. Alveolar Fluid Lining (surfactant)
 4. Tissue Barrier
 - Alveolar Epithelium
 - Basal Lamina
 - Interstitium
 - Endothelium (Vessel wall)
 5. Blood Plasma
 6. RBC Membrane
 7. Uptake by Haemoglobin
- **Convection: Blood Pumped Around Body (Active)**
- **Diffusion: Air Diffuses From Blood → Peripheral Cells (Passive)**



Creative Commons: <https://guides.hostos.cuny.edu/bio140/7-29>



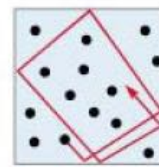
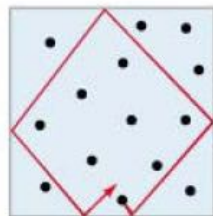
Delmalani18, CC BY-SA 4.0 <<https://creativecommons.org/licenses/by-sa/4.0/>>, via Wikimedia Commons

- **Boyle's Law:**

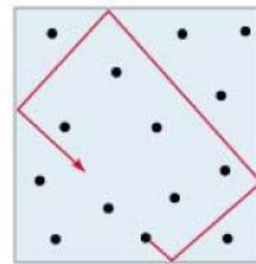
- **At a constant temperature, the pressure of a gas is *inversely proportional* to its volume.**
- I.e: Gases move from High Pressure → Low Pressure
- (inverse relationship between gas volume & pressure)
 - $P_1/V_1 = P_2/V_2$

Boyle's Law

$$P = 1 / V$$



(a) Decrease volume, pressure rises

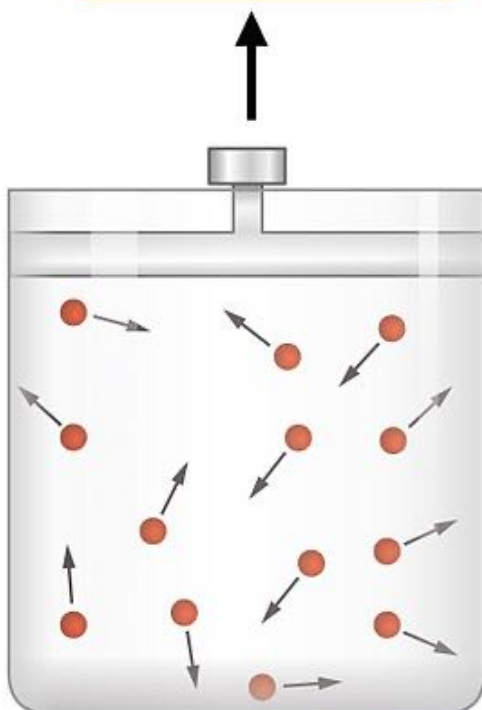


(b) Increase volume, pressure falls

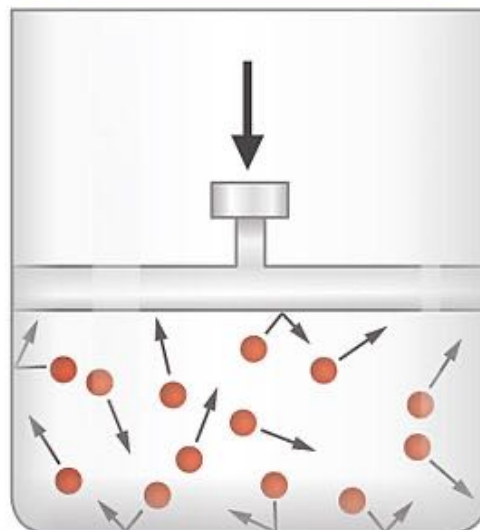
Pressure created by molecules colliding with walls of container

Key Point: Gases move from area of high pressure to area of low pressure

As volume increases, pressure decreases



As volume decreases, pressure increases



$$\begin{aligned} pV &= k \\ p_1V_1 &= p_2V_2 \\ p &= k/V \end{aligned}$$

- **Dalton's Law (of partial pressures):**

- The total pressure of a mixture of gasses is equal to the sum of each gas's partial pressure.
 - Eg. Atmospheric Pressure (sea) = 760mmHg = sum of P_{Nitrogen} , P_{Oxygen} , P_{Water} & $P_{\text{CarbonDioxide}}$
- Also, the proportion (%age) of a gas in a mixture =
 - The %age of the total pressure that it contributes =
 - Its partial pressure.
- Simply: *Each gas in a solution exerts a pressure exactly proportional to its abundance.*

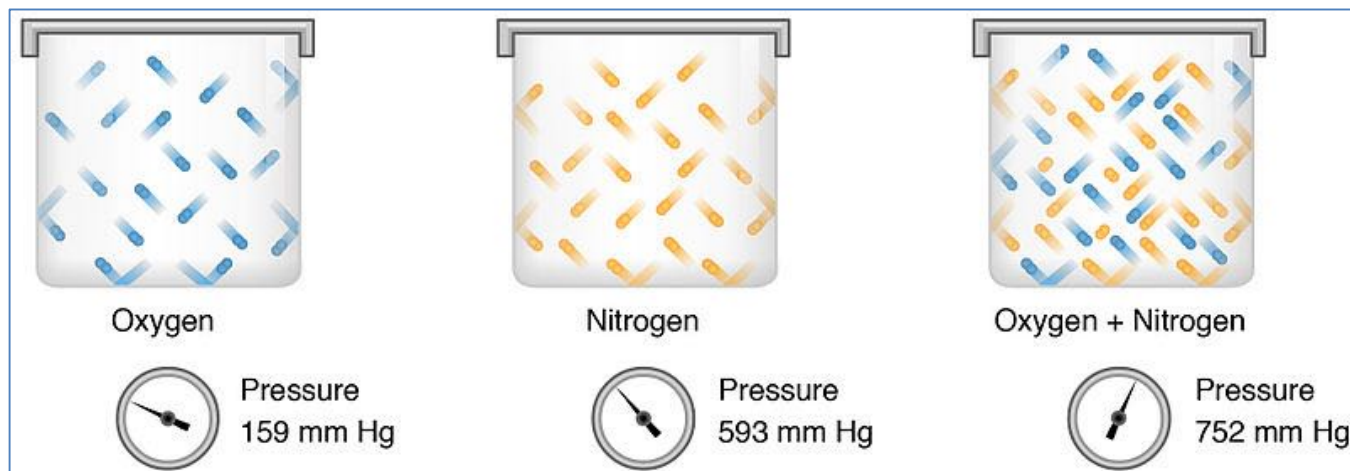
Remember: Atmospheric pressure is 760 mmHg (at sea level)

Abundance in air:	Collisions:	Partial Pressure:
78.6% N ₂ (Nitrogen)	78.6% N ₂	N ₂ 78.6% x 760 = 597 mmHg
20.9% O ₂ (Oxygen)	20.9% O ₂	O ₂ 20.9% x 760 = 159 mmHg
0.5% H ₂ O (Water)	0.5% H ₂ O	H ₂ O 0.5% x 760 = 3.8 mmHg
0.04% CO ₂ (Carbon dioxide)	0.04% CO ₂	CO ₂ 0.04% x 760 = 0.3 mmHg

$$P_{\text{N}_2} + P_{\text{O}_2} + P_{\text{H}_2\text{O}} + P_{\text{CO}_2} = 760 \text{ mmHg}$$

Gases exert a pressure proportional to their abundance

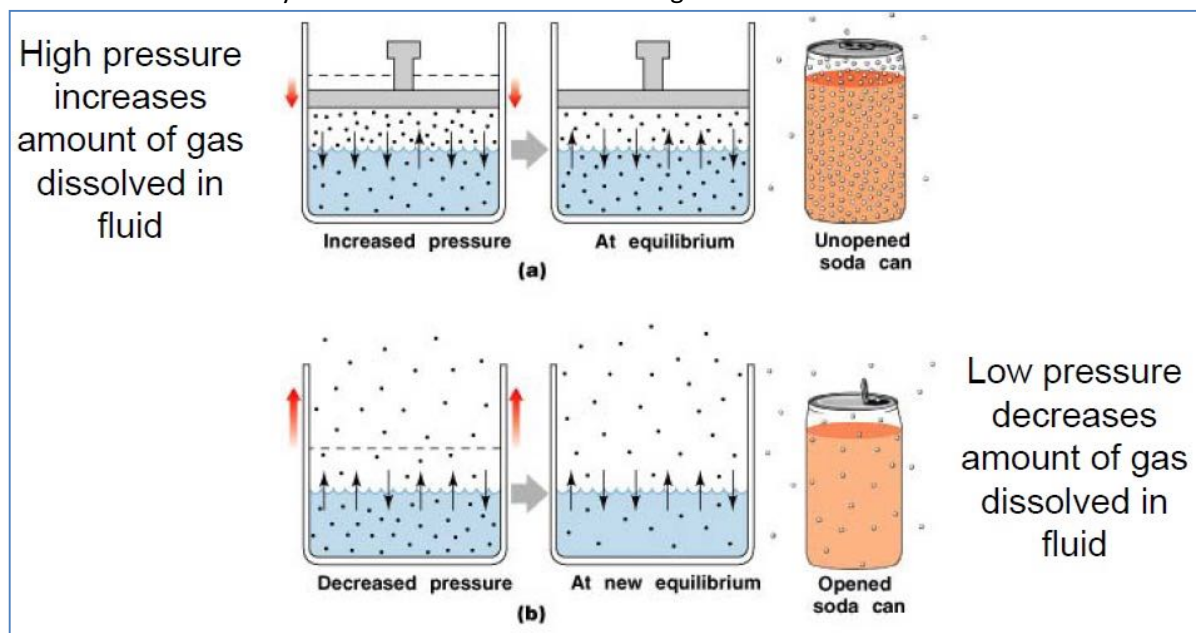
More gas = higher pressure



OpenStax College, CC BY 3.0 <<https://creativecommons.org/licenses/by/3.0/>>, via Wikimedia Commons

- **Henry's Law (of dissolved gases):**

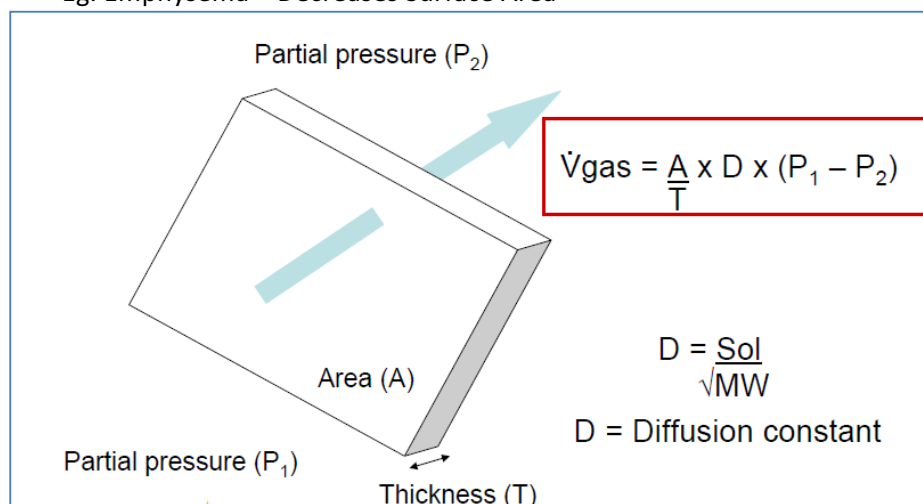
- 'The amount of gas in solution is proportional to the partial pressure of that gas'
 - More gas dissolves in a solution when pressure (and hence partial pressure) is increased.
 - The only other factor is how *soluble* the gas is in that solvent.



Source: Unattributable

- **Fick's Law (of gas diffusion)**

- Diffusion increases with:
 - Increased Surface Area
 - Decreased Membrane Thickness
 - Increased Partial Pressure *Gradient* (Difference between P_{Outside} & P_{Inside})
 - Increased Diffusion Constant (D) ($D = \text{Gas Solubility} / \sqrt{\text{Molecular Weight}}$)
 - I.e: The more soluble, the better the diffusion.
 - I.e: The smaller the molecule, the better the diffusion.
- **Implications for Lung Design:**
 - Alveolar Surface Area must be large as possible
 - Basal laminae of alveoli & capillaries are fused → minimises thickness of membrane.
- **Note: In a healthy resting lung,** the only factor that significantly changes with each breath is the *Partial Pressure Gradients*.
 - Therefore - the main determinant of the rate of gas diffusion across the alveolar membrane.
- **Note: In a pathological lung,** other factors (Surface area/membrane thickness/Gas Solubility (due to surfactant composition)) may determine the rate of gas diffusion across the alveolar membrane.
 - Eg. Pneumonia – Increases Thickness
 - Eg. Emphysema – Decreases Surface Area



Source: Unattributable

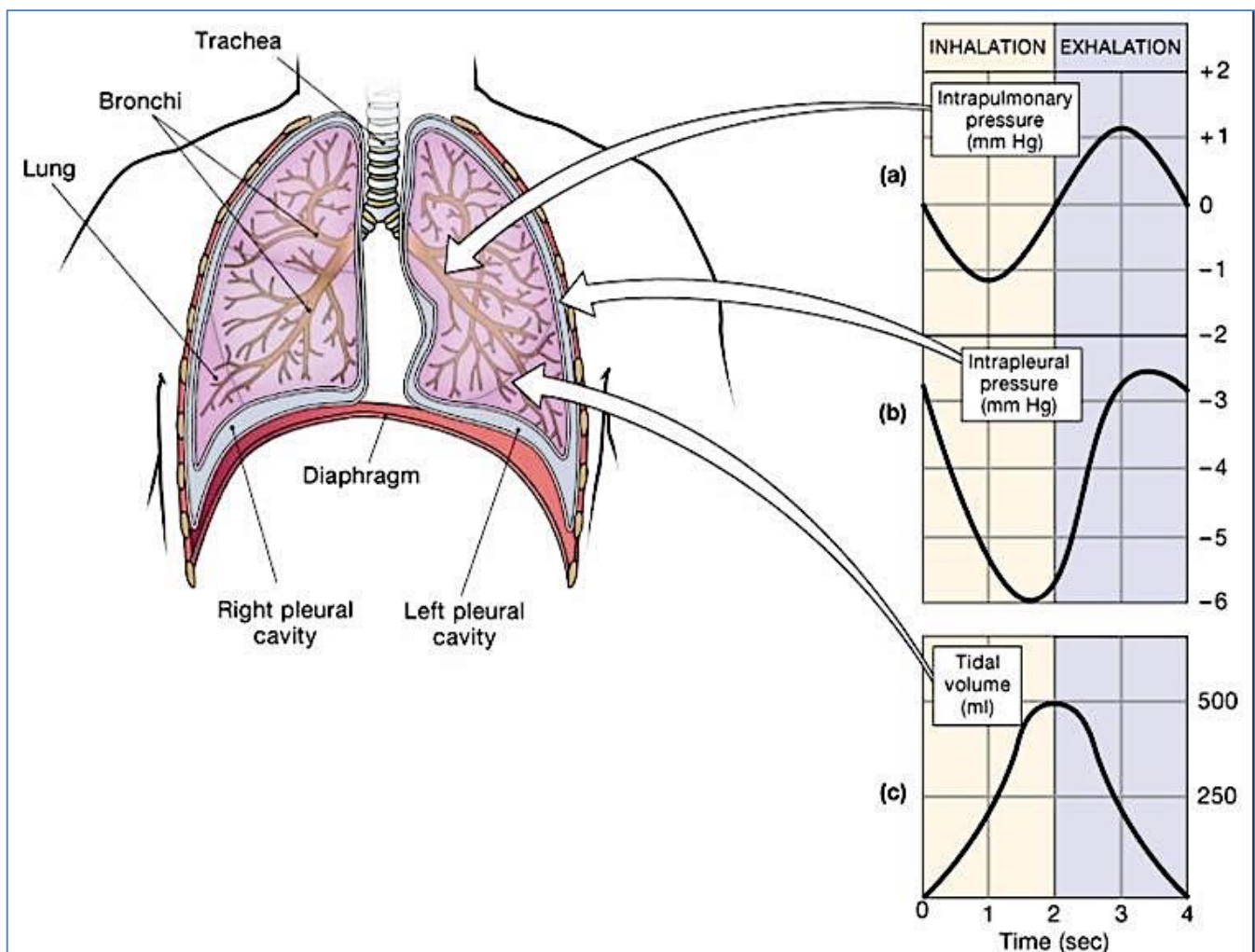
- **Pressure Changes:**

○ **Intrapleural Pressure:**

- Negative Pressure between Visceral & Parietal Pleural Membranes....*Due To 2 Forces:*
 - Elastic Recoil of The Lungs
 - Surface Tension of *Alveolar Fluid* – acts to shrink alveoli to smallest possible.
- **Always Subatmospheric (Negative):**
 - Becomes *more* subatmospheric during inhalation
 - Becomes *less* subatmospheric during exhalation
- Due to serous fluid in Intrapleural space:
 - Secures the pleurae together + Anchors lungs to thoracic wall.
 - ****Ensures *negative pressure* gradient in pleural cavity. (INTRAPLEURAL PRESSURE)**
 - ****Prevents lung from collapsing**
 - *Sucks the lung outwards towards chest wall*
 - *Aids in passive recoil of the lung during expiration.*
 - **Note: Pneumothorax:** Accumulation of air in the pleural cavity → Intrapleural pressure dissipates → lung collapses.
 - Traumatic (Penetrating/Non-penetrating)
 - Spontaneous (Disease complication)

○ **Intrapulmonary Pressure:**

- Pressure in the Alveoli
- **Alternates between Positive & Subatmospheric (Negative) Pressures.**
 - *Negative* pressure during Inhalation
 - *Positive* pressure during Exhalation

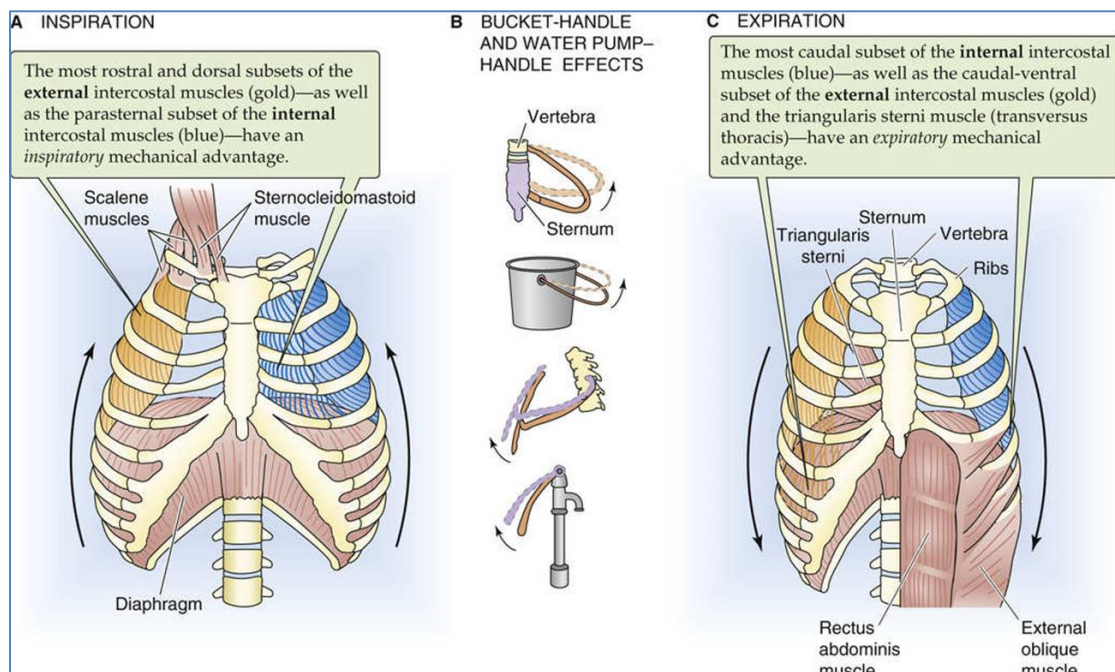


- **Inhalation:**

- **Diaphragm:**
 - Contracts
 - Moves inferiorly
- **External Intercostals:**
 - Contract
 - Move ribs out & up (bucket & pump handle movements.)
- **Accessory Muscles (If Forced):**
 - Scalenes
 - Sternocleidomastoids
 - Pectoralis Minors
- **Lung Volume:**
 - Increases
- **IntraPleural Pressure:**
 - Becomes *more* subatmospheric (more negative)
- **IntraPulmonary Pressure:**
 - Becomes *negative*. (relative to P_{atm})
- **Air:**
 - Flows In

- **Expiration:**

- **Diaphragm:**
 - Relaxes
 - Moves superiorly
- **External Intercostals:**
 - Relax
 - Rib cage descends due to recoil of costal cartilages
- **Accessory Muscles (If Forced):**
 - Abdominal Wall Muscles (Transverse & Oblique)
 - Internal Intercostals
- **Lung Volume:**
 - Decreases
- **IntraPleural Pressure:**
 - Becomes *less* subatmospheric (more positive)
- **IntraPulmonary Pressure:**
 - Becomes *positive*. (relative to P_{atm})
- **Air:**
 - Flows Out



Respiratory Rates:

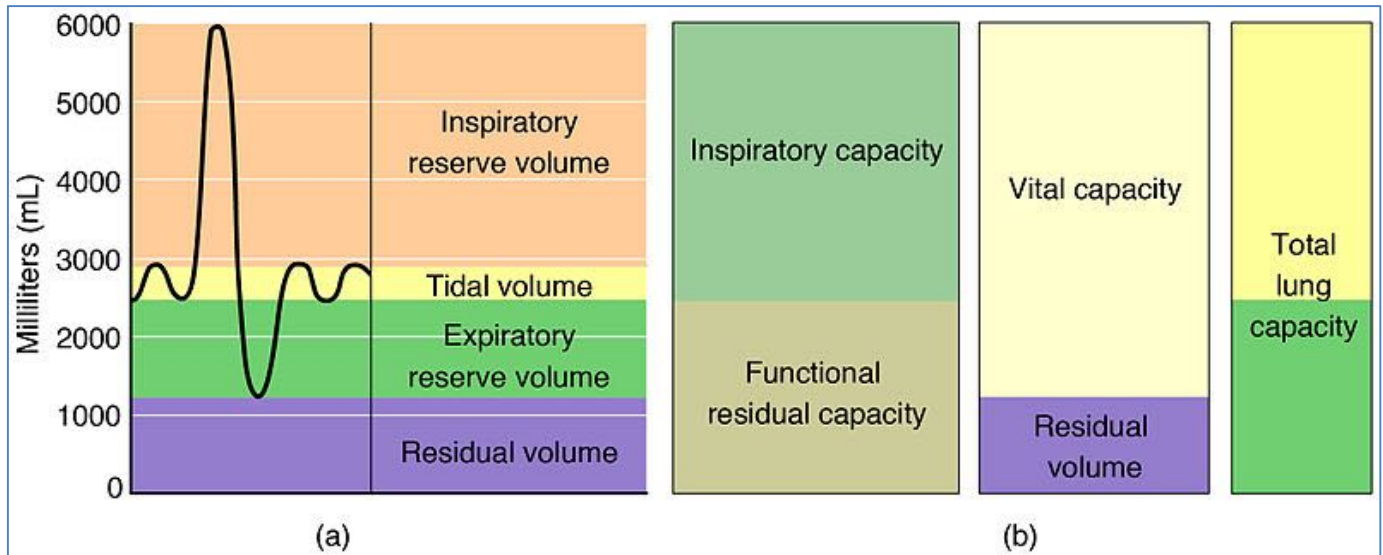
- **Respiratory Rate:** (f)
 - o Breathing Frequency
- **Respiratory Minute Volume (Minute Ventilation Rate):** ($\dot{V}_E$)
 - o Amount of air moved *via Tidal Ventilation* Each Minute.
 - o $\dot{V}_E = V_T \times f$
 - Minute Ventilation Rate = Tidal Volume x Respiratory Rate
 - o Note: the dot means a 'rate'.
 - o Note: The Same Minute Ventilation Rate can be achieved by Different Combos of V_T & f .
- **Alveolar Ventilation:** ($\dot{V}_A$)
 - o Amount of air *reaching the Alveoli each minute*
 - o $\dot{V}_A = (V_T - V_D) \times f$
 - Alveolar Ventilation = (Tidal Volume – Dead Space) x Frequency
 - o Note: High V_T & Low f . *maximizes Alveolar Ventilation*, but is Energetically Expensive.
 - High f . & Low V_T are Energetically Cheap, but have *low Alveolar Ventilation* due to Increased Dead Space.

Respiratory Volumes:

- **Tidal Volume:** (V_T)
 - o Volume of air *inhaled OR exhaled during 1x Normal Breath*.
- **Dead Space:** (V_D)
 - o Amount of air in *Conducting Zone* that doesn't take part in Gas Transfer.
 - o There is always a small volume of air from the previous breath that will re-enter the alveoli.
- **Expiratory Reserve Volume:** (ERV)
 - o Volume of Additional air that can be EXPIRED *After A Normal Quiet Expiration*
 - o Ie: Beyond Tidal Volume.
- **Inspiratory Reserve Volume:** (IRV)
 - o Volume of Additional air that can be INSPIRED *After A Normal Quiet Inhalation*
 - o Ie: Beyond Tidal Volume.
- **Residual Volume:** (RV)
 - o Air left in lungs after *Maximum Forced Expiration*.
 - o Ie: Air that *can't* be breathed out (Therefore Cannot be seen/measured on a Spirometer)

Respiratory Capacities:

- **Inspiratory Capacity:** (IC)
 - o Volume of air that can be INSPIRED *After A Normal Quiet Expiration*
 - o Ie: Tidal Volume + Inspiratory Reserve Volume
 - o $IC = V_T + IRV$
- **Functional Residual Capacity:** (FRC)
 - o Total Air Remaining *After A Normal Quiet Expiration*
- **Vital Capacity:** (VC)
 - o Max Air you can Move Into OR Out of your lungs.
 - o Ie: Expiratory Reserve + Tidal Volume + Inspiratory Reserve
 - o $VC = ERV + V_T + IRV$
- **Total Lung Capacity:** (TLC)
 - o Total Air in Lungs *After A Forced Inspiration*
 - o Ie: Residual Volume + Expiratory Reserve Volume + Tidal Volume + Inspiratory Reserve Volume.
 - o $TLC = RV + ERV + V_T + IRV$



OpenStax College, CC BY 3.0 <<https://creativecommons.org/licenses/by/3.0/>>, via Wikimedia Commons

THE PHYSIOLOGY OF BREATHING

THE PHYSIOLOGY OF BREATHING

Bronchial Circulation:

- O₂/Nutrient-Rich blood supply → Lung Tissues
- Originates from *Systemic Circulation*
- Via small *Bronchial Arteries*.
- Blood empties into *Pulmonary Veins* → Left Atrium
 - o Note: because of this, Venous Return to L-Atrium is *More* than R-Ventricular Output.

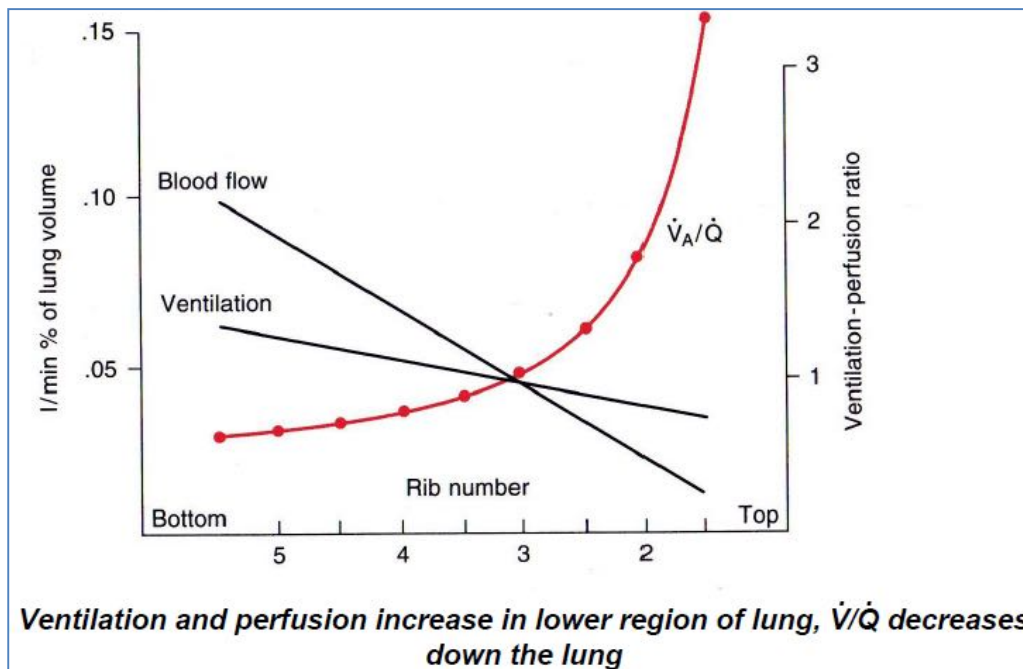
Pulmonary Circulation:

- **Features:**
 - o Much shorter than systemic circulation
 - o Artery lumens are larger than systemic..Due to:
 - *Thin* walls (1/3 of aorta)
 - *Distensible* Walls:
 - Stretch to maintain low BP
 - Prevents rupture of microvessels.
- **Pressures:**
 - o *Pressure & Resistance - MUCH LOWER THAN SYSTEMIC CIRCUIT*
 - o *Prevents rupture of Microvessels*
 - o **Pulmonary Artery:**
 - Systolic = 25mmHg
 - Diastolic = 8mmHg
 - o **Pulmonary Capillary Pressure:**
 - ≈ 7mmHg
 - o **Pulmonary Vein:**
 - ≈ 2mmHg
- **Resistance:**
 - o Much lower than systemic
 - o Extra-Alveolar vessels have Low Resistance @ High Lung Volumes – Due to Radial Traction
 - o However, as lung approaches full volume, Transmural Pressure overcomes Radial Traction → Decreases Vessel Diameter → Increases Resistance
- **Blood Volume:**
 - o Average = 450-500mL (9% Total Blood Volume)
 - o Only 70-100mLs of this is involved in Gas-Exchange at any one time.
 - o ≈70mLs of blood is replaced each heart-beat
 - o Varies Greatly in different Circumstances:
 - **Physiologically:**
 - **Eg. Blowing Trumpet** → ↑Thoracic Pressure → may expel 250mL blood
 - **Eg. Acute blood loss** → ↓ Total Blood Volume → affects Pulmonary Blood Volume
 - **Eg. Posture:** Supine or Standing – standing ↓ B-Volume by 10%
 - **Eg. Systemic Vascular Tone:** ↑Systemic Vascular Tone – Squeezes blood into Pulmonary Circuit.
 - **Pathologically:**
 - **Eg. L-Heart Failure** → Blood backs up in Pul-Circuit → ↑Pulmonary Blood Volume.
- **Blood Flow:**
 - o Pulmonary Blood Flow = Cardiac Output
 - o Anything affecting CO, will equally affect P-Blood Flow (HR/SV/Contractility/Venous Return)
- **PO₂ Vs. Blood Flow: (HYPOXIC VASOCONSTRICTION):**
 - o Alveolar Blood is *Directed To* areas of the lungs that are *Best Oxygenated (Best Ventilated)*.
 - Opposite to systemic circulation
 - o If P_{A-O2} (Alveolar O₂ Partial Pressure) Decreases, adjacent blood vessels constrict for a few minutes.
 - -Restricts blood flow to poorly ventilated alveoli.
 - o **le: Automatic Mechanism** that **Matches Blood Flow with Ventilation**

Regional Pulmonary Blood Flow:

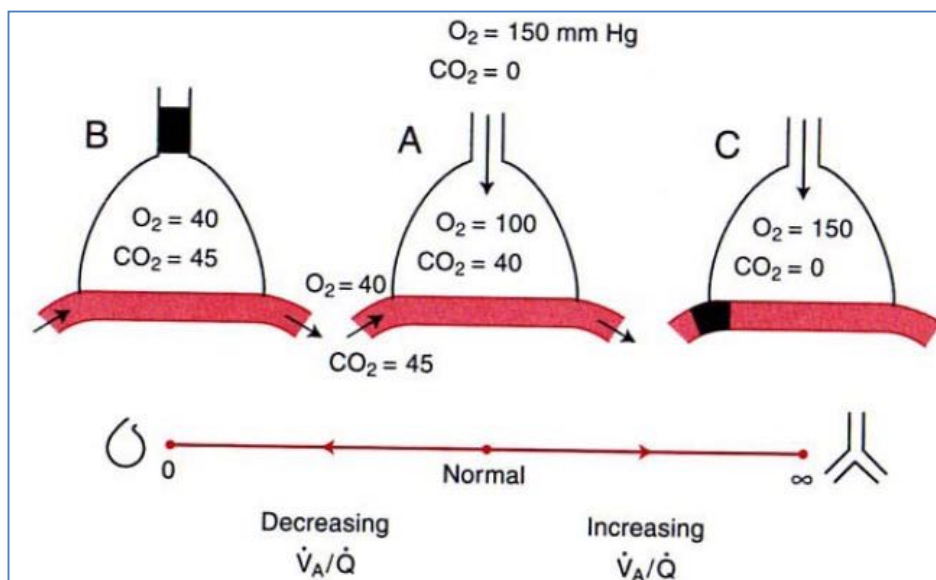
- Ventilation-Perfusion Matching:

- Not All Alveoli are Perfused or Ventilated equally.
- Even in a normal lung, there will be an imbalance of Perfusion & Ventilation.
- This Imbalance is **quantified by a Ventilation-Perfusion Ratio**.
- This ratio **changes depending on location within lung**.
- *Blood Flow Rate increases toward Base of Lung.*
- *Ventilation Rate increases toward Base of Lung.* (To a Lesser Degree than Blood Flow)



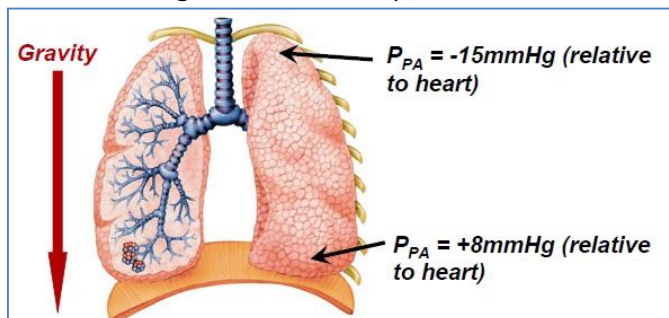
- Ventilation-Perfusion Ratios:

- $\dot{V}_A/\dot{Q}$ (Alveolar Ventilation Rate / Blood Flow Rate)
- **Approaching Zero:**
 - When Ventilation = 0, but there's still Perfusion, $\dot{V}_A/\dot{Q} = 0$
 - No fresh gas coming into lung →
 - Partial Pressures in Alveolus will equalise with Capillary Partial Pressures.
- **Approaching Infinity:**
 - When Adequate Ventilation, but *No Perfusion*, $\dot{V}_A/\dot{Q} = \text{Infinity}$
 - No blood supply to alveolus →
 - Partial Pressures in Alveolus will equalise with Atmospheric Partial Pressures.



- **Effect of Gravity:**

- **P_a (Arterial Pressure):**
 - Pressure at top – Lower
 - Pressure at bottom – Higher
 - Pressure difference $\approx 23\text{mmHg}$ between Top & Bottom.
- **When Standing:** Little flow @ Top, 5x Flow @ Bottom.

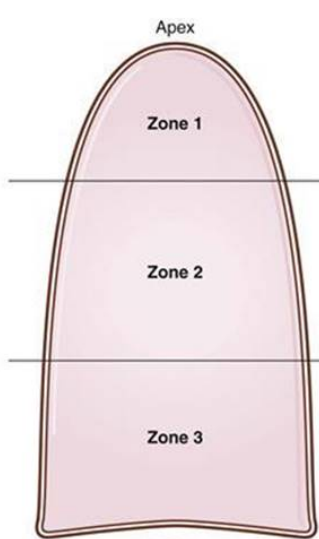


- **Blood Pressure Vs. Alveolar Air Pressure (TRANSMURAL PRESSURE):**

- Capillaries are distended by internal BP.
- But, are also compressed by Alveolar Air Pressure
 - The **net result** between these 2 pressures, **determines perfusion** (vessel patency)
- Lung is divided into 3 zones based on the 3 possible **net results** of BP vs. Alveolar Air Pressure.

- **3 Zones:**

- **Zone 1:**
 - Capillary Pressure never exceeds Alveolar Air Pressure.
 - No Blood Flow at all.
 - V/Q Ratio $\rightarrow$ Infinity
- **Zone 2:**
 - Capillary Pressure only exceeds Alveolar Air Pressure during Systole.
 - Intermittent Blood Flow (Flow during systolic pressure)
 - V/Q Ratio = Normal
- **Zone 3:**
 - Capillary Pressure always exceeds Alveolar Air Pressure.
 - Constant Blood Flow.
 - V/Q Ratio $\rightarrow$ Still Normal, but lower.
- Note: Zone 1 is generally only seen under Abnormal Conditions. Normal Lung = usually Zone 2 & 3.

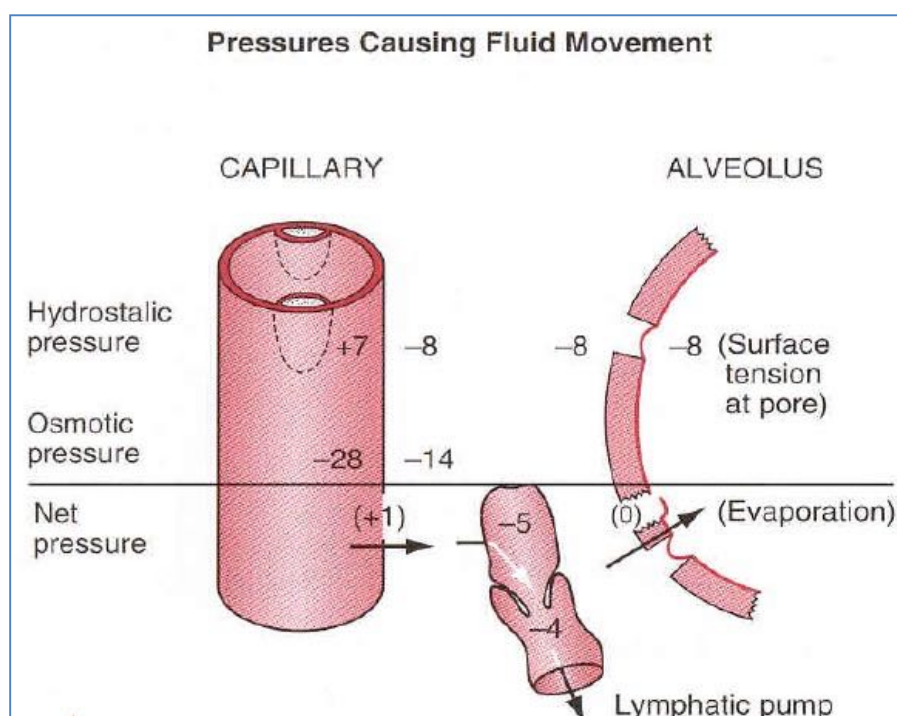
BLOOD FLOW DISTRIBUTION IN THE LUNG		
	Blood Flow	Pressures Driving Blood Flow
 Apex Zone 1 Zone 2 Zone 3 Base	Lowest	$P_A \geq P_a > P_v$
	Medium	$P_a > P_A > P_v$
	Highest	$P_a > P_v > P_A$

Effect of Exercise on Blood Flow:

- Blood Flow through lungs increases 4 – 7 Times!
- However, Rise in $BP_{\text{Pulmonary}}$ Must be kept low to stop microvessel rupture.
- Extra Flow is Accommodated for in 3 Ways:
 - o Distension of Capillaries.
 - o Increasing *number* of Open Capillaries
 - o For Blood Flow to be maintained, Pulmonary Arterial Pressure must Be Increased.

Pulmonary Capillaries:

- **Dynamics:**
 - o Blood Transit Time through Capillaries ≈ 0.8 sec.
 - o During Exercise, $CO \uparrow \rightarrow$ Capillary Transit time shortens to ≈ 0.3 sec.
 - Note: If less than 0.3 sec. Blood doesn't have time for adequate Gas Exchange.
- **Fluid Exchange:**
 - o **Same Forces as Systemic Capillaries:**
 - Hydrostatic Pressure
 - Colloid-Osmotic Pressure
 - o **But Different Magnitudes:**
 - **Hydrostatic:**
 - $\approx 7\text{mmHg}$
 - Pulmonary Capillary Hydrostatic Pressure Lower Than Systemic
 - Slightly Negative Interstitial Hydrostatic Pressure
 - o Keeps alveoli 'dry'
 - **Colloid-Osmotic Pressure**
 - Capillary Osmotic Pressure Higher than Interstitium.
 - o $\approx -28\text{mmHg}$
 - o Draws Fluid back into Capillaries
 - o Opposes Hydrostatic Pressure.
 - Pulmonary Capillaries – relatively leaky to proteins $\rightarrow$ some proteins move to Interstitium.
 - o Interstitial Osmotic Pressure $>$ Alveolar Osmotic Pressure
 - o Draws Alveolar Fluid $\rightarrow$ Interstitium
 - o Prevents build-up of fluid in alveoli.
 - o Note: if too leaky, proteins in Interstitium can cause Oedema.



Preventing Pulmonary Oedema:

- Negative Interstitial Pressure:

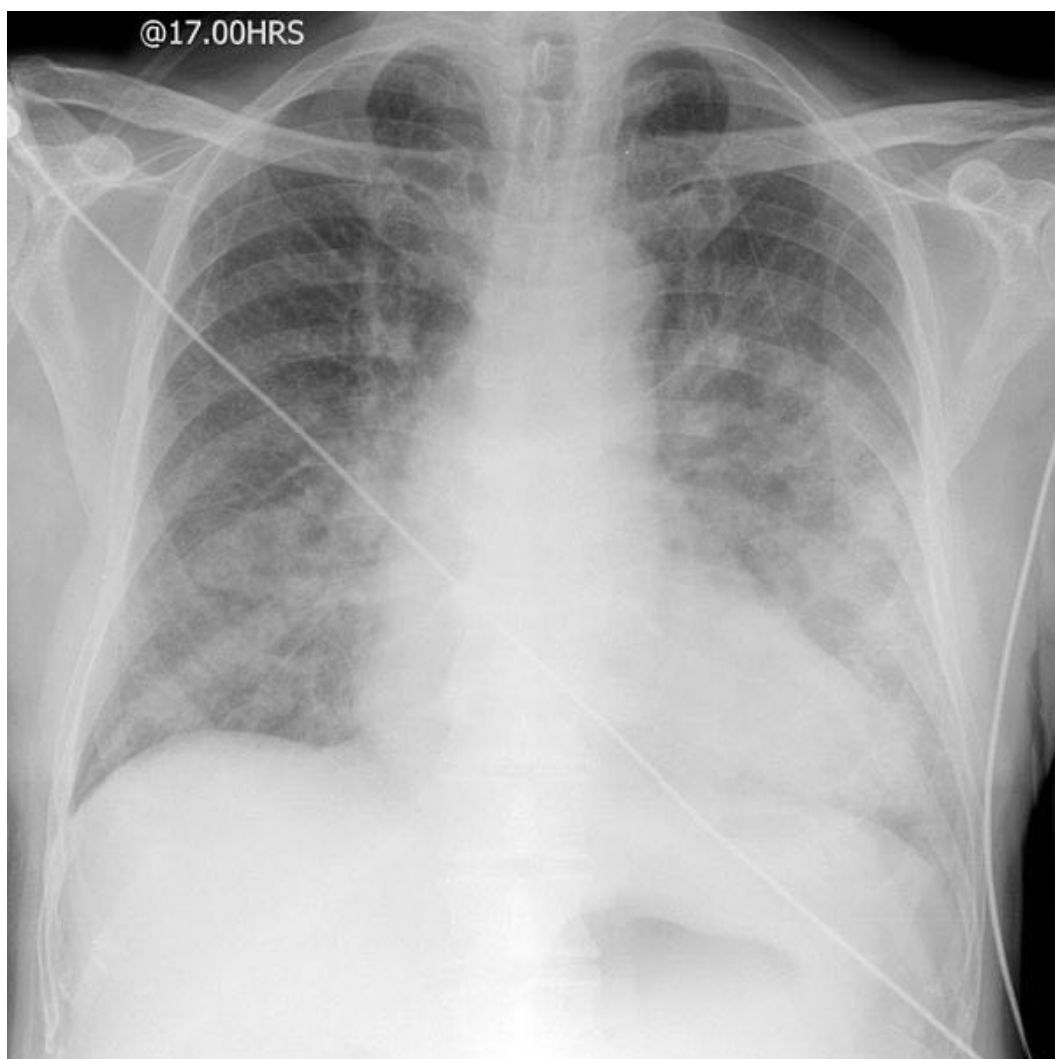
- Slightly Negative Interstitial Hydrostatic Pressure
- Keeps alveoli 'dry'
- Fluid in Alveoli is sucked into Interstitium → Lymphatics
- Note: If Interstitial Fluid Pressure was +ve, Oedema would occur.

- Lymphatic Vessels:

- Extend from all tissues of lungs.
- Actively pump Interstitial Fluid → Blood Vessels

- Oedema Safety Factor:

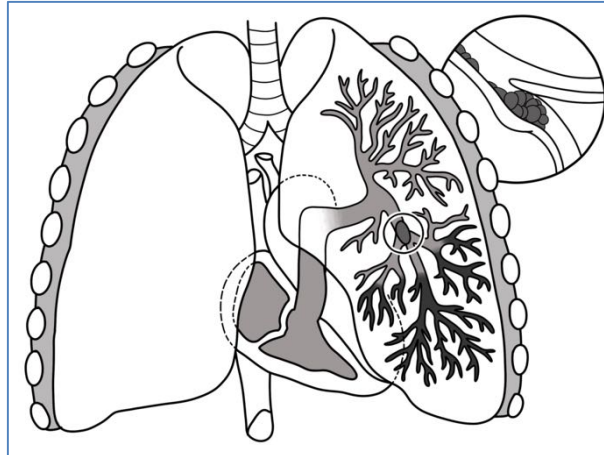
- For oedema to occur, Pulmonary Capillary Pressure must rise above Colloid Osmotic Pressure.
 - Pulmonary Cap-Pressure $\approx$ 7mmHg
 - Capillary Osmotic Pressure $\approx$ -28mmHg
- Therefore a +21mmHg rise in Pulmonary Capillary Pressure is needed.
 - If this occurs, lethal oedema may occur within *hours!*
- Chronic elevated Pulmonary Capillary Pressure → Expanded Lymph Vessels.



Frank Gaillard, Jeremy Jones, CC BY-SA 3.0 <<https://creativecommons.org/licenses/by-sa/3.0/>>, via Wikimedia Commons

Pulmonary Embolism:

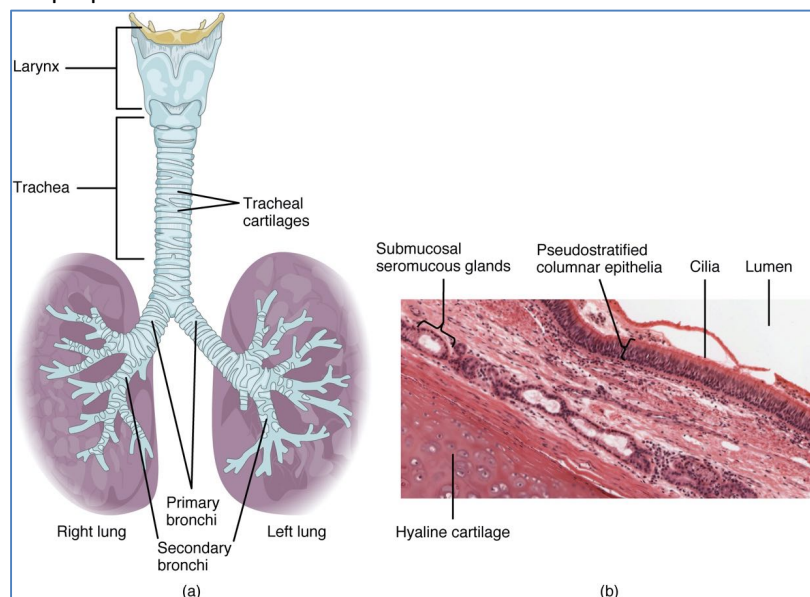
- Foreign fragments blocking a blood pulmonary vessel.
- Often due to Blood Clot (Thrombus)
- Blockage of vessel in lung will impact/prevent effective oxygenation of blood.



Baedr-9439, CC0, via Wikimedia Commons

Pulmonary Ventilation (Breathing):

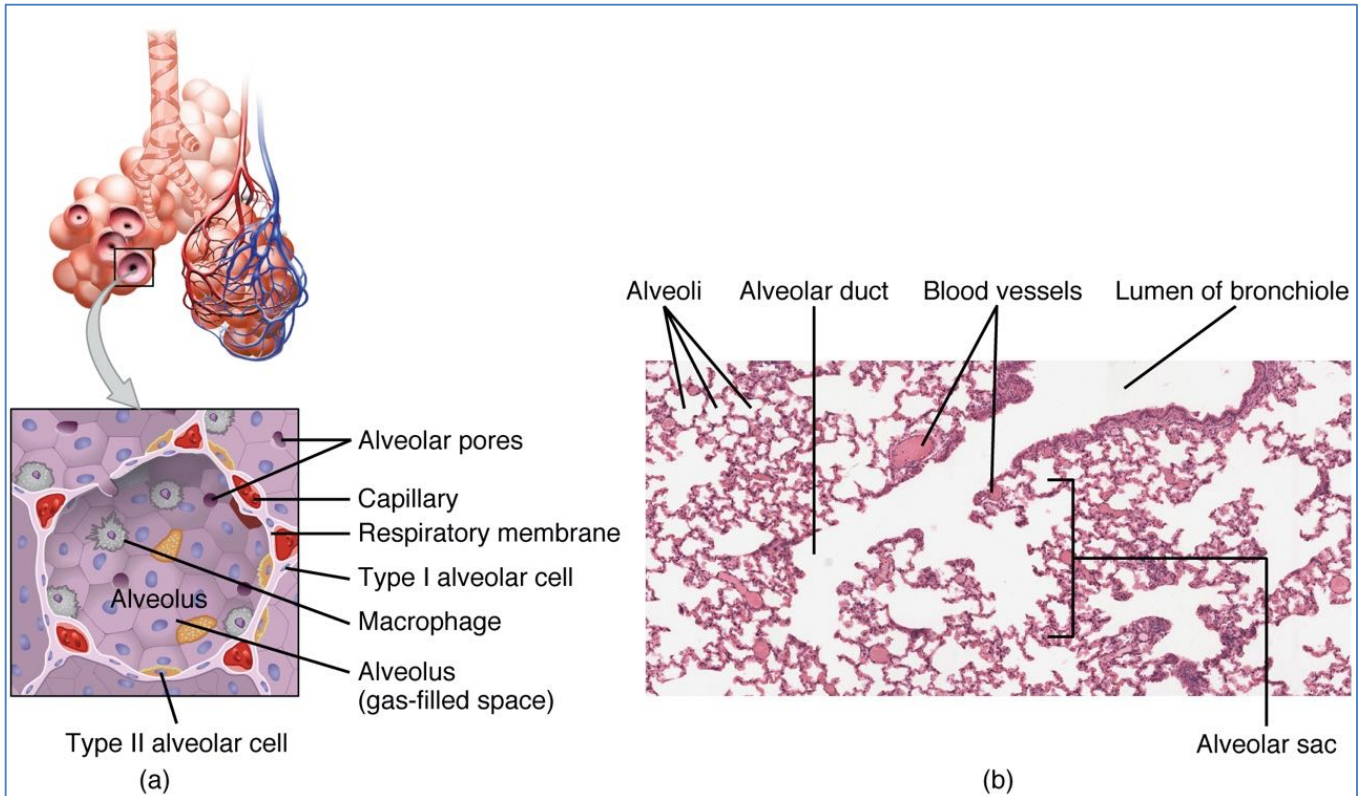
- The physical *movement* of air into/out of respiratory tract.
- Different from respiration.
- AIM: To Maintain adequate *alveolar* ventilation
- **Conducting Zone:**
 - o No Gas Exchange
 - o Allow Air movement → Alveoli
 - o Filter particulate matter in air (such matter in alveoli ↓-effectiveness of gas transfer)
 - Nose/mouth/nasopharynx
 - Trachea
 - Bronchi
 - Bronchioles
 - o Must be kept open (Patent) to breath.
 - o **Patency Maintained by:**
 - Cartilage Rings – Trachea
 - Cartilage Plates – Bronchi
 - Transmural Pressure + Radial Traction of surrounding tissues – Bronchioles
 - o **Patency Problems:**
 - Snoring
 - Sleep Apnoea



Creative Commons: <https://guides.hostos.cuny.edu/bio140/7-29>

- **Respiratory Zone:**

- Gas Exchange
 - Respiratory Bronchioles
 - Alveolar Ducts
 - Alveoli (Extensive blood supply)
- **Patency Maintained by:**
 - $\bar{v}$ e Intrapleural Pressure
 - Surface Tension of Alveolar Fluid
- **Patency Problems:**
 - Emphysema
 - Asthma
 - Infant Respiratory Distress Syndrome



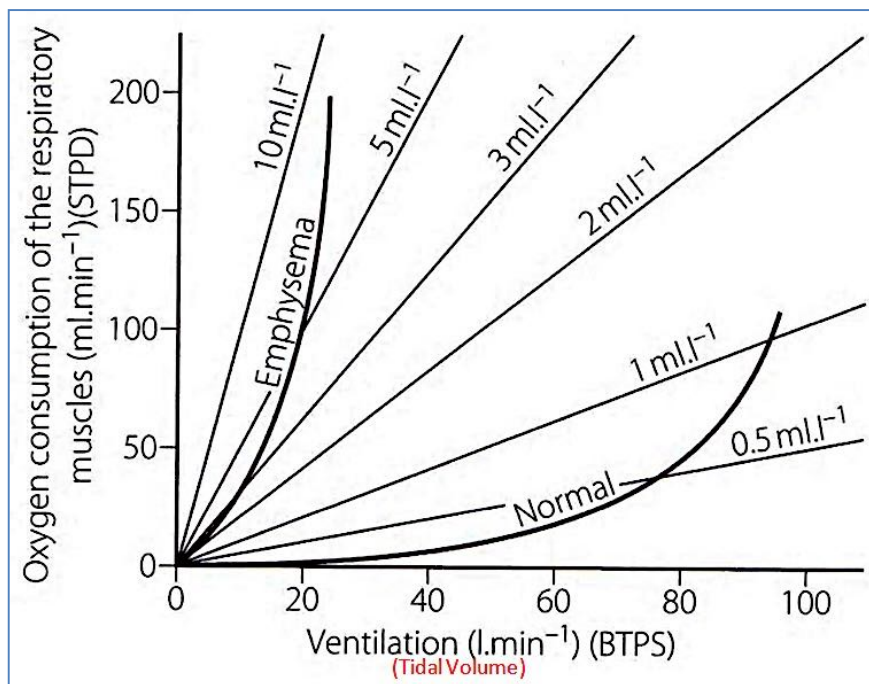
Creative Commons: <https://guides.hostos.cuny.edu/bio140/7-29>

Lung Compliance:

- Ease of Expansion of Lungs
- I.e: Pressure change needed to *create* Volume Change
 - Low Compliance – Requires greater force to expand lungs (opposite for high compliance)
- **Factors:**
 - **Pulmonary Conn. Tissue:**
 - Increase in Conn. Tissue (scarring/cirrhosis) → Decreases Compliance
 - **Surfactant:**
 - Prevents pulmonary epithelial surfaces from adhering during exhalation.
 - Low Surfactant → Decrease Compliance
 - **Thoracic Cage Mobility:**
 - Decrease in Thoracic Cage Mobility → Lower Compliance
 - I.e: Arthritis/scoliosis/etc.

Energetic Cost of Ventilation:

- Resting Ventilation $\approx$ 3-5% of Body's Energy Demands
- **Cost of Ventilation Increases When:**
 - **V_T Increases:**
 - I.e: Exercise
 - **Compliance Decreases:**
 - From: Decrease surfactant/
 - Increase in Conn. Tissue (scarring/cirrhosis) $\rightarrow$ Decreases Compliance
 - Decrease Thoracic Mobility
 - **Airway Resistance Increases:**
 - From broncho-constriction
 - Eg. Asthma/Emphysema
 - Note: Airway Diameter is an EXTREMELY POWERFUL FACTOR OF FLOW RATE.
- **Note:** Some Path. Conditions Increase the cost of breathing so significantly that the respiratory muscles alone consume more oxygen than they bring in.

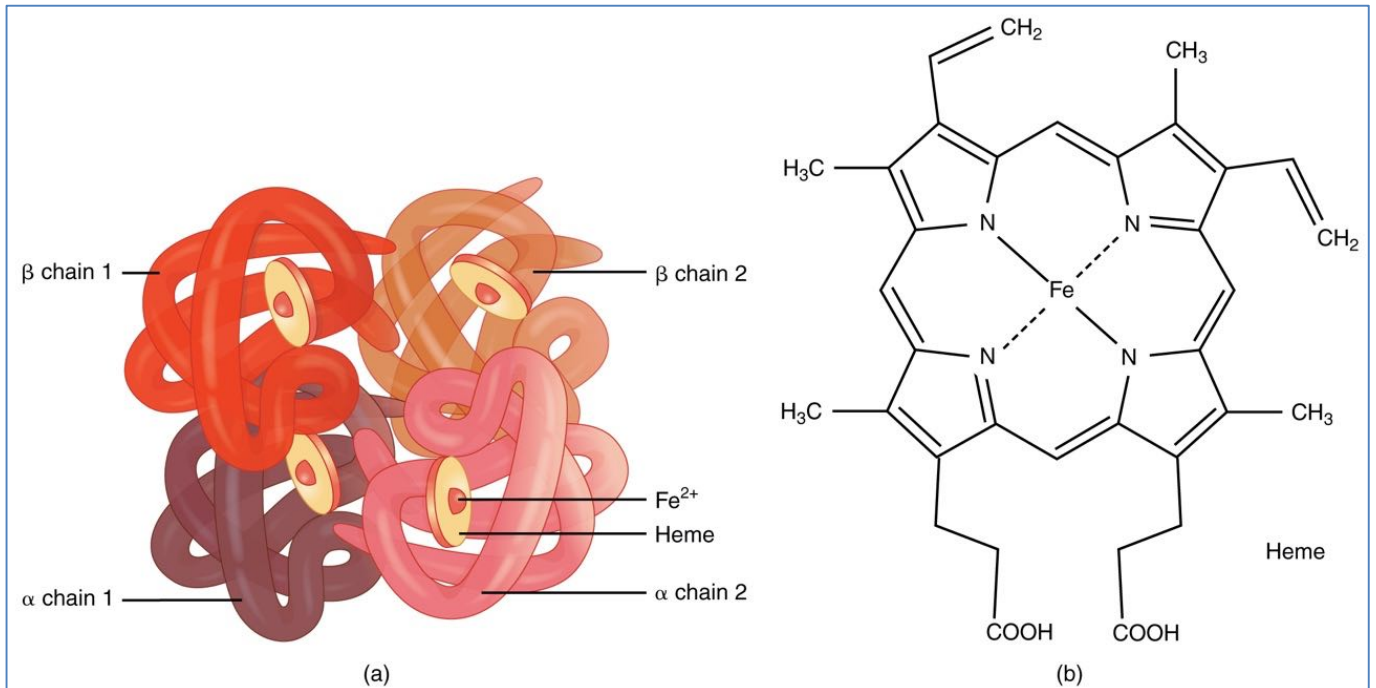


HEMOGLOBIN / HAEMOGLOBIN

HEMOGLOBIN (HB):

- What is it?:

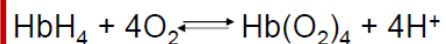
- A 4-Protein-Subunit Molecule
- Each Protein-Subunit has a *Heme Unit* with a *Central Iron Molecule*.



OpenStax College, CC BY 4.0 <<https://creativecommons.org/licenses/by/4.0/>>, via Wikimedia Commons

- Role in O₂ Transport:

- Each *Heme Unit* can carry 1xOxygen Molecule (O₂)
- Therefore 1xHaemoglobin can carry 4xOxygen Molecules.
- Transports O₂ from Lungs → Tissues (Note: Also Transports CO₂ from Tissues → Lungs)

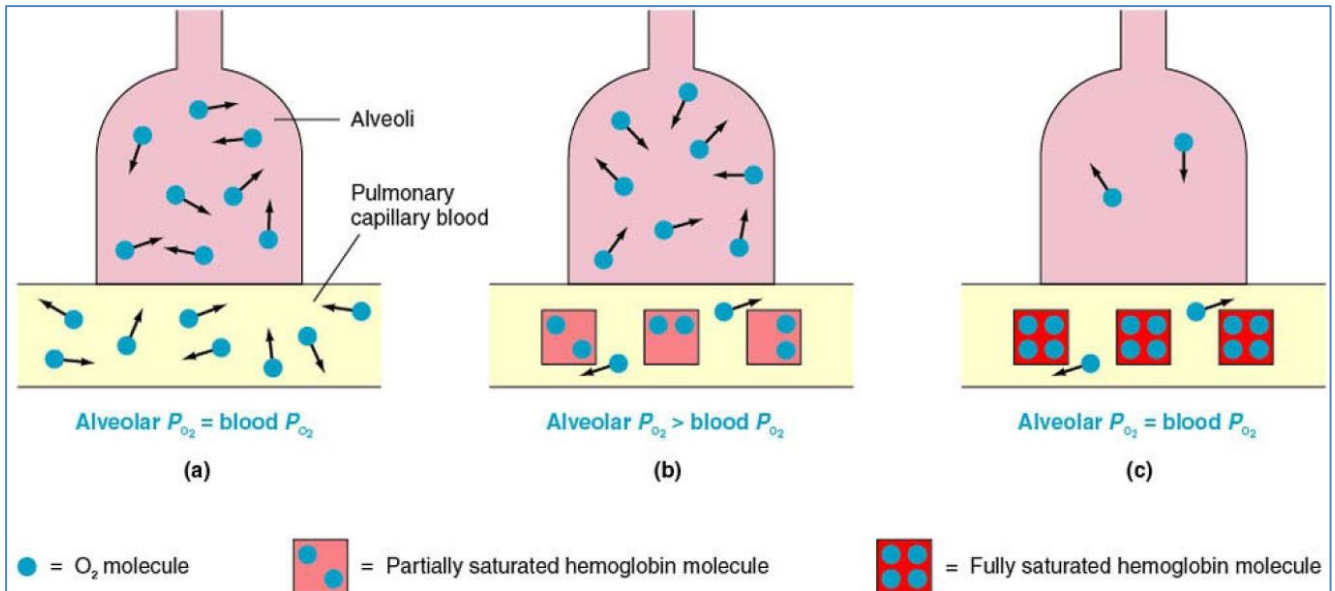


- Factors Altering Hb Affinity for O₂:

- **Things Changing its Shape/Functional Properties:**
 - Hb Saturation: % of Heme units containing bound O₂
 - Therefore also P_{O2}
 - P_{CO2}
 - Blood pH
 - Temperature
 - 2,3-BisPhosphoGlycerate (or DPG – disphosphoglycerate) (By-product of Glycolysis.)

- The Physics Behind Hb's Function:

- **1. Greatly Increases O₂-Carrying Capacity of Blood:**
 - By binding O₂, Hb effectively removes the dissolved O₂ from solution.
 - Acts as an O₂ buffer
 - → More of the Alveolar O₂ can diffuse into the blood (→ & Haemoglobin) before the *Partial Pressure Gradient* is equalized.
 - Hence, *Blood-O₂ Content* = *Dissolved O₂* + *Hb-Bound O₂*
 - As P_{O2} ↑, %Hb-saturation ↑



○ **2. Binds O₂ Co-Operatively:**

- The more O₂ Molecules bound to Hb, the *easier* it becomes to bind another. (up to 4)

- Due to Hb's conformational change between **2 States (isoforms)**:

○ **T-State (Tense):**

- Low O₂-Hb Saturation
- Low affinity for O₂

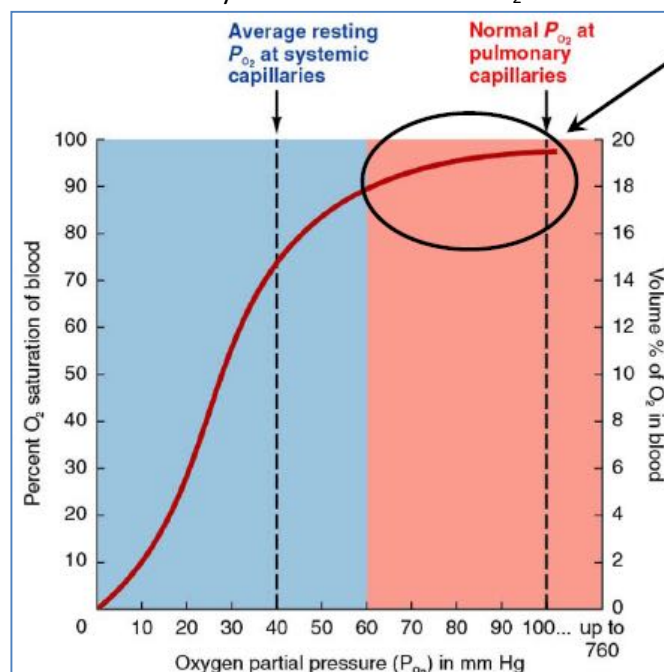
○ **R-State (Relaxed):**

- High O₂-Hb Saturation
- High affinity for O₂

○ **3. O₂-Hb-Dissociation Curve:**

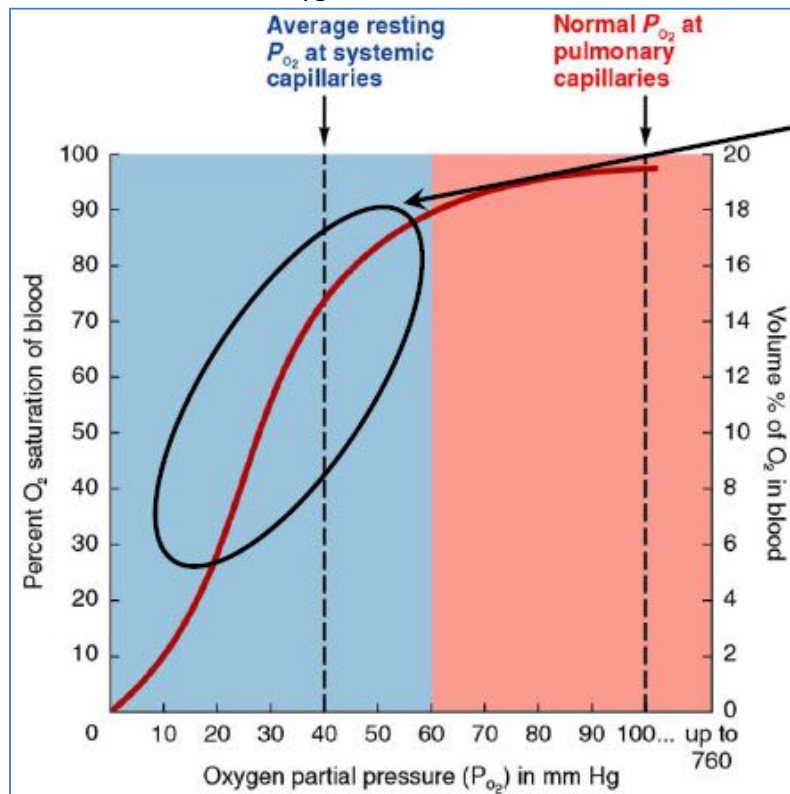
▪ **Plateau Region (O₂ Loading Zone):**

- In the lungs (P_{O_2} = high)
- The P_{O_2} Range where pulmonary capillaries are *Loaded* with O₂.
- Note: Normal P_{O_2} in pulmonary capillaries $\approx$ 100mmHg, however the plateau region extends way below that (to $\approx$ 60mmHg).
 - This allows blood from lungs $\rightarrow$ Systemic circulation $\rightarrow$ Tissues, before releasing its oxygen.
 - I.e: The Plateau = safety margin for O₂ *Carrying*.
 - Enables you to maintain blood-O₂ saturation even when P_{O_2} falls markedly.



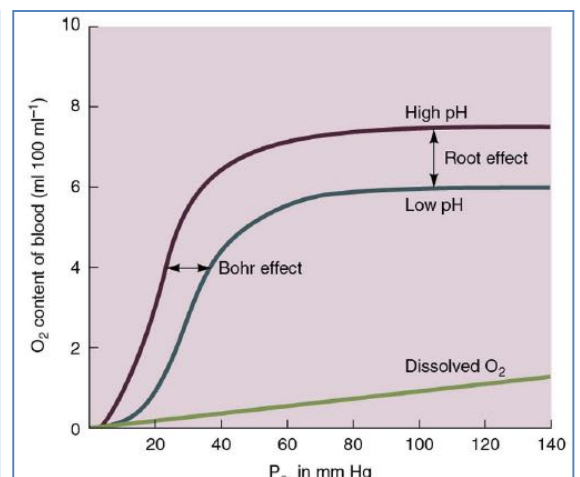
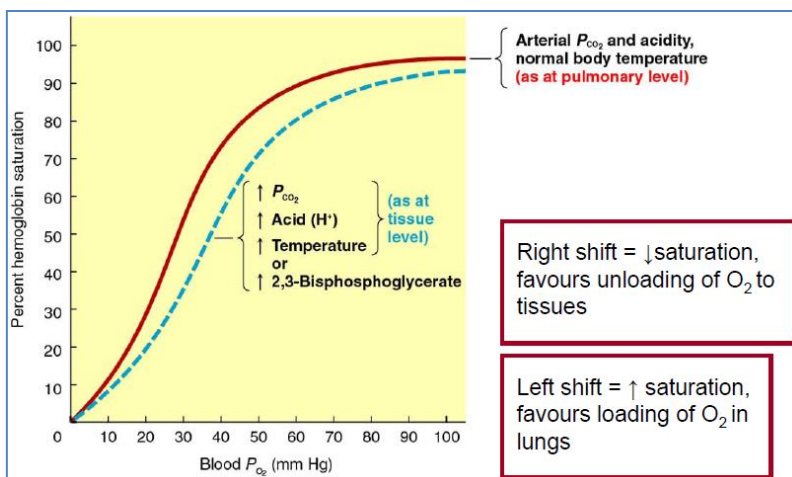
▪ **Steep Region (O₂ Un-Loading Zone):**

- In Systemic Capillary Beds (P_{O_2} = low)
- The P_{O_2} Range where Capillary beds *Unload* their O₂ → Tissue cells.
- Note: As soon as P_{O_2} drops below ≈ 60 mmHg, Hb begins to 'Dump' its O₂.
 - I.e: Small $\downarrow$ in Capillary P_{O_2} → Large $\downarrow$ in Blood-O₂ Saturation.
 - Allows Oxygenated blood → Pass O₂ to Metabolizing Tissues.



▪ ***Shifting The Curve:**

- *Remember the *Things that Change Hb's Shape/Functional Properties* (see above)
 - Such changes shift the O₂-Hb-Dissociation Curve.
- **Right Shift:**
 - Favours *Unloading* of O₂ to Tissues
 - Reduces Hb's Affinity for O₂ → Stabilises 'T-Conformation'.
 - Hb-Saturation Decreases
 - **Causes:**
 - $\uparrow$ Temperature (eg. exercising muscles)
 - $\uparrow$ DPG (from Glycolysis)
 - $\uparrow P_{CO_2}$ (causes $\uparrow$ Carbonic Acid → $\downarrow$ affinity for O₂) → Bohr Effect
 - $\uparrow$ Acid (H^+) ($\downarrow$ ability (not affinity) to bind O₂) → Root Effect



- **Left Shift:**
 - Favours *Loading* of O_2 to Tissues
 - Increases Hb's Affinity for O_2 → Stabilises 'R-Conformation'.
 - Hb-Saturation Increases.
 - **Causes:**
 - Opposites of Above
- **Way to Remember:**
 - Curve shifts to *Right* in an *Exercising Muscle*.

An exercising muscle is:

- hot
- acidic (lactic acid)
- has high P_{CO_2}
- undergoing rapid glycolysis (lots of DPG)

• Factors that favour a right shift:

- Increasing temperature
- Decreasing pH
- Increasing P_{CO_2}
- Increasing DPG

Opposite of these will
Left shift the curve

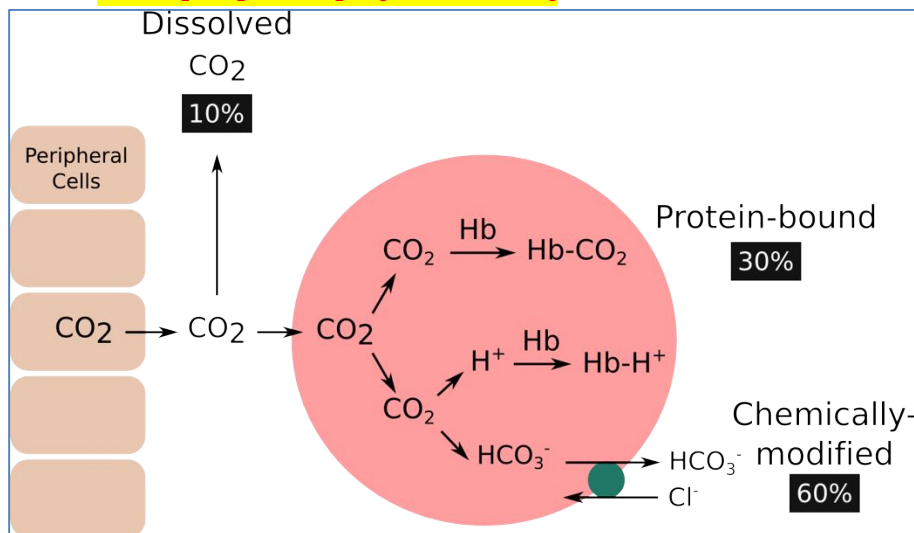
Right shifts favours unloading

Left shift favour loading

Opposites of these factors → Left Shift

Mechanisms of CO₂ Transport

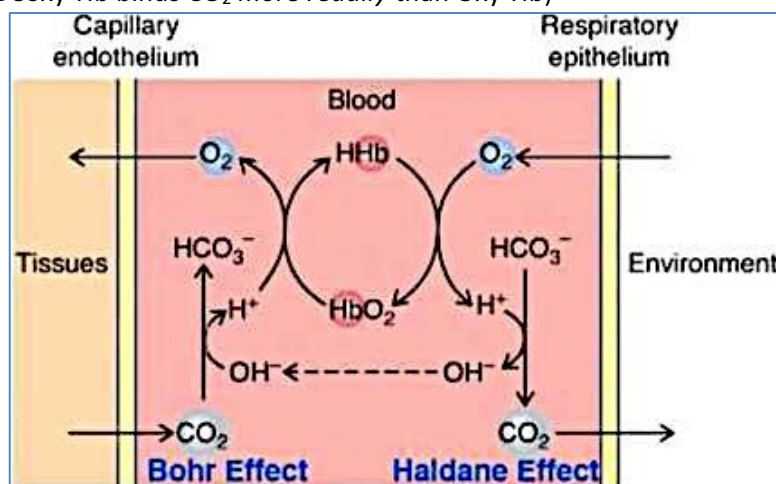
- CO₂ produced by metabolising cells
- Produced in Mitochondria
- Diffuses into blood.
- **3 Routes To The Lungs:**
 - **1. Dissolved In Plasma:**
 - Tissue CO₂ → Dissolved Plasma CO₂ → Pulmonary Capillaries → Diffusion to Alveoli
 - 5-10% of Total Body-CO₂
 - **2. Bound to Hb:**
 - Tissue CO₂ → Dissolved RBC CO₂ → CO₂ + Hb → HbCO₂ → Pulmonary Capillaries (P_{CO2} ↓ as dissolved CO₂ diffuses to Alveoli) → Dissolved RBC CO₂ → Diffusion to Alveoli
 - 25-30% of Total Body-CO₂
 - Note: at a different site to O₂
 - **3. In Bicarbonate-Ion Form:**
 - Tissue CO₂ → Dissolved RBC CO₂ → H₂CO₃ → HCO₃⁻ → Exits RBC to Plasma → Pulmonary Capillaries → Re-Enters RBC from Plasma → Dissolved RBC CO₂ → Diffusion to Alveoli
 - 60-70% of Total Body-CO₂
 - Converted to Bicarb by **Carbonic Anhydrase:**
 - $\text{CO}_2 + \text{H}_2\text{O} \leftrightarrow \text{H}_2\text{CO}_3 \leftrightarrow \text{H}^+ + \text{HCO}_3^-$



<https://www.seekpng.com/ima/u2t4o0e6r5y3o0o0/>

Factors Altering CO₂ Transport Efficiency:

- **Bohr Effect:**
 - Not only does ↑ P_{CO2} cause ↑ Carbonic Acid → ↓ affinity for O₂ → Unloading of O₂.....
- **Haldane Effect:**
 - But Unloading of O₂ also Favors binding of CO₂.
 - (Deoxy-Hb binds CO₂ more readily than Oxy-Hb)



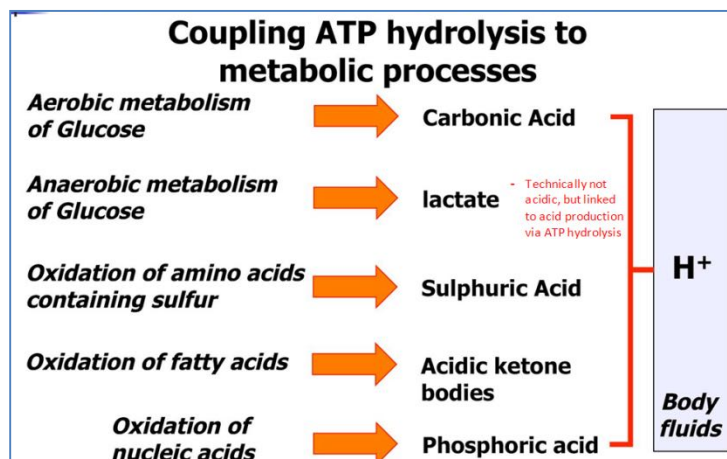
Source: Unattributable

BODY ACID-BASE BALANCE

BODY ACID-BASE BALANCE

Acid Production:

- The Body turns over up to 150 Moles of H^+ per day – THAT'S A LOT!!
- Where does it come from?
 - o Metabolic Processes:
 - Most H^+ comes from *Hydrolysing ATP* (Ie: *Aerobic Metabolism*)
 - $ATP + H_2O \rightarrow ADP + P_i + H^+$
 - Note: The Body turns over ≈ 40 kg of ATP *per day*!
 - Much H^+ also comes from:
 - Anaerobic Glucose Metabolism
 - Amino Acid Metabolism
 - Fatty Acid B-Oxidation.
 - Nucleic Acid Metabolism.



- Despite LOADS OF H^+ produced, *Body pH is Finely Regulated*.
 - o Ie: Very small pH changes observed in body.

What is pH?:

- pH = the *negative log* of the H^+ ion concentration.
- $pH = -\log [H^+]$
- Remember:
 - o **Acid:** Proton (H^+) Donor
 - o **Base:** Proton (H^+) Acceptor

Physiological pH Values:

- **Arterial pH = 7.40**
 - o Note: pH of < 6.9 can be lethal
- **Venous pH = 7.35** - more acidic due to *higher HCO_3^-* (Ie: Higher P_{CO_2})
- **Urine pH = 4.5 to 8.0**
- **Stomach pH = 0.8** - requirement of chemical digestion & activation of digestive enzymes.
- **Bile pH = 7.8 to 8.6** - needs to be alkaline to break down fats.

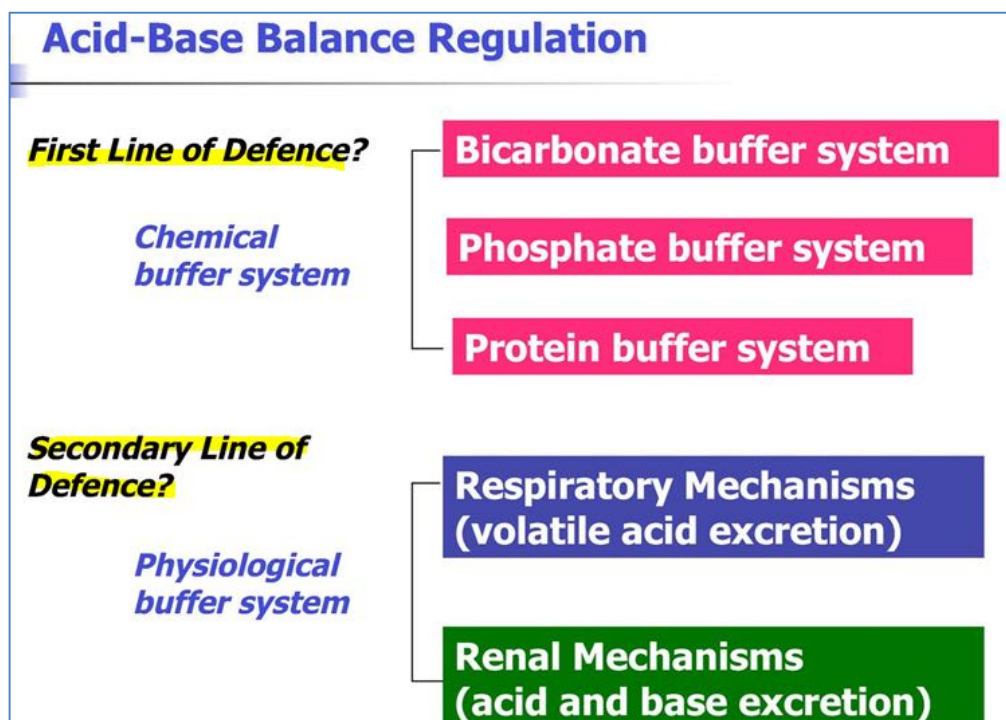
Acid-Base Homeostasis Regulated By:

- Buffers:

- **What are they?:**
 - Solutions of A Weak Conjugate Acid & A Weak Conjugate Base.
 - Resist changes in pH
- **pK of A Buffer:**
 - Mathematically → The $-\log$ of the Equilibrium Constant ($K_{eq} = \frac{[Products]}{[Reactants]}$)
 - The pH of the Buffer Solution where both the Conjugate Acid & Base are at **50% dissociation**.
 - It is the pH that the Buffer Solution wants to be at.
 - Hence – yields Max. Buffering Power.
 - I.e: If an experiment required a pH of 7.4, you would conduct it in a buffer of $pK=7.4$
- **Isohydric Principle:**
 - There are many H^+ buffers in the body...
 - All *Buffers* in a *Common Solution* are in *Equilibrium* with the *Same $[H^+]$*
 - Thus, All Buffers in the Body Work Together.
 - **Isohydric Principle =**
“When $[H^+]$ Changes, the *Balance* of all buffer systems changes at the same time.”

- Acid-Base Balance Lines of Defence:

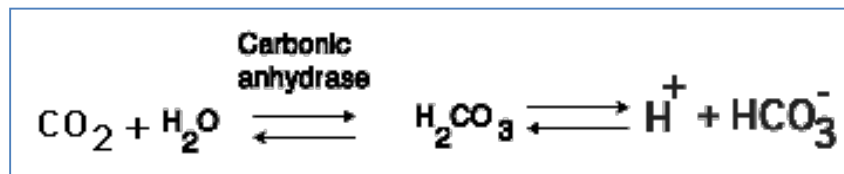
- **1. Chemical Buffer Systems:**
 - #1. Bicarbonate Buffer System
 - Phosphate Buffer System
 - Protein Buffer System
- **2. Physiological Buffer Systems:**
 - Respiratory Mechanisms
 - Renal Mechanisms



-1st Line Of Defence: Chemical Buffer Systems:

- #1. Carbonic-Acid-Bicarbonate Buffer System:

- The most important Body Buffer System
- Occurs **within the Red Blood Cell**
 - **Carbonic Anhydrase** (in RBC) catalyses: $(\text{CO}_2 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{CO}_3)$
- Operates **in conjunction with the respiratory system.**
 - I.e: Blowing off CO_2 shifts eq. To the left $\rightarrow$ Less $[\text{H}^+]$ $\rightarrow$ pH increases.
- **Clinical Assessment of Acid/Base:**
 - **3 Factors Required:**
 - 1. Blood pH
 - 2. Blood P_{CO_2}
 - 3. Plasma Bicarbonate
- **When the ratio of $[\text{HCO}_3^-]/[\text{H}_2\text{CO}_3] = 20:1$, The blood pH will be normal = pH 7.4**
 - I.e: The [Bicarbonate] : [Carbonic Acid] = 20:1
 - I.e: The [Bicarbonate] : [Carbon Dioxide] = 20:1
 - **Changing this ratio – Changes Blood pH:**
 - **pH $\uparrow$ When:**
 - [Bicarbonate] $\uparrow$ (Pushes Equation to the Left)
 - [Carbon Dioxide] $\downarrow$ (Pushes Equation to the Left)
 - **pH $\downarrow$ When:**
 - [Bicarbonate] $\downarrow$ (Pushes Equation to the Right)
 - [Carbon Dioxide] $\uparrow$ (Pushes Equation to the Right)

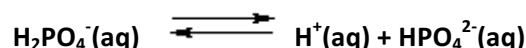


Simplified Equation



- #2. Phosphate Buffer System:

- Second most important Body Buffer System
- Operates in the internal fluid of all cells.



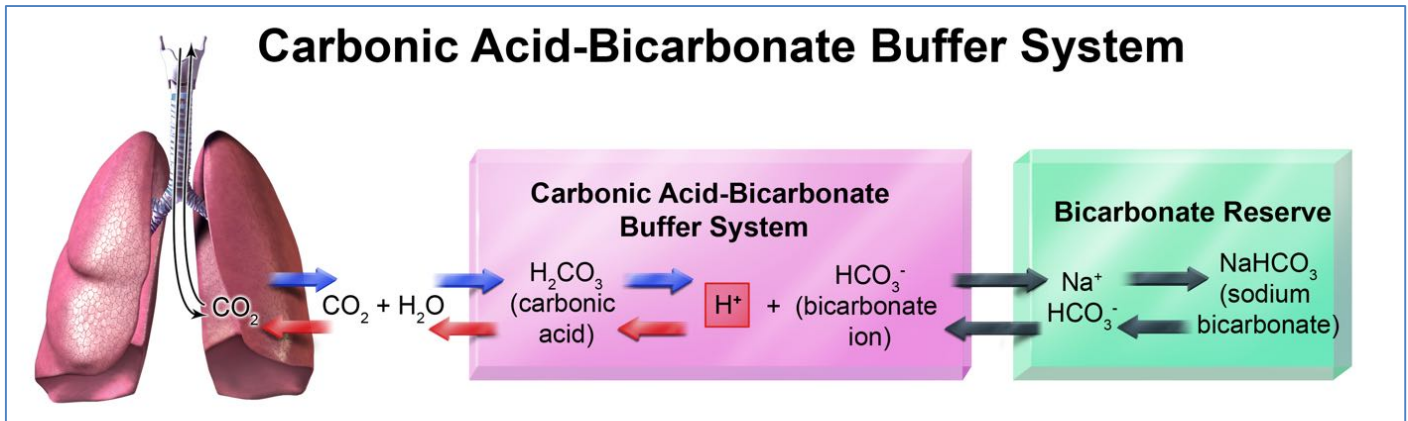
- #3. Protein Buffers (in RBCs & Intracellular Buffers)

- Both intracellular and extracellular **proteins have negative charges and can serve as H^+ buffers.**
- However, because most proteins are inside cells, this primarily is an **intracellular** buffer system.
 - Eg. Haemoglobin (Hb) is an excellent intracellular buffer because of its ability to bind H^+ .
 - Forms a *weak acid + carbon dioxide (CO_2)*.
 - After O_2 is released (in the peripheral tissues), Hb binds CO_2 and H^+ ions
 - As blood reaches the lungs these actions reverse themselves $\rightarrow$ Hb binds O_2 , releasing the CO_2 and H^+ ions
 - The H^+ combines with bicarbonate (HCO_3^-) $\rightarrow$ carbonic acid (H_2CO_3). The H_2CO_3 breaks down to form water (H_2O) and carbon dioxide (CO_2) which are excreted via expiration through the lungs. Therefore respirations help maintain pH.

- 2nd Line Of Defence: Physiological Buffer Systems:

- Respiratory System – Short Term:

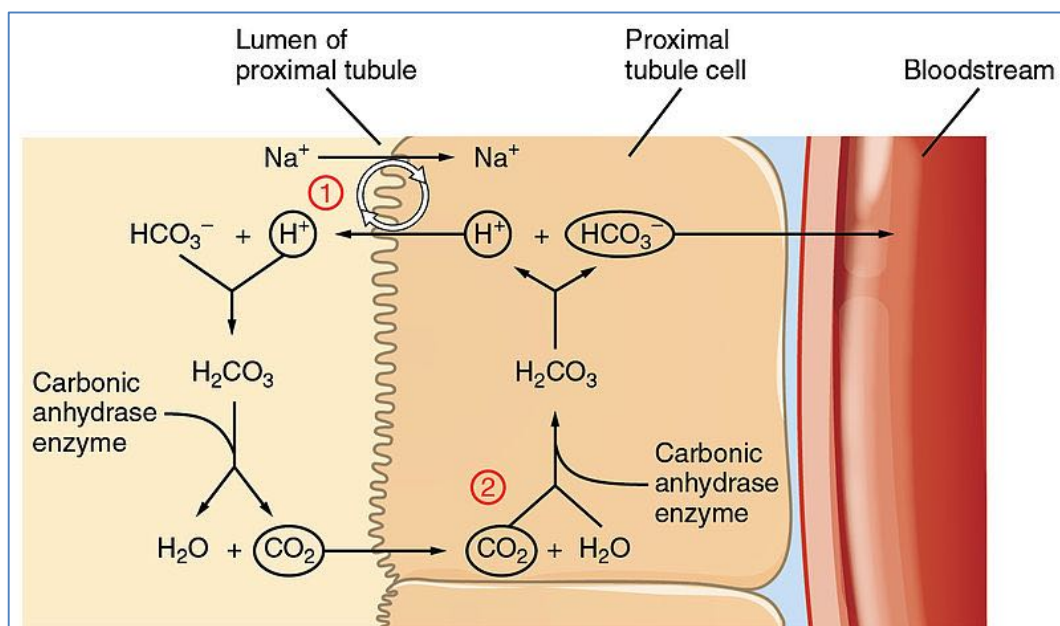
- (CO₂ Excretion)
- CO₂ constantly produced during Metabolic Processes
- Eliminated by lungs.
- If not eliminated from body, pH would quickly become *Acidic* (Bicarb-Buffer Eqn. Shifts to Right)
- **CO₂: The Controller Of Ventilation:**
 - CO₂ is the main controller because H⁺ can't cross the *Blood-Brain-Barrier*.
 - **ΔP_{CO2} → ΔpH of Cerebro-Spinal Fluid → Sensed by Medulla (respiratory centres) → ΔResp's**
 - ↑P_{CO2} Increases Ventilation Rate + Depth (eg. Exercise)
 - ↓P_{CO2} Decreases Ventilation Rate + Depth (eg. After Hyperventilating)



BruceBlaus, CC BY-SA 4.0 <<https://creativecommons.org/licenses/by-sa/4.0>>, via Wikimedia Commons

- Kidneys – Long Term:

- **Kidneys Control Acid/Base by excreting either:**
 - Acidic Urine
 - Basic Urine
- **Mechanism:**
 - **HCO₃⁻ Filtered → Renal Tubules → Combined with H⁺ → Carbonic Acid → H₂O + CO₂ → Blood**
 - **H⁺ Filtered → Renal Tubules → Combines with HCO₃⁻ → Carbonic Acid → H₂O + CO₂ → Blood**
→ Combines with HPO₄²⁻ or NH₃ → Excreted in Urine.
 - In Short:
 - Carbonic Acid is recovered → CO₂ & H₂O → Blood
 - Ammonium & Hydrogen Phosphate → Excreted in Urine.



OpenStax College, CC BY 3.0 <<https://creativecommons.org/licenses/by/3.0>>, via Wikimedia Commons

Metabolic Vs. Respiratory pH Disturbances:

- Metabolic –

○ – Acidosis:

- Due to ↓[HCO₃]
- (Due to inability of the body to form bicarbonate (HCO₃⁻) in the kidney)
- (Or, Due to Lactic/Keto Acid build-up)

○ – Alkalosis:

- Due to ↑[HCO₃]
- (Due to Loss of H⁺ in Urine or Vomiting)
- (Or, Due to Retention of Bicarbonate (HCO₃))

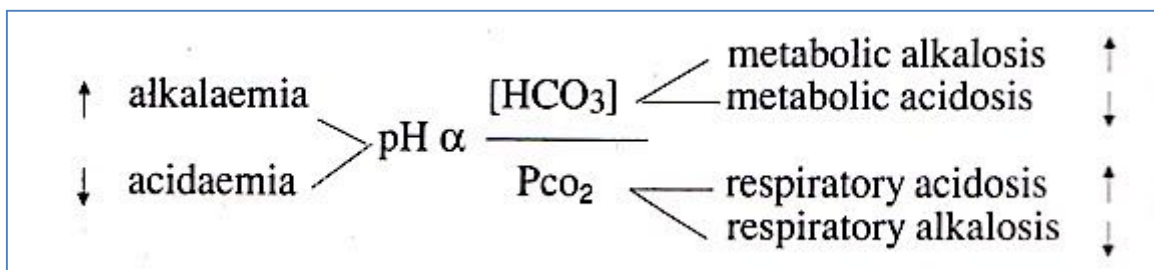
- Respiratory –

○ – Acidosis:

- Due to ↑P_{CO2}
- (Due to decreased ventilation of the pulmonary alveoli, →elevated P_{CO2}).

○ – Alkalosis:

- Due to ↓P_{CO2}
- (Due to increased alveolar respiration (hyperventilation) →decreased plasma [CO₂])



- Compensatory Mechanisms:

- In either Metabolic or Respiratory Acidosis/Alkalosis, the compensatory mechanism will always be the other system.
 - I.e: If Metabolic Acidosis, the Compensatory Mech. Will be the Respiratory System (vice versa)
- **Note: Regulation of breathing** – normally via P_{CO2} (because H⁺ can't cross Blood-Brain Barrier). However, in *Metabolic Acidosis*, the P_{CO2} is already lower than normal (due to right-shift in equilibrium) and therefore can't stimulate breathing. *Instead*, the Primary Factor would be *Blood pH* on *Peripheral Chemoreceptors*.

Anion Gap:

- What is it?:

- The Difference between Plasma Concentrations of *Measured* Anions & Cations, Minus *Unmeasured* Cations
- Anion Gap = (Na⁺ – (HCO₃⁻ - Cl⁻)) – Unmeasured Cations

- Background Info:

- Plasma concentrations of Anions & Cations must be *Equal* – to maintain electrical neutrality.
- However, only certain ions (Na⁺, Cl⁻ & HCO₃⁻) are measured in the Lab.
- Therefore, there are other *Unmeasured Ions* that contribute charge to the solution, resulting an 'imbalance' in the Anion Gap Equation.
 - This magnitude of this 'Imbalance' can indicate concentrations of *Unmeasured Ions*:
 - Eg. Lactic Acid
 - Eg. Keto Acids

- An important tool for Evaluating The Type of Metabolic Acidosis:

- **If Patient is in Metabolic Acidosis, and Their Anion Gap is Normal:**
 - The ↓[HCO₃⁻] has likely been balanced by ↑[Cl⁻]
 - Therefore – *Hyperchloraemic* Metabolic Acidosis
- **If Patient is in Metabolic Acidosis, and Their Anion Gap is High:**
 - Due to ↑[Lactic Acid] → *Lactic Acidosis*.....OR
 - Due to ↑[Keto Acids] → *Keto Acidosis*

CONTROL OF BREATHING

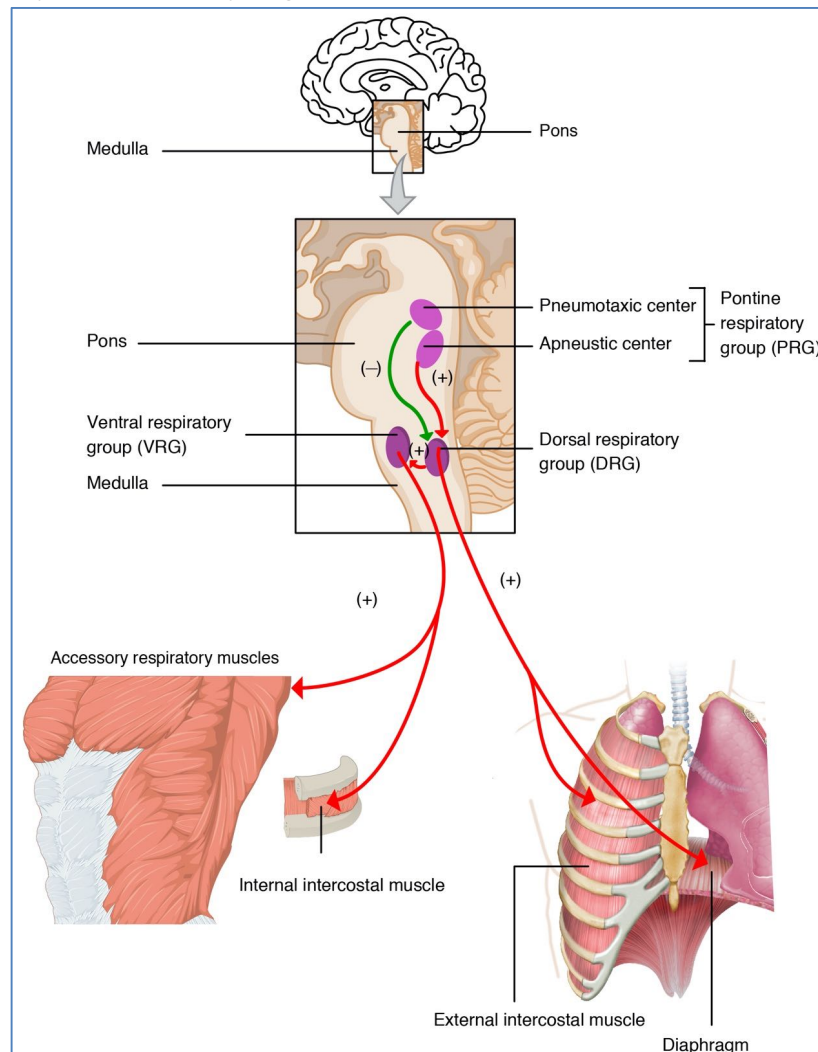
CONTROL OF BREATHING

Control Of Breathing:

- **Why?**
 - To Match Alveolar Ventilation with Metabolic Demand.
 - Keeps Arterial P_{O_2} & P_{CO_2} Very Stable – Even During Severe Exercise.
 - Note: *Venous* gas levels do change markedly.
- **Where?**
 - “Respiratory Centre” of the Brain. (Medulla & Pons of Brainstem)
- **What it Does:**
 - Produces *Uninterrupted Rhythmic* Breathing Activity Throughout Life.
 - Automatically Adjusts Breathing to meet Changing Demands.

Overview of Central Regulation:

- **Input From:**
 - Chemoreceptors (Chemosensitive area of Medulla, Aortic Bodies, Carotid Bodies) +
 - Mechanoreceptors (Stretch)(Lung)
- **The Respiratory Centre:**
 - Medulla
 - DRG – Dorsal Respiratory Group
 - VRG – Ventral Respiratory Group
 - Pons
 - PRG – Pontine Respiratory Group
 - Cerebral Cortex
 - Voluntary Override
- **Output To:**
 - Respiratory Muscles – (Diaphragm, Intercostals, Abdominals)



The Respiratory Centre (Brainstem – Medulla/Pons):

- Medulla:

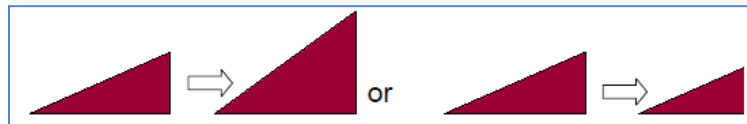
○ ****Dorsal Respiratory Group (DRG):**

▪ **Location:**

- Bilaterally in Dorsal Portion of Medulla
- In Close Relation to the '*Nucleus Tractus Solitarius*'.

▪ **Main Function = INSPIRATION:**

- Initiates the *Respiratory Rhythm*.
- Determines *Timing* of Respiratory Cycle
- Via "*Inspiratory Ramp Signals*":
 - (Mechanism Unclear)
 - Signal isn't On/Off
 - Instead it Starts Weak, then Steadily *Increases* for ≈ 2sec.
 - Signal Ceases for ≈ 3sec. → Allows passive Exhalation. (Elastic Recoil)
 - **Signal Can Be Modified** to Alter *Speed & Rate* of Inspiration:
 - ↑Speed – By Increasing Ramp-Signal *Gradient*.
 - ↑Rate – By Terminating Ramp-Signal Early. (Pneumotaxic Centre)



▪ **Input From:**

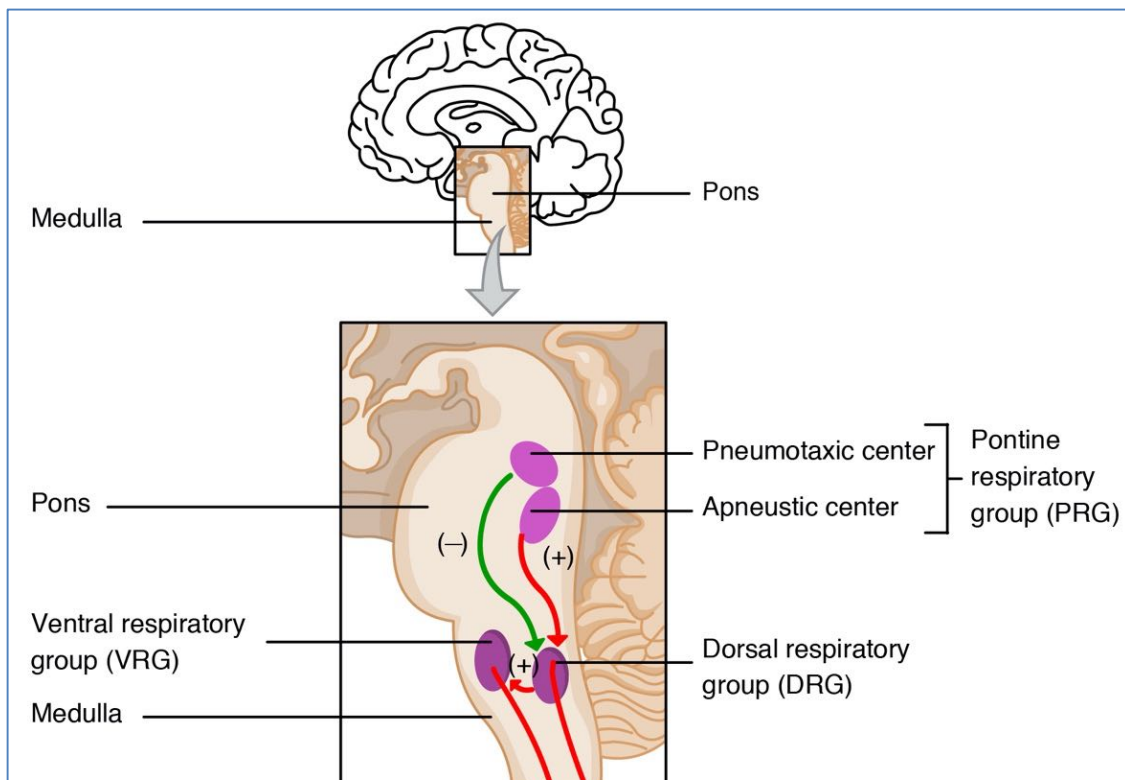
• **Peripheral Chemo/Baro/Mechano-Receptors:**

- Glossopharyngeal Nerves (CN-IX)
- Vagus Nerves (CN-X)

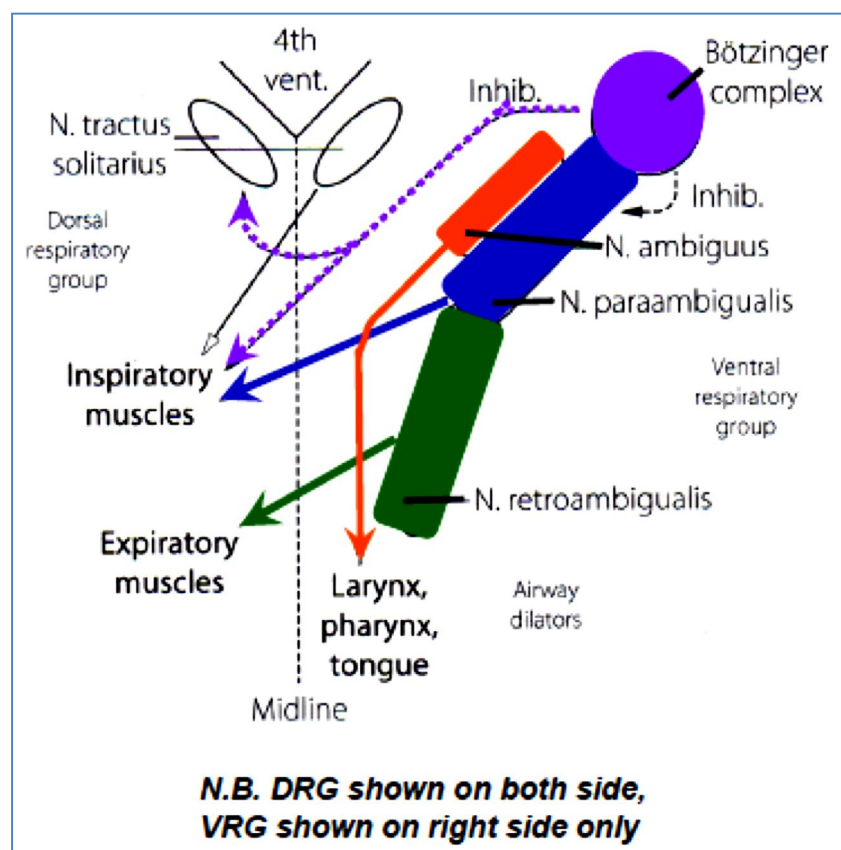
▪ **Output From DRG:**

• **Motor Neuron Cell Bodies (in Spinal Cord):**

- Impulse → Phrenic Nerve (C3-C5) → Diaphragm
- Impulse → Intercostal Nerves (T1-T11) → Intercostal Muscles

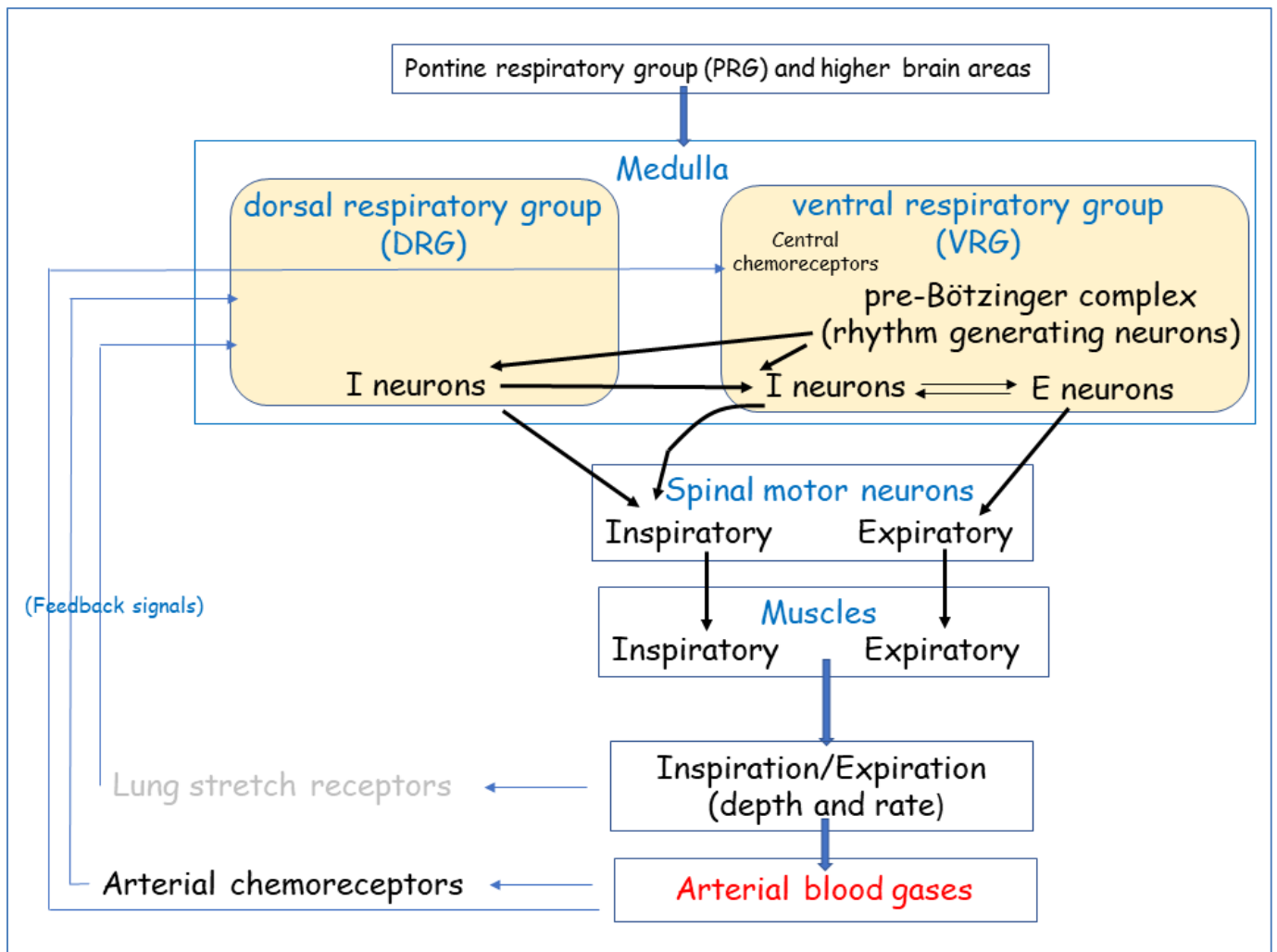


- **Ventral Respiratory Group (VRG):**
 - **Location:**
 - In Each Side of Medulla, Lateral to DRG.
 - **Main Overall Function = EXPIRATION_(FORCED):**
 - ****Important for Active Expiration**
 - – (Abdominal Muscles + Internal+Innermost Intercostals).
 - Inactive during Normal Quiet Breathing.
 - Has NO ROLE in Respiratory Rhythm
 - Contributes *Extra* Respiratory Drive during ↑↑Demand.
 - Inspiratory &
 - Expiratory
 - **4 Functional Parts:**
 - **Botzinger Complex:**
 - Both Inspiratory & Expiratory Functions (Much Unknown)
 - **Nucleus Ambiguus:**
 - Controls Airway Patency
 - (Dilator Functions of Larynx, Pharynx & Tongue)
 - **Nucleus Paraambigualis:**
 - Contraction of *Inspiratory Muscles* on Opposite side of body.
 - ****Nucleus Retroambigualis:**
 - Contraction of *Expiratory Muscles* on Opposite side of body.



Source: Unattributable

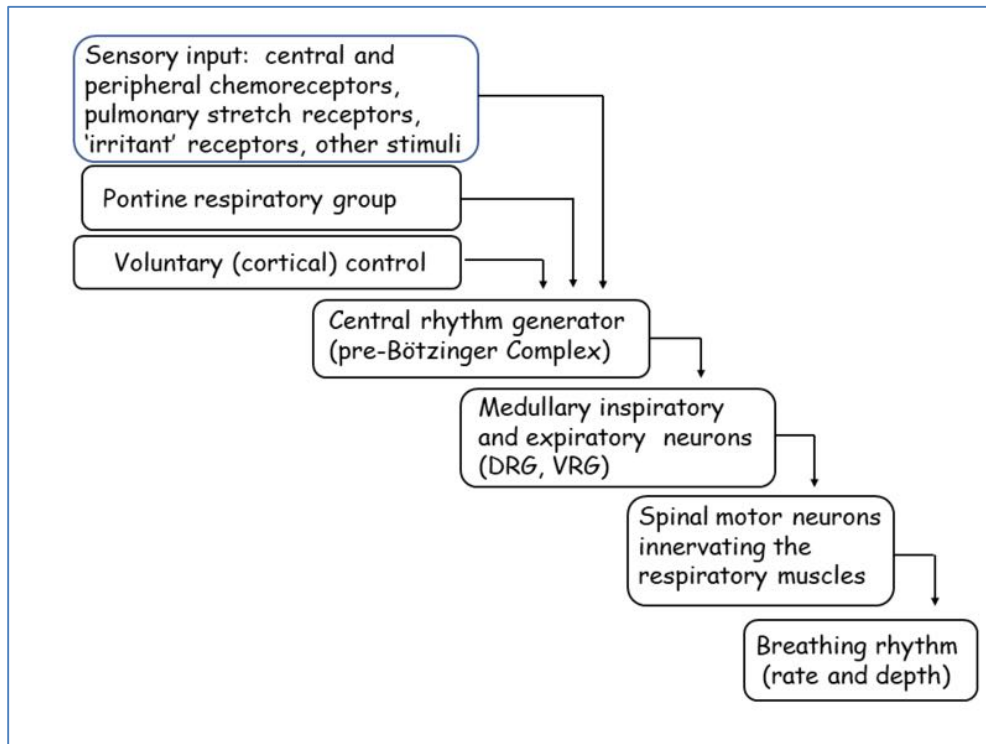
- **Pons:**
 - **Pontine Respiratory Group (PRG):**
 - **2 Parts:**
 - **Pneumotaxic Centre:**
 - **Location:**
 - Upper Pons
 - **Function:**
 - Controls “Off-Switch” for DRG-Inspiratory-Ramp-Signal.
 - Ie: Terminates *Ramp-Signal* Early
 - - **Limits Inspiration
 - - Increases Breathing Frequency
 - - Decreases Tidal Volume
 - **Apneustic Centre:**
 - **Location:**
 - Lower Pons
 - **Function:**
 - Prolongs *Ramp-Signal* (Ie: Terminates Signal Later)
 - - **Prolongs Inspiration
 - - Decreases Breathing Frequency
 - - Increases Tidal Volume



Alan Sved, CC BY-SA 4.0 <<https://creativecommons.org/licenses/by-sa/4.0/>>, via Wikimedia Commons

- **Cerebral Cortex:**

- **Conscious Override:**
 - Of All of the Above. (To an extent)
 - - By bypassing Medulla-Respiratory Centres → Act *Directly* on Respiratory Muscles.
- **Important For:**
 - Speech
 - Singing
 - Sniffing
 - Coughing
 - Wind Instruments
- Mechanism Is Very Complicated → Don't need to know.



Alan Sved, CC BY-SA 4.0 <<https://creativecommons.org/licenses/by-sa/4.0>>, via Wikimedia Commons

Peripheral Input:

- **Upper Respiratory Tract Reflexes:**
 - Eg. Cough/Sneeze Reflexes. – Don't Know Details
 - Receptors in Nose/Pharynx/Larynx
 - Respond to Toxins/Irritants/Temperature
- **Lung Reflexes:**
 - **Pulmonary Stretch Receptors:**
 - **Slowly Adapting Stretch-Receptors (SARs):**
 - Sensitive to Inflation/Deflation.
 - Ie: Lung-Volume Sensors
 - Located in Airways
 - **Rapidly Adapting Stretch-Receptors (RARs):**
 - Sensitive to Tidal Volume, Frequency, Or Lung Compliance.
 - Also Nociceptive & Chemosensitive.
 - ***Inflation Reflex: ("Hering Breuer Reflex"):**
 - Prevents Over-Inflation
 - - By Turning Off the Inspiratory-Ramp-Signal. (Similar to Pneumotaxic Centre)
 - - Activated in response to ↑ Pulmonary 'Stretch'
 - - Signal Carried Via VAGUS-Nerve → DRG
 - **Deflation Reflex:**
 - Prevents Lung Collapse (Over-Deflation)
 - - Stimulates Inspiration when Lung-Volume is too Low.

- **Chemical Control of Respiration:**

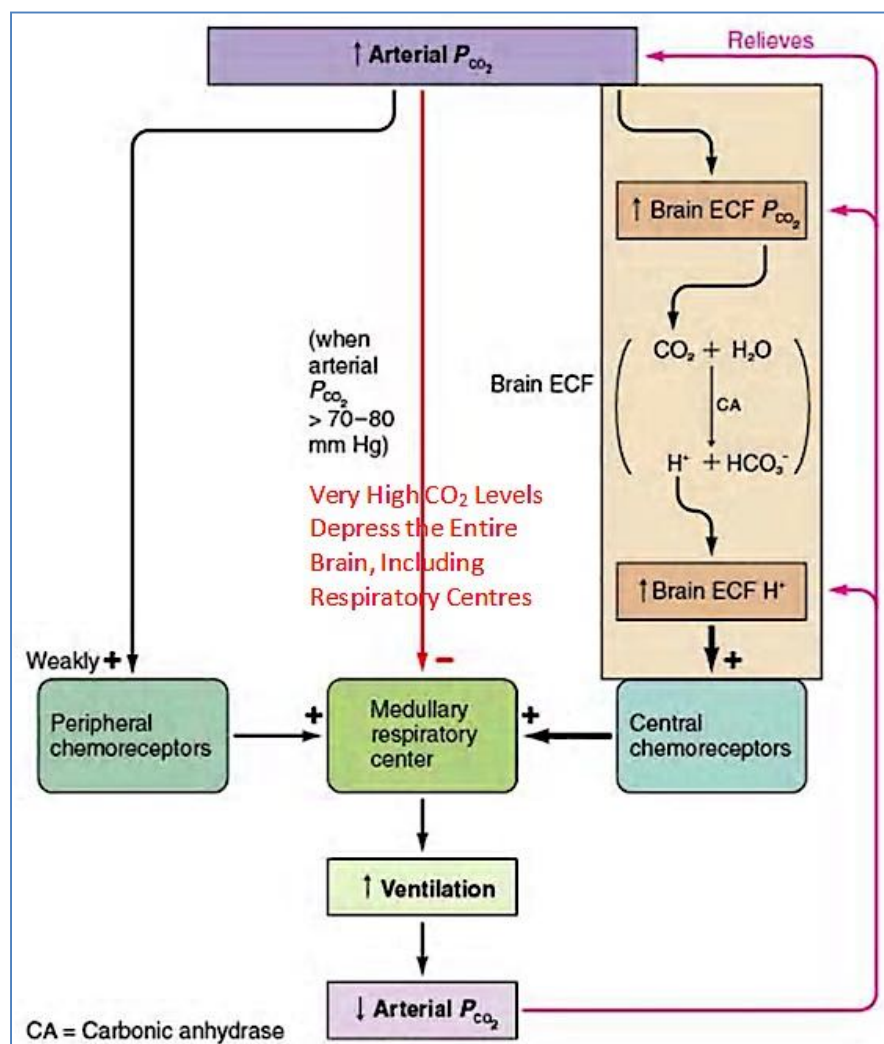
○ ***↑Arterial PCO_2 :**

▪ **Central Chemoreceptors – (Chemosensitive Area of Medulla):**

- ****↑Arterial $PCO_2 \rightarrow \uparrow CSF-[H^+]$** (Cerebro-Spinal Fluid)
- $\uparrow CSF-[H^+]$ Acts Directly on *Chemosensitive Area on Medulla*.
- $\uparrow CSF-[H^+]$ Stimulates Respiratory Centre
 - Note: H^+ = the *Only* important Stimulus, However, H^+ can't cross Blood Brain Barrier
 - Therefore, $[H^+]$ is determined by Arterial PCO_2 (which can diffuse into CSF)
 - $CO_2 \rightarrow HCO_3 + H^+ \dots$ Via the Bicarbonate-Buffer System.
- ***POTENT, BUT INDIRECT***

▪ **Peripheral Chemoreceptors – (Aortic & Carotid Bodies):**

- **↑Arterial $PCO_2 \rightarrow \uparrow Arterial-[H^+]$**
- $\uparrow Arterial-CO_2 \rightarrow HCO_3 + Arterial-H^+ \dots$ Via the Bicarbonate-Buffer System.
 - $\uparrow H^+$ Stimulates Ventilation
 - $\downarrow H^+$ Depresses Ventilation
- Very Important in Acid-Base Control $\rightarrow$ Respiratory Compensation for ΔpH .



Source: Unattributable

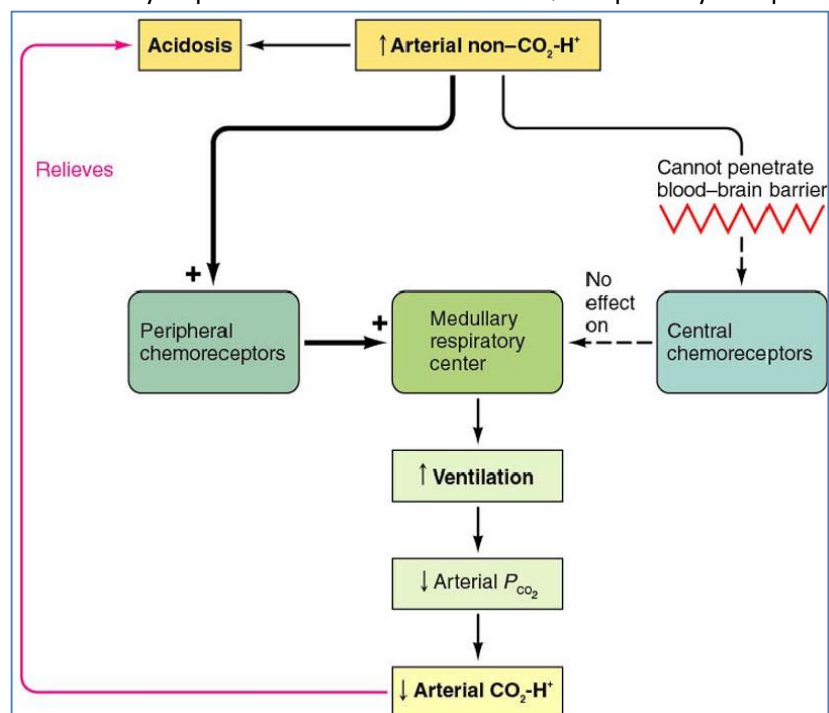
- **Arterial Non-CO₂ [H⁺]:**

- **Peripheral Chemoreceptors – (Aortic & Carotid Bodies):**

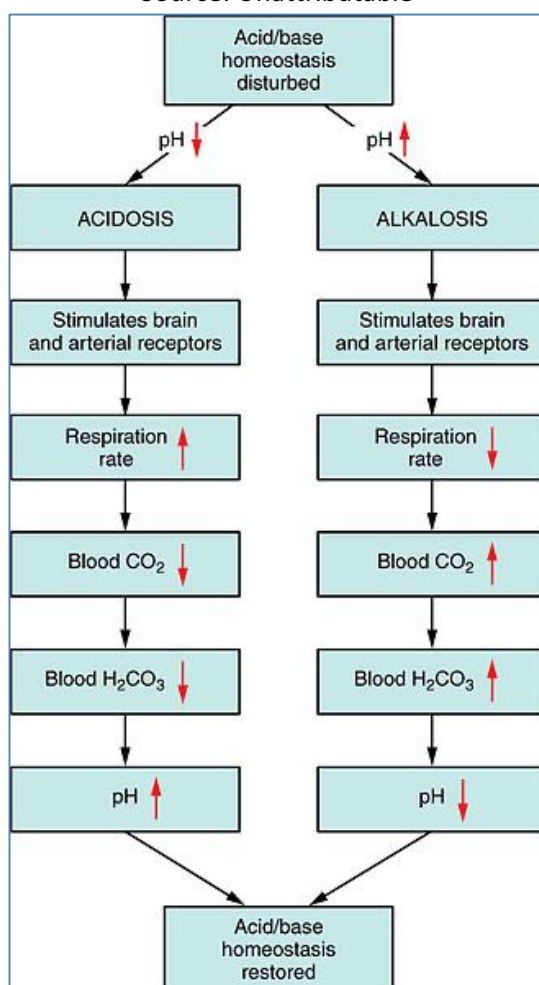
- **↑ Non-CO₂-Generated [H⁺] → ↑ Arterial-[H⁺]**

- **Note:** Non-CO₂-Generated [H⁺] = Lactic-Acid/Keto-Acids/Etc.
 - ↑ H⁺ Stimulates Ventilation
 - ↓ H⁺ Depresses Ventilation

- Very Important in Acid-Base Control → Respiratory Compensation for ΔpH.



Source: Unattributable



○ **↓ Arterial O₂:**

▪ **Peripheral Chemoreceptors – (Aortic & Carotid Bodies):**

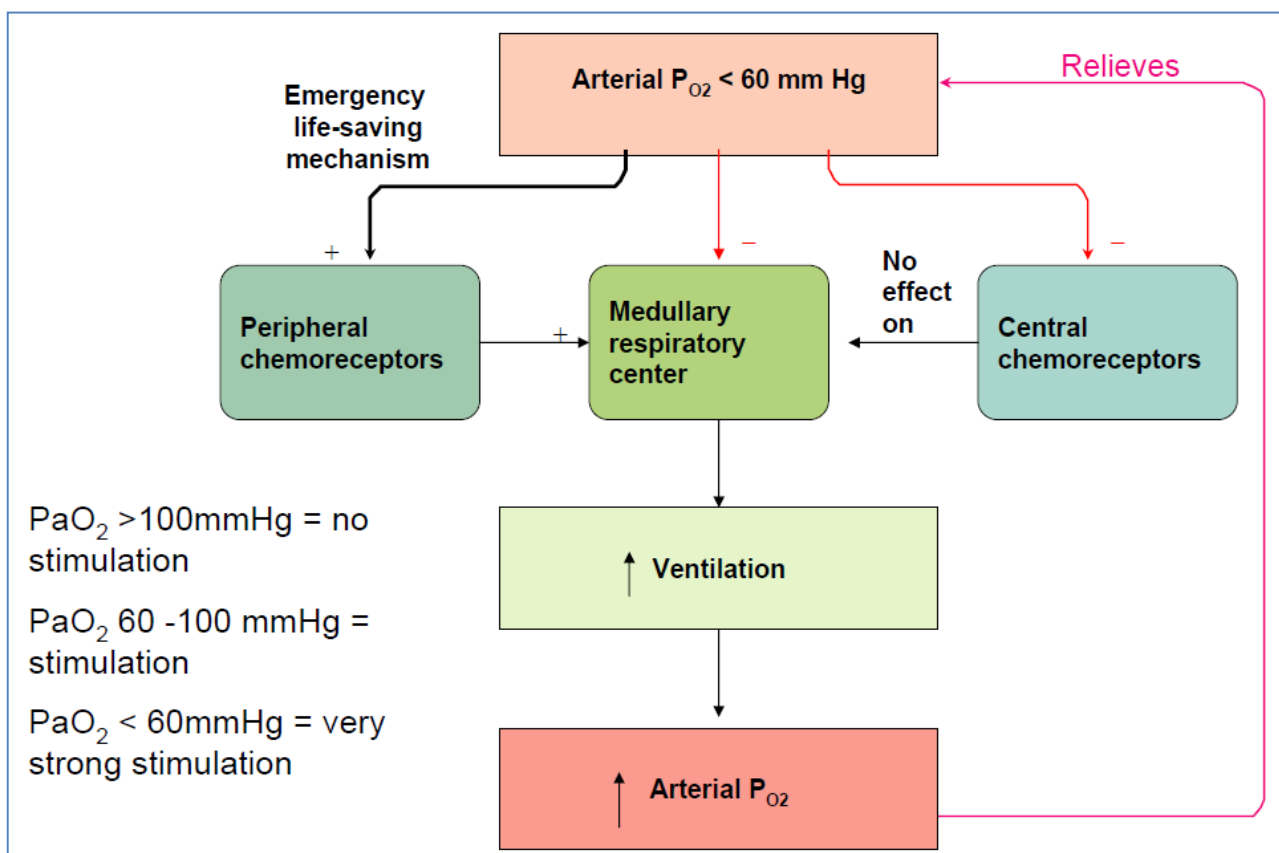
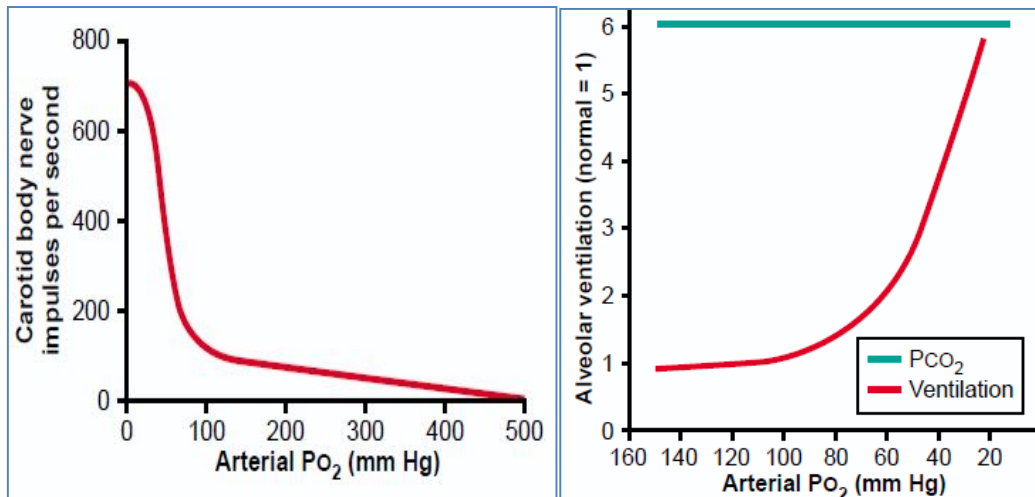
• **↓ Arterial-O₂ (to below ≈100mmHg) → Strong Respiratory Stimulation**

- Increased Breathing Rate
- Increased Breathing Depth

- Note: these receptors have their own blood supply, with *extreme* blood flow → Ensures that the receptors don't alter the gas levels of the arterial blood that they're measuring.

• **Note: Acclimatization:**

- In Low O₂ environments (mountain climbing), the Central Respiratory Centres lose sensitivity for CO₂. Therefore, Low-O₂ takes over as the #1. Respiratory Driver.



Source: Unattributable

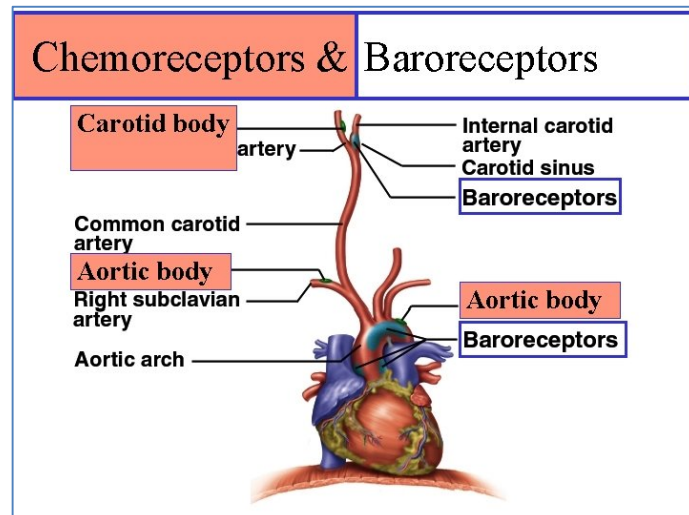
- **Mechanical Stimulation:**

○ **Baroreceptor Reflexes:**

- $\downarrow \text{BP} \rightarrow \uparrow \text{Respiratory Rate}$
- $\uparrow \text{BP} \rightarrow \downarrow \text{Respiratory Rate}$

○ **Lung 'J'-Receptors:**

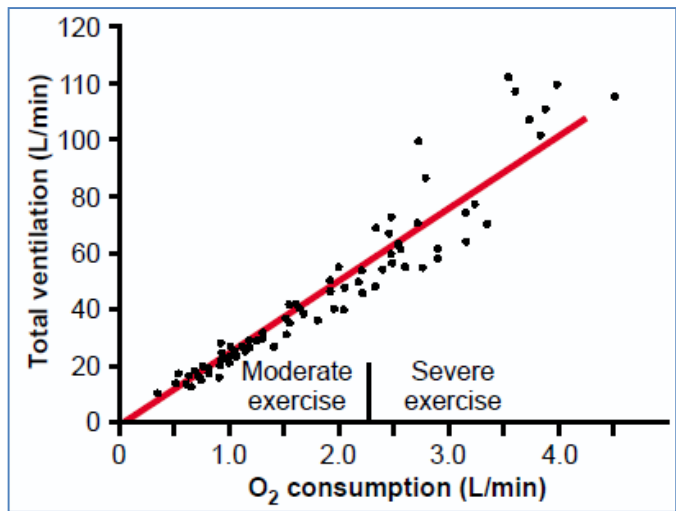
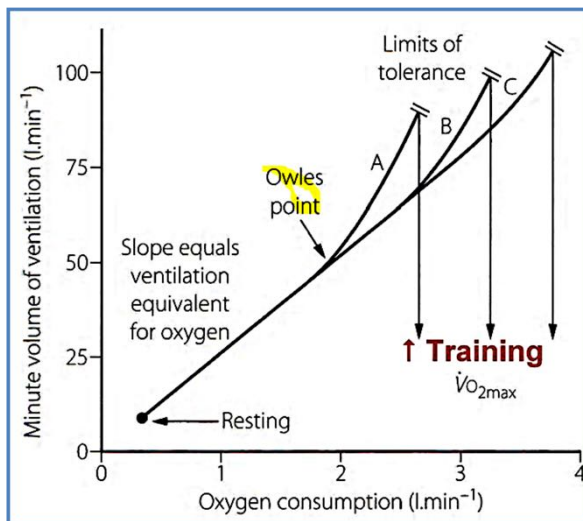
- Sensory Nerves in Alveolar Walls, Juxtaposed to the Pulmonary Capillaries.
 - (hence 'J'-receptors)
- Stimulated when Pulmonary Capillaries are engorged with blood & During Pulmonary Oedema.
- Function – Unclear!



<https://slidetodoc.com/chapter-20-blood-vessels-and-circulation-general-anatomy/>

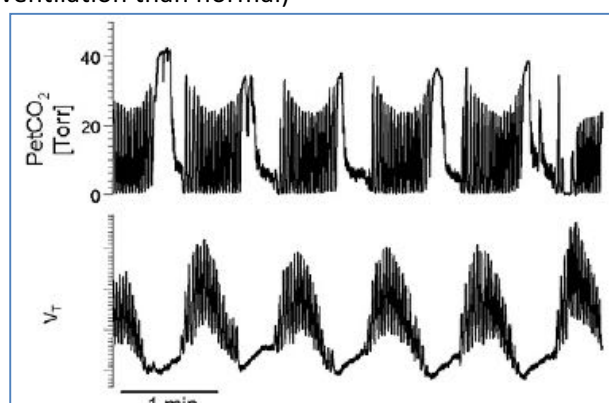
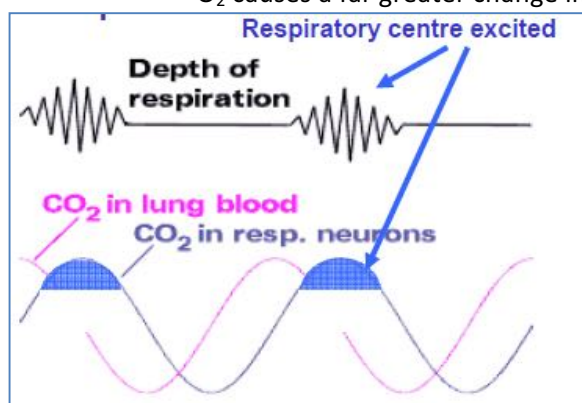
Ventilatory Response To Exercise:

- Note: Gas levels remain stable during exercise – (Ventilation is well matched to O_2 Consumption)
- **During Light-Moderate Exercise:** – Linear Relationship between O_2 Demand & Ventilation.
- **During Severe Exercise:** - O_2 Consumption Exceeds Body's ability to supply it → Anaerobic Metabolism:
 - o Lactic Acid Buildup → Lactic-Acidosis → Hyperventilation.



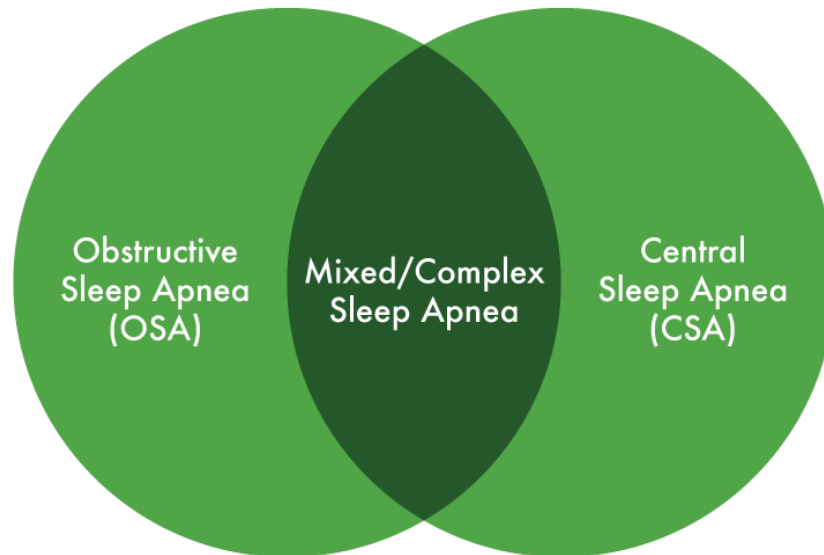
Respiratory Regulation Disorders:

- **Odine's Curse:**
 - o AKA: "Primary Alveolar Hypoventilation Syndrome"
 - o Patient has to **REMEMBER to Breathe**.
 - Due to Defect in Automatic Respiratory Control.
 - Require Ventilation when Sleeping to Keep Breathing.
- **Cheyne-Stokes Respiration:**
 - o 1. Person Overbreathes,
 - -Blowing off too much CO_2 from Pulmonary Blood
 - -At the same time, increasing O_2
 - o 2. Something delays time for oxygenated pulmonary blood to reach brain
 - (which delays inhibition of ventilation.)
 - o 3. By this time, person has already overventilated.
 - o 4. Oxygenated blood finally reaches brain, over-depressing respiratory centre.
 - o 5. Person Underbreathes,
 - - Blood CO_2 increases
 - - Blood O_2 decreases
 - o 6. This 'depleted' blood takes a while to reach brain and stimulate ventilation.
 - o 7. When this blood finally reaches brain, it over-stimulates respiratory centre → Cycle starts again.
 - o **Factors leading to Cheyne-Stokes Breathing:**
 - When a long delay occurs between transport of blood from lungs to brain. – (Eg. Cardiac Failure)
 - Increased Negative Feedback Gain in Respiratory Control areas (I.e: A change in blood CO_2 or O_2 causes a far greater change in ventilation than normal)



- **Sleep Apnea:**

- Note: Apnoea = Absence of Spontaneous Breathing
- **Obstructive Sleep Apnea:**
 - Temporary Cessations of breathing due to airway obstruction or compromised airway patency. (Respiratory effort continues)
- **Central Sleep Apnea:**
 - Temporary cessations of breathing due to lack of respiratory drive from Dorsal Respiratory Group. (No Respiratory Effort)

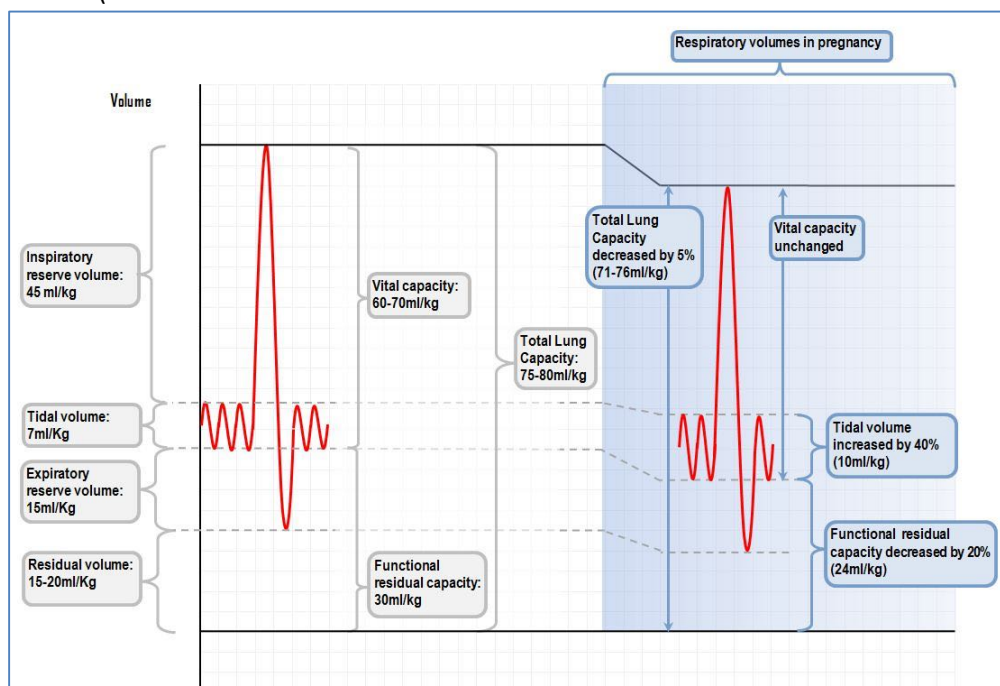


FOETAL LUNG DEVELOPMENT & TRANSITION TO EXTRAUTERINE LIFE

FOETAL LUNG DEVELOPMENT & TRANSITION TO EXTRAUTERINE LIFE

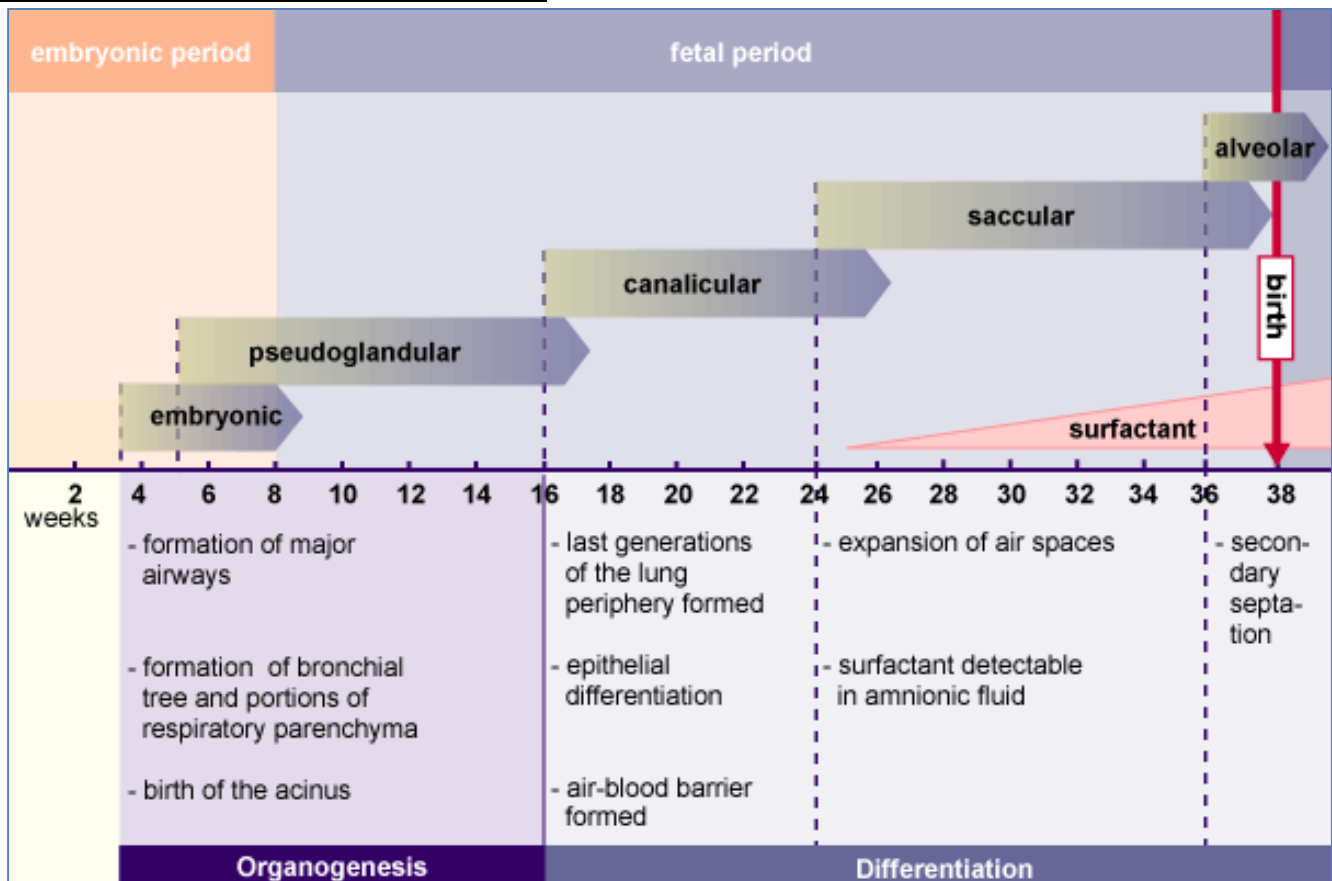
Effects of Gestation on the Maternal Respiratory System:

- **High Oestrogen Levels:**
 - Causes Fluid Retention – Increases Blood Volume
 - Causes Oedema of Airway Mucosa
 - Stimulates Mucous Gland Proliferation & Growth
- **High Progesterone Levels: (6x normal)**
 - **Hyper-Sensitizes Central (CO₂) Chemoreceptors:**
 - (I.e: Chemosensitive Area of Medulla becomes hypersensitive to CO₂ Levels)
 - The 'normal' P_{CO₂} is reset to a lower P_{CO₂} → Stimulates Relative Hyperventilation (40% ↑ V_{Tidal})
 - "Relative Hyperventilation" = Where the mother breaths more than what *her* metabolic rate would dictate.
 - Why? – To aid clearance of Foetal CO₂
 - How? – Via a 40% increase in Tidal Volume (but no change in Frequency)
 - **↑Respiratory Drive Results in:**
 - ↓Arterial P_{CO₂}
 - No change in Arterial P_{O₂}
 - ↑pH (Alkalosis).
- **15-30% Increase in Metabolic Rate:**
 - Increase in Oxidative Phosphorylation → ↑O₂ Consumption → ↑Respiratory Rate.
- **Uterus Invades Thoracic Cavity:**
 - Compression of Lungs & Diaphragm by Enlarged Uterus →:
 - ↓Lung Compliance
 - ↓Residual Volume
 - ↓Functional Residual Capacity
 - ↓Expiratory Reserve Volume
 - ↓Inspiratory Reserve Volume
 - Compensated for by:
 - Increased AP-Diameter
 - Increased Transverse-Diameter
 - (I.e: Chest Volume increases to combat the invasion of Uterus into Thoracic Cavity)



Yeomans, Edward R., and Larry C. Gilstrap III. "Physiologic changes in pregnancy and their impact on critical care." *Critical care medicine* 33.10 (2005): S256-S258.

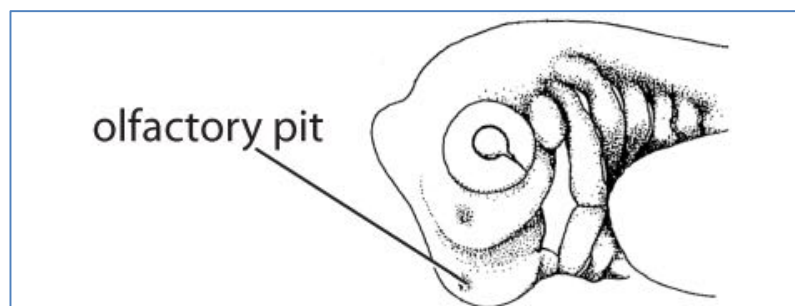
Development of the *Foetal* Respiratory System:



<https://www.embryology.ch/anglais/rrespiratory/phasesn01.html>

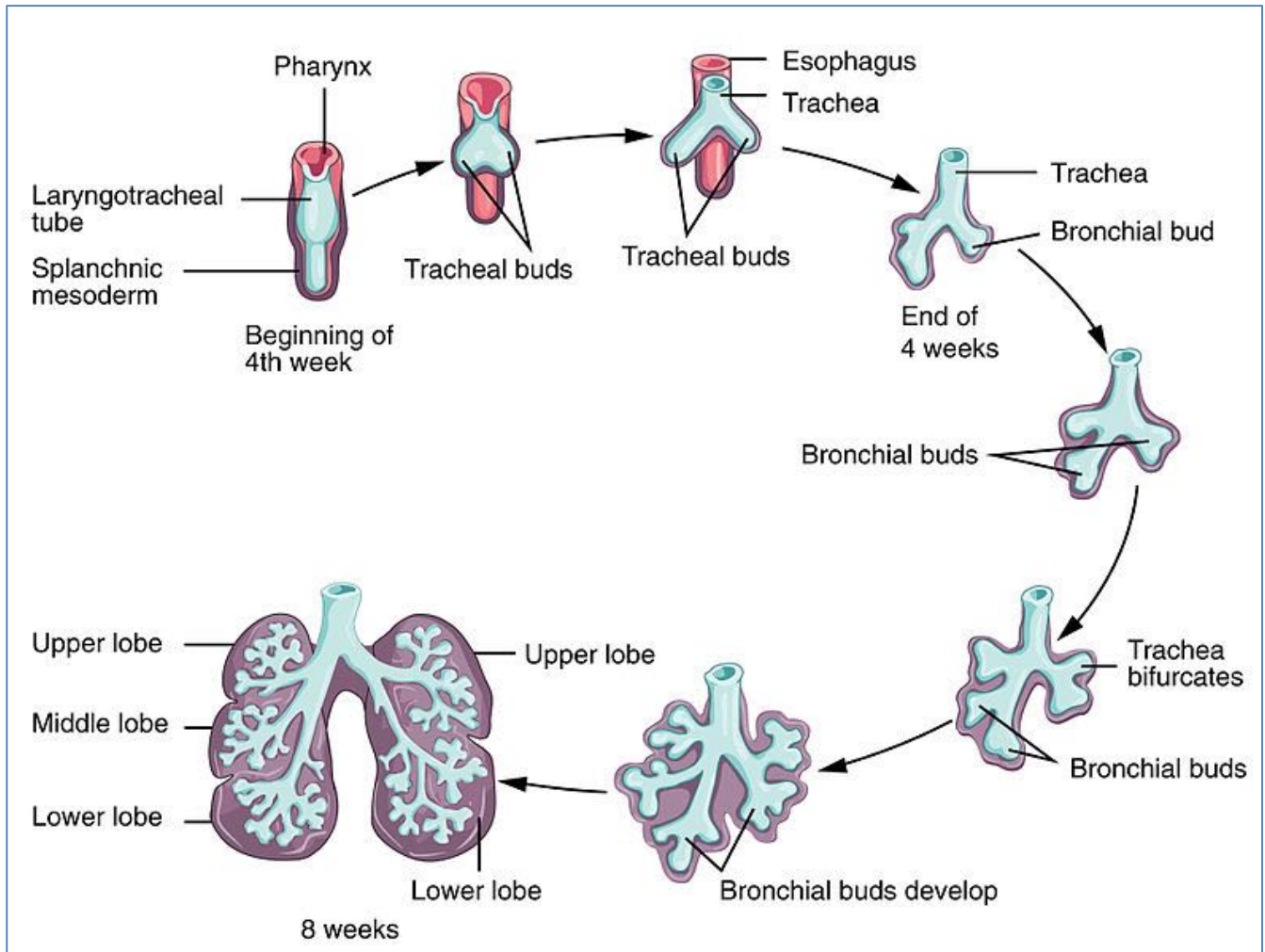
1. Embryonic Stage:

- Upper Respiratory Structures Develop First, Then Lower Respiratory Structures.
- **Wk 4: Olfactory Pits/'Placodes':**
 - Form from Ectoderm (The outermost of the 3 Primary Germ Layers)
 - Olfactory Placodes Invaginate to form Nasal Cavities.
 - Nasal Cavities Extend Posteriorly → Connect with Developing Foregut.



Parexocoetus, CC BY-SA 4.0 <<https://creativecommons.org/licenses/by-sa/4.0/>>, via Wikimedia Commons

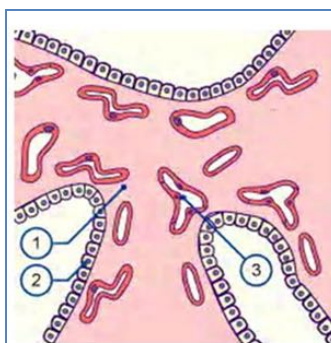
- **Wk 5: "LaryngoTracheal Bud" (Respiratory Epithelium):**
 - Forms from Endoderm (The innermost of the 3 Primary Germ Layers)
 - Develops as an 'outpocket' of the Foregut Mesoderm → Becomes Respiratory Mucosa:
 - Pharyngeal Mucosa
 - Tracheal Lining
 - Bronchial Mucosa
 - Bronchiole Mucosa
 - Alveolar Membrane
 - By wk 6/7, all Basic Upper Respiratory Tree is laid down.



OpenStax College, CC BY 3.0 <<https://creativecommons.org/licenses/by/3.0>>, via Wikimedia Commons

- **2. Pseudoglandular Stage:**

- **Wk 8: Formation of Airway Smooth Muscle/Cartilage/Blood Vessels/Interstitium:**
 - Mesoderm covers the Endoderm-Derived Respiratory Epithelium with Associated Structures.
- **Wk 8-16: Rapid Proliferation of the Airways:**
 - By wk16, all airway divisions are complete. (Down to the terminal bronchioles)



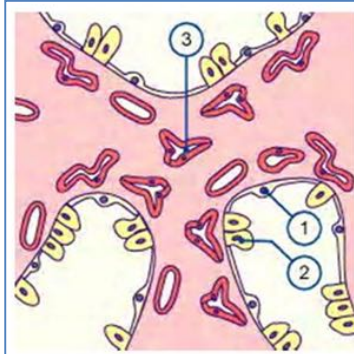
Lung tissue in the pseudoglandular stage.

- 1 = Lung mesenchyma
- 2 = Type II pneumocytes
- 3 = Capillaries

Adapted from: <http://www.embryology.ch/dutch/rrespiratory/fasesen03.html>

- **3. Canalicular Stage:**

- **Wk 16-26: Rapid Angiogenesis (Vascular Proliferation):**
 - Respiratory Capillaries forming around Potential Airspaces.
- **Thinning of the Acinar Walls by Fibroblast Apoptosis:**
 - Fibroblasts undergo Apoptosis → Reduces Acinar Wall Thickness.
 - Brings the Capillaries in closer association with developing air-spaces
 - Note: Both of the above are required for Sufficient Gas-Exchange from Alveoli → Blood.
- **First Appearance of Surfactant:**
 - Type-II Surfactant-Producing cells (Pneumocytes) appear.
 - ↓Surface Tension of Fluid in Lungs → Draws Acinar Walls Apart → ↑Lung Volume



Lung tissue in the canalicular stage.

1 = Type 1 pneumocytes

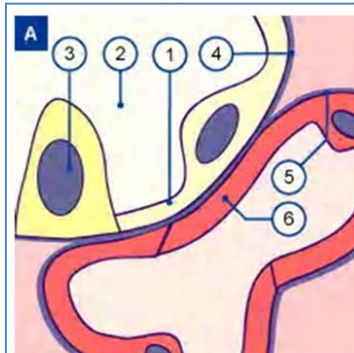
2 = Type II pneumocytes

3 = Capillaries

Adapted from: <https://www.embryology.ch/anglais/rrespiratory/phasen04.html>

- **4. Saccular Stage:**

- **Wk 25-35: Proliferation of Air-Spaces:**
 - Airspaces develop a sac-like appearance = Become 'Saccules'.
 - Airspaces contain both Type-I & Type-II Epithelial Cells (Type 2 = Surfactant-Producing Cells).
 - Note: By 27 Weeks, the Air-Spaces are sufficient for gas exchange (Ie: A 9 wk premature baby's lungs can support gas exchange)



Lung tissue in the saccular stage.

1 = Type 1 pneumocytes

2 = saccular space

3 = Type II pneumocytes

4 = Basement membrane of air passage

5 = Basement membrane of capillaries

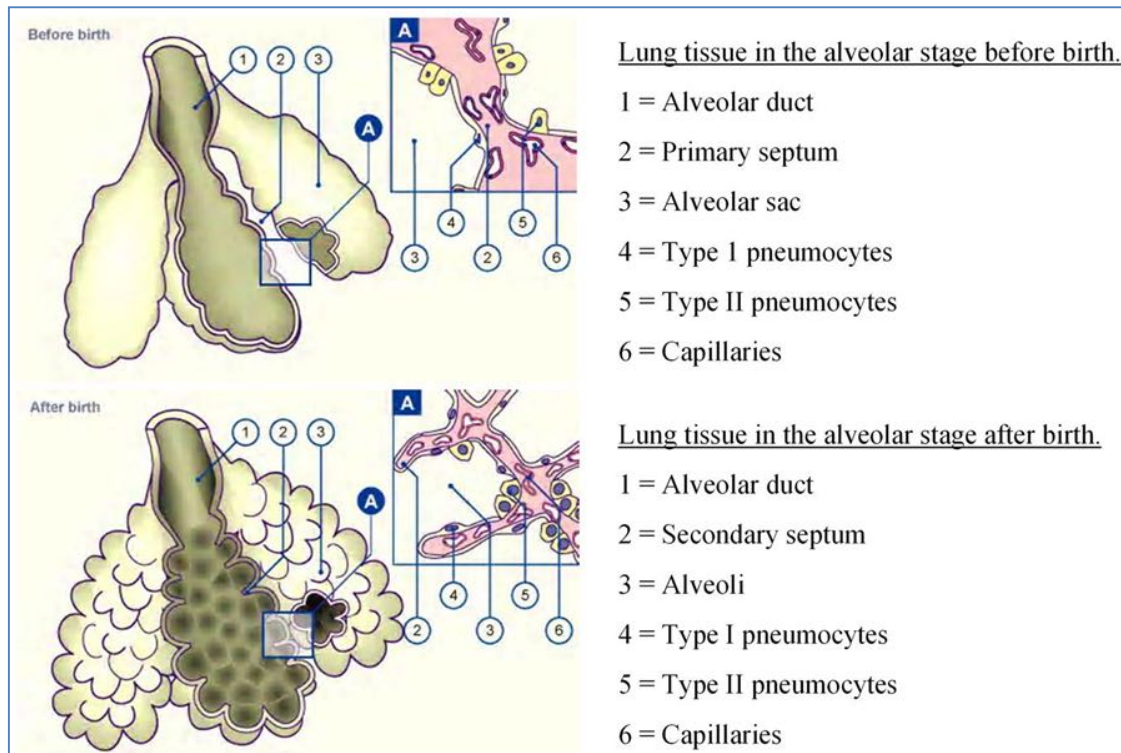
6 = Endothelium of the capillaries

Adapted from: <https://www.embryology.ch/anglais/rrespiratory/phasen05.html>

- 5. Alveolar Stage:

○ Wk 35 – Term: Rapid Development of Alveoli:

- Terminal Saccules Invaginate into Cup-Like Structures = Primitive Alveoli.
- Thinning of Septal Wall by Type-II Cells giving rise to thinner Type-I Cells.
- Note: The Above two developments Dramatically Increase Surface Area of Lung.
- Note: By birth, ≈50Million Alveoli have developed. This increases to around 300million when fully developed at 3yrs old.



Adapted from: <https://www.embryology.ch/anglais/rrespiratory/phasen06.html>

Foetal Lung Fluid:

- Type-I Epithelial Cells continuously secrete fluid into the Air-Spaces.
- This fluid flows from airway → Amniotic Fluid (Flushes out debris from the lung)
- Maintains positive pressure within the Air-Spaces Relative to Amniotic Fluid (Prevents Lung Collapse)
 - Also causes lung *expansion* → Stimulates Cell Division/Growth
 - Stimulates Differentiation of Type-II Cells to Type-I Cells

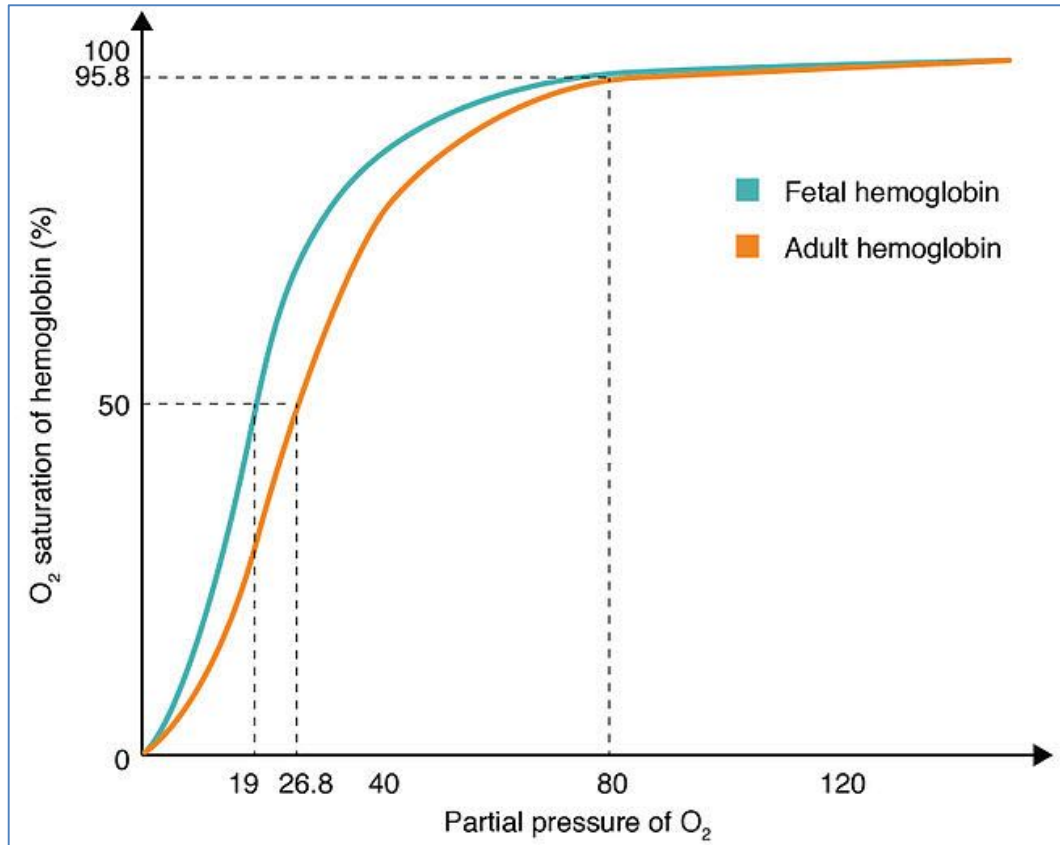
Foetal Respiratory Movements:

- Attempted Respiratory Movements occur *In-Utero*. – Increase in frequency from 22-35wks of gestation.
- However, During the Last Week of pregnancy, Respiratory Movements are Inhibited:
 - To Prevent Lungs Filling with Fluid
 - To Prevent Aspiration of Meconium in the Amniotic Fluid.

Foetal Haemoglobin:

- Foetal-Hb has a different Saturation Curve to Adult-Hb:
 - o Left-Shifted Compared to Adult-Hb (Favours O_2 Loading)
 - o Foetal-Hb has a *Higher Affinity for O_2* (Ie: It's designed to operate at lower P_{O_2})
- Note: These properties are critical for Foetal-Hb to pull O_2 off the Maternal-Hb
 - o (Ie: At a particular P_{O_2} , the Maternal-Hb will favour *Unloading*, while the Foetal-Hb favours *Loading*.)

Foetal Hb Replaced by 6mths of Age.



Creative Commons: Anatomy & Physiology, Connexions Web site. <http://cnx.org/content/col11496/1.6/>

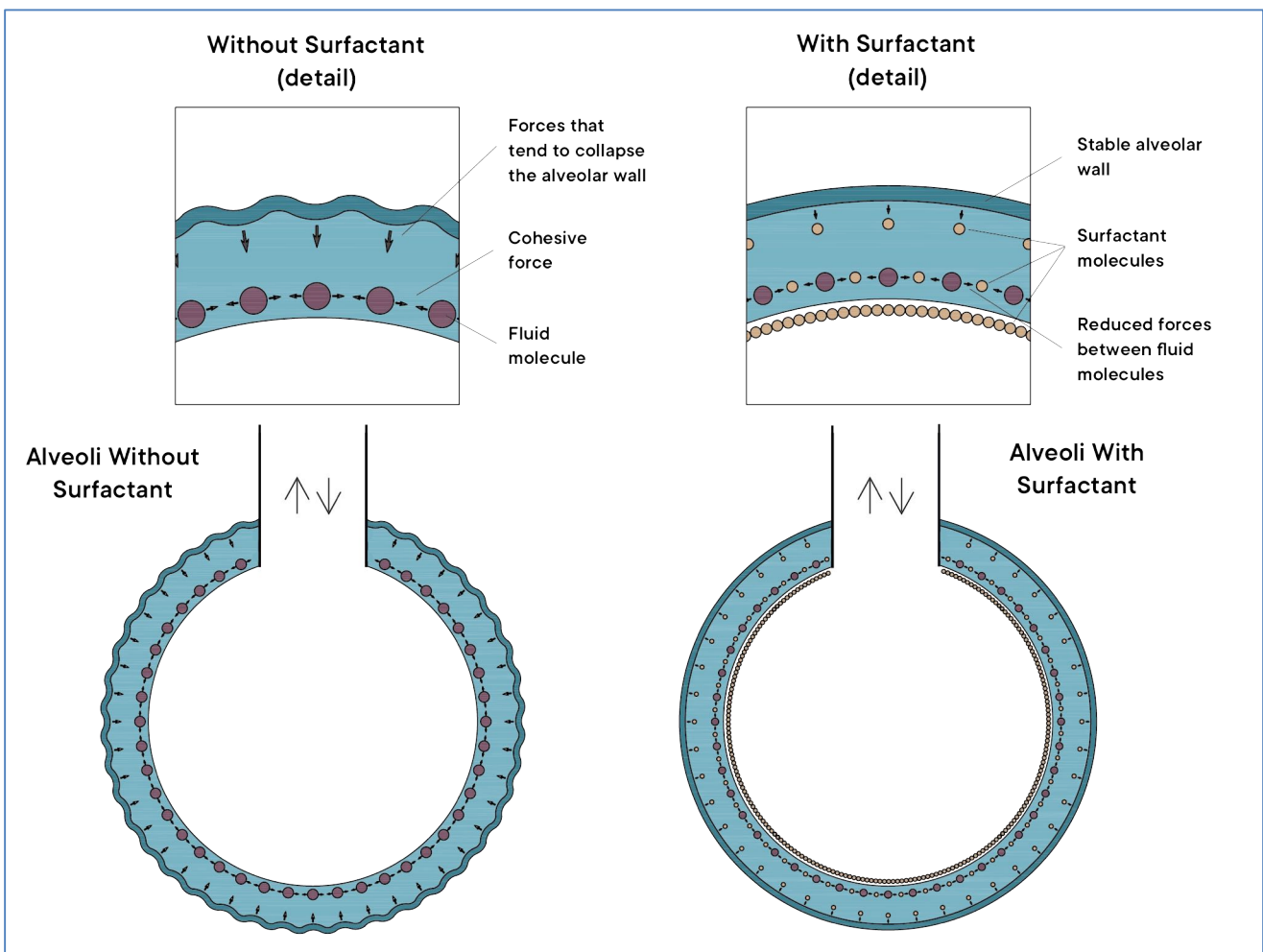
Pulmonary Surfactant:

- Without Surfactant:

- Surface Tension of the Fluid lining the lungs causes the Alveoli to Recoil.
- – caused by the Attraction between Adjacent H₂O Molecules acting to ↓ Surface Area & Alveolar Size
- Note: Surface Tension of H₂O is so strong that the Alveoli would collapse if the fluid was just H₂O
 - Without Surfactant, Lung Compliance is 10-20% of Normal.
 - → Requires huge Negative Pressures to maintain Patent Alveoli.
 - Lack of Alveolar Surfactant = Primary Cause of *Infant Respiratory Distress Syndrome* (IRDS)
- **Note: The *Inward Pressure* of Surface Tension INCREASES as Radius DECREASES.**
 - Therefore, if two alveoli of different sizes were joined by a common airway, the smaller alveolus would collapse into the bigger alveolus (Due to the pressure gradient)

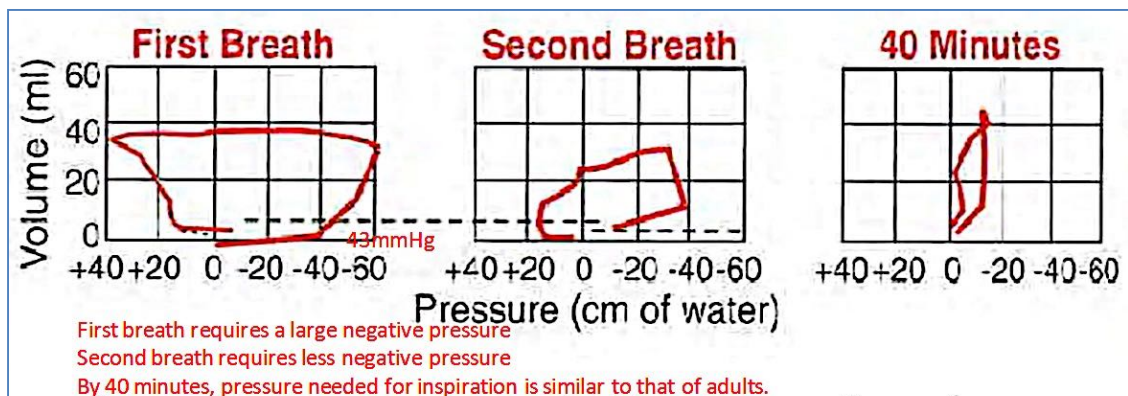
- With Surfactant:

- Note: Pulmonary Surfactants = Mainly Lipids 90% (DPPC), Some proteins & carbohydrates
- Surfactant Produced by Type-II Alveolar Cells
- DPPC is responsible for Surfactant's Effect on Surface Tension:
 - ↓ Alveolar Pressure
 - ↓ Energetic Cost of Ventilation
 - ↓ Lung's Tendency to Collapse
- **With Surfactant, the *Inward Pressure* of Surface Tension DECREASES as Radius DECREASES.**
 - I.e: It Decreases the Surface Tension in Smaller Alveoli MORE THAN the Larger Alveoli.
 - - This Solves the Problem of Small Alveoli Collapsing into Larger Alveoli.



Transition to Extra-Uterine Life:

- **Loss of Placenta:**
 - o Loss of Metabolic Support
 - o Loss of Respiratory Support → Hypoxia/Hypercapnia → Respiratory Centre → Breathing.
- **Need to Begin Breathing to Survive:**
 - o Breathing Triggered by:
 - Hypoxia
 - Touch/Skin-Cooling
 - Possible Chemical/Hormonal Stimuli
 - o When?
 - Most take their first-breath within 20s & are breathing normally by 90s after birth.
 - Note: Baby may survive for up to 10mins without breathing, however, after 8mins brain damage occurs.
 - Note: Maternal Anaesthetics can delay Initiation of Breathing.
- **Removing Fluid From Lungs:**
 - o At Birth, Alveolar Epithelial Cells switch from Secretion to Absorption of Fluid.
 - o Also, Thoracic Compression of Vaginal Delivery → Squeezes some of the fluid in the lung.
- **The First Breaths:**
 - o Remaining Fluid in lungs creates High Surface Tension – Requires large –ve Pressure to Inflate.
 - o **First Breath:**
 - Between -25mmHg and -40mmHg is needed. (Compared to an Adult's -15mmHg)
 - It is important that the baby doesn't expire fully after taking its first breath – as this would cause the alveoli to collapse again.
 - o **Subsequent Breaths:** Require less negative pressure.
 - o **By 40 Minutes:** Inspiratory force is similar to Adults.



Circulatory Readjustments:

- **Closure of Foramen Ovale:**
 - o Huge Loss of Vasculature (Through loss of placenta) → Doubles Systemic Vascular Resistance.
 - → ↑Aortic Pressure → ↑L-Atrial & L-Ventricular Pressures.
 - o Mechanical Inflation of Lungs → Dilates Pulmonary Vasculature → ↓Pulmonary Vascular Resistance.
 - → ↓Pulmonary Artery Pressure → ↓R-Atrial & R-Ventricular Pressures.
 - o – The Combination of the above two pressure changes causes the valve-like Foramen Ovale to Close.
- **Closure of Ductus Arteriosus:**
 - o Decrease in Prostaglandins cause muscular wall of Ductus Arteriosus to Contract → Duct Occludes.

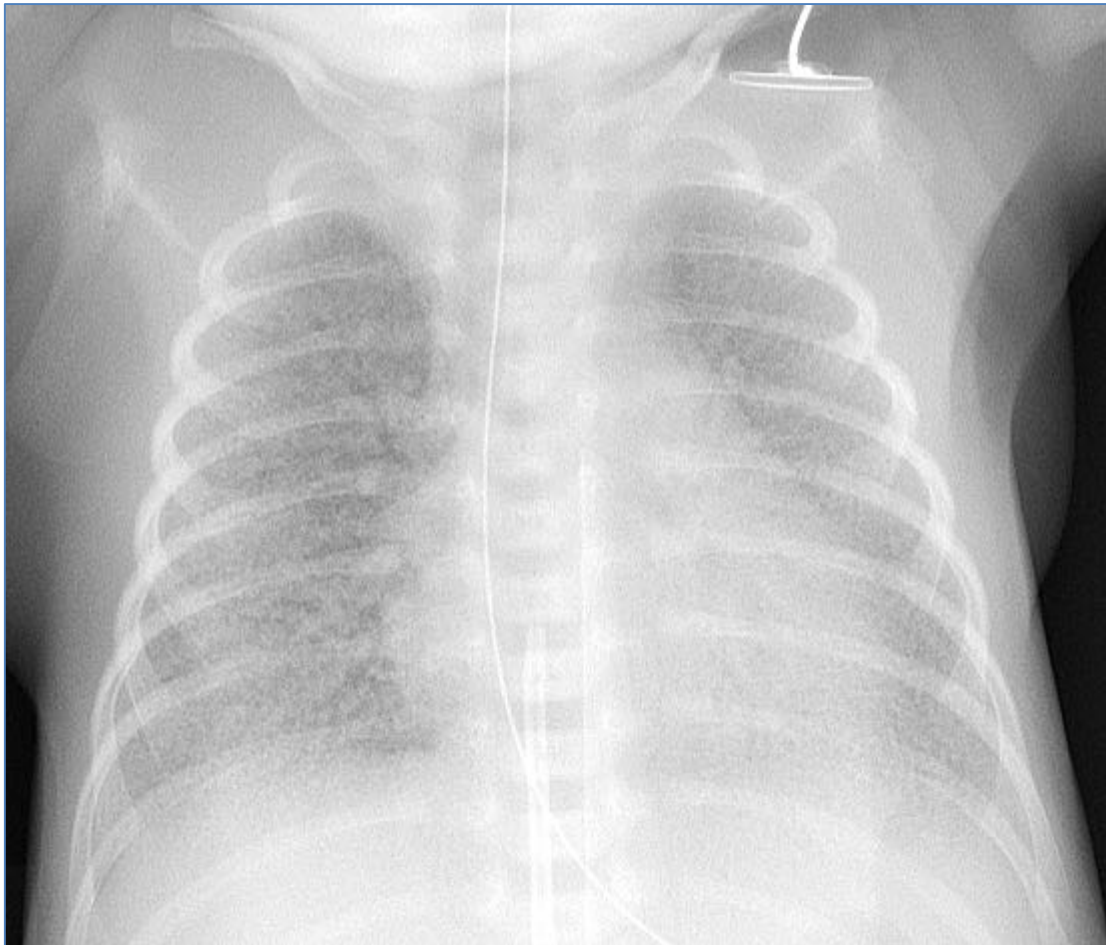
Postnatal Lung Growth & Lung Size:

- Foetal Hb Replaced by Adult Hb by 6 months.
- Alveolar number increases (From 50 million @ birth to 300 million in adulthood)
- Alveolar Diameter Increases (From 0.15mm @ birth to 0.3mm in Adult)
- Surface Area Increases (From 4m² @ birth to 100m² in Adult)
- Lung Volume Increases (From 250mL @ birth to 5000mL in Adult)
- Lung Weight Increases (From 50g @ birth to 800g in Adult)

When Things Go Wrong:

- **INFANT RESPIRATORY DISTRESS SYNDROME:**

- Typically seen in Premature Infants (around 25wks or less)
- **Where the baby is born with Insufficient Pulmonary Surfactant:**
 - → ↑Alveolar Surface Tension
 - → ↓Lung Compliance (10-20% of Normal)
 - -- Requires Large Negative Pressures to Keep Alveoli Open.
- I.e: Premature Infants (with poorly developed inspiratory muscles) must maintain High Negative Pressures just to Breathe.
- Also: The Alveolar Membranes may be damaged due to the sheer stress of trying to breath.
- **Results in:**
 - Alveolar Collapse ("Atelectasis")
 - ↑Energetic Cost of Breathing
 - Hypoxia & Hypercapnia (↑Energetic Cost of Breathing – Respiratory Muscles consume a lot of the Inspired O₂)
 - Hyaline (Glassy) Membranes
 - Periodic (Cheyne-Stokes) Breathing – due to significant fluctuations in Blood-Gas Levels & due to premature Central-Respiratory Controllers.
- **Treatment:**
 - Mechanical Ventilation:
 - CPAP – Constant Positive Airway Pressure
 - PEEP – Positive End-Expiratory Pressure
 - Note: Requires a fine balance of Mechanical Ventilation as it may damage the lung lining.
 - Exogenous Surfactant – (Artificial/Natural)



Mikael Häggström, M.D. - Author info - Reusing images- Conflicts of interest: NoneMikael Häggström.Consent note:
Written informed consent was obtained from a parent of the individual, including for online publication., CCO, via
Wikimedia Commons

- **SUDDEN INFANT DEATH SYNDROME:**

- **Definition:**
 - “Death of an infant under 1year which remains unexplained – usually occurs while asleep”
- **Peak Incidence** = 2-3 Months
- **Contributing Factors:**
 - Premature Birth
 - Low Birth Weight
- **Causes – Probably Multifactorial:**
 - Central Apnoea – No Respirator Signals due to Poorly Developed Central Control.
 - Abnormal Temperature Control – Overheating may be a cause of SIDS.
 - Prolonged Apnoea during Cheyne-Stokes Breathing (Ie: The stress associated with the period of apnoea prevents re-initiation of breathing)

- **Mother’s Water Breaks at 27 Weeks:**

- Risk of Foetal Infection
- Lack of Mechanical Protection of baby
- Baby may be contorted – as there is no room to move with no amniotic fluid.
- **Effects on Respiratory Development:**
 - Amniotic Fluid Aids in the Development of Lungs – If no fluid → Lung membranes become dry → Don’t grow properly → ↓Surfactant Production → IRDS.

- **MECONIUM ASPIRATION:**

- If the baby defecates in-Utero, then inhales it.
- → Infection
- (Covered in more detail in Obstetrics subject)

HYPERBARIC & HYPOBARIC CONDITIONS

HYPERBARIC & HYPOBARIC CONDITIONS

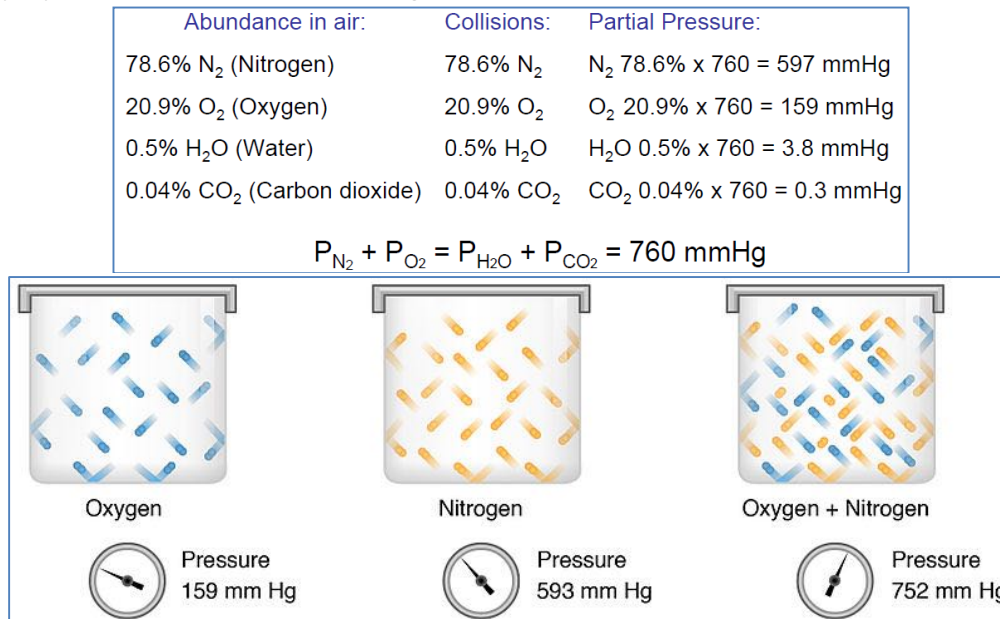
Basics of Hyper/Hypo-Baric Conditions:

- **Barometric Pressure Decreases Greatly With Altitude**
 - Note: The *Percentage* of Oxygen is the same at any Altitude (21%)
 - Note: The *Partial Pressure* of Water Vapour is the same at any Altitude (47mmHg)
- **Barometric Pressure Increases Greatly With Depth (Underwater)**
 - Note: Every 10 metres you go down, the pressure increases by 1xAtmosphere (760mmHg).

Revision of The Gas Laws:

- Dalton's Law of Partial Pressures:

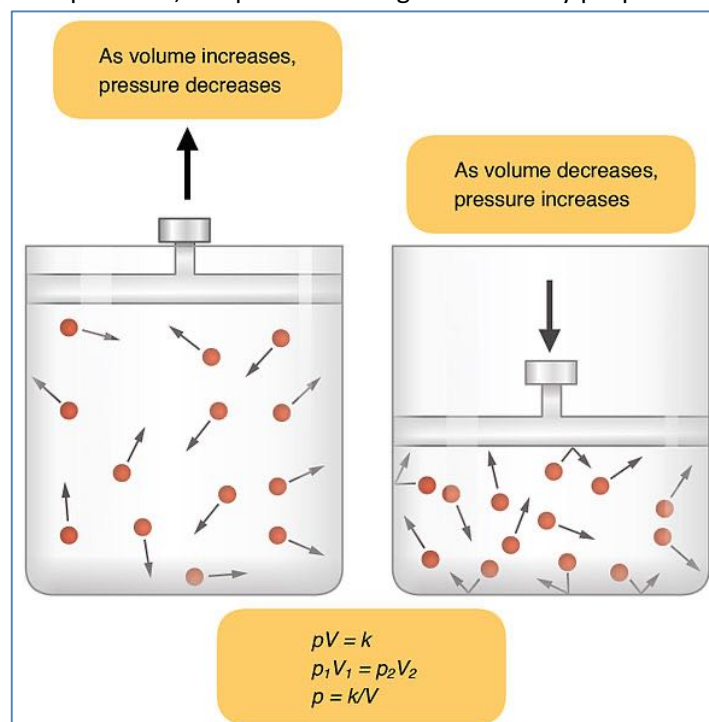
- The total pressure of a mixture of gasses is the combined sum of each gas's partial pressure, which is proportional to its abundance (%age) in the mixture.



OpenStax College, CC BY 3.0 <<https://creativecommons.org/licenses/by/3.0>>, via Wikimedia Commons

- Boyle's Law of Pressure Vs. Volume:

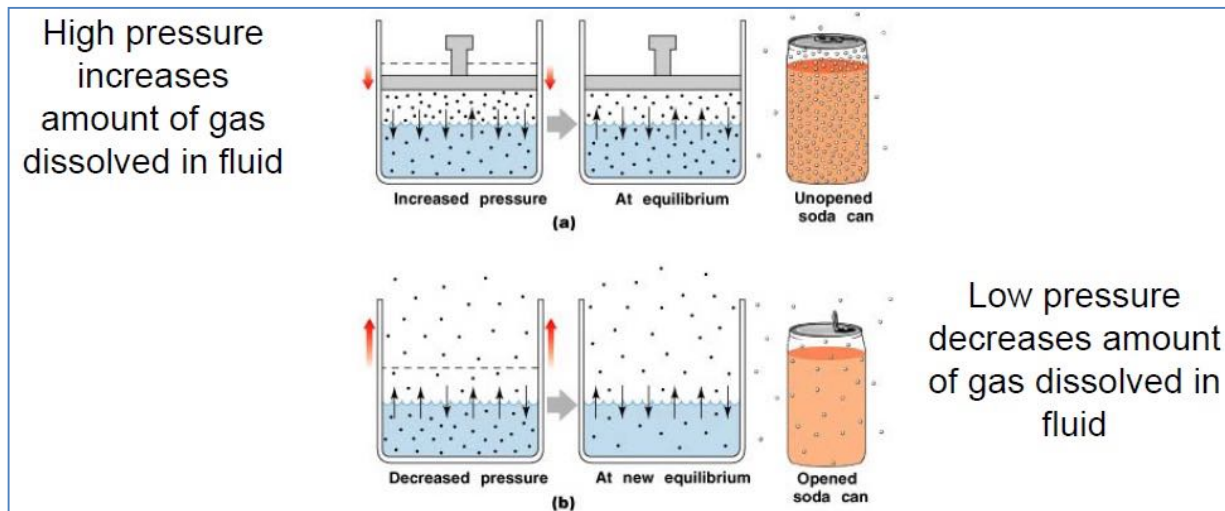
- At a constant temperature, the pressure of a gas is inversely proportional to its volume.



OpenStax College, CC BY 3.0 <<https://creativecommons.org/licenses/by/3.0>>, via Wikimedia Commons

- **Henry's Law of Dissolved Gasses:**

- At a constant temperature, the amount of gas dissolved in a solution is proportional to the Partial Pressure and solubility of that gas in that liquid.



Source: Unattributable

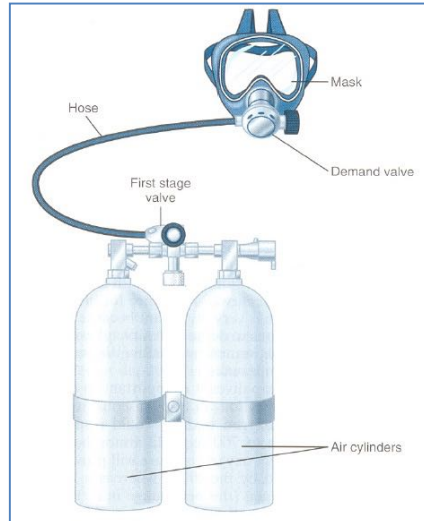
HYPERBARIC Conditions (Ie: SCUBA Diving):

Relevant Gas Laws:

- **Dalton's Law:** Increase in Pressure → Increases the Partial Pressures by the same amount.
- **Boyle's Law:** Increase in Pressure Decreases Gas Volumes.
- **Henry's Law:** Increase in Pressure Forces More Gas into Solution.

The Breathing Apparatus:

- Tanks of Compressed Air (Needs to be compressed to inflate the lungs underwater)
- Regulator 'steps down' the pressure.
- Mouthpiece + Valve
- Hose – Must have the smallest possible Dead-Space (to avoid re-breathing expired CO₂)



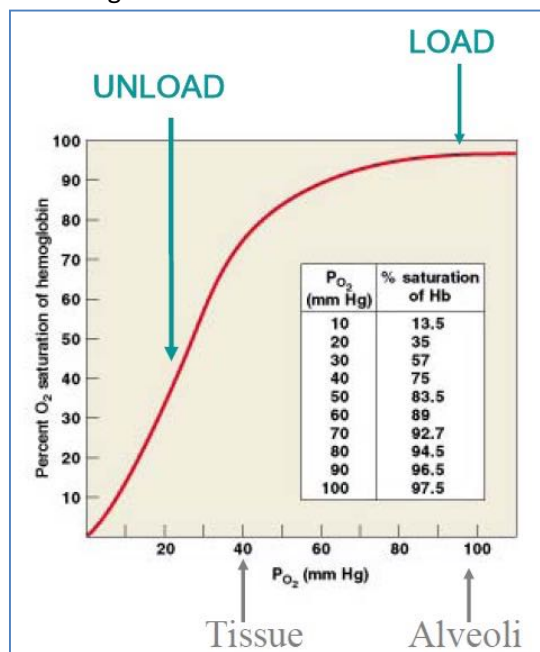
Effects of High Partial Pressures (of O₂):

Acute Oxygen Poisoning:

- If the diver descends more than ≈ 80m, he will be breathing O₂ at 1500mmHg.
- A P_{O₂} of 1500mmHg may lead to Oxygen Poisoning.
- A P_{O₂} of 3000⁺mmHg will cause Seizures & Coma. (Brain is particularly sensitive)
- Also, excess oxygen converts to O₂-Free Radicals → Cellular Damage

Hemoglobin Saturation Curve:

- Normally, all of the Blood's O₂ is bound to Haemoglobin (little/none in simple solution)
- However, since SCUBA air is delivered at high pressure, lots of O₂ will dissolve in the blood.
- As a result, the O₂ Partial Pressures in the Blood far exceed Hb's Functional Range.
- This renders Haemoglobin useless and it remains 100% saturated.

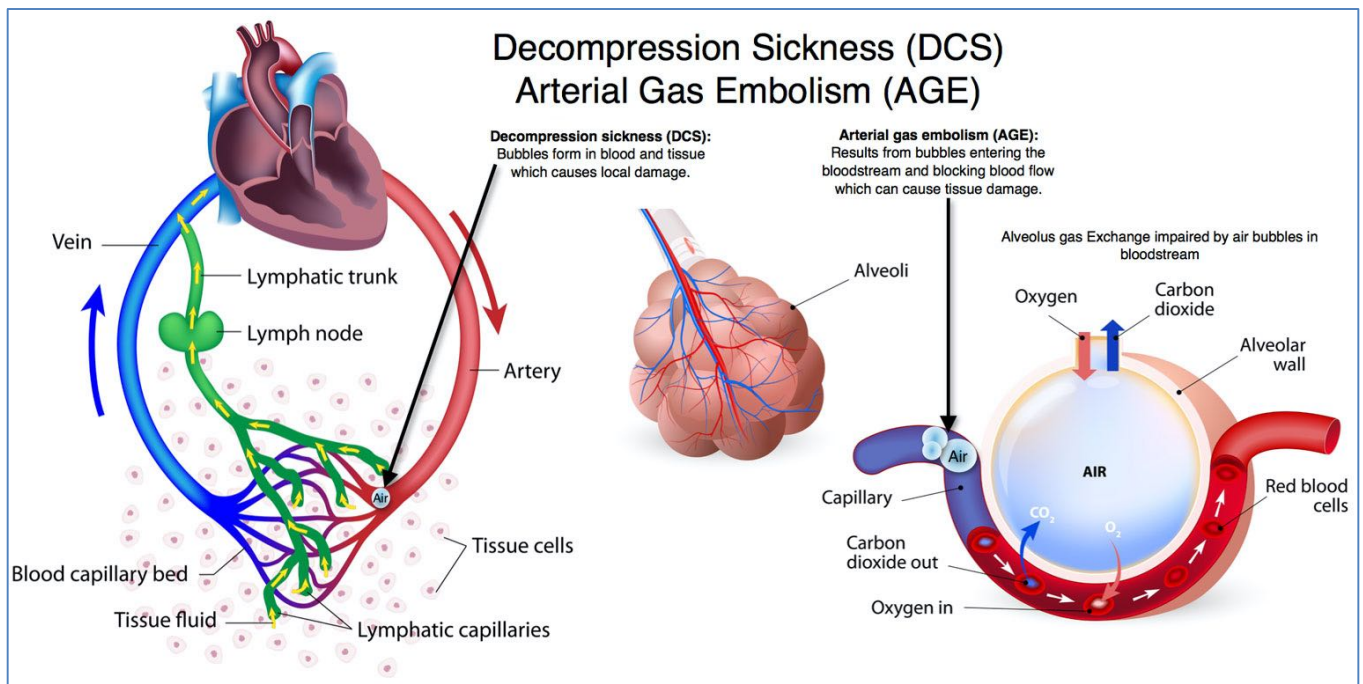


- **Effects of High Partial Pressures (of CO₂):**

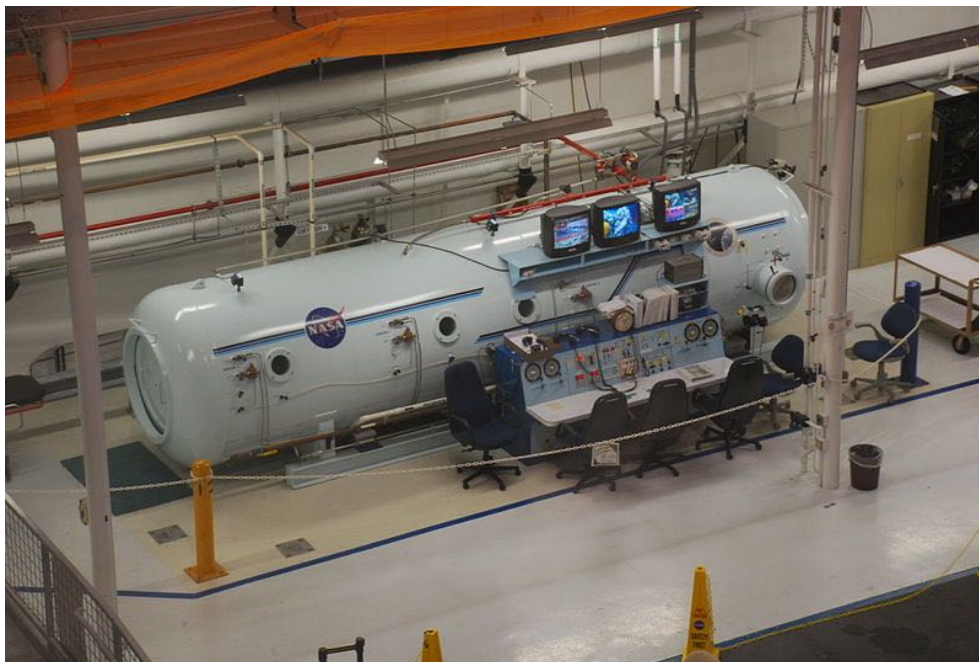
- CO₂ is Less of a Problem than O₂ & N₂.
 - Because [blood CO₂] is determined by metabolism, not the Environment.
 - Therefore, P_{CO₂} will be constant provided the Diver exhales normally.
- However, if CO₂ builds up in the “Dead Space” of the Apparatus & is Re-Breathed, CO₂ Levels will rise.
 - Diver may become Hypercapnic (High CO₂)
 - A Diver may tolerate a P_{CO₂} of up to 80mmHg by increasing Respiration.
 - Beyond 80mmHg, Respiratory Acidosis results.

- **Effects of High Partial Pressures (of N₂):**

- Note: N₂ is not metabolised. Therefore it remains in the body until P_{aN₂} decreases → Removed by Lungs
- **Nitrogen Narcosis:**
 - Nitrogen is Highly Lipid Soluble.
 - High Amounts of Dissolved N₂ → ‘Nitrogen Narcosis’ (aka. “Raptures of the Deep”)
 - Similar to Alcohol Intoxication
 - Effects increase dramatically with increased depth. (Similar Stages to Alcohol Intoxication)
- **Decompression Sickness (The Bends):**
 - If a diver ascends too quickly, the pressure decreases, causing N₂ to fall out of solution.
 - → N₂ Bubbles
 - These bubbles amalgamate → Bigger & Bigger → Block Larger Vessels (“Air Emboli”)
 - **May Lead to:**
 - Tissue Ischaemia/Death
 - Pains in Joints
 - Pains in Muscles
 - Paralysis
 - Unconsciousness
 - The Chokes (Some Air-Emboli can rupture lung capillaries → Bleeding into the lungs)
 - **Treatment:**
 - Diver placed in Hyperbaric Decompression Chambers (Pressurized Tank)



<https://www.pedilung.com/pediatric-lung-diseases-disorders/diving-medicine/decompression-sickness-dcs-arterial-gas-embolism-age/>



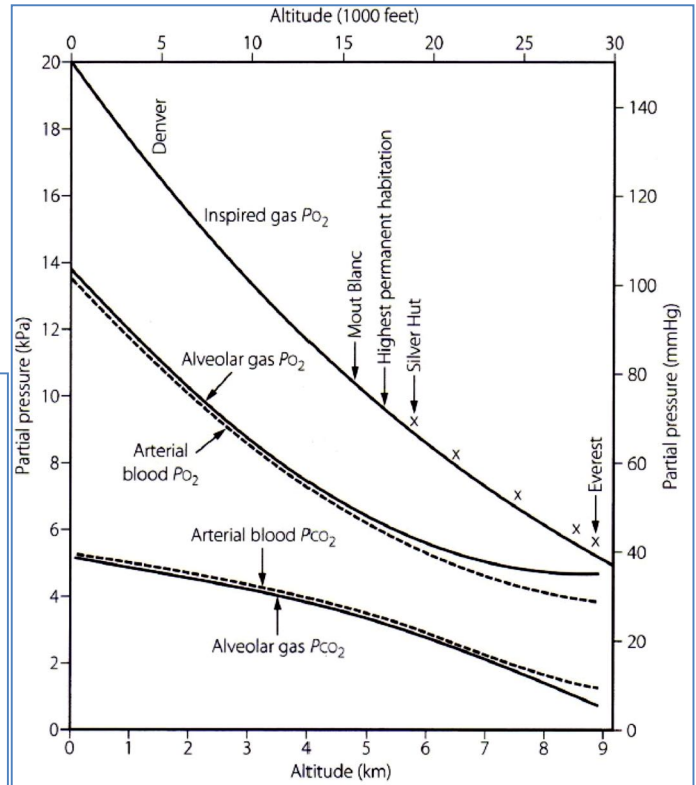
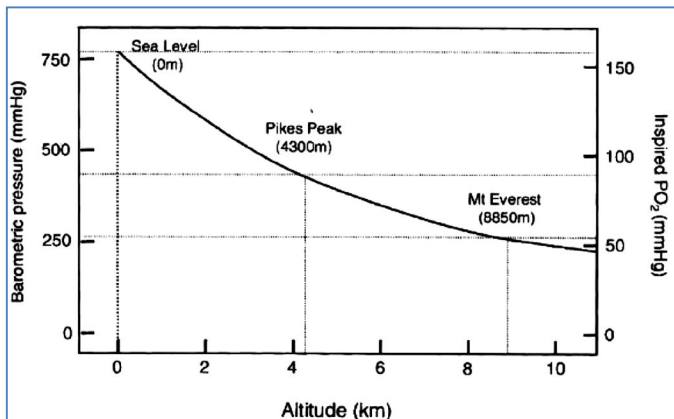
Mike, CC BY 2.0 <<https://creativecommons.org/licenses/by/2.0>>, via Wikimedia Commons

HYPOBARIC Conditions (Ie: Mountain Climbing):

- Challenges to Altitude:

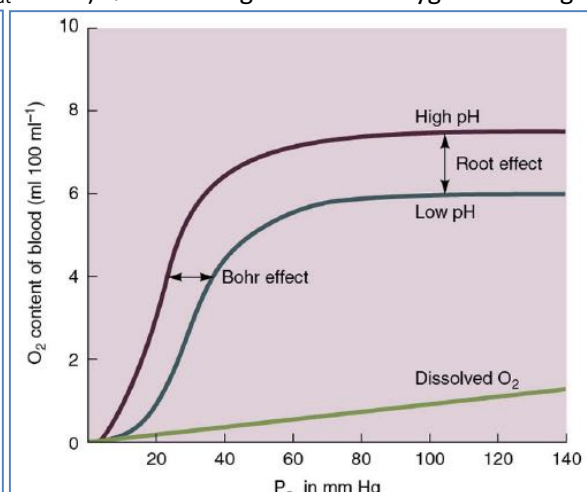
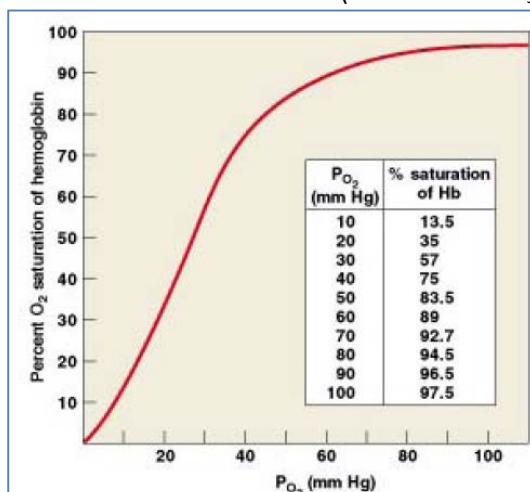
o ****Hypoxia:**

- Note: At High Altitude, the Air Pressure (& therefore the Oxygen Partial Pressure) Decreases.
- Therefore by decreasing *Inspired P_{O_2}* , The *Alveolar & Arterial O_2* pressures decrease as well.
- This leads to Hypoxia.
- Note: Alveolar & Arterial P_{CO_2} decreases slightly @ High Altitude.



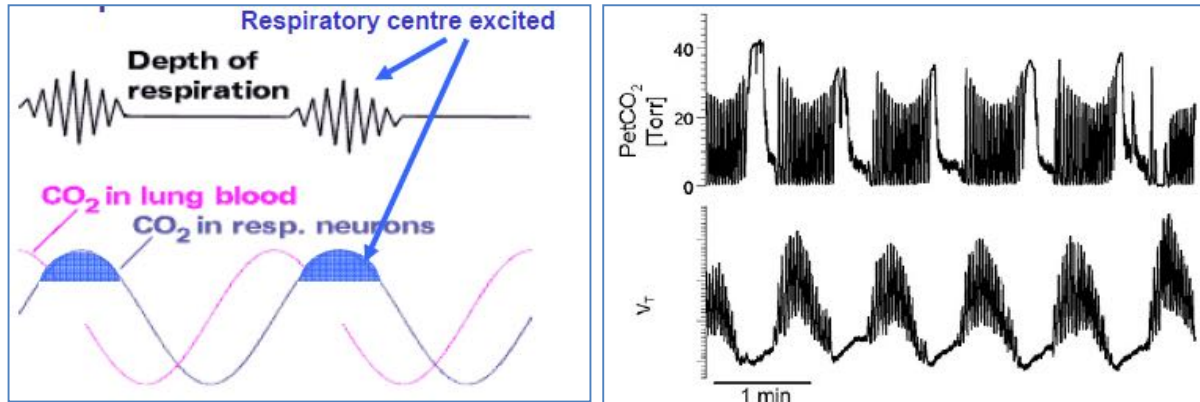
■ Hb-Oxygen Saturation:

- At Sea-Level, $P_{O_2} \approx 160\text{mmHg}$ → $Hb_{\text{sat}} = 95\%$
- At 10,000ft, $P_{O_2} \approx 110\text{mmHg}$ → $Hb_{\text{sat}} = 90\%$
- Any higher, P_{O_2} Falls Rapidly → Hb_{sat} Falls Rapidly
- **Note:** At high altitudes, **water vapour** accounts for a *relatively larger proportion* of the inspired gas than at sea-level → Decreases the proportion of pressures of the other gasses → Further decreases the P_{O_2} as Altitude Increases.
- **Note:** Once Hb_{sat} reaches $\approx 60\text{-}65\%$, it remains fairly stable despite further increases in altitude (And subsequent decreases in P_{O_2}).
- Why? Because the increased ventilation → Respiratory Alkalosis ($\uparrow\text{pH}$) → “Root Effect” (Left Shift of Hb_{sat} Curve) → Favouring Increased Oxygen-Loading.



▪ **Cheyne-Stokes Breathing @ Altitude:**

- 1. Low Oxygen Levels (<60mmHg) Stimulate Peripheral O₂ Chemoreceptors.
 - Stimulates the Respiratory Centre to hyperventilate.
- 2. Increased respirations blow off too much CO₂ → Respiratory Alkalosis.
 - Depresses Central Chemoreceptors (On the Medulla)
 - → Depresses Respiratory Centre
- 3. Arterial Oxygen Levels fall to <60mmHg again → Cycle Starts Again.

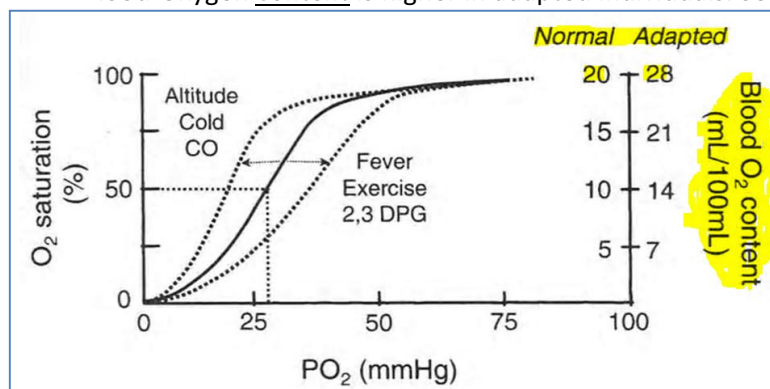


▪ **Effects of Hypoxia:**

- **Begin @ ~12,000ft:**
 - Impaired Mental Function/Judgement
 - Impaired Memory
 - Impaired Motor Function
 - Drowsiness
 - Lassitude
 - Fatigue
 - Nausea
 - Euphoria
- **@ ~18,000ft:**
 - Twitching
 - Seizures
- **@ ~23,000ft:**
 - Coma
 - Death

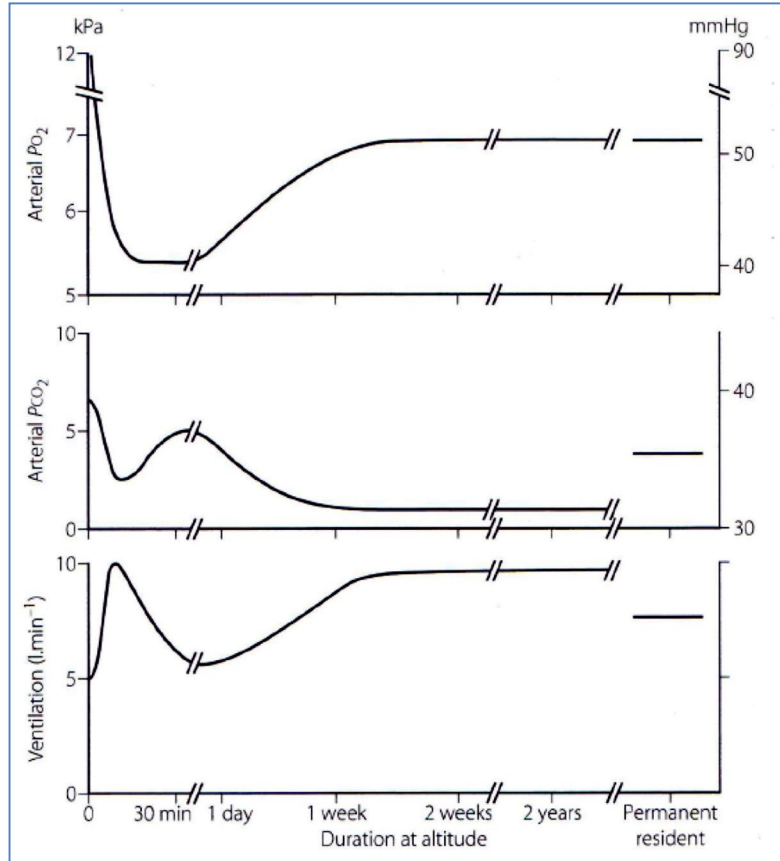
▪ **Acclimatisation:**

- Body Compensates for Low O₂ Levels over a Prolonged Time by:
 - Increasing Ventilation
 - Increasing RBC Production (Therefore ↑ O₂ Carrying Capacity of Blood)
 - Rapid Angiogenesis in Lungs → Increase Diffusing Capacity
 - Rapid Angiogenesis in Tissues → Allows ↑ Perfusion.
 - Fine-Tuning the Efficiency of Aerobic Metabolic Processes.
- Blood-Oxygen Content is higher in adapted individuals. See below:



▪ **Process of Acclimatization:**

- Initially, there is a sharp increase in Respiratory Rate, decreasing P_{CO_2} and Slowing the onset of Hypoxia. This initial Hyperventilation quickly subsides & P_{CO_2} returns to normal.
- Within a few days, Hyperventilation returns and continues for the long term. This partially restores the P_{O_2} , however P_{CO_2} falls again → Respiratory Alkalosis.
- Over the long term, the Kidneys compensate for the Respiratory Alkalosis by excretion of Bicarb.



▪ **Mountain Sickness:**

• **(AMS) Acute Mountain Sickness:**

- Occurs in those who ascend Too High Too Fast.
- Will die if not given O_2 or Removed to Lower Altitude.
- Requires Immediate Descent.

○ **Manifestations:**

▪ **(HACE) High Altitude Cerebral Oedema:**

- Build-Up of Fluid In the Brain – Due to Vasodilation of the Cerebral Vessels in response to Hypoxia
- → Headache
- → Decreasing Consciousness
- → Hallucinations
- → Psychotic Behaviour

▪ **(HAPE) High Altitude Pulmonary Oedema:**

- Pulmonary Vessel Constriction due to Hypoxia → Increased Pulmonary Pressure → Fluid Build-up in the Lungs.
- → Persistent Productive Cough

• **(CMS) Chronic Mountain Sickness:**

- Occasionally occurs in those at altitude for a long time.
- **Symptoms:** Due to ↑ Red Blood Cells – Sluggish Blood:
 - **Pulmonary Hypertension**
 - **Myocardial Hypertrophy → Heart Failure**

AIRWAY HYPERSENSITIVITY & ASTHMA

AIRWAY HYPERSENSITIVITY & ASTHMA:

Airway Smooth Muscle:

- Regulates airway diameter by Bronchoconstriction/Bronchodilation
- Exists in all airways.
- Makes up most of non-cartilaginous airways. (eg. Bronchioles = almost entirely smooth muscle)
- If this muscle spasms, airway diameter will Decrease.

Autonomic Effects on Smooth Muscle:

- **Sympathetic: → Bronchodilation**
 - o **β -Adrenergic** Receptors (on Smooth Muscle) → Bronchodilation
 - (Ie: Ventolin = β -adrenergic **AGONIST**)
 - o Most of the Sympathetic drive comes from **Adrenaline** as Innervation of Airways is Sparse
- **Parasympathetic: → Bronchoconstriction**
 - o **M_3 -Muscarinic Cholinergic** Receptors (on Smooth Muscle) → Bronchoconstriction
 - o Most of the Parasympathetic drive comes from Vagus Innervation.

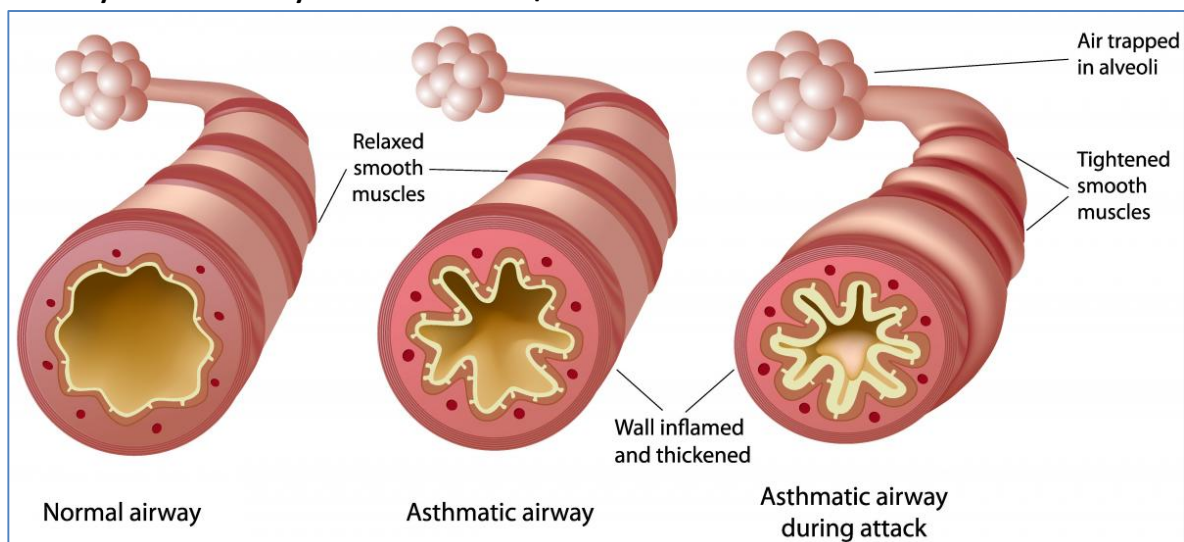
Immune-System Effects on Smooth Muscle:

- **Inflammatory Chemicals Can → Bronchoconstriction**
 - o (Leukotrienes, Histamines, etc.)
- **Inhaled Irritants Can Directly → Bronchoconstriction**
 - o (Dust, haydust, sawdust, perfume, smoke, etc.)

Asthma:

What is Asthma?:

- Hypersensitivity of Airways to Various Stimuli → Inflammation → Constriction of Airways.
- Ie: A chronic **Inflammatory Disorder** → Damage to Airway Epithelium → Amplifies Neural, Inflammatory & Immune responses → Episodic, Reversible Constriction. (Ie: A **Variable Obstructive PD**)
- **Changes in the Airway:**
 - o Narrowed Airway
 - o Swollen Mucosa (Mucosal Oedema)
 - o Hypertrophied Mucosal Glands → Excess Mucus Production
 - o Thicker Mucus
 - o Hypertrophied Smooth Muscle → Stronger Spasms
 - o Constriction of Smooth Muscle
 - o Thickened Smooth Muscle Layer
- **Inevitably leads to Airway **OBSTRUCTION** & ↑Resistance to Airflow.**



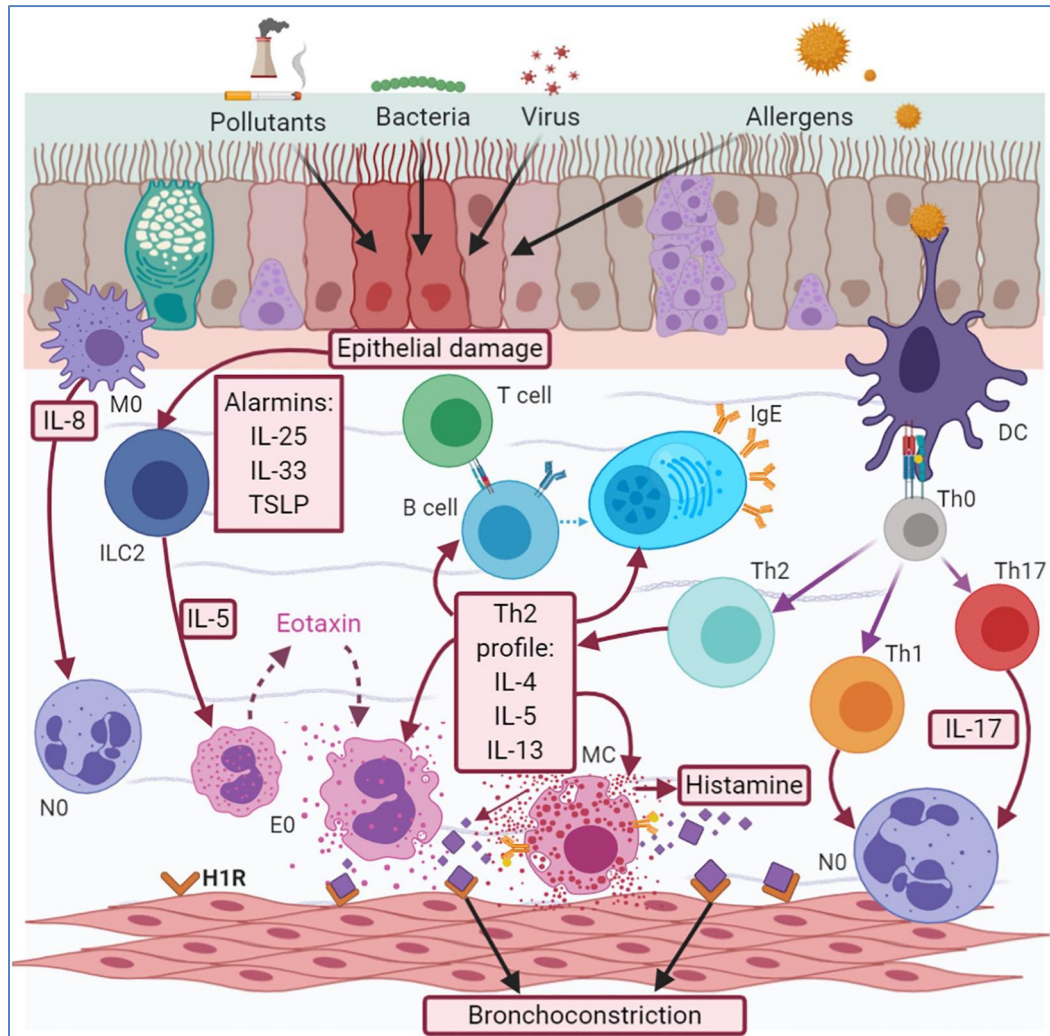
<https://www.asthmafoundation.org.nz/your-health/living-with-asthma>

Aetiology:

- **Types:**
 - **1. Atopic** (Allergic) Asthma (**Type 1 Hypersensitivity Reaction - IgE.**)
 - **2. Non-Atopic** Asthma (Viral-Induced/Drug-Induced (Eg. Aspirin)/Occupational)
- **Environmental Triggers** – (Dust/Pollen/Dander/Mould/Smoke/Pollution/Perfume/Cold Air)
- **Genetic** – (FamHx is Common)

Pathophysiology: **Type 1 Hypersensitivity Reaction:**

- **Rapid Immune Reaction to a Previously-Sensitised Antigen → Mast-Cell/Basophil Degranulation → Release Inflammatory Mediators →**
 - **Sensitization:**
 - 1. Antigen enters the body.
 - 2. 'Antigen-Presenting Cell' Presents the Antigen to 'Type-2 Helper-T-Cells' (TH2-Cells)
 - 3. TH2-Cells Produce Cytokines → Activate B-Cells
 - 4. B-Cells Produce IgE-Antibodies
 - 5. IgE-Antibodies attach to Mast-Cells.
 - **Re-Exposure:**
 - 6. Re-Exposure of Antigen → Attaches to Antibody on Mast-Cell → Mast-Cell Degranulates
 - 7. Degranulation Releases Mediators → Type-1-Hypersensitivity Reaction Occurs.
 - **Mediators Include:**
 - Histamine
 - Leukotrienes
 - Prostaglandins
 - **Initial (Early) Phase:**
 - Re-Exposure → Mast-Cell Degranulation → Release of Mediators (incl. Histamine)→...
 - ↑Mucus Secretion
 - 'Loosens' the Tight-Junctions between Mucosal Cells → Antigen enters Submucosa.
 - Submucosal Mast-Cells Stimulated→Degranulate.
 - Degranulated Mediators Directly Stimulate Nerve Terminals →...
 - Smooth Muscle Spasm → Bronchoconstriction
 - Vasodilation
 - ↑Vascular Permeability
 - Smooth Muscle Spasm
 - **Late Phase:**
 - Release of Inflammatory Mediators Lead to:
 - Influx of Leukocytes, Basophils, Neutrophils & Eosinophils
 - Eosinophils release '*Major Basic Protein*' → Epithelial Damage.
 - Epithelial Damage → Causes Localized Oedema
 - ↓Mucociliary Function → Accumulation of Mucus
 - ↑Airway Responsiveness



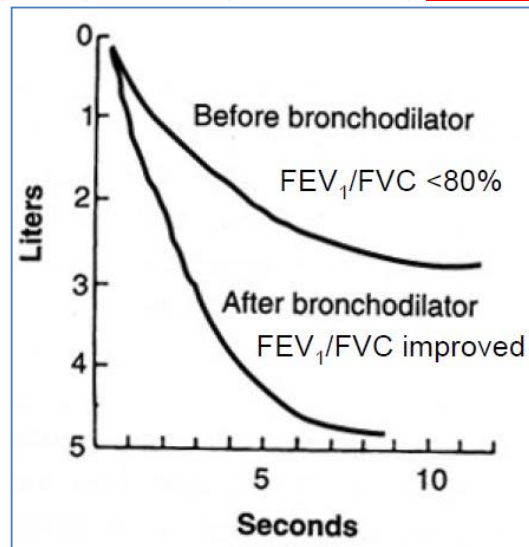
Creative Commons: Pavón-Romero, Serrano-Pérez, García-Sánchez, Ramírez-Jiménez and Terán.
<https://www.frontiersin.org/articles/10.3389/fcell.2021.663535/full>

Clinical Signs of Asthma:

- Asymptomatic between 'Attacks'.
- **'Attacks' of Severe "Dyspnoea"** (Shortness of Breath) Due to Bronchospasm. (Constriction)
 - Coughing
 - Wheezing
- **'Attacks' Triggered by:**
 - Exposure to Allergen (Pollens/Dust/Animal Dander)
- **Dynamic Airway Compression:**
 - Bronchoconstriction + Oedema + Inflammation of Airway → ↑Reliance on Forced Expiration
 - Equal Pressure Point moves into lower (Unsupported) airways.
- **↓FEV₁** (Forced Expiratory Volume in 1 sec) – Due to being an Obstructive condition.
- **↓PEFR** (Peak Expiratory Flow Rate) – Due to ↑Frictional Resistance
- **↑RV** (Residual Volume) – Due to EPP moving lower → Airway Compression → Gas Trapping → Hyperinflation of Lungs.
- **↓Arterial P_{O2}** - Due to Poor Ventilation (↓V/Q Ratio)
- **Response to Bronchodilators:**
 - Asthma **IS** responsive to Bronchodilators
 - However, Chronic Bronchitis & Emphysema are **NOT** responsive
 - I.e: Expiratory Flow Measurements Increase with Bronchodilators.
 - **This is a useful Diagnostic Tool for Determining Chronic & Variable Obstructive Conditions.**
 - **Note: Bronchodilators may also have a Vasodilator Effect:**
 - Leading to slightly ↓ P_{aO2} – due to ↑perfusion of poorly ventilated areas.
 - However the benefits of ↑Ventilation outweigh the slight ↓ P_{aO2}
- **Note: "Status Asthmaticus":** Acute Asthma Unresponsive to Bronchodilators/Corticosteroids. (Can be Fatal)

Diagnosis:

- **Clinical Features:**
 - Dyspnoea, Wheeze, Cough
 - Chest Tightness
 - Tachypnoea, Hyperinflation, ↑ Respiratory Effort,
- **Spirometry:**
 - ↓FEV₁ (Forced Expiratory Volume in 1 sec) – Due to being an Obstructive condition.
 - ↓PEFR (Peak Expiratory Flow Rate) – Due to ↑ Frictional Resistance
 - ↑RV (Residual Volume) – Due to Gas Trapping → Hyperinflation of Lungs.
 - ↓Arterial P_{O2}
 - **Response to Bronchodilators:**
 - Asthma **RESPONDS** to Bronchodilators; COPD's **DO NOT**.
 - *This is a useful Diagnostic Tool for Determining Chronic & Variable Obstructive Conditions.*



Non-Atopic Asthma:

- (Non-Allergic Asthma)
- (Therefore, No Family History & IgE Levels are Normal.)
- **Viral-Induced Asthma:**
 - Asthma triggered by Respiratory-Tract Infections (Mostly Viral)
 - **Pathogenesis:**
 - Believed that Viral-Induced Inflammation of Respiratory Mucosa *Low*ers the Threshold for Stimulation of Sub-Epithelial Vagal (Parasympathetic) Receptors.
 - → ↑ Parasympathetic Stimulation
 - → Bronchoconstriction.
- **Drug-Induced Asthma:**
 - Asthma provoked by Pharmacological Agents
 - The Most Common:
 - Aspirin-Sensitive Asthma – (Stimulates Production of Leukotrienes → Bronchoconstriction)
 - Others:
 - Codeine & Morphine – (Stimulate Mast Cells)
 - Mellitin (Bee Venom) – (Stimulates Mast Cells)
- **Occupational Asthma:**
 - Triggered by Minute Amounts of inhaled pollutants (Fumes, Gases, Chemicals, Dusts)
 - Mechanism Varies with Substance:
 - Either: Hypersensitivity Reactions (Similar to Atopic Asthma)
 - Or: Direct release of Bronchoconstrictors (Without a Hypersensitivity Response)
- **Note: Exercise-Induced Asthma:**
 - Believed to be due to Cooling & Drying of the airway.
 - However, the mechanism is still unclear.

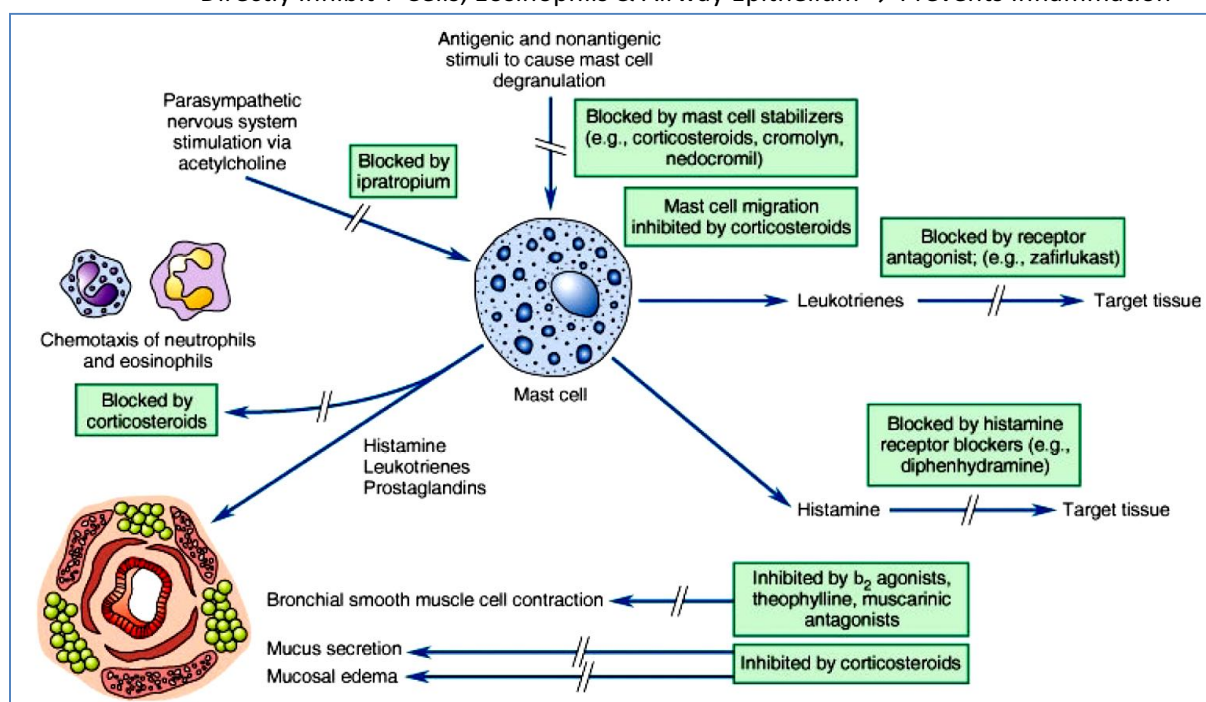
Drugs Used In Asthma: (2 General Types)

- 1. Bronchodilators:

- To Reverse/Prevent Bronchoconstriction
- **β₂-Agonists:**
 - Stimulate β₂-Adrenergic Receptors on:
 - **Airway Smooth Muscle:**
 - Activated β₂-Adrenergic Receptors → Mimic the Physiological Actions of Adrenaline (A Sympathetic Response) → Bronchodilation
 - **Mast Cells/Neutrophils/Eosinophils:**
 - Activated β₂-Adrenergic Receptors → Inhibits Mediator Release
- **Anticholinergics:**
 - Inhibit Muscarinic Receptors on:
 - **Airway Smooth Muscle:**
 - Acts to inhibit the Parasympathetic effect (Constriction) on Airways.
 - (Blocks Parasympathetic NS Stimulation by blocking Acetylcholine Receptors.)
 - Therefore causes Bronchodilation.
 - Note: they're less effective than β₂-Agonists.
 - But may be useful in conjunction with β₂-Agonists.
(ie: Acting on both Para- & Sympathetic Pathways)
- Note: β-Agonists are more effective than Anticholinergics because there is more sympathetic innervations in the lung (& heart). This feature is part of the body's failsafe – so that during rest (where parasympathetic NS should dominate, leading to bronchoconstriction), there is enough residual sympathetic innervations to keep airways dilated. Because of this, there are more sympathetic receptors for potential drug action → Equates to ↑ Effectiveness of β-Agonists.

- 2. Anti-Inflammatory Drugs:

- To halt Inflammatory Response
- **Corticosteroids:**
 - Stabilize Mast-Cell Membrane → Prevents Degranulation.
 - Reduce Chemotaxis (Migration) of Mast-Cells, Neutrophils & Eosinophils.
 - Inhibits Mucus Secretion
 - Inhibits Mucosal Oedema
 - Enhances β-Receptor Expression/Function (Amplifies Sympathetic Responses)
 - Disrupt Production of Inflammatory Mediators (Cytokines) from Neutrophils & Eosinophils.
 - Directly Inhibit T-Cells, Eosinophils & Airway Epithelium → Prevents Inflammation



Source: Unattributable

Nebulizer Vs. Inhaler:

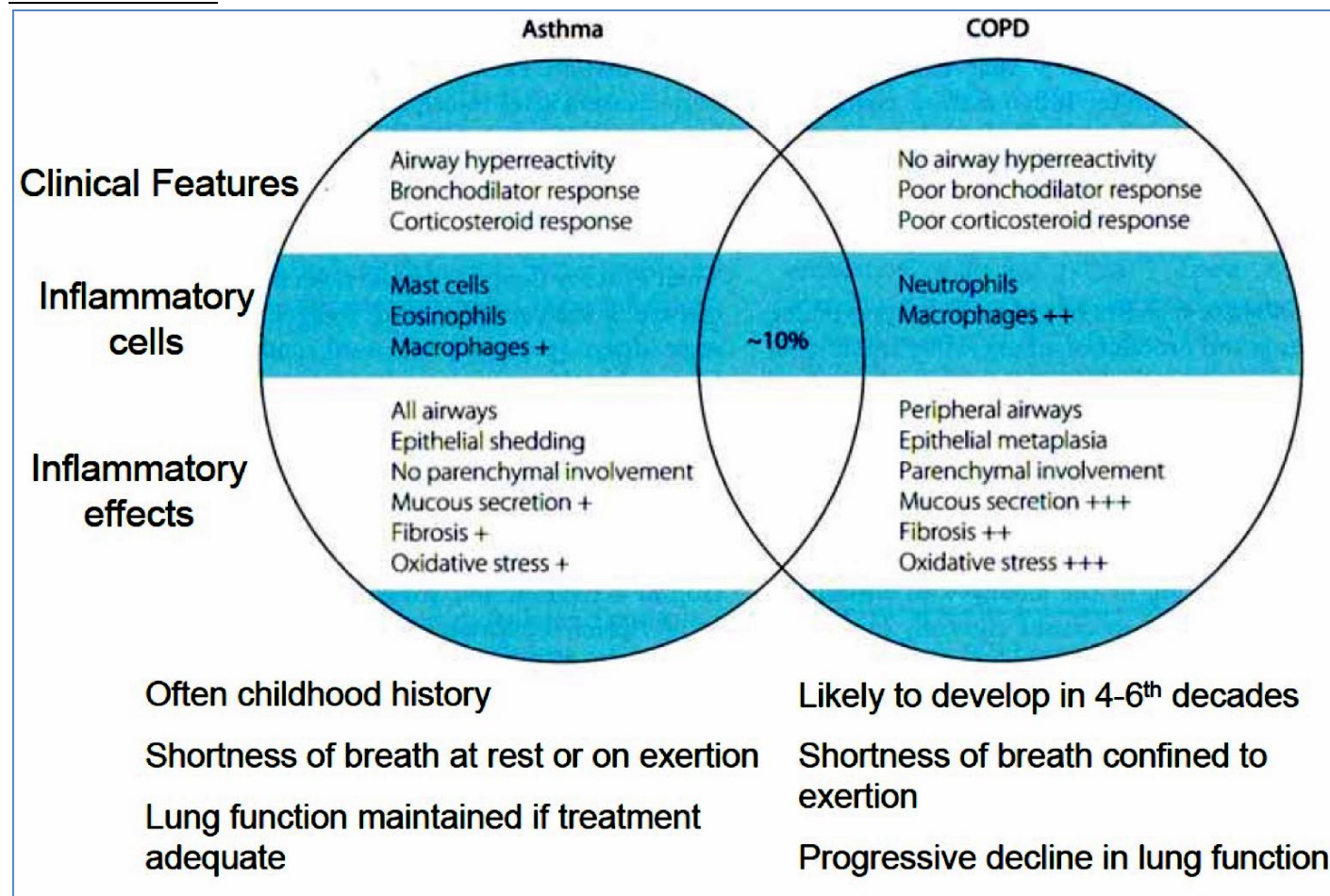
- Nebulizers allow higher doses of β -Agonists (Bronchodilators) than a puffer.
- Nebulizers are also easier for the patient during an acute attack.

Clinical Management:

- **Prevention:**
 - **Mild Asthma:** Inhaled Corticosteroids (*Budesonide* or *Fluticasone*)
Or Inhaled Antimuscarinic (*Ipratropium Bromide*) – If ICS-Intolerant.
 - **Moderate Asthma:** LABA + Inhaled Corticosteroid Combinations
 - Symbicort [*Budesonide* + *Eformoterol*]
 - or Seretide [*Fluticasone* + *Salmeterol*]
 - **Severe Asthma:** Oral Leukotriene Inhibitors (Singulair [*Montelukast*])
- **Acute Attack:**
 - **First Aid (Where *Salbutamol* is the only Rx):**
 - **“4x4x4 Rule” – 4xPuffs, 4xBreaths/Puff, Wait 4 Mins....Then Repeat if Necessary.**
 - **Paediatric:**
 - (Brief History & Examination)
 - O2 if Necessary – (Distressed or SpO₂<92%)
 - 1. Ventolin(***Salbutamol***) Via Spacer <6puffs (<6yo) or <12puffs (>6yo) q20mins in 1st Hour;
 - 2. (If SEVERE)+/- ***Ipratropium Bromide*** 2puffs (<6yo) or 4puffs (>6yo) q20mins in 1st hr
 - (Note: Spacer should only be loaded with 1x puff/drug at a time)
 - (Note: If no improvement after 1st Hour → Call Ambulance → ED)
 - 3. (1^o HC Setting) add Systemic PO-***Prednisolone*** – (Continue OD for 3-5days);
 - 4. (If SEVERE & Still no improvement, add IV-***Magnesium Sulfate***)
 - 5. (If STILL SEVERE → ICU Admission → IV-***Aminophylline***)
 - **Adult:**
 - (Brief History & Examination)
 - O2 if Necessary – (Distressed or SpO₂<92%)
 - 1. Ventolin(***Salbutamol***) Via Spacer <6puffs (<6yo) or <12puffs (>6yo) q20mins in 1st Hour;
 - 2. (If SEVERE)+/- ***Ipratropium Bromide*** 2puffs (<6yo) or 4puffs (>6yo) q20mins in 1st hr
 - (Note: Spacer should only be loaded with 1x puff/drug at a time)
 - (Note: If no improvement after 1st Hour → Call Ambulance → ED)
 - 3. (1^o HC Setting) add Systemic PO-***Prednisolone*** – (Continue OD for 7-10days);
 - 4. (If SEVERE & Still no improvement, add IV-***Aminophylline***)

Example of Asthma Management Plan:	
Prophylaxis:	Singulair (Montelukast Sodium), 1x 4mg Tablet, Every Night
Mild Asthma Symptoms: (Mild wheeze, tight chest, shortness of breath)	Ventolin (Salbutamol), 2x Puffs (via spacer if available), Repeat 3-4x daily as necessary.
Moderate Asthma Symptoms: (Moderate wheeze, tight chest, shortness of breath)	Ventolin (Salbutamol), up to 6x Puffs (via spacer if available), Repeat 3-4x daily as necessary. + Prednisolone, 1x 20mg dose (4mLs of 5mg/ml liquid), Immediately & then every morning for at least the next 3 days.
Severe Asthma Symptoms: (Severe wheeze, tight chest, shortness of breath. Eg if...: - Requiring Ventolin >3hrly - No relief from Ventolin - Persistent wheeze >24hrs - Or severe attack)	Ventolin (Salbutamol), up to 6 Puffs (via spacer if available), Repeat dose every 15-30mins if not improving. Call an ambulance if worried or not improving. (Continue to give Ventolin as above while waiting for ambulance)

Asthma Vs. COPD:

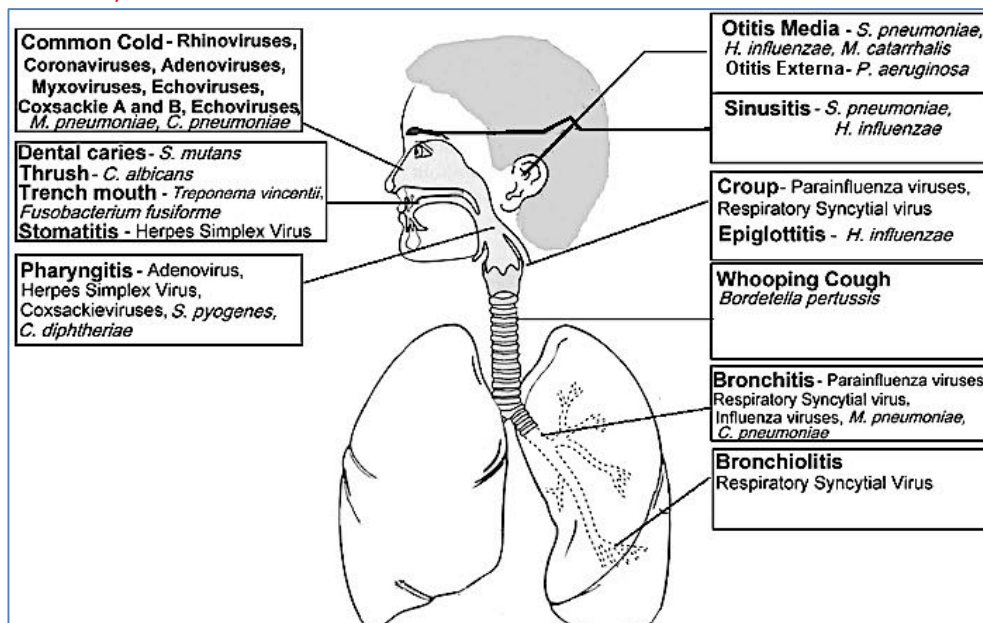


UPPER RESPIRATORY TRACT INFECTIONS (URTI'S)

UPPER RESPIRATORY TRACT INFECTIONS (URTI'S)

Pathogens at different levels of respiratory tract

- **Nasopharynx:**
 - Rhinovirus
 - Adenovirus
 - Coronavirus
 - RSV (Respiratory Syncytial Virus)
 - Influenza Virus
- **Sinuses & Ears:**
 - Strep. pneumoniae
 - H. Influenza
 - Staph. Aureus
 - Moraxella Catarrhalis
- **Oropharynx:**
 - Adenovirus
 - Rhinovirus
 - Epstein-barr Virus
 - Group A B-Haemolytic Streptococcus
- **Epiglottitis:**
 - H-Influenzae
- **Trachea/Bronchi:**
 - RSV (Respiratory Syncytial Virus) (**Croup**)
 - Rhinovirus
 - Coronavirus
 - Adenovirus
 - Parainfluenza
 - Influenza virus
 - Strep pneumonia
 - H. Influenza
- **Lower Respiratory Tract:**
 - Strep. pneumonia
 - Haem. Influenza
 - Staph. Aureus
 - Mycoplasma
 - Chlamydia pneumonia
 - (**Pseudomonas**)
 - (**Klebsiella**)



Viruses Causing Common Colds – (Typically Rhinoviruses):

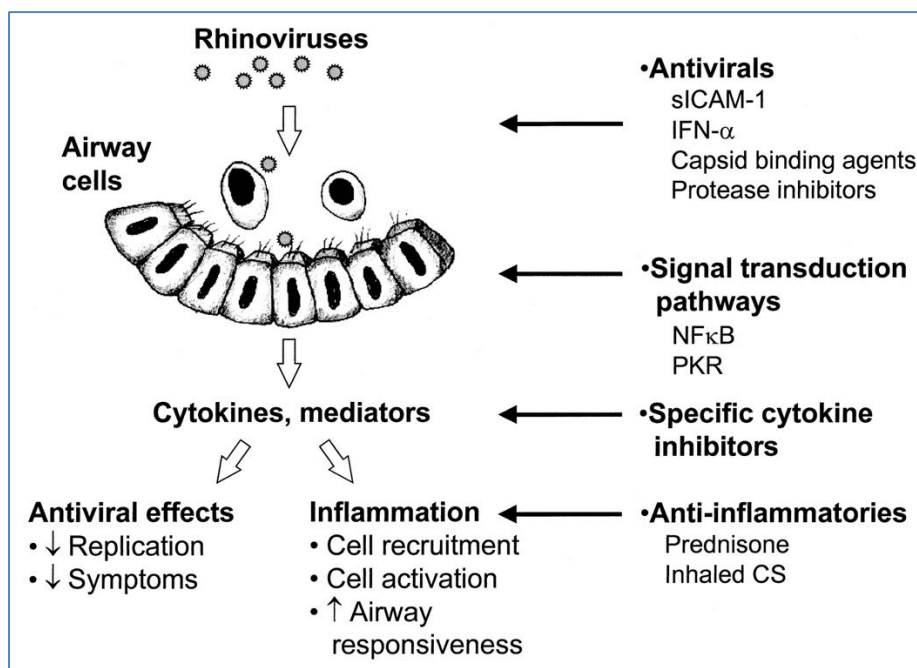
- **#1 Rhinoviruses:**
 - Typically restricted to URT
 - **Many Serotypes:**
 - Endemic throughout the year
 - No Cross-Protection between Serotypes
 - → **Possibility of Repeated Infections**
 - Short Incubation Period (2-3 days) with Inflammation, Oedema and Copious Exudate
 - Resolution due to Immune System. (Self-Limiting)
- **Adenovirus:**
 - Most infections occur in early-life <5yrs
 - Rarely causes disease
 - Symptoms = Nasal Congestion, Cough, Pharyngitis (Sore Throat)
- **Coronaviruses (Including SARS):**
 - Can infect URT & LRT
 - Replication is confined to the Epithelial Layer
 - Infection is usually Mild
- **Coxsackie Virus A**
- **Influenza Viruses (Orthomyxoviruses) (may also cause LRTI)**
 - **Prevalence:**
 - Up to 20% of the population may be infected in any one year
 - Majority of Deaths are Infants and Elderly
 - **Pathogenesis:**
 - **Short Incubation Period (2-3 days)**
 - **Abrupt Onset of Symptoms:**
 - Shivering/Fever (39C), Malaise, Headache, Myalgia, & Sometimes Pneumonia
 - **Few Complications:**
 - Mainly Secondary Bacterial Infections
 - **Vaccination:**
 - Indication = Anyone over 6mths who wants to ↓ Risk of Catching Influenza.
 - Especially those over 50yrs
- **Paramyxoviruses:**
 - **Parainfluenza Viruses (1-4)**
 - Causes 30% of all RTIs
 - Causes 50% of RTIs in Preschool Children
 - Can be Asymptomatic
 - Transmission is by Respiratory Secretions (Eg. Toys in Childcare Centres)
 - → Major Manifestations =
 - Necrotising Bronchiolitis
 - Respiratory Syncytia
 - Diagnosis: Viral Isolation or RT-PCR.
 - **Respiratory Syncytial Virus**
 - Highly Contagious
 - Transmission is by Respiratory Secretions (Eg. Toys in Childcare Centres)
 - → Major Manifestations:
 - Initial Cold-Like Symptoms
 - ***Necrotizing Bronchiolitis**
 - (+/- Pneumonitis)
 - Respiratory Syncytia – (Because viruses enter via fusion proteins → Which join cells together)
 - Within 24hrs - Severe Illness, Cyanosis & Distress
 - Significant Mortality
 - Reinfection later in life is frequent.
 - Diagnosis: Viral Isolation or RT-PCR.
 - **Morbillivirus (Measles Virus)**
 - **Rubulavirus (Mumps Virus)**

DIFFERENTIATING BETWEEN URTI SYNDROMES – (Their Distinguishing Features):

- **Typical URTI Symptoms:**
 - Fever, Malaise, Headache, Cough
- **+ Specific Symptoms:**
 - *Adolescent + Sore Throat + Lymphadenopathy* = **EBV (Glandular Fever)**
 - *Toddler + Drooling Saliva* = **Epiglottitis (Haemophilus Influenza)**
 - *Toddler + Barking Cough* = **Croup**
 - *58yr + Type 2 Diabetes* = **Atypical Organism (Due to Immunocompromise)**
 - *Rash + Myalgia* = **Non-Specific Viral**
 - *Photophobia + Rash + Neck Stiffness* = **Meningitis (Eg. Neisseria Meningitidis)**
 - *Pharyngitis → Arthritis + New Cardiac Murmur after 2wks* = **GAB-Strep Pharyngitis → Rh-Fever**
 - *Hx of Recurrent Pneumonia + SOB* = **Cystic Fibrosis/Immunocompromised/Smoker**
 - *SOB + Weight Loss* = **Tuberculosis (Mycobacterium Tuberculosis)**
 - *Community Outbreak + Travel History* = **SARS (SARS-Associated Coronavirus)**

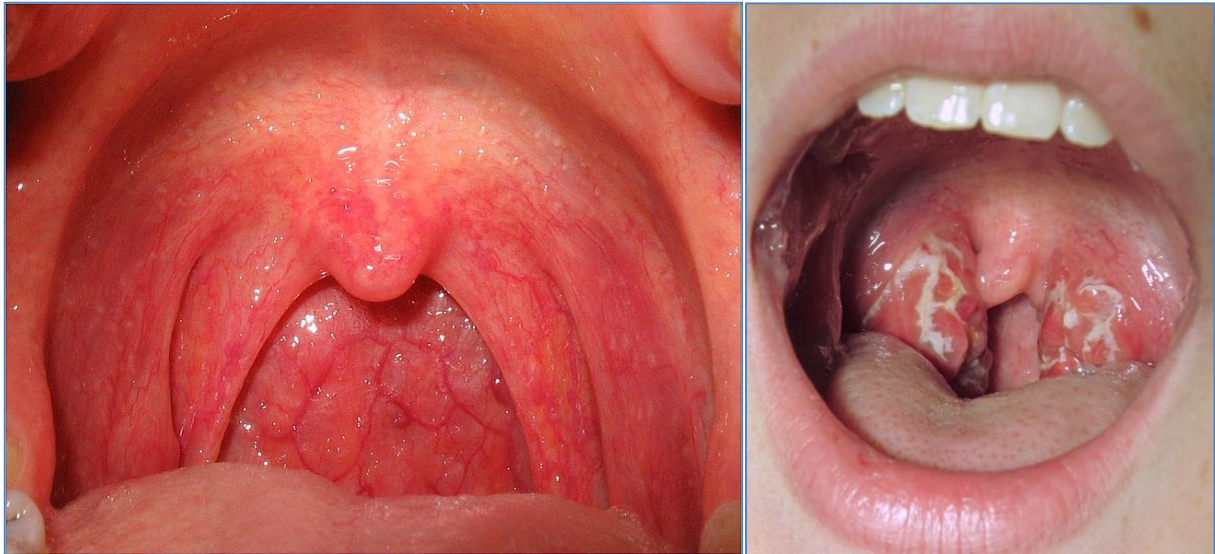
COMMON COLD (ACUTE RHINITIS)

- **Aetiology:**
 - **Rhinoviruses**, Adenoviruses, Paramyxoviruses, Influenza viruses, Myxoviruses,
- **Pathogenesis:**
 - **Transmission** – (Droplet Transmission/Contact Secretions)
 - **Viral Infection of URT Mucosa** → URT Inflammation → Mucous Hypersecretion
 - (Note: No Cross-Protection between Serotypes → Possibility of Repeated Infections)
- **Clinical Features**
 - **Short Incubation Period (2-3 days)**
 - **1wk Of Symptoms:**
 - **Local** - Nasal Congestion, Sneezing, Sore Throat, Hoarseness, Cough, Conjunctivitis
 - **General** - Malaise, Headache, Myalgias, Mild Fever
 - **Signs**
 - Rhinorrhoea
 - Inflamed Nasal/Oropharyngeal Mucosa
 - Lymphadenopathy
 - **Note: Normal Chest Exam**
 - **Complications**
 - **Secondary Bacterial Infection:** (Otitis Media, Sinusitis, Tonsillitis, Bronchitis, Pneumonia)
 - **Asthma/COPD Exacerbation**
 - **Benign Inflammatory Nasal Polyps**
- **Diagnosis:**
 - **Differentials:**
 - Allergic Rhinitis, Pharyngitis, Influenza, Laryngitis, Croup, Sinusitis, Bacterial Infections
 - **Clinical Diagnosis** – (Symptoms + Nasal Exam + Inflamed Mucosa + Watery Discharge)
 - **Laboratory Diagnosis** – ONLY if *Other Conditions are Suspected*.
- **Management:**
 - **Patient Education**
 - **No Antibiotics Indicated Because Of Viral Aetiology**
 - **Consider 2° Bacterial Infection** if NO Resolution after 3-10 Days
 - ***Symptomatic Relief:**
 - **Paracetamol**
 - **Decongestants** (*Phenylephrine/Pseudoephedrine*), **Antihistamines**
 - + Rest, Hydration, Gargling Warm Salt Water, Steam
 - +(↑ Dependence On Bronchodilators/Inhaled Steroids For Asthmatics & COPD)



PHARYNGITIS (SORE THROAT)

- **Definition**
 - = Inflammation of the Oropharynx (*Without* inflammation of the tonsils)
- **Aetiologies:**
 - **Viral (40-60%) – Most Common:**
 - **Adenovirus**, Coxsackie, HSV, **EBV**, **Influenza** Virus (Orthomyxovirus),
 - **Bacterial**
 - **"Strep. Pyogenes" (GABH-Streptococcus)** – (*Rh-Heart Disease, PSGN & Scarlet Fever)
 - *Neisseria gonorrhoea*, *Chlamydia pneumoniae*, *Mycoplasma pneumoniae*, ***Corynebacterium diphtheria***
- **Morphology:**
 - Red, Inflamed Oropharynx
 - May have white lesions
 - May have pus



Viral: <https://commons.wikimedia.org/wiki/File:Pharyngitis.jpg>

Strep: https://commons.wikimedia.org/wiki/File:Pos_strep.JPG

- **Clinical Features**
 - Typically a self-limited infection with no significant sequelae
 - **Bacterial - Group A Beta-Hemolytic Streptococcus**
 - **Absence Of Cough**, Pharyngitis, + Flu-Like Illness
 - **Signs: Fever + Tonsil Exudate + Lymphadenopathy + <15yo + NO Cough**
 - **Complications!!**
 - **Rheumatic Fever**
 - **Glomerulonephritis**
 - **Meningitis**
 - **Viral - Adenovirus**
 - **Cough** – (Due to Rhinorrhoea), Pharyngitis, + Flu-Like Illness
 - **Viral - Ebv (Infectious Mononucleosis)**
 - Pharyngitis, Fever, **Lymphadenopathy, Fatigue, Rash**
- **Investigations**
 - **Suspected GABH-Strep:**
 - **Throat Culture = Definitive** (But TOO SLOW in the real world!!)
 - **RDT For Streptococcal Antigen**
 - **ASOT (Anti-Streptolysin-O-Titres)** – (But only shows recent infection).
 - **Suspected EBV (Infectious Mononucleosis):**
 - **Peripheral Blood Smear** – (Reactive Lymphocytes)
 - **"Monospot" Test** (I.E. The Latex Agglutination Assay, Or "Monospot")
 - **EBV Serology**

- **Management**

- If ?GABH-Strep:

- ****Throat Swab if:** **Fever + Tonsil Exudate + Lymphadenopathy + <15yo**
 - **Antibiotics!!:** **Penicillin-V/G or Erythromycin if Penicillin Allergic**

- If ?Viral Pharyngitis:

- Antibiotics NOT indicated
 - *Paracetamol/NSAIDs*
 - Decongestants (*Phenylephrine*)

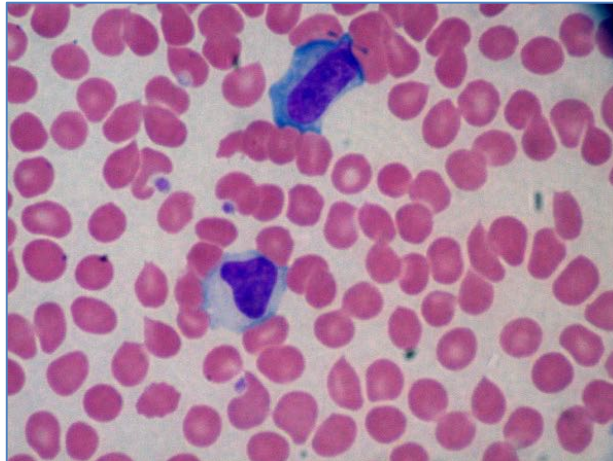
- If ? Infectious Mononucleosis (EBV):

- Antibiotics NOT indicated; **Note: Penicillin will → Rash (Pathognomonic)**
 - **Self-Limiting Course;** Rest During Acute Phase Is Beneficial
 - **Supportive Treatment:** NSAIDS for fever, sore throat, malaise

Other Notable Pharyngitis's:

- (EPSTEIN BARR VIRUS) – INFECTIOUS MONONUCLEOSIS (GLANDULAR FEVER):

- **Aetiology:**
 - Epstein Barr Virus
- **Pathogenesis:**
 - Transmitted through Saliva (Ie: Kissing Disease)
 - Incubation period <8wks.
 - **Preferentially Infects B-Cells → Reactive B-Lymphocytes → “Mononucleosis”**
- **Morphology:**
 - Tender Cervical Lymphadenopathy
 - Blood Smear – Lymphocytosis with *Atypical Lymphocytes*



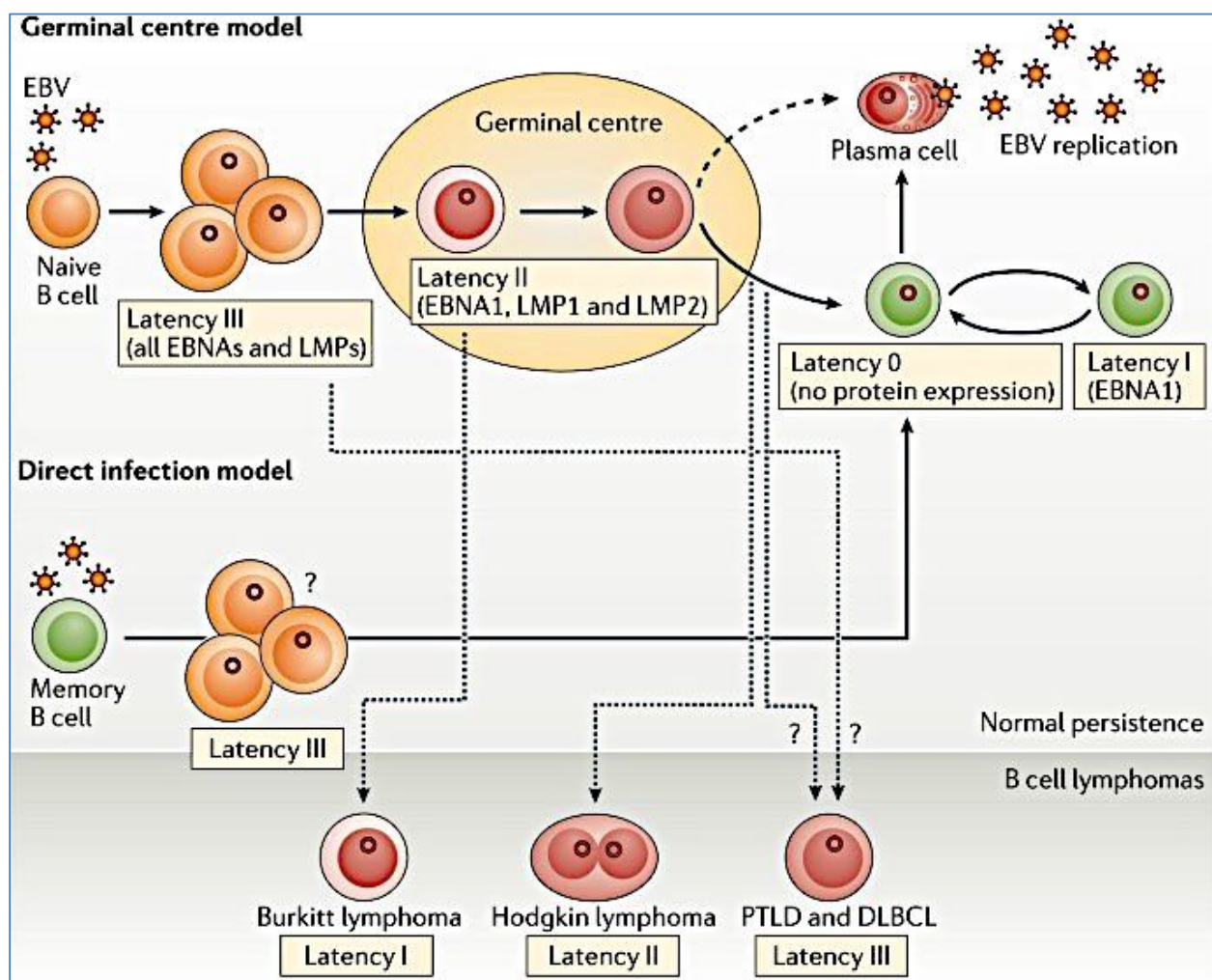
Creative commons: <https://www.flickr.com/photos/euthman/145052721>

- **Clinical Features:**
 - **Signs/Symptoms:**
 - Fever +
 - **Glandular Fever Triad:**
 - **Fatigue**/Malaise (Anorexia/Lethargy)
 - **Pharyngitis** (Sore Throat)
 - **Lymphadenopathy** (Especially Cervical)
 - **Others** – (Splenomegaly, Hepatitis, Haemolysis, Jaundice)



James Heilman, MD, CC BY-SA 3.0 <<https://creativecommons.org/licenses/by-sa/3.0/>>, via Wikimedia Commons

- **Diagnosis:**
 - Typically Clinical
 - Peripheral Blood Smear – (Reactive Lymphocytes)
 - “Monospot” Test (I.E. The Latex Agglutination Assay, Or "Monospot")
 - EBV Serology
 - + LFTs
- **Treatment:**
 - Antibiotics NOT indicated; Note: Penicillin will → Rash (*Pathognomonic*)
 - Self-Limiting Course; Rest During Acute Phase Is Beneficial
 - Supportive Treatment: NSAIDS for fever, sore throat, malaise
- **Complications:**
 - EBV is an Oncogenic Herpesvirus → Tumours:
 - →Burkett's Lymphoma
 - →Hodgkin's Lymphoma
 - →Nasopharyngeal Carcinoma



Creative Commons: https://media.springernature.com/lw685/springer-static/image/art%3A10.1038%2Fncr.2016.92/MediaObjects/41568_2016_Article_BFncr201692_Fig3_HTML.jpg

- **DIPHTHERIA:**

- **Aetiology:**
 - Gram Positive Bacterium – *Corynebacterium Diphtheriae*
- **Pathogenesis:**
 - **Transmission** – Aerosol, Physical Contact.
- **Morphology:**
 - Adherent Whitish Pseudomembrane Over Pharynx & Tonsils (May → Obstruction)
- **Clinical Features:**
 - High Fever, Sore Throat, Fatigue, Nausea & Vomiting
 - **Pseudomembrane on Tonsils & Pharynx** - May have Airway Obstruction & Dysphagia
- **Complications:**
 - **Systemic Exotoxin →**
 - **Myocarditis** (Potentially fatal toxigenic Cardiomyopathy → Heart Failure)
 - Peripheral Neuritis.
 - Chronic Non-Healing Ulcers
- **Diagnosis:**
 - Swab M/C/S
 - + Toxin Detection
- **Treatment:**
 - **Penicillin or Erythromycin** (if Penicillin Allergic)



User:Dileepunnikri, CC BY-SA 3.0 <<https://creativecommons.org/licenses/by-sa/3.0/>>, via Wikimedia Commons

- **SCARLET FEVER ("STRAWBERRY TONGUE"):**

- **Aetiology:**
 - Certain strains of **GABH-Strep "Pyogenes"** (Which are infected with a "Bacteriophage" [Virus] → Produce an Erythrogenic toxin)
- **Pathogenesis:**
 - **GABH-Strep Infection → Exotoxin → Local effect on Tonsils/Pharynx/Skin**
 - → **Tongue**
 - Initially covered with white exudate
 - Exudate is shed
 - inflammation of underlying tissue
 - → **Skin**
 - Diffuse, Erythematous Rash
- **Complications:**
 - Rheumatic Heart Disease
 - PSGN
- **Diagnosis:**
 - ****Throat Swab if: Fever + Tonsil Exudate + Lymphadenopathy + <15yo**
- **Treatment:**
 - **Antibiotics!:** **Penicillin-V/G or Erythromycin if Penicillin Allergic**



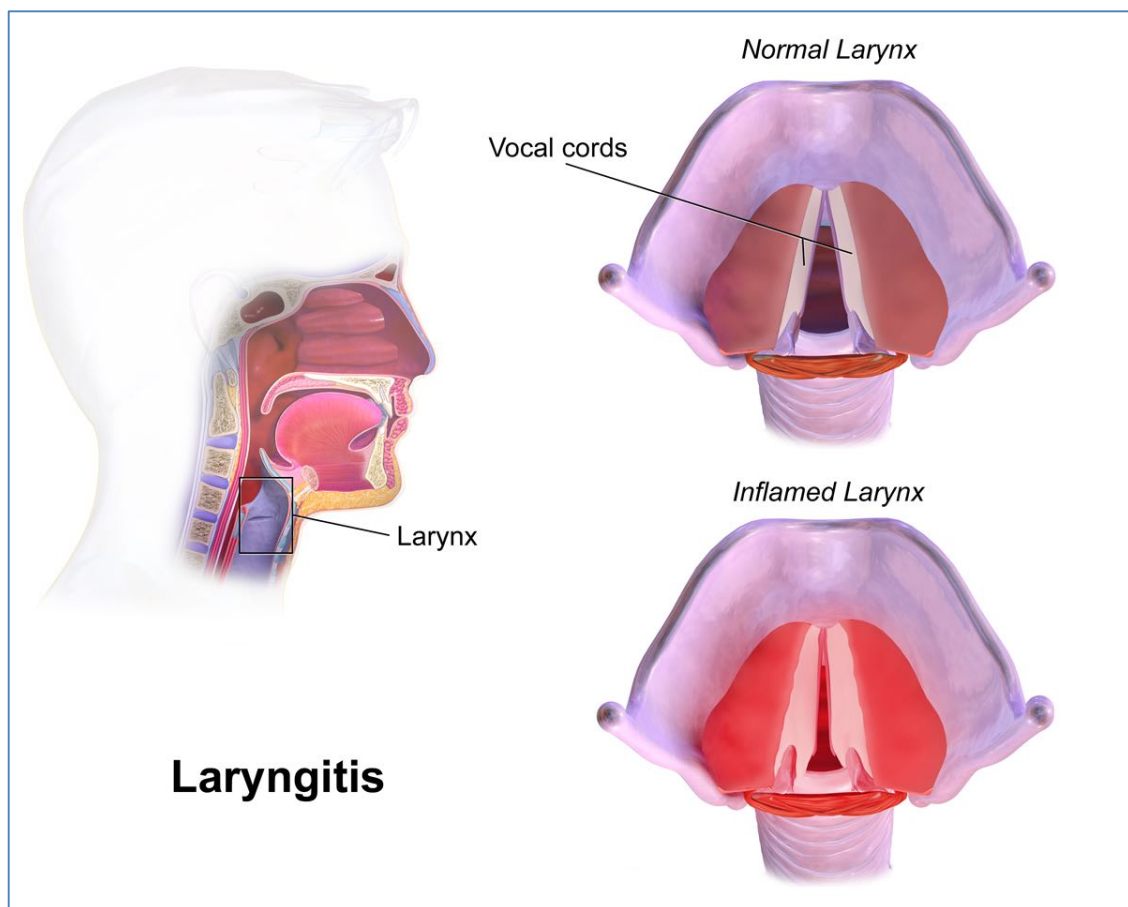
- 1 - Estreya at English Wikipedia., CC BY 2.5 <<https://creativecommons.org/licenses/by/2.5>>, via Wikimedia Commons
2 - www.badobadop.co.uk, CC BY-SA 3.0 <<https://creativecommons.org/licenses/by-sa/3.0>>, via Wikimedia Commons



Afag Azizova, CC BY-SA 3.0 <<https://creativecommons.org/licenses/by-sa/3.0>>, via Wikimedia Commons

ACUTE LARYNGITIS

- **Aetiology (Note: Infective aetiologies similar to pharyngitis):**
 - **Viral:** *Adenovirus*, Influenza
 - **Bacterial:** *Group A Streptococcus*
 - **Acute Voice Strain** → Submucosal Haemorrhage → Vocal Cord Oedema → Hoarseness
 - **Toxic Fume Inhalation**
- **Clinical Features**
 - URTI Symptoms, **Hoarseness, Aphonia, Cough Attacks, ± Dyspnea**
- **Morphology:**
 - True Vocal Cords Erythematous/Oedematous With Vascular Injection And Normal Mobility
- **Treatment**
 - Self-Limited, Resolves Within -1 Week
 - Voice Rest
 - Humidification, Hydration
 - Avoid Irritants (E.G. Smoking)
 - Treat With Antibiotics If There Is Evidence Of Coexistent Bacterial Pharyngitis



Laryngitis

BruceBlaus, CC BY-SA 4.0 <<https://creativecommons.org/licenses/by-sa/4.0>>, via Wikimedia Commons

CHRONIC LARYNGITIS

- **Definition**
 - Long Standing Inflammatory Changes In Laryngeal Mucosa
- **Aetiology**
 - Repeated Attacks Of Acute Laryngitis
 - Chronic Irritants (Dust, Smoke, Chemical Fumes)
 - Chronic Voice Strain
 - Chronic Sinusitis With Postnasal Drip (PND)
 - Chronic Alcohol Use
 - Oesophageal Disorders: eg. GORD, Hiatus Hernia
- **Clinical Features**
 - Chronic Dysphonia – (Note: Rule Out Malignancy)
 - Cough, Globus Sensation, Frequent Throat Clearing 2° To GORD
- **Morphology:**
 - Cords Erythematous, Thickened With Ulceration / Granuloma Formation And Normal Mobility
- **Treatment**
 - Remove Offending Irritants
 - Treat Related Disorders Eg. Antisecretory Therapy For GORD
 - Speech Therapy With Voice Rest
 - ± Antibiotics, ± Steroids To Decrease Inflammation



Creative commons: <https://www.voicedoctor.net/media/structural-injury-inflammation-bacterial/chronic/chronic-bacterial-laryngitis>

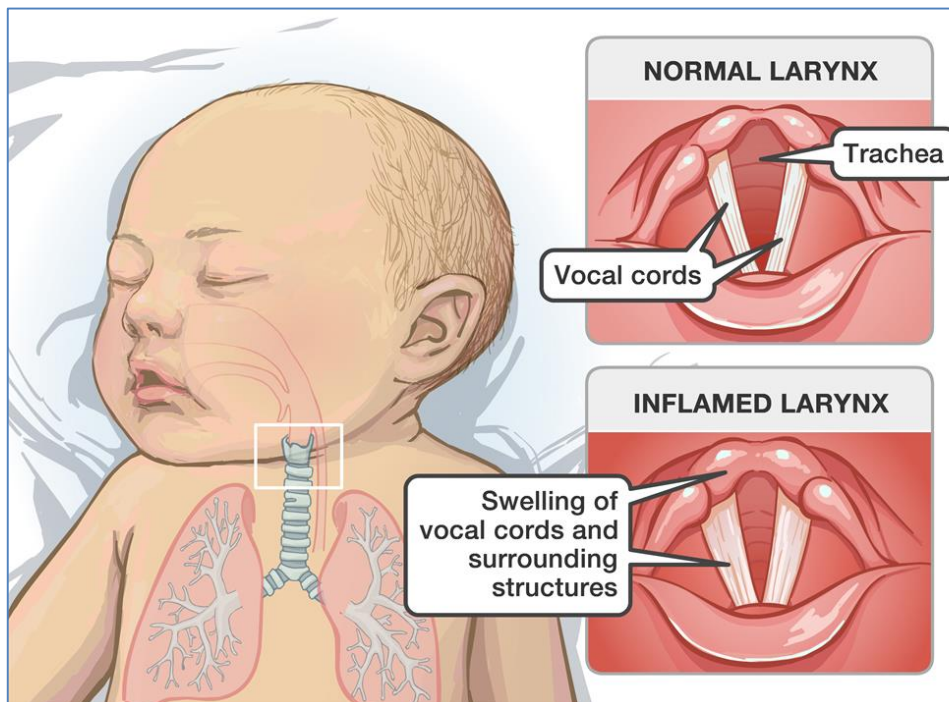
ACUTE TONSILLITIS

- **Aetiology**
 - **GABH-Strep (Pyogenes)**
 - **Or S-pneumoniae, S-aureus, H-Influenzae, M-Cararrhalis**
 - **Or Epstein-Barr virus (EBV)**
- **Clinical Features**
 - **Typically Children (5-10yrs) & Adolescents (15-25yrs)**
 - **Symptoms:**
 - Pharyngitis
 - Referred Ear Pain
 - Headache
 - Dysphagia, Odynophagia, Trismus
 - Malaise, Fever
 - **Signs:**
 - Fever
 - Reddened Throat
 - Tonsils – (Enlarged, Inflamed ± Exudates / White Follicles)
 - Swollen, Tender Cervical Lymphadenopathy
 - **(If Scarlet Fever → Strawberry Tongue & Scarletiform Rash)**
 - **(If EBV → Palatal Petechiae)**
- **Differentials:**
 - **Strep Pharyngitis**
 - Viral Pharyngitis
 - **EBV**
 - Peritonsillar Abscesses
- **Investigations**
 - **FBC – (↑WCC + Differentials)**
 - **Throat Swab MCS – (?GABH-Strep Pyogenes)**
 - **(Suspected if Fever + Tonsillar Exudate + Lymphadenopathy + NO Cough + <15yo)**
 - **ASOT – Anti-Streptolysin ‘O’ Titre – (?GABH-Strep Pyogenes).**
 - **Monospot Test –(?EBV)**
- **Treatment/Management:**
 - (Bed Rest, Soft Diet, Fluids)
 - **Paracetamol**
 - **Antibiotics –(if Fever + Tonsillar Exudate + Lymphadenopathy + NO Cough + <15yo):**
 - **Penicillin** Or **Amoxicillin** (**Erythromycin** If Penicillin Allergy) X 10 Days
- **Complications**
 - **Abscess:** Peritonsillar (a “Quinsy”), Intra Tonsillar
 - **Post-Streptococcal:**
 - **Glomerulonephritis**
 - **Rheumatic Heart Disease**
 - **Scarlet Fever**



ACUTE LARYNGOTRACHEOBRONCHITIS (CROUP)

- **What is it?**
 - Inflammation Of Tissues In Subglottic Space ± Tracheobronchial Tree
 - + Thick, Viscous, Mucopurulent Exudates Which Compromises Upper Airway → Barking Cough
- **Aetiology – Viral:**
 - *RSV or Parainfluenzae (Most Common), II, III, Influenza A And B
- **Pathogenesis:**
 - URTI
 - → Inflammation Of Tissues In Subglottic Space
 - → Thick, Viscous, Mucopurulent Exudates Which Compromises Upper Airway → **Barking Cough**
- **Morphology:**
 - Inflamed Upper Airways + Larynx
- **Clinical Features**
 - Typically Children <5yrs
 - **Signs of Croup - the 3 S's**
 - 1. Stridor
 - 2. Subglottic swelling
 - 3. Seal bark cough
 - +/- Cyanosis & Respiratory Distress
- **Treatment**
 - (Note: Viral ∴ NO Antibiotics)
 - Oral/IM Corticosteroids (**Dexamethasone** / **Prednisone**)
 - Nebulised **Epinephrine**
 - Humidified **O2**
 - +/- Intubation If Severe



<https://www.aboutkidshealth.ca/croup>

ACUTE EPIGLOTTITIS

- **Aetiology**
 - **HiB – (*Haemophilus Influenzae* type B)** (Uncommon due to HiB vaccine)
 - (Gram neg coccobacillus)
- **Clinical Features**
 - Typically Children 1-4yo
 - High Fever & Unwell
 - Sore Throat, Dysphagia, Anorexia
 - **Obstructive Symptoms – MEDICAL EMERGENCY → INTUBATE:**
 - **Difficulty Swallowing, DROOLING, cyanotic/pale, inspiratory stridor, slow breathing,**
- **Investigations:**
 - **Preparations For Intubation Or Tracheotomy** Must Be Made Prior To Any Manipulation
 - **Lateral Neck XR** - Cherry-Shaped Epiglottic Swelling ("Thumb Sign") - Only If Stable
 - **WBC (Elevated)**
 - **Blood And Pharyngeal Cultures After Intubation**
- **Treatment**
 - ***Admit to ICU**
 - **Urgent Intubation → Secure Airway**
 - + Humidified O2
 - **Antibiotics – (*Ceftriaxone* + *Clindamycin*)**
 - **Extubate When Afebrile**
 - **Watch For Meningitis**



https://commons.wikimedia.org/wiki/File:Epiglottitis_endoscopy.jpg

PERTUSSIS - WHOOPING COUGH:

- **Aetiology:**
 - ***Bordetella pertussis*** – (Only a human pathogen)
- **Pathogenesis:**
 - Infection of Trachea & Bronchi → **Toxins** → Widespread Trachea/Bronchi Inflammation
 - Pertussis toxin → increased secretions → Cough
 - Dermonecrotic toxin → vasoconstriction, ischaemia
 - Tracheal cytotoxin → inhibition of cilia movement
- **Clinical Features:**
 - Severe childhood disease
 - → Dyspnoea
 - → Chronic, Severe Coughing Fits
 - Highly Contagious (infants <12mths)
- **Investigations:**
 - Diagnosed on Clinical Suspicion
 - (Culture takes <2wks – TOO Long!)
- **Management:**
 - Empirical Antibiotics – (**Azithromycin** / **Clarithromycin** / **Erythromycin**)
 - + Booster Vaccination (Unvaccinated / Adolescents / Adults)
 - + Vaccinate close contacts (DTP Vaccine)
 - +/- Post-Exposure Prophylaxis in Close Contacts (**Azithromycin**)

Pertussis vaccine immunity wanes over time	Young infants are most at risk	How to prevent infection
Pertussis, or whooping cough, is a respiratory tract infection that can be prevented with vaccination. PROTECTION  However, protection takes several shots to establish and then protection wanes over time.	Infants less than three months of age have only received one dose of the vaccine.  Infants get the most sick A recent study found most infants less than three months with pertussis had severe disease and of those¹: • 92% required hospitalization • 28% intensive care admission	Make sure you and your family are up to date with your vaccines²: 1. ROUTINE CHILDHOOD IMMUNIZATION SCHEDULE - 2, 4, 6, 12-23 months and 4-6 years  2. ADOLESCENT BOOSTER - at 14-16 years 3. ADULT BOOSTER 4. WITH EVERY PREGNANCY - Protective antibodies are transferred to the baby - Best between 27-32 weeks 

<https://www.canada.ca/en/public-health/services/reports-publications/canada-communicable-disease-report-ccdr/monthly-issue/2018-44/issue-9-september-6-2018/article-1a-pertussis-whooping-cough-still-danger-infants-infographic.html>

MEASLES VIRUS:

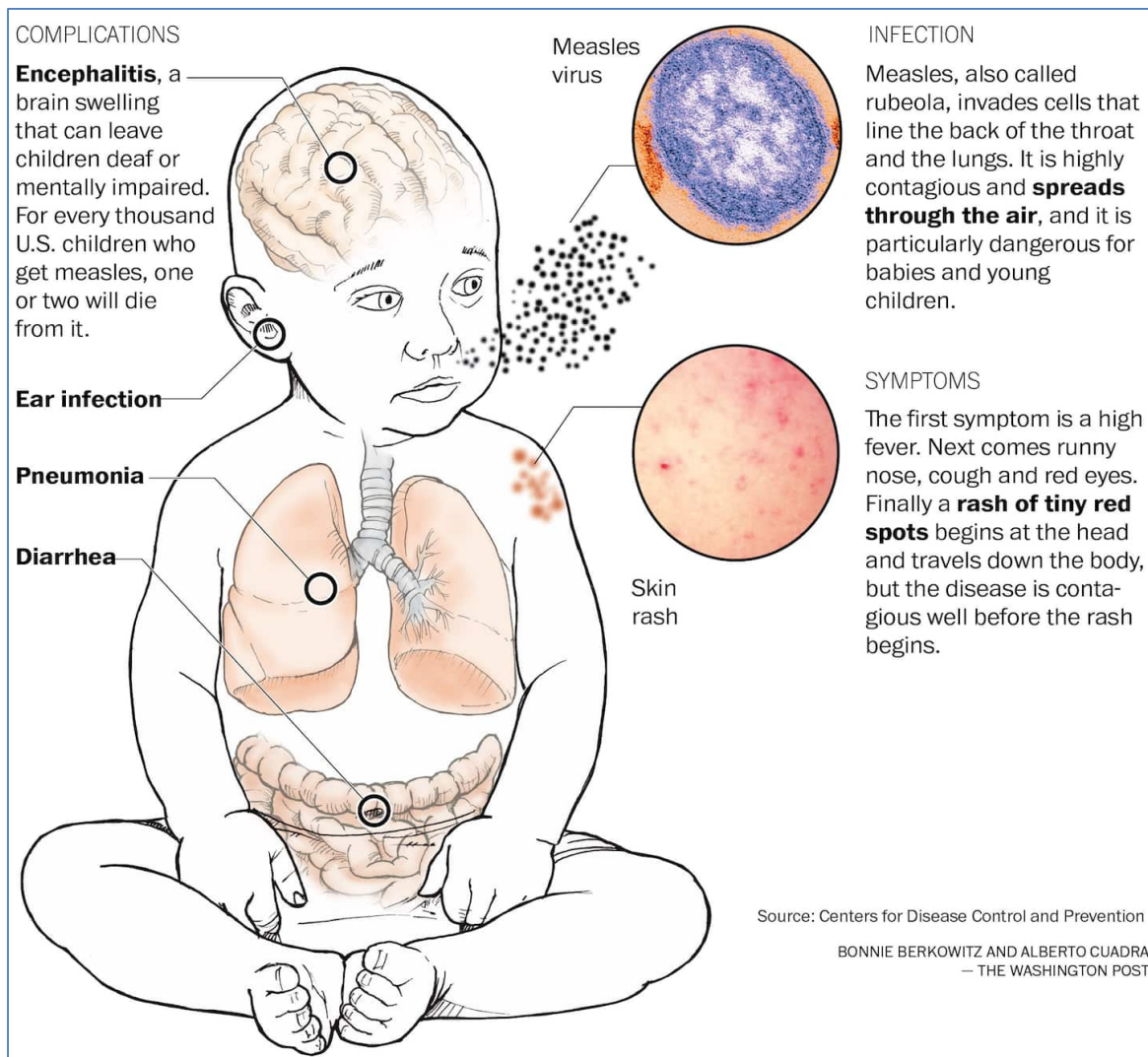
- **Aetiology:**
 - Measles Virus
- **Pathogenesis:**
 - **HIGHLY CONTAGIOUS - Aerosol/Contact Transmission**
 - Typically a Respiratory Infection; Also → Produces a Viraemia → Rash
- **Presentation:**
 - **Fever**
 - **URTI** - Cough, Rhinorrhoea, Red Eyes
 - **Rash** - Maculopapular Erythematous (Morbilliform)
 - **“Koplik’s Spots”** – Seen on the Inside of the Mouth
- **Diagnosis:**
 - **Clinical Diagnosis** (Generalised Maculopapular Rash + Fever)
 - **Serology**
 - **PCR**
- **Treatment:**
 - Supportive Mx.
 - **Vitamin A Supps.**
 - **+/- Ribavirin (Antiviral)**
 - **Prevented by MMR Vaccine –(Note: Contraindicated in Pregnancy)**
- **Complications Include:**
 - Croup, Otitis Media, Gastroenteritis
 - Febrile convulsions
 - **Subacute Sclerosing Panencephalitis (very rare)**
 - (Progressive Encephalitis due to Chronic Measles Infection)
 - No Cure; Fatal



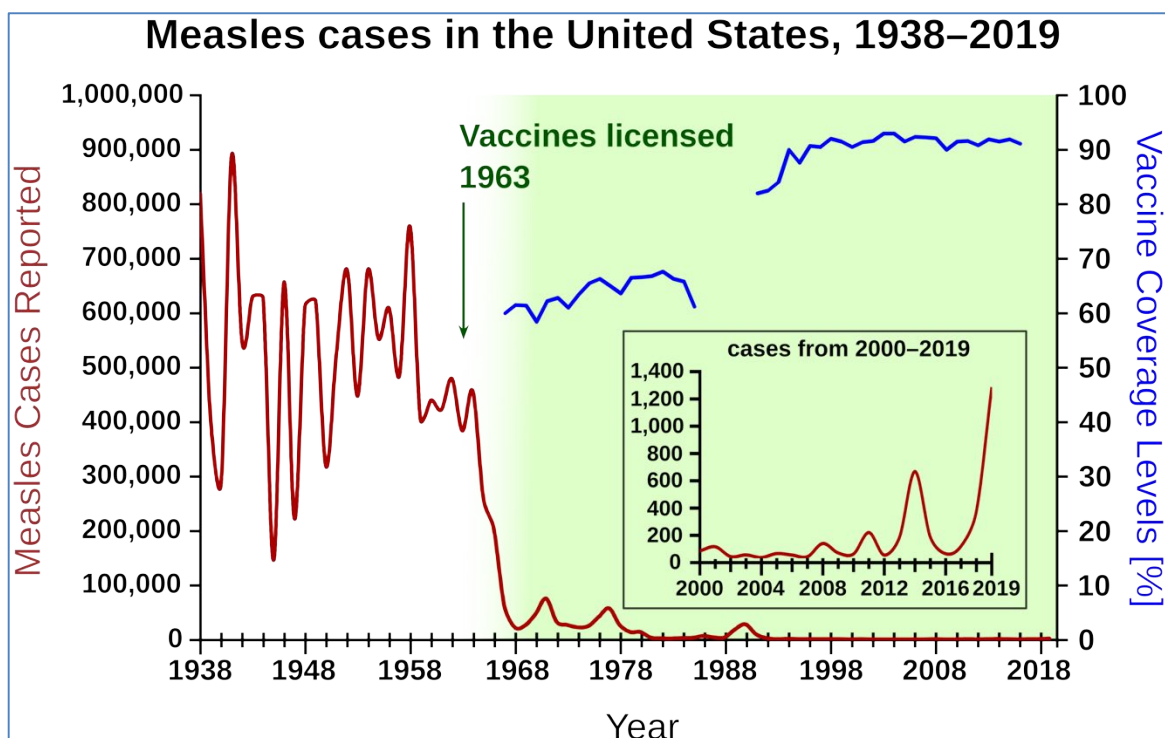
Photo Credit:Content Providers(s): CDC/Dr. Heinz F. Eichenwald, Public domain, via Wikimedia Commons



CDC, Public domain, via Wikimedia Commons



Public domain: CDC



Julius Senegal, CC BY-SA 4.0 <<https://creativecommons.org/licenses/by-sa/4.0/>>, via Wikimedia Commons

MUMPS VIRUS:

- **Aetiology:**
 - Mumps Virus
- **Pathogenesis:**
 - Aerosol Transmission / droplets / saliva
 - Respiratory Tract Infection → Lymph Nodes & Salivary Glands (+Viraemia)
- **Presentation:**
 - Fever & Malaise
 - Muscle pain
 - Headache
 - Loss of appetite
 - Painful Enlargement of Parotid Salivary Glands → Parotitis
 - (Can also cause orchitis)
- **Diagnosis:**
 - Serology
 - PCR
- **Treatment:**
 - Usually Self-Limiting
 - No specific antiviral treatments exist
 - Treatment is symptomatic (bed rest, fluids, saline gargles, antipyretics).
- **Prevention:**
 - MMR Vaccine (Admin at least 3x in Childhood)



Photo Credit: Content Providers: CDC/NIP/Barbara Rice, Public domain, via Wikimedia Commons

RUBELLA VIRUS (Aka "German Measles):

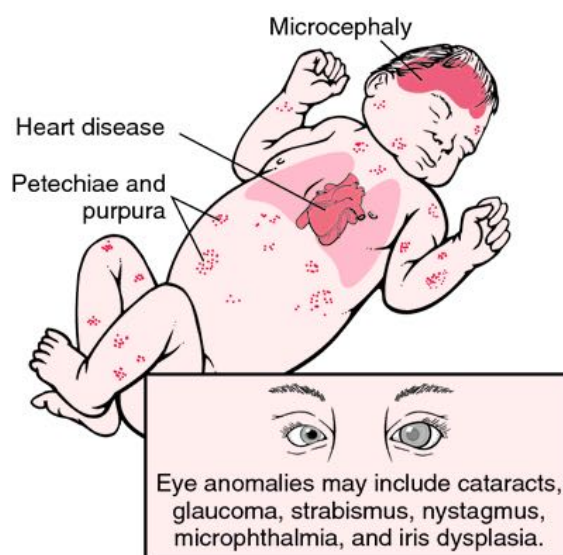
- **Organism:**
 - Rubella Virus
- **Transmission:**
 - **Aerosol Transmission**
- **Presentation:**
 - Initial Flu-Like Symptoms
 - * **Generalised Rash – (Red & Itchy)**
 - Low-grade Fever, Lymphadenopathy, Joint Pains, Headache, Conjunctivitis.
- **Diagnosis:**
 - Clinical Diagnosis
 - Presence of Virus-Specific IgM Antibodies
- **Treatment:**
 - No Specific Treatment
 - Controlled in Australia by vaccination (MMR Vaccine)
 - **Test pregnant women for immunity early.**
- **Prevention:**
 - (Note: Rubella *Itself* is relatively Benign, so why bother Vaccinating?)
 - **MMR Vaccine:**
 - **(Live Attenuated)**
 - **#1 Aim:** Prevent Rubella in Pregnant Women → ↓ Congenital Rubella Syndrome.
 - Aimed at *BOTH* Males & Females to ↓ Male Transmission to Pregnant Females
- **Prognosis:**
 - **Typically Benign – (Self-Limiting [1-3 Days])**
- **Complications:**
 - Complications may include arthritis, thrombocytopenia purpura, and encephalitis
 - *****HOWEVER, Maternal Infection in PREGNANCY can be SERIOUS!!**
 - **CONGENITAL RUBELLA SYNDROME – (If Infected in the 1st 20wks of Pregnancy)**
 - → Miscarriage
 - → **Serious Malformations – (Cardiac/Cerebral/Blindness/Deafness)**



commons.wikimedia.org, CC BY-SA 4.0 <<https://creativecommons.org/licenses/by-sa/4.0/>>, via Wikimedia Commons



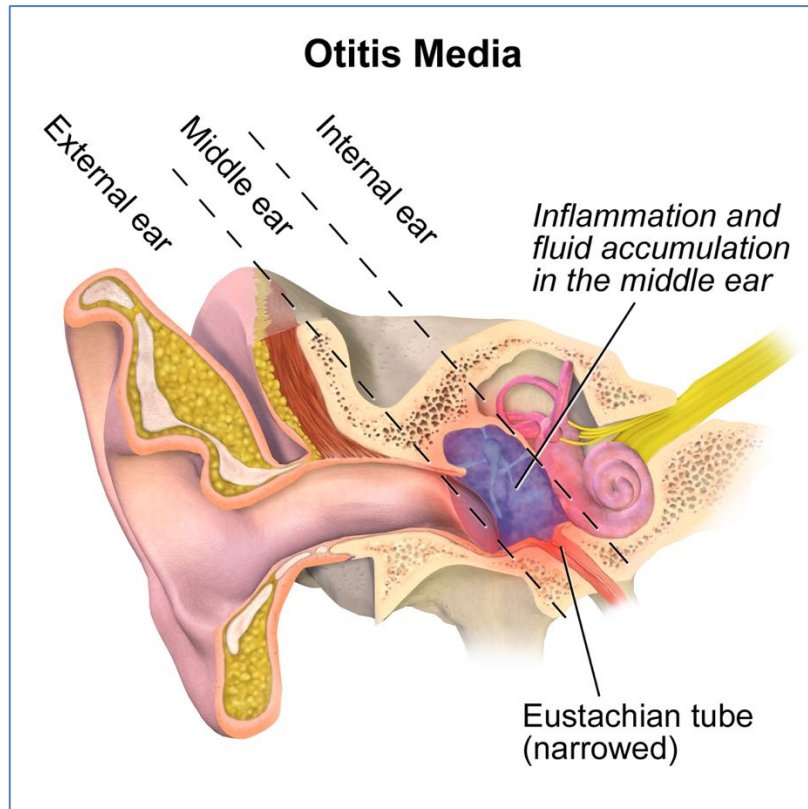
<https://www.cdc.gov/rubella/about/photos.html>



http://phil.cdc.gov/phil_images/20030724/28/PHIL_4284_lores.jpg

ACUTE OTITIS MEDIA (AOM)

- **Aetiology**
 - ***S. pneumoniae* - 35%** of cases (incidence decreasing due to pneumococcus vaccine)
 - *H. Influenzae* - 25% of cases
 - *S. aureus* and *S. pyogenes* (all beta-lactamase producing)
- **Predisposing Factors**
 - Eustachian Tube Dysfunction / Obstruction – (Eg. Down's Syndrome, Tumour, etc)
 - Upper Respiratory Tract Infection (URTI)
 - Allergies / Allergic Rhinitis
 - Chronic Sinusitis
- **Pathogenesis**
 - **Obstruction Of Eustachian Tube** → Middle Ear Stasis → Infection
- **Clinical Features**
 - **Epidemiology**
 - **70% of Children have AOM before 3yo**
 - Typically Children <6yo
 - **Classic Triad:**
 - **Otalgia**
 - **Fever** (especially in younger children)
 - **Hearing Loss**
 - (+ Rarely Tinnitus, Vertigo)
 - (+Otorrhoea If Tympanic Membrane Perforated)
 - **Infants / Toddlers**
 - Ear-Tugging
 - Irritable, Poor Sleeping
 - Vomiting And Diarrhoea
- **Investigations:**
 - **Otoscopy Of Tympanic Membrane**
 - Hyperaemia
 - Bulging TM
 - Loss Of Landmarks: Handle And Short Process Of Malleus Not Visible
 - **Swab MCS if Perforated & Exudative**
 - **Audiometry**
- **Treatment**
 - **Medical:**
 - **Antibiotics –(Amoxicillin +/- Ciprofloxacin Ear Drops)**
 - **"Sofradex" Aural Toilet**
 - **Symptomatic Therapy**
 - *Paracetamol*
 - Nasal Decongestants – (*Phenylephrine / Pseudoephedrine*)
 - **Surgery – (If Medically Unresponsive)**
 - Tympanotomy
 - Grommets Insertion
 - +/- Tonsillectomy
 - +/- Adenoidectomy
- **Complications of AOM**
 - **CHRONIC (>2wks) SUPPURATIVE OTITIS MEDIA**
 - ***Pseudomonas* (Tobramycin) or MRSA (Rifampicin)**
 - **MASTOIDITIS**
 - **CHOLESTEATOMA**
 - **MENINGITIS**
 - **FACIAL NERVE PARALYSIS**
 - **DEAFNESS** –(+/- Learning Delays)



BruceBlaus, CC BY-SA 4.0 <<https://creativecommons.org/licenses/by-sa/4.0/>>, via Wikimedia Commons



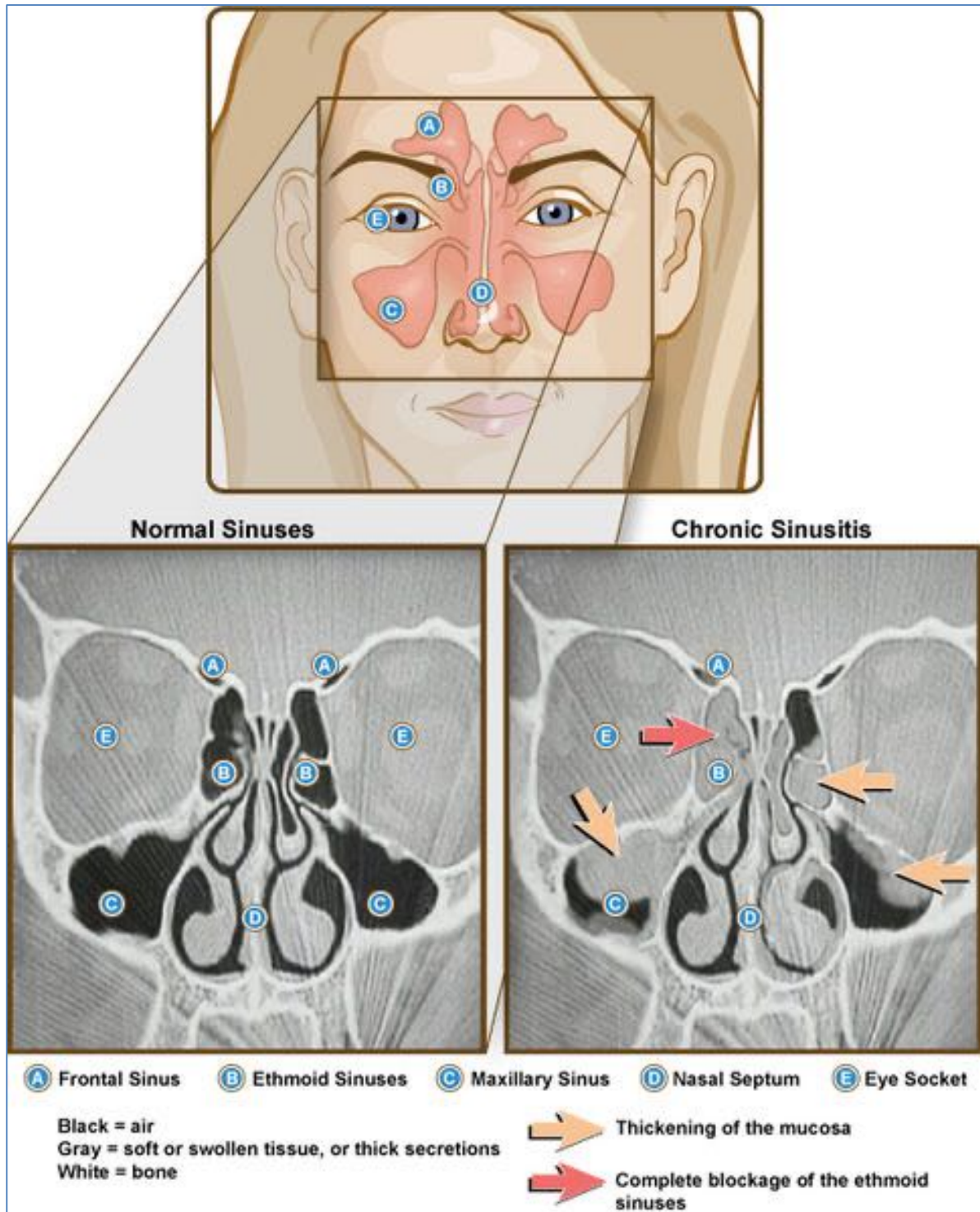
Michael Hawke MD, CC BY 4.0 <<https://creativecommons.org/licenses/by/4.0/>>, via Wikimedia Commons

ACUTE SUPPURATIVE SINUSITIS (<4wks):

- **Definition**
 - Acute Infection And Inflammation Of The Paranasal Sinuses – Up to 4wks in Duration
- **Aetiology**
 - ****Viral (Most Common):**
 - ****Rhinovirus, Influenza, Parainfluenza**
 - Bacterial:
 - *S. Pneumoniae* (35%), *H. Influenzae* (35%), *M. Catarrhalis*, Anaerobes (Dental)
- **Clinical Features:**
 - **Symptoms:**
 - Facial Pain / Pressure
 - Nasal Congestion
 - Purulent Nasal Discharge
 - Fever
 - **(Signs More Suggestive Of A Bacterial Etiology):**
 - >10day Duration
 - Mucopurulent Discharge
- **Investigations:**
 - Clinical Diagnosis
 - Transillumination of Sinuses
 - +/- Skull XR – (Opaque Sinuses & Fluid-Levels)
- **Management**
 - *Paracetamol*
 - Decongestants – (*Phenylephrine / Pseudoephedrine*)
 - Intranasal Corticosteroid – (Nasonex [*Mometasone*])
 - **+/- Antibiotics – (Augmentin [*Amoxicillin + Clavulanate*] or Rulide [*Roxithromycin*])**
 - (+ Supportive Mx)

CHRONIC SINUSITIS (>3MTHS):

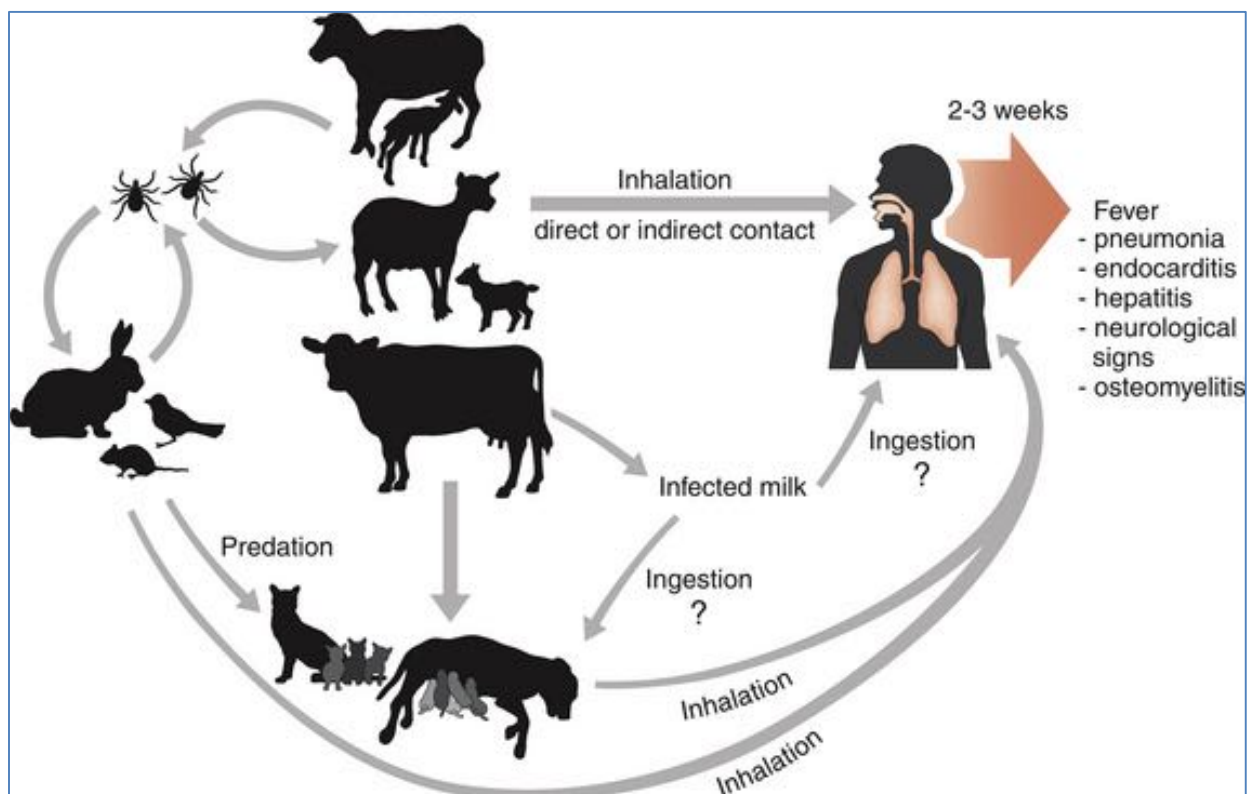
- **Definition**
 - Inflammation Of The Paranasal Sinuses Lasting >3 Months → Irreversible Changes in Epithelium
- **Aetiology – Any of the Following:**
 - Progression from Acute Sinusitis (Viral/Bacterial)
 - Untreated Nasal Allergy
 - Chronic Inflammatory Disorder E.G. Wegener's
- **Clinical Features (Similar To Acute, But Less Severe)**
 - Facial Pain / Pressure
 - Chronic Nasal Congestion
 - + Halitosis
- **Investigations:**
 - Clinical Diagnosis
 - + Head CT –(Pre-Surgical)
- **Management:**
 - **Antibiotics 3-6wks – (Augmentin [*Amoxicillin + Clavulanate*] or Rulide [*Roxithromycin*])**
 - Intranasal Corticosteroid – (Nasonex [*Mometasone*])
 - Decongestants – (*Phenylephrine / Pseudoephedrine*)
 - Surgery –(If Medical Therapy Fails)



NIAID, CC BY 2.0 <<https://creativecommons.org/licenses/by/2.0>>, via Wikimedia Commons

Q-FEVER

- **Aetiology:**
 - o *Coxiella Burnetii* - (found in cattle, sheep, goats and other domestic mammals, (cats and dogs))
- **Transmission:**
 - o Inhalation of Endospores / Contact with Unpasteurised Milk, Urine, Faeces of infected animals.
- **Pathogenesis:**
 - o **2-3wk Incubation**
 - o **Two-Stage Disease:**
 - **Acute Stage** (Headaches, chills, and respiratory symptoms)
 - **Chronic Stage** (Asymptomatic, Insidious)
- **Clinical Features:**
 - o **Acute Symptoms:**
 - **Flu-Like Symptoms:** Abrupt Onset Fever, Chills, Sweats & Malaise
 - **Respiratory** - Dry Cough, Pleuritic Pain
 - **GI Symptoms** - Nausea, Vomiting And Diarrhoea.
 - **Neuro:** +/- Severe Headache & Confusion
 - **MSK:** +/- Myalgia & Arthralgia
- **Diagnosis:**
 - o Serology
 - o PCR
 - o **TOEcho** – (If Suspected Endocarditis)
 - o **LFT** – (↑ALT & AST)
- **Treatment:**
 - o Antibiotics – **Doxycycline**
- **Complications:**
 - o Progression to **Atypical Pneumonia** → life threatening **ARDS**
 - o Rarely **Granulomatous Hepatitis** which can → hepatomegaly and RUQ pain.
 - o Chronic form of Q fever → **Endocarditis**
- **Prevention:**
 - o **Q-Vax** (Whole-cell, killed vaccine via intradermal injection)
 - o (Note: Skin and blood tests should be done first to identify pre-existing immunity; vaccinating subjects who already have an immunity can result in a severe local reaction.)



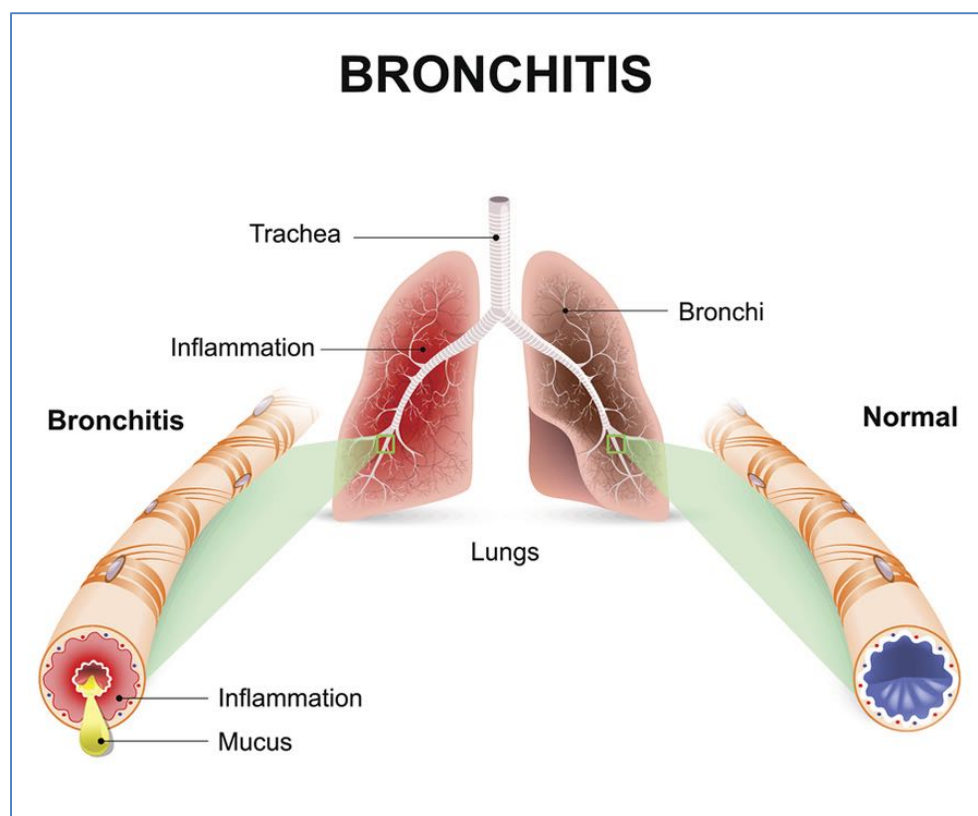
<https://tophealthdoctors.com.au/q-fever-vaccine/>

LOWER RESPIRATORY TRACT INFECTIONS

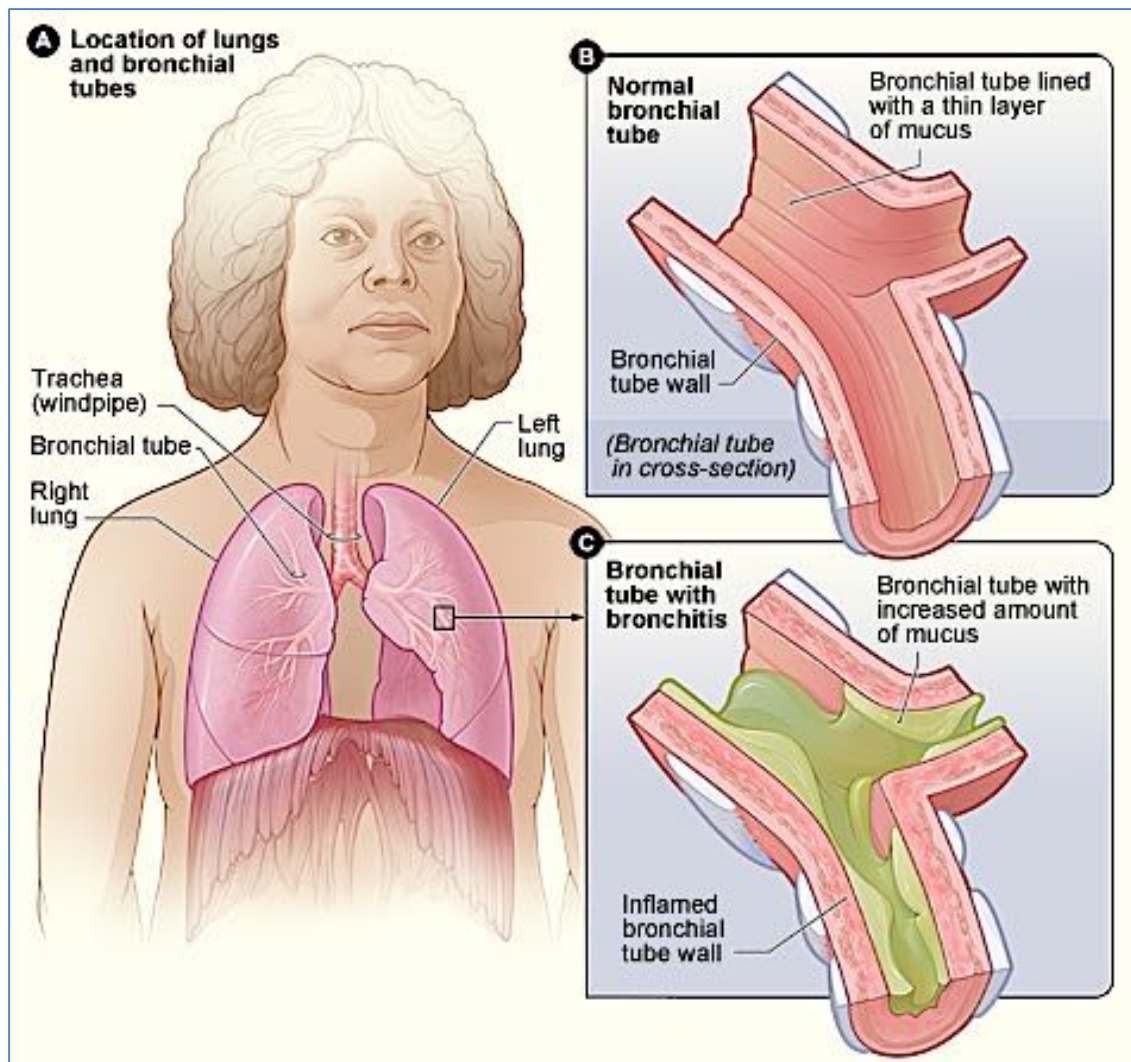
LOWER RESPIRATORY TRACT INFECTIONS

BRONCHITIS (ACUTE)

- **Definition**
 - Acute Infection Of The Tracheobronchial Tree → Inflammation With Resultant Bronchial Oedema And Mucus Formation
- **Aetiology**
 - **80% viral: rhinovirus**, corona virus, adenovirus, influenza, parainfluenza, RSV
 - **20% bacterial: *S-pneumoniae***
- **Pathogenesis:**
 - Acute Infection Of The Tracheobronchial Tree → Inflammation With Resultant Bronchial Oedema And Mucus Formation → Airway Obstruction → Cough/Wheeze
- **Clinical Features:**
 - URTI Symptoms
 - Productive Cough (Esp. @ Night)
 - Wheezing
 - (Note: Lower-Lung Examination Normal; Suspect Pneumonia if Crackles)
- **Investigations**
 - **Typically a Clinical Diagnosis**
 - **CXR** – (Rule out Pneumonia/CHF – if cough >3 weeks, abnormal vital signs, localized chest findings)
 - **Spirometry + Bronchodilatory** – (Rule out Asthma)
- **Differential Diagnosis**
 - **URTI/Asthma/Exacerbation of COPD/Sinusitis/Pneumonia/Bronchiolitis/Pertussis**
 - **Others:** reflux esophagitis, CHF, bronchogenic CA, aspiration syndromes, CF, foreign body
 - **Bacterial?** - Higher Fevers + Excessive Purulent Sputum.
- **Management**
 - **Symptomatic Relief:** Paracetamol, Rest, Fluids (3-4 L/ Day When Febrile), Humidified O2.
 - **Bronchodilators [*Salbutamol*]** – (May ↓Symptoms)
 - **Antibiotics [*Doxycycline / Erythromycin*]** If: Elderly/Comorbidities/Suspected Pneumonia.



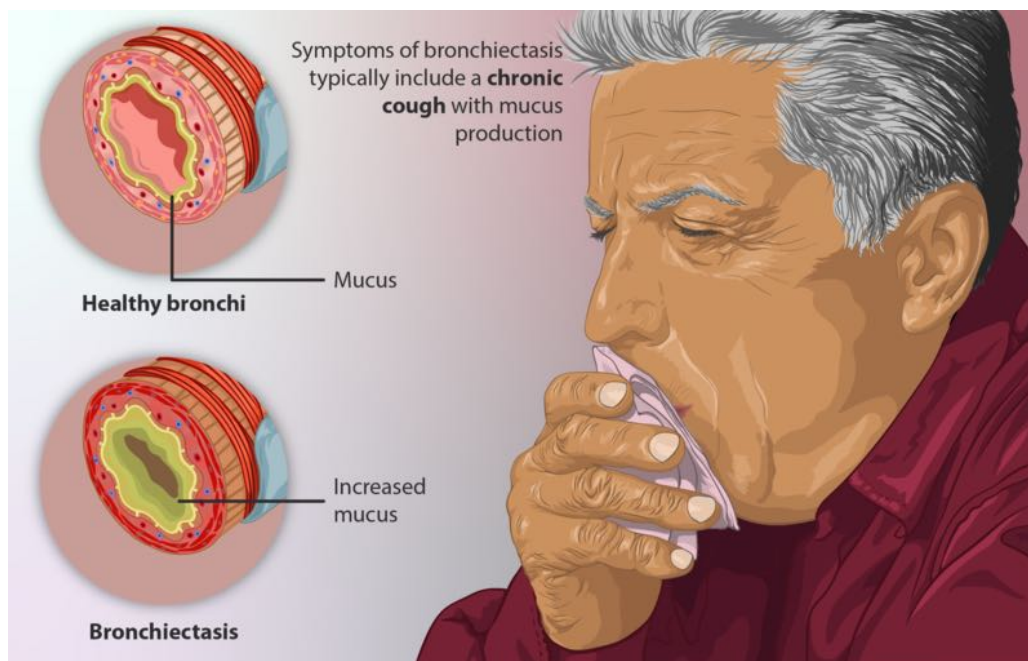
<https://www.cdc.gov/antibiotic-use/bronchitis.html>

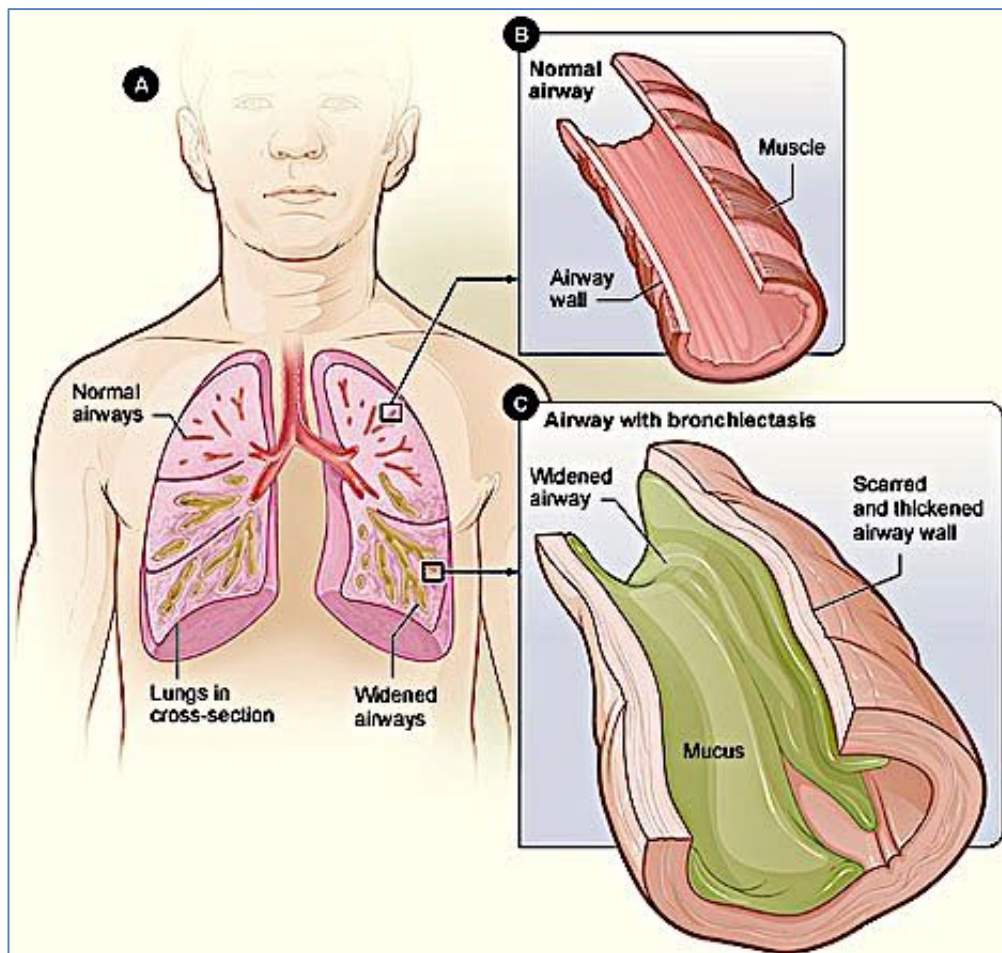


National Heart Lung and Blood Institute, Public domain, via Wikimedia Commons

BRONCHIECTASIS:

- = Localized, Permanent Dilation of part of the Bronchial Tree due to Destruction of Muscle & Elastic Tissue.
- **Aetiology:**
 - A Result of Chronic or Severe Necrotizing Lung Infections
 - Often seen in:
 - Cystic Fibrosis
 - Post-infectious Conditions
 - Bronchial Obstruction (Eg tumour/foreign bodies)
 - HIV
- **Pathogenesis:**
 - Chronic **AND/OR** Severe **Bacterial** Lung Infections →
 - →Bronchial Inflammation, (Often with Necrosis)
 - →Damage to Airway Walls (Destruction of Supporting Smooth Muscle & Elastic Tissues)
 - →Fibrosis & Eventually Dilation of Airways.
 - → Irregular, Permanently Dilated Bronchus filled with Pus.
- **Morphology:**
 - Usually in lower lobes
 - Permanent Dilatation of Bronchi (Often 4x Normal)(*Extending to the Pleura*)
 - CXR – Bronchial Markings are Visible more than 1/3 the way across a lung x-ray
- **Clinical Features:**
 - Symptoms:
 - Cough
 - Copious Purulent Sputum (Green/Yellow), mixed infections
 - Fever
 - Complications:
 - Pneumonias (Staph/Strep/Enterococci/Haemophilus/Pseudomonas)
 - Empyema
 - Septicaemia
 - Meningitis
- **Investigations:**
 - CXR – (Bronchial Markings out towards Periphery)
 - FBC – (If ?Current Infection – Typically *Pseudomonas*)
- **Management:**
 - Physiotherapy – (Postural Drainage & Cupping)
 - If Symptoms of Infection – Anti-Pseudomonal Antibiotics (**Tobramycin**)

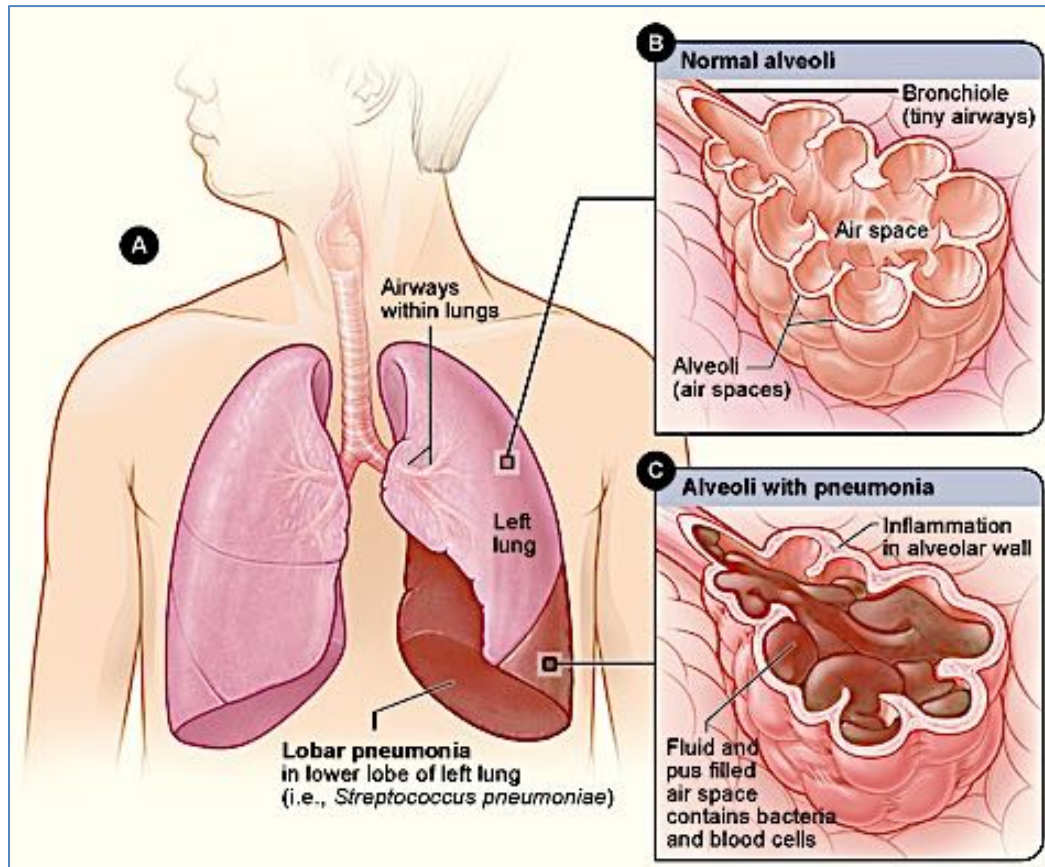




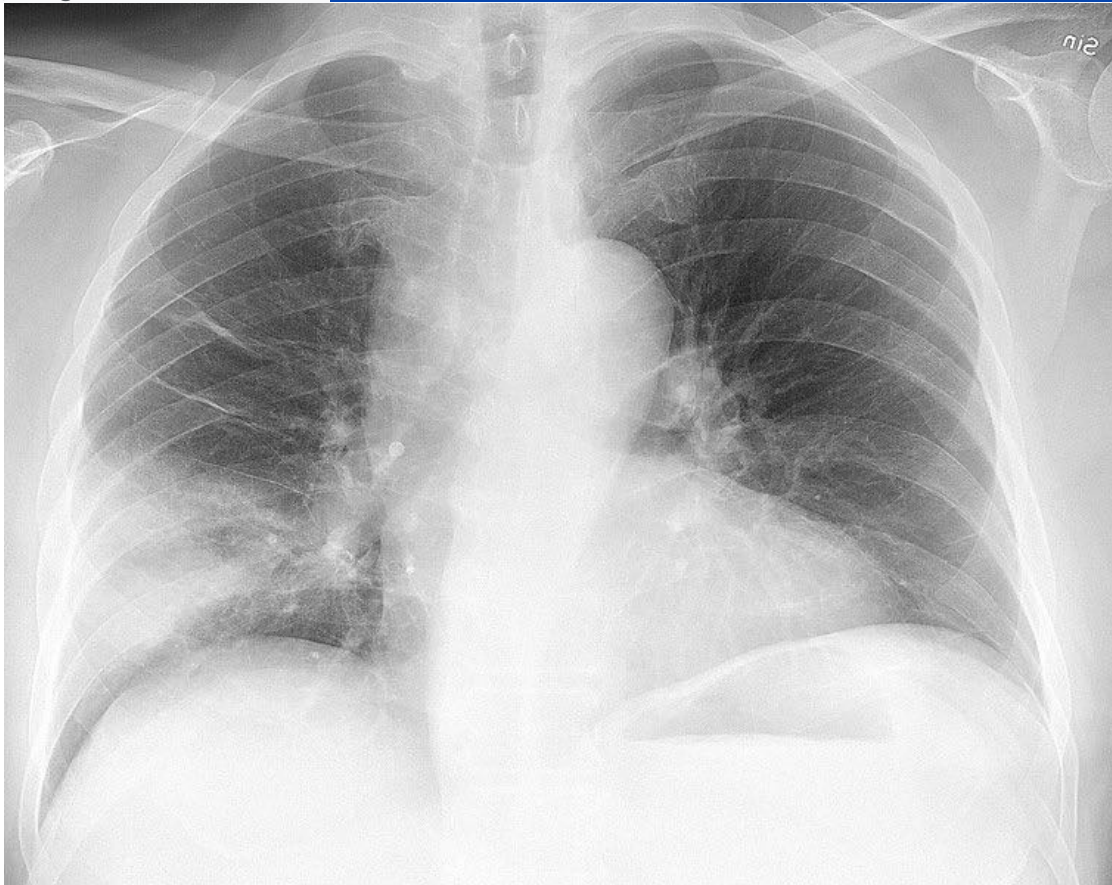
National Heart Lung and Blood Institute, Public domain, via Wikimedia Commons

PNEUMONIAS (“Infections of the Lung”):

- **Aetiology:**
 - **Community Acquired:**
 - Usually Gram-Positive – (**Strep pneumonia** [90%])
 - Occasionally Gram-Negative – (**H_Influenzae**)
 - **Hospital Acquired (Nosocomial - >48hrs POST Admission):**
 - Usually Gram-Negative – (**Pseudomonas aeruginosa**, **E coli**, Klebsiella)
 - **Atypical/Interstitial Pneumonia (“Walking Pneumonia”):**
 - Intracellular Bacteria – (**Mycoplasma**, Chlamydia, Legionella, Coxiella Burnetii)
 - **In Immunocompromised:**
 - **Cytomegalovirus**
 - **Pneumocystis jirovecii**
 - Fungal (Candida/Aspergillus)
- **Clinical Features:**
 - **General Pneumonia Triad (WHO):**
 - **Fever**
 - **Tachycardia**
 - **Tachypnoea (+/- Breathlessness)**
- **Types of Pneumonias - Based on Morphology:**
 - **LOBAR-PNEUMONIA (Well Defined; One Lobe):**
 - **Aetiology:**
 - Typically **Strep Pneumoniae** (Gram Positive Diplococci)
 - (Or *Klebsiella* in Aged)
 - **Pathogenesis:**
 - Whole Lobe Involvement
 - Exudate *Within Alveolar Spaces* → Alveolar Consolidation
 - **Morphology:**
 - Follows Anatomical Boundaries (Physically & on CXR)
 - Entire Lobe Consolidation/Opacity on CXR
 - **Clinical Features:**
 - **Symptoms:**
 - Abrupt onset High Fever + Chills
 - Productive Cough (Occasionally Rusty Sputum &/or Haemoptysis)
 - Pleuritic Chest pain + Pleural Rub.
 - **Signs:**
 - Usually Unilateral
 - Exudation – Entire Lobe Consolidation
 - **Cardinal Pneumonia Signs –(Fever, Tachycardia, Tachypnoea)**



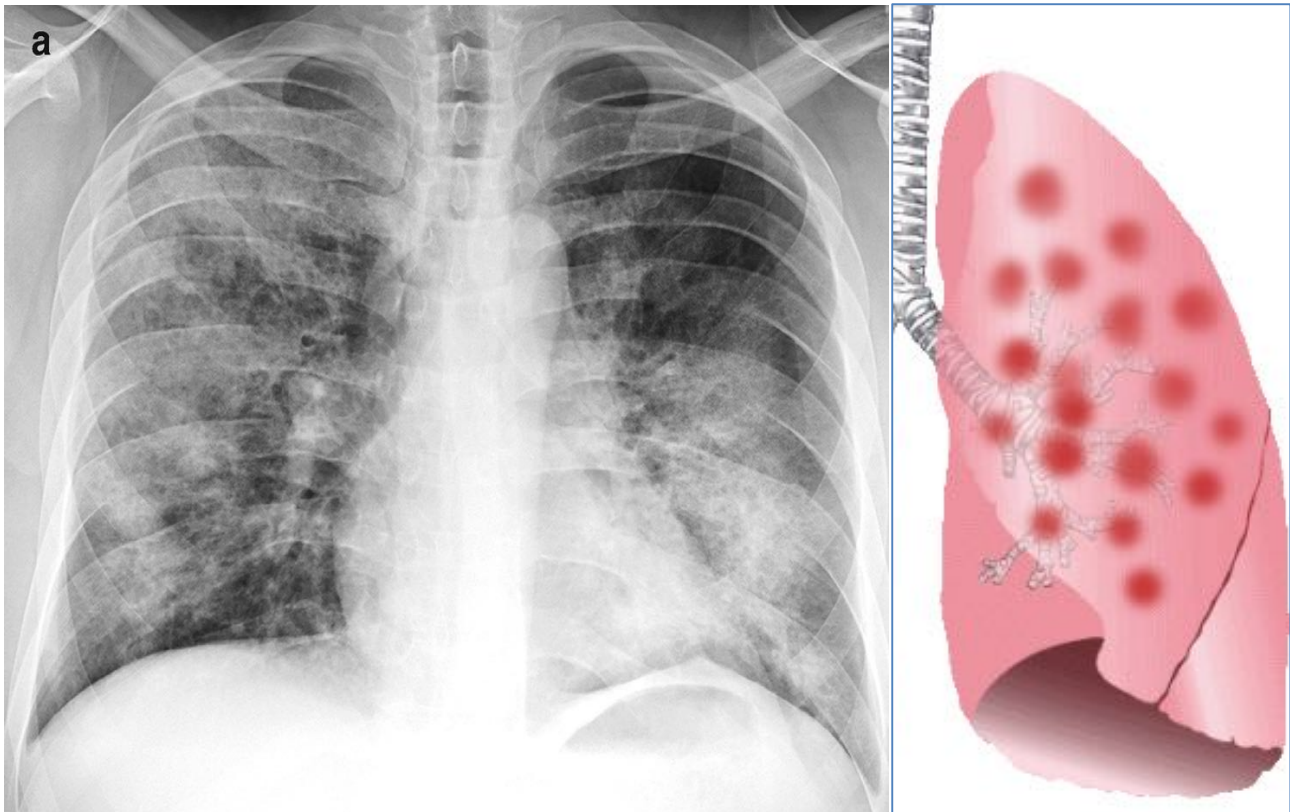
Heart, Lung and Blood Institute - <http://www.nhlbi.nih.gov/health/health-topics/topics/pnu/causes.html>



Mikael Häggström, M.D. - Author info - Reusing images- Conflicts of interest: NoneMikael HäggströmConsent note: Written informed consent was obtained from the individual, including online publication., CC0, via Wikimedia Commons

○ **BRONCHO-PNEUMONIA (Patchy; Multiple Lobes):**

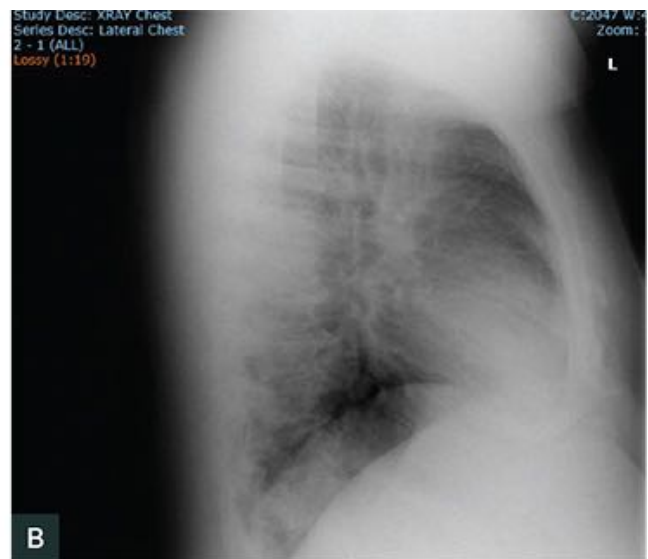
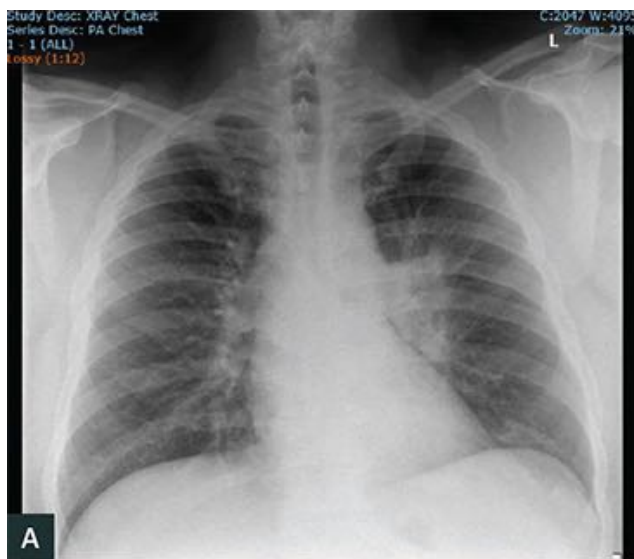
- **Aetiology:**
 - **Secondary to Debilitating Diseases, Extremes of Age, or Post-Surgery:**
 - Gram Positive - **Strep Pneumoniae, Staph Aureus**
 - Or Gram Neg - **H. Influenzae**
- **Pathogenesis:**
 - Patchy Areas of Acute Suppurative Inflammation → Patchy Consolidation
 - Basal Lower Lobes Common (Due to gravity – bacteria settle in the lower lungs)
- **Morphology:**
 - Doesn't follow anatomical boundaries – Often *Multi-Lobar & Bilateral*.
 - Usually Bilateral Patchy Consolidation → Scattered Opacities on CXR
- **Clinical Features:**
 - **Symptoms:**
 - Abrupt onset High Fever + Chills
 - Productive Cough (Occasionally Rusty Sputum &/or Haemoptysis)
 - Pleuritic Chest pain + Pleural Rub.
 - **Signs:**
 - Usually Bilateral
 - Patchy Consolidation – Usually Bilateral
 - **Cardinal Pneumonia Signs –(Fever, Tachycardia, Tachypnoea)**



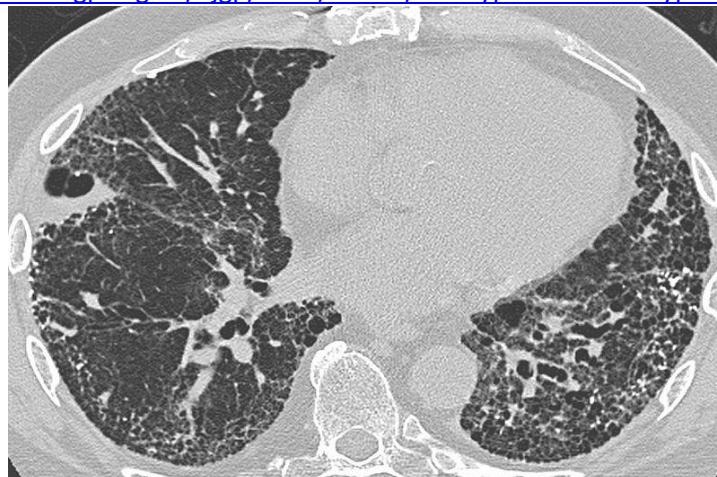
1. Franquet T., Chung J.H., CC BY 4.0 <<https://creativecommons.org/licenses/by/4.0/>>, via Wikimedia Commons
2. Suraj at Malayalam Wikipedia, Public domain, via Wikimedia Commons

○ **ATYPICAL, INTERSTITIAL PNEUMONIA (“Walking Pneumonia”):**

- **Aetiology:**
 - **Typically Intracellular Bacteria:**
 - **Mycoplasma**, Chlamydia pneumonia, Legionella, Q-Fever (Coxiella burnetii)
 - **Or Viral:**
 - **Influenza A/B, RSV – Respiratory Syncytial Virus**, Coronaviruses (SARS/Covid)
- **Pathogenesis:**
 - Interstitial Inflammation (NOT within the Alveolar Spaces)
 - Note: 2° Bacterial Pneumonia (Typically Strep/Staph) may follow.
- **Morphology:**
 - Inflammation localised to Alveolar Wall/Septa (Interstitium); NO Alveolar Exudate
 - Typically Bilateral.
- **Clinical Features:**
 - **Symptoms:**
 - Initial URTI → SLOW Onset (Days-Weeks)
 - Symptoms more *General* & ‘Flu-like’.
 - Few Localizing Symptoms:
 - Often NO Cough
 - Wheezing (Not seen in other pneumonias)
 - **Signs:**
 - No *Physical Signs* of Consolidation
 - *Unresponsive* to Common Antibiotics



<https://www1.racgp.org.au/ajgp/2018/march/an-atypical-case-of-typical-pneumonia>

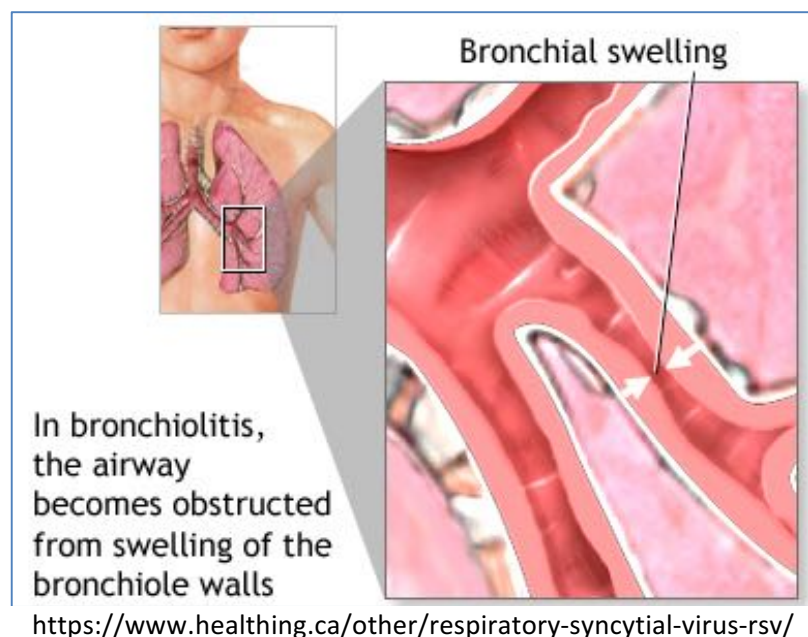


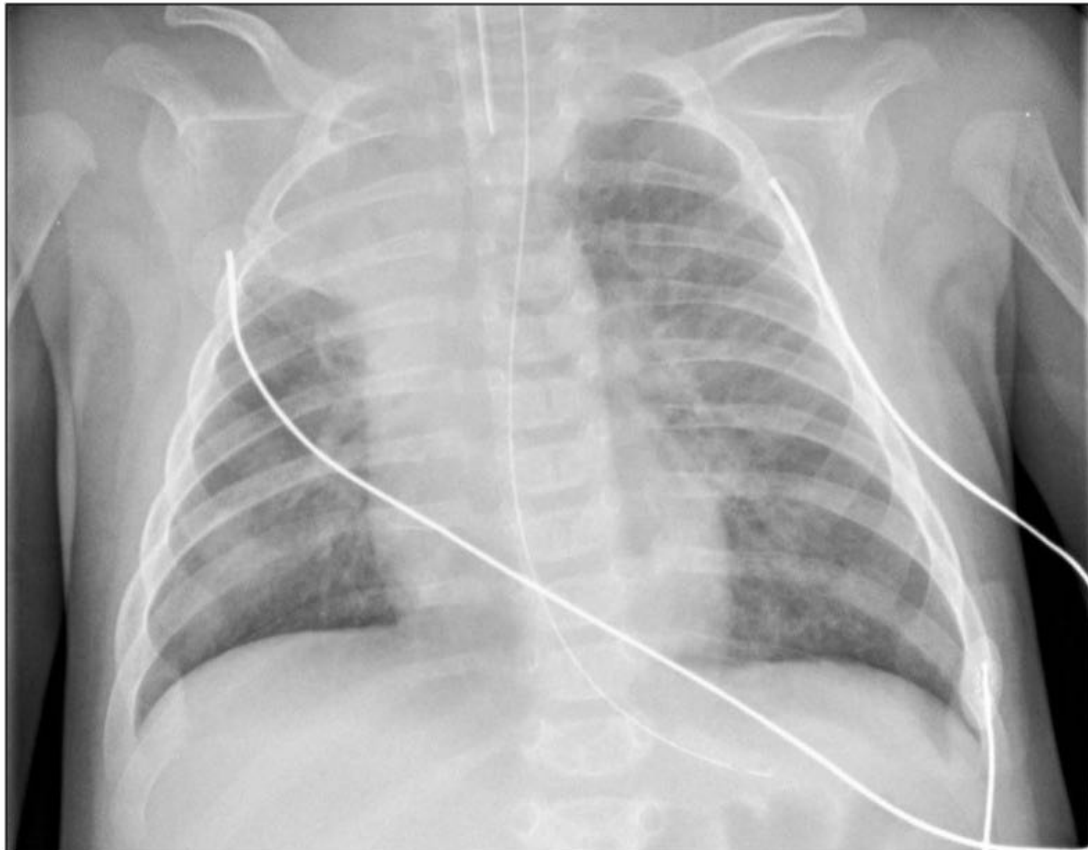
Yale Rosen from USA, CC BY-SA 2.0 <<https://creativecommons.org/licenses/by-sa/2.0/>>, via Wikimedia Commons

- **Investigations For Pneumonia:**
 - **CXR** – (Consolidation Lobar/Broncho/Interstitial)
 - **Sputum MCS** – (Sputum / NPA – Nasopharyngeal Aspirate / BAL – Bronchio-Alveolar Lavage)
 - **Blood Culture** if ?Septic
 - **Serological Testing** – (If ?Atypical Pneumonias)
- **Management:**
 - ?Admit to ICU? – **CURB-65** – (Score >3 → ICU):
 - **Confusion**
 - **Uraemia**
 - **Resp Rate >30**
 - **BP <90/60**
 - **>65yo**
 - **Antibiotics:**
 - **Empirical:**
 - ?G-Positive: **Amoxicillin** / **Benz-Penicillin-V** / **Doxycycline** / **Clarithromycin**
 - ?G-Neg: **Gentamicin** / **Ceftriaxone**
 - Severe: + **Meropenem** / **Imipenem**
 - **But Ultimately Dictated by MCS.**
 - **Fluids**
 - **O2** if Sats <92%
 - **+/- Ventilation**
- **Possible Complications of Pneumonia:**
 - **ARDS – Acute Respiratory Distress Syndrome:**
 - Severely Impaired Gas Exchange → Hypoxia & Confusion.
 - **Rx. Mechanical Ventilation and ICU.**
 - **Lung Abscesses**
 - **Pleuritis/Pleural Effusion/Empyema**
 - Inflammation of the pleura (Strep Pneumoniae)
 - Blood Rich Exudate/Pus in Pleural Space
 - **Rx. Drainage + MCS → IV Antibiotics**
 - **Septicaemia, Meningitis**
 - **Fibrosis, Scarring, Adhesions**
 - **Rarely Adenocarcinoma**

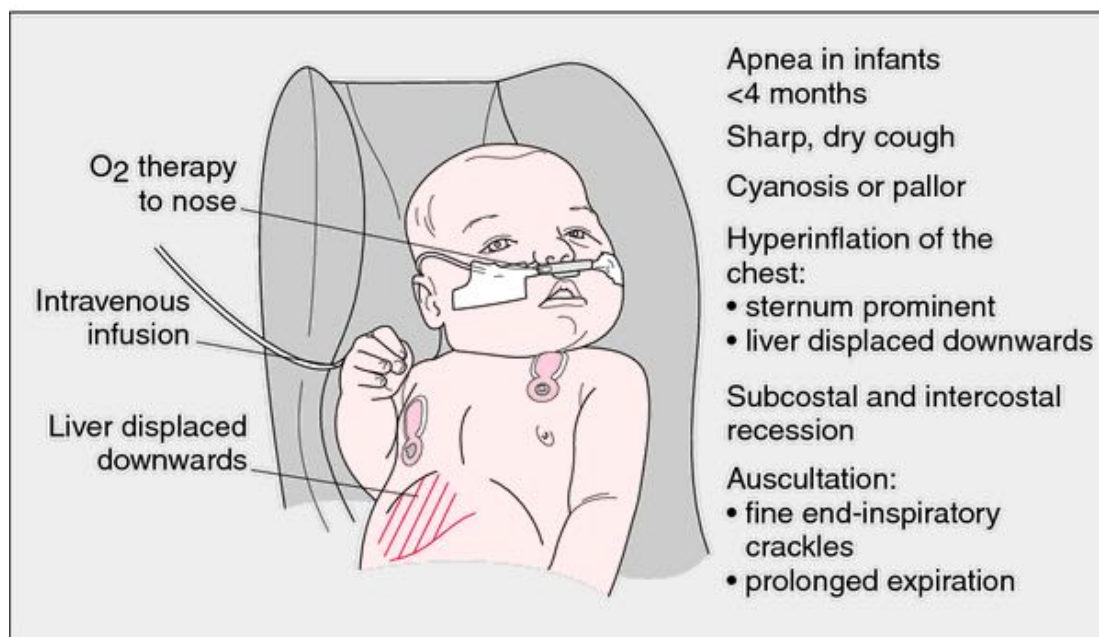
BRONCHIOLITIS:

- **Aetiology:**
 - **Respiratory Syncytial Virus (RSV)** (>50%)
 - parainfluenza, influenza, rhinovirus, adenovirus, rarely *M. pneumoniae*
- **Clinical Presentation**
 - Common, **affects 50% of children in first 2 years of life**
 - **Initial URTI with cough and fever → Respiratory Distress**
 - **Wheezing, Tachypnea, Tachycardia**
 - **Intercostal Recessions, Tracheal Tug, Supraclavicular Recessions, Rib Flaring**
 - **+ Feeding difficulties, irritability**
- **Investigations**
 - **CXR** (Air trapping, peribronchial thickening, atelectasis, increased linear markings)
 - **NPA for PCR**
 - **FBC** (Lymphocytosis)
- **Treatment**
 - **Fluid** Rehydration
 - **Paracetamol** (fever)
 - **Humidified O2**
 - Bronchodilator (Ventolin [**Salbutamol**])
 - **If Severe → Intubation and Ventilation**
 - **Indications For Hospitalization**
 - **Hypoxia:** SpO2 <92%
 - **Resting Tachypnea** >160/minute
 - **Respiratory Distress even after Salbutamol**
 - **<6 months old**
 - **Feeding Problems**





<https://www.ncbi.nlm.nih.gov/books/NBK442240/figure/ch5.f3/>



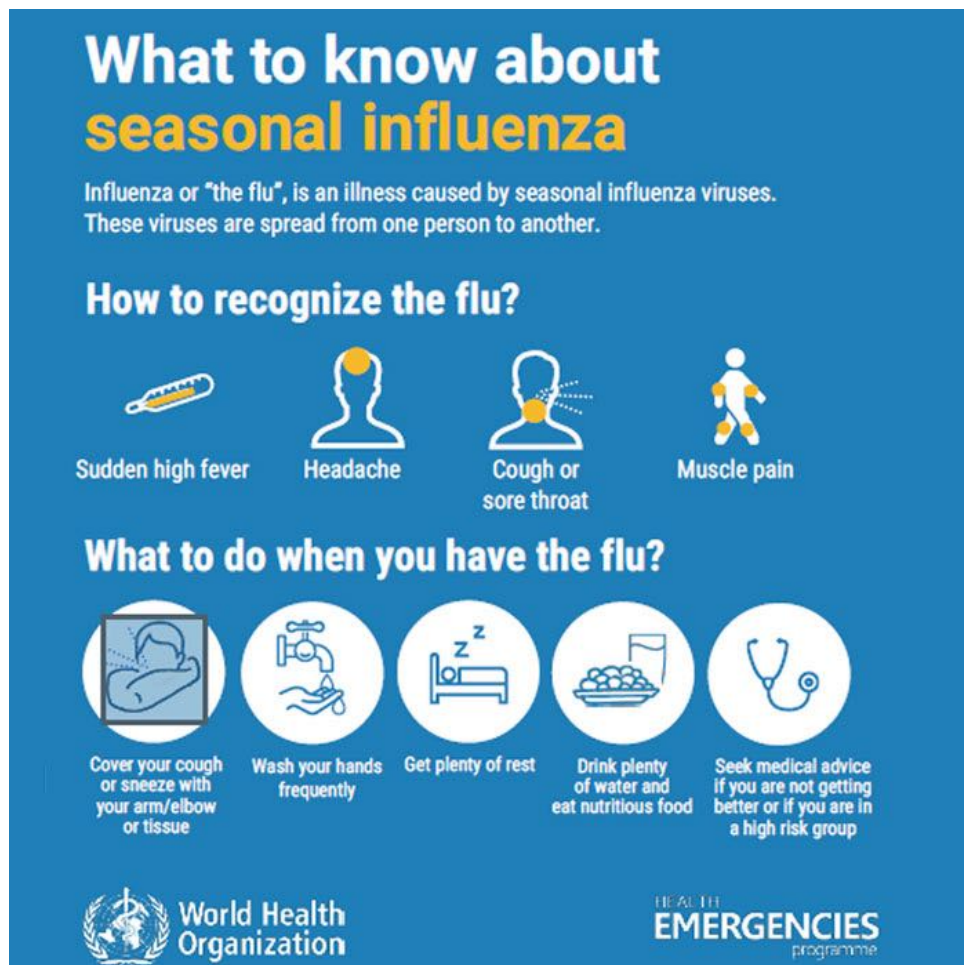
<https://m.theindependentbd.com//magazine/details/159068/Bronchiolitis:-Not-to-be-treated-lightly>

GLOBAL RESPIRATORY VIRUSES

GLOBAL RESPIRATORY VIRUSES

SEASONAL FLU (INFLUENZA A & B):

- **Aetiology:**
 - Influenza Virus A & B
- **Pathogenesis:**
 - **Transmission:** airborne spread. droplet
 - **Incubation Period:** 1-4 days
 - **Contagious for:** 1 day Before Sym Onset, and the next 7 days.
 - Viral-Induced Epithelial Dysfunction & Destruction
- **Clinical Features:**
 - **Symptoms:** Chills, Fatigue, Cough, Myalgias, Arthralgias, Headache
 - **Signs:** High Fever (<42°C); But Chest Clear (Unless 2° Bacterial Pneumonia)
 - **Complications:** 2° Bacterial Pneumonia, Otitis Media, Sinusitis
- **Diagnosis:**
 - **Clinical Diagnosis (Signs & Symptoms)**
 - +/- Nasopharyngeal Swabs
 - +/- Serology
 - **Note:** CXR is usually *Normal*.
- **Treatment and Prevention**
 - **Primarily Supportive Treatment:**
 - Bed Rest, Fluid, Paracetamol/Analgesics, Antitussives, Decongestants
 - **+/- Antivirals (Effective within 48 hours of onset):**
 - *Oseltamivir* (Tamiflu TM) / *Zanamivir* (Relenza TM) → **Reduce <24hrs of Symptoms**
 - **Vaccine:**
 - **FluVax** is recommended **Annually** for **Everyone**
 - (Note: Vaccine is reformulated each year to include current serotypes)



Source: WHO

BIRD FLU (H5N1):

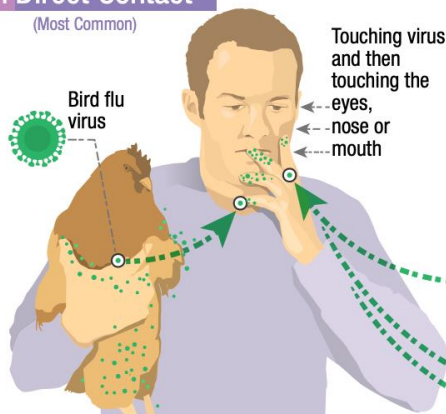
- **Aetiology:**
 - Influenza H5N1
- **Pathogenesis:**
 - **Transmission** – Aerosol/Direct Contact
 - **Incubation Period Generally 2-8 Days**
 - **Infection with Influenza H5N1** → Viral Replication → Virus-induced Epithelial Dysfunction
 - **Mortality Rate** ≈63%
- **Clinical Features:**
 - **Symptoms:** High Fever (>38°C), Headache, Myalgias, Cough (± Sputum), Dyspnoea + Others
 - **Pneumonia:** Consolidation, Tachypnoea, Tachycardia
 - **Often Progresses To ARDS** → Multi-Organ Failure → Death
- **Investigations:**
 - **NPA** → PCR
 - **CXR** - (Infiltrates +/- Pleural Effusions)
- **Treatment**
 - **ICU** - (Ventilation, Fluids)
 - **Antivirals** – (*Oseltamivir* (Tamiflu TM) / *Zanamivir* (Relenza TM))
- **Prevention**
 - No Vaccine
 - Hygiene Precautions
 - Post-Exposure-Prophylaxis – (*Oseltamivir* / *Zanamivir*)
 - **Notify Public Health**
 - **Contact Tracing and Quarantine**

How Infected Backyard Poultry Could Spread Bird Flu to People

Human Infections with Bird Flu Viruses Rare But Possible

1 Direct Contact

(Most Common)

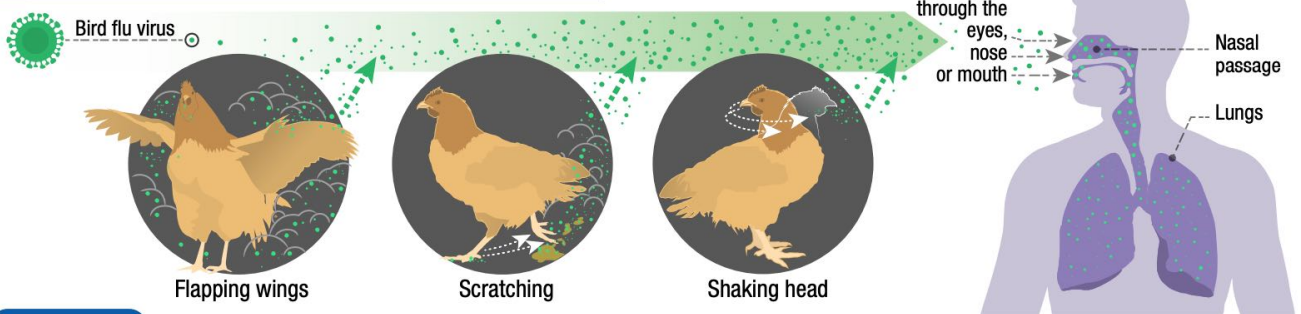


Infection can occur without touching poultry.

2 Contaminated Surfaces



3 Bird Flu Virus in the Air (in Droplets or Dust)



U.S. Department of
Health and Human Services
Centers for Disease
Control and Prevention

www.cdc.gov/flu/avianflu/avian-in-humans.htm

CS261152

Source: CDC

SWINE FLU (H1N1):

- **Epidemiology**
 - HUMAN to HUMAN - NOT by pigs; documented mass pig slaughtering was unnecessary
 - Incubation Period 24--48 Hours
- **Aetiology:**
 - H1N1 – (A Novel strain - genes from 5 different flu viruses)
- **Pathogenesis:**
 - Droplet Transmission – Human to Human.
 - Respiratory Tract Infection
- **Clinical Features:**
 - **Low Mortality Rate** - 2 deaths in first 600 cases in the US
 - **Infects The Young (<5yrs) And Old (>65yrs)**
 - **Transmission:** Aerosol/Contact (Human:Human)
 - **Symptoms:** Fever, Cough, Sore throat, N/V/Dia (25%), Myalgia/Arthralgia, Headache

Emergency warning signs	
In children	In adults
• Laboured breathing • Cyanosis • Dehydration • Irritability • Fever with rash • Quiet, not interacting	• Shortness of breath • Pain in chest or abdomen • Confusion • Persistent or severe vomiting

- **Diagnosis:**
 - **Clinical Suspicion**
 - **PCR** – (Nasal/Nasopharyngeal/Oropharyngeal)
 - **Notify Public Health**
 - **Contact Tracing and Quarantine**
- **Treatment**
 - **Antivirals** – (*Oseltamivir* (Tamiflu TM) / *Zanamivir* (Relenza TM))
 - **+Supportive.**

Swine Flu

Swine flu is a respiratory disease which infects pigs but also people, typically those who have been in contact with pigs.

Symptoms in humans
Person-to-person transmission is through coughing, sneezing

The virus
Influenza A subtypes: H1N1, H1N2, H3N1, H3N2, H2N3
New "reassortant" virus
Forms when genes from different viruses begin to mix

When flu spreads
person-to-person, rather than from animals to humans, it can continue to mutate, making it harder to treat or fight off
Incubation time 5-10 days

There are no vaccines that contain current swine flu virus causing illness in humans.

Labels on the diagram:
Fever
Lethargy, lack of appetite
Loss of consciousness, ending in death
Coughing
Nausea, vomiting
Diarrhea
Runny nose
Sore throat

SARS – SEVERE ACUTE RESPIRATORY SYNDROME:

- **Definition**
 - Rapidly progressing viral pneumonia caused by a SARS-associated coronavirus
- **Aetiology:**
 - **SARS-Associated Coronavirus**
 - **MERS-Associated Coronavirus**
 - Incubation: 2-7 days
- **Pathophysiology**
 - Droplet Transmission – Human to Human.
 - Respiratory Tract Infection with SARS-Associated Coronavirus
 - → Atypical Pneumonia +/- Respiratory Distress Syndrome
- **Clinical Features**
 - **Difficult To Differentiate SARS from other Community-Acquired Pneumonias Because:**
 - **Initial Symptoms Are Not Specific:**
 - Fever, Chills, Malaise,
 - Headache, Myalgia,
 - Cough, Sore Throat, Productive Cough
 - **However, Some Patients Deteriorate with:**
 - Persistent Fever,
 - ↑SOB & Desaturation
 - **Critically ill patients Require ICU Admission and Mechanical Ventilation**
- **Complications**
 - Respiratory failure
 - Liver failure
 - Heart failure
- **Diagnosis:**
 - **Clinical Suspicion – Symptoms, Hx of Travel, Hx of Contact**
- **Investigations:**
 - **CXR** – Features of Atypical Pneumonia
 - **Lab** – Neutrophilia, Lymphopenia, ↑CRP, & ↑LDH
 - **RT-PCR** – from Blood/Sputum/NPA/Swabs.
 - **Serology** – (antibody detection via ELISA)
 - **RAT** – Rapid Antigen Test
- **Treatment**
 - **Follow local public health protocols**
 - **Quarantine** (negative-pressure room, N95 Mask, gown, gloves, eye protection)
 - **Antivirals** – (*Ribavirin, others*)
 - **Steroids** - (To prevent immune mediated lung damage. Eg. Dexamethasone)
 - **Supportive management +/- ventilation**



<https://www.cdc.gov/ncidod/eid/vol9no6/03-0264.htm>

OBSTRUCTIVE & RESTRICTIVE LUNG DISORDERS

OBSTRUCTIVE & RESTRICTIVE LUNG DISORDERS

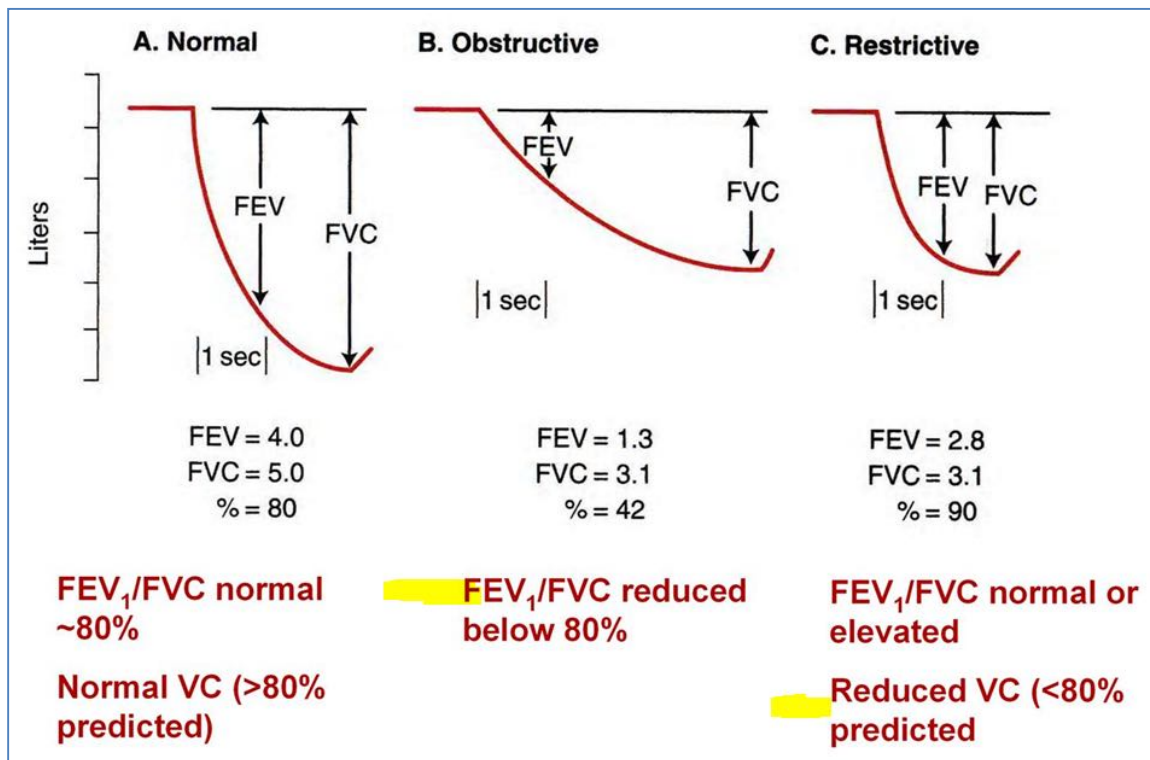
Obstructive Vs. Restrictive Pulmonary Diseases:

- Obstructive:

- Involves **Airway** Obstruction → ↑ **Airway** Resistance
- **Effects on Lung Capacities/Volumes:**
 - ↑TLC (Total Lung Capacity)
 - ↑RV (Residual Volume)
 - ↑FRC(Functional Residual Capacity)
 - ↓VC (Vital Capacity) - Because They Can't Expel All the Gas in their Lungs
 - ↓FEV₁ (Forced Expiratory Volume in 1 Sec) – Because of Dynamic Airway Compression
- **Key Diagnostic Feature:**
 - If their **FEV₁ is Less Than 80% of FVC**
 - (FEV₁ = Forced Expiratory Volume in 1 Second)
 - (FVC = Forced Vital Capacity = Max Air Expired After Full Inspiration)

- Restrictive:

- Involves **Lung** Restriction → ↑Resistance to **Lung Expansion**
- (I.e: ↓Chest or Lung Compliance / Obesity → Weight on Chest / Pregnancy → ↑Abdominal Pressure)
- Note: *Normal* Airway Resistance.
- **Effects on Lung Capacities/Volumes:**
 - ↓TLC (Total Lung Capacity)
 - ↓VC (Vital Capacity)
 - ↓IC (Inspiratory Capacity)
- **Key Diagnostic Feature:**
 - If their **Measured VC is Less Than 80% of their Predicted VC.**
 - (Measured Vital Capacity = Patient's VC Measured by Spirometry)
 - (Predicted Vital Capacity = Average Healthy VC based on Age/Sex/Size)

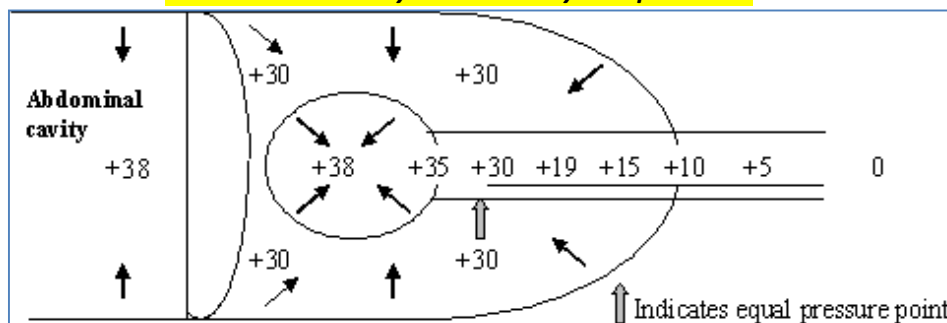


Source: Unattributable

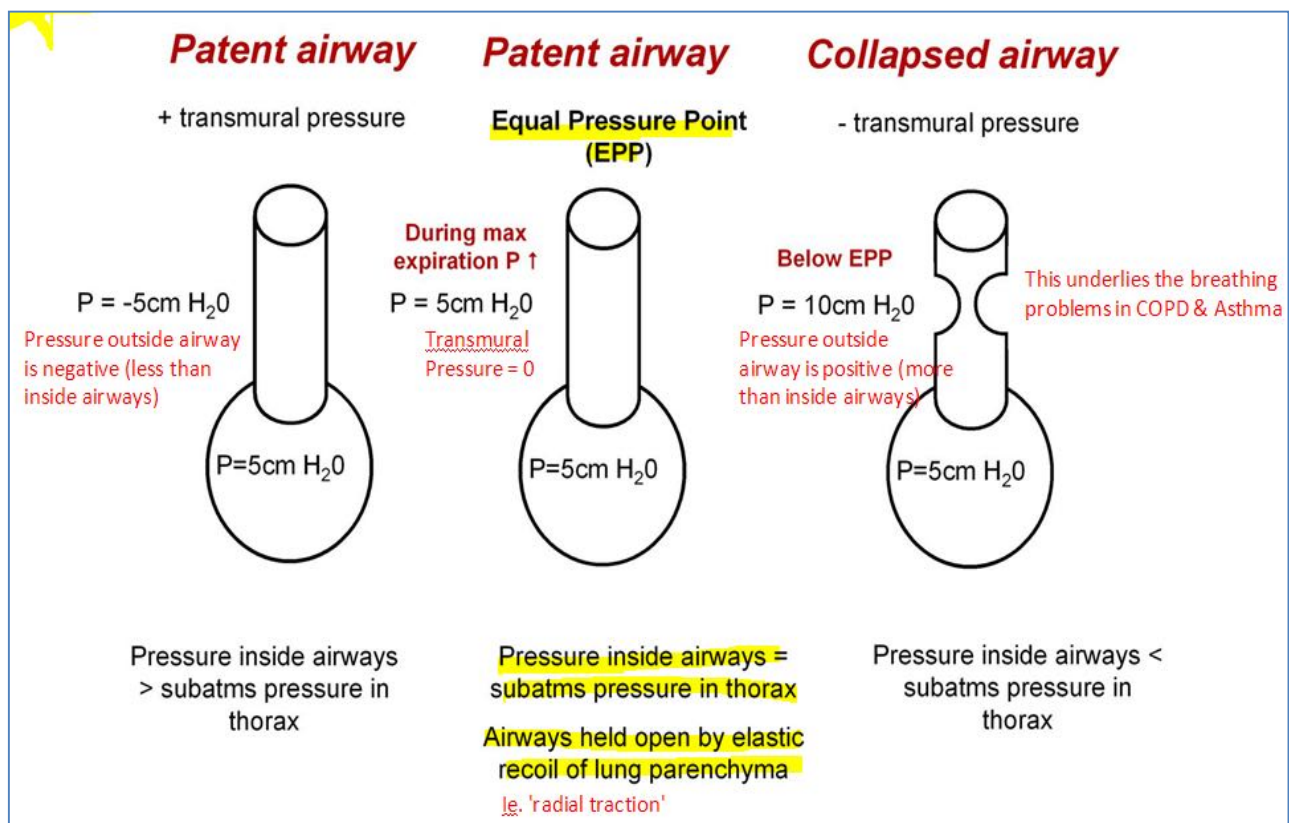
Dynamic Airway Compression:

- Equal Pressure Point:

- **EPP:** Is The Location in an Airway where Intrapleural (Thoracic) Pressure = The Intra-Airway Pressure.
 - If EPP occurs in Larger, Cartilaginous Airways, the Airways Remain Open.
 - However, If EPP Occurs in Smaller, Unsupported Airways, the Airways will Collapse.
 - This is Known as "**Dynamic Airway Compression**"



- **During Passive Expiration:**
 - The Alveolar Pressure is Mostly due to The Elastic Recoil of The Lungs (& Partly due to the Recoil of the Thoracic Cage.)
 - Since the Highest Proportion of the Alveolar Pressure is due to the Lung's Elastic Recoil, The Thoracic Pressure is Relatively Low.
 - Therefore, the EPP will occur *High Up* in the Larger, Cartilaginous Airways.
 - → Airways Remain Patent
- **During Forced Expiration:**
 - (I.e: IN OBSTRUCTIVE CONDITIONS)
 - The Alveolar Pressure is Mostly due to The Expiratory Muscles → ↑ Thoracic Pressure. (& Partly due to Elastic Recoil of Lungs.)
 - Since the Highest Proportion of the Alveolar Pressure is due to the ↑ Thoracic Pressure, The Pressure of the Lung's Elastic Recoil is Relatively Low.
 - Therefore, The EPP will occur *Lower Down* in the Smaller, less-supported Airways.
 - → Airways Collapse.



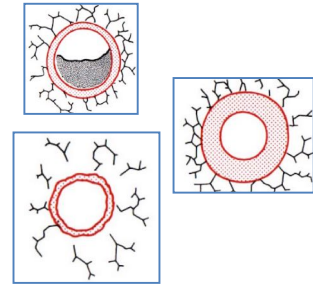
CHRONIC OBSTRUCTIVE PULMONARY DISEASE (COPD):

- What Are They?:

- **Permanent** NARROWING/OBSTRUCTION of the AIRWAY.
- **ie: Increased Resistance to Airflow.**
- – Is an 'Umbrella Term' – Usually Refers to **Chronic Bronchitis**, **Emphysema**, or Mixture of BOTH.

- 3 Causes:

- **1. Conditions With The Lumen** - (Eg. Excessive Mucous)
- **2. Conditions Within The Wall of the Airway:**
 - Inflammation & Oedema (Chronic Bronchitis or Asthma)
 - Bronchoconstriction (Asthma)
- **3. Conditions Outside The Airway:**
 - Destruction of Lung Parenchyma (eg. Emphysema)
 - Localised Compression of Airway
 - Peribronchial Oedema

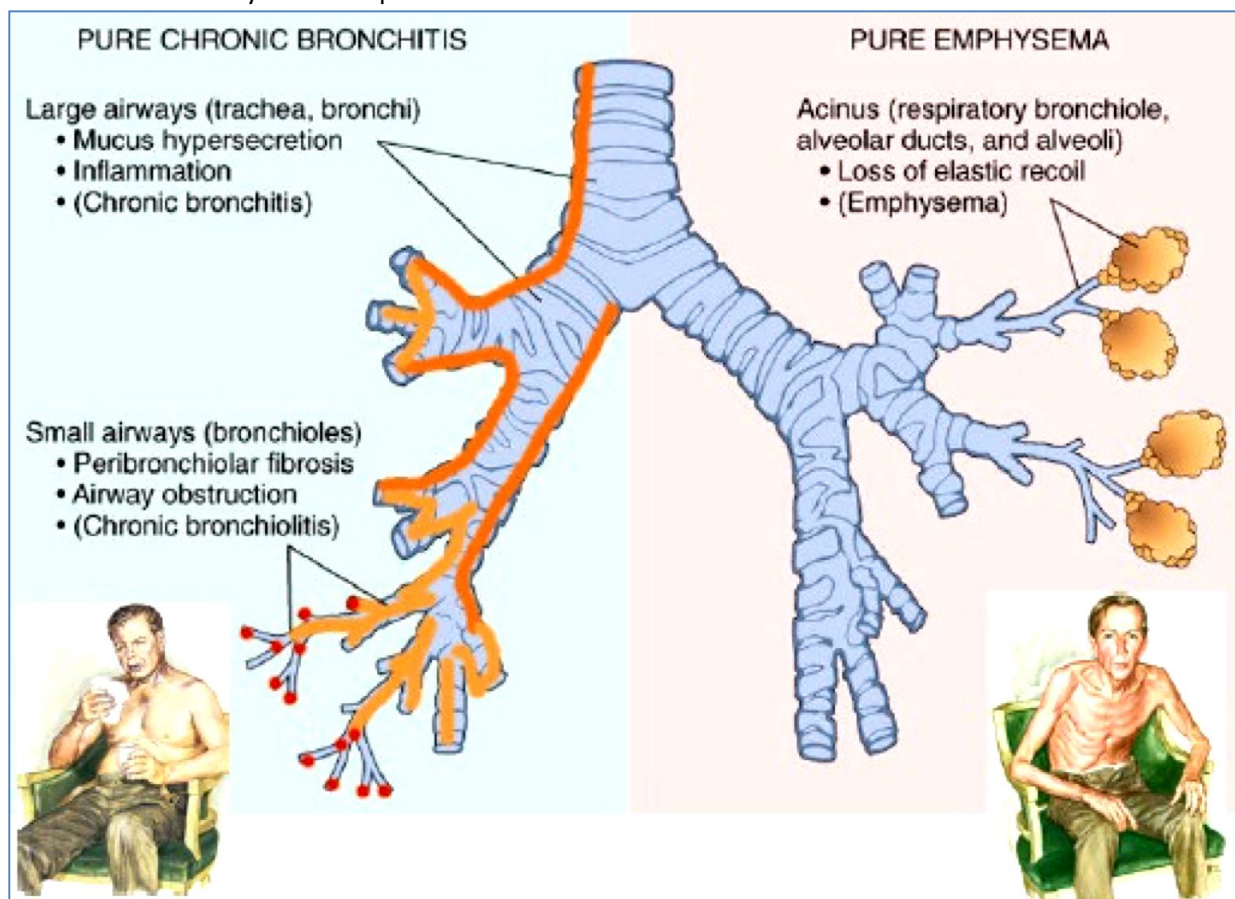


- Aetiology:

- **#1. Smoking**
- (Genetic – α 1-Antitrypsin Deficiency → Congenital Emphysema)

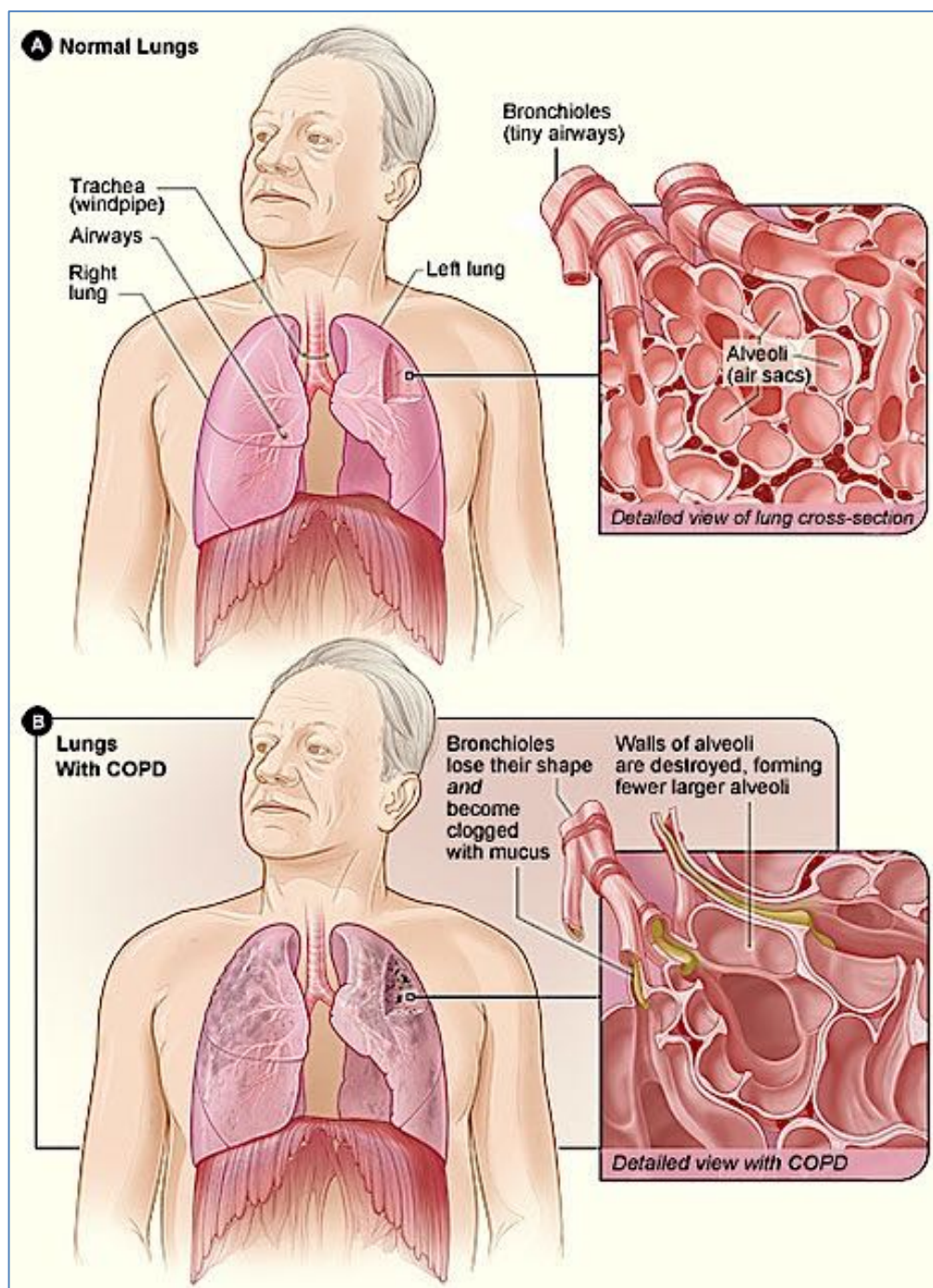
- Clinical Features:

- **Type A – Pinker 'Puffer' – (Emphysema):**
 - Normal Blood Gasses
 - Little/No Cough
 - Breathless
 - Quiet Breath Sounds
 - No Peripheral Oedema
- **Type B – Blue 'Bloater' (Chronic Bronchitis):**
 - Low O_2 + High CO_2 + Cyanosis → Blue (hence name)
 - Chronic Productive Cough
 - Breathless
 - Loud, Abnormal 'Crackling' Breath Sounds ("Crepitations"/"Rales")
 - May Have Peripheral Oedema



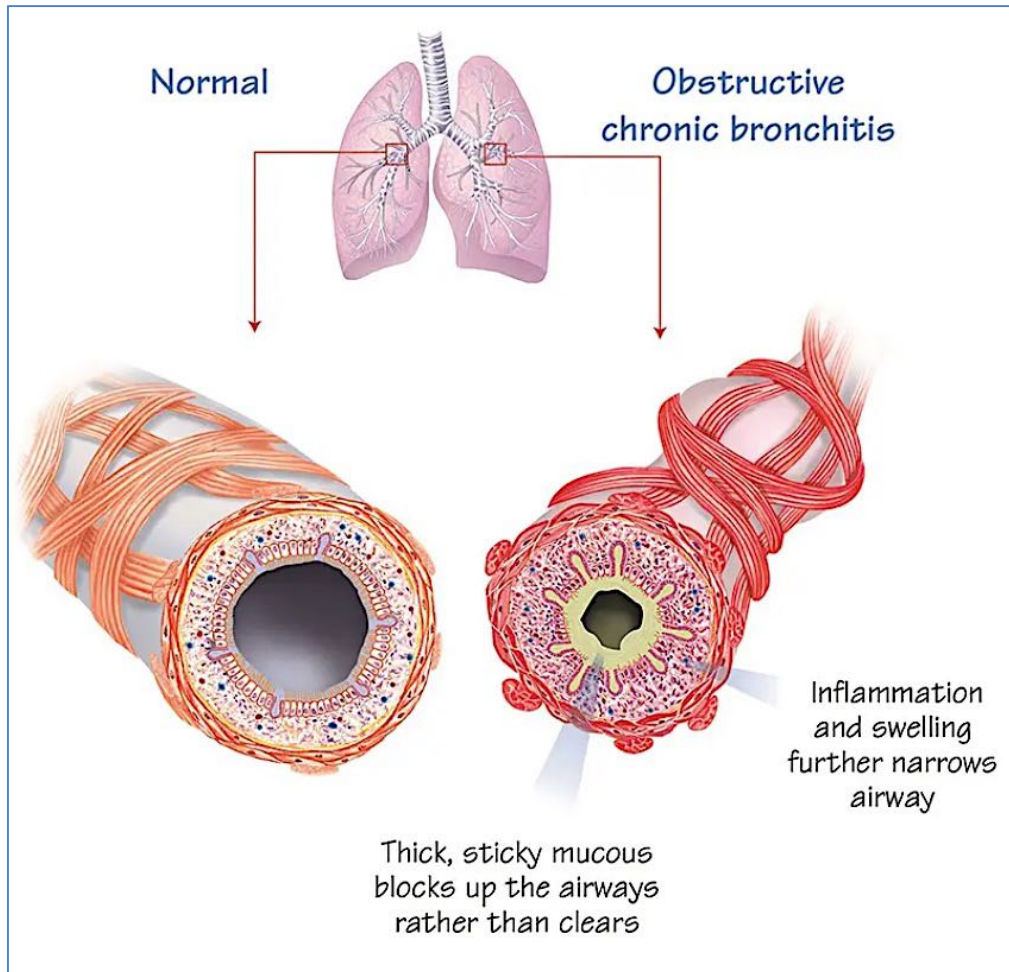
Source: Unattributable

- **Complications of COPD:**
 - Acute Infective Exacerbations
 - Cor-Pulmonale – (RV-Failure 2° to Pulmonary HTN):
 - Polycythaemia (Due to Hypoxia) → high Hb
 - Bronchiectasis
 - End Stage Lung Disease (due to extensive lung fibrosis) → Palliative O2 Therapy.
 - Lung Cancer (Indirectly – due to smoking)
- **Investigations:**
 - ↓ Decreased VC
 - ↓ Decreased FEV1:VC Ratio (FEV1 <80% of Predicted)
 - Mild – FEV1 60-80% → Cough, Exertional Dyspnoea.
 - Mod – FEV1 40-60% → Above + Wheeze, Sputum.
 - Severe – FEV1 <40% → Above + Right Heart Failure (Cor-Pulmonale).
 - ↓ PEF



CHRONIC BRONCHITIS:

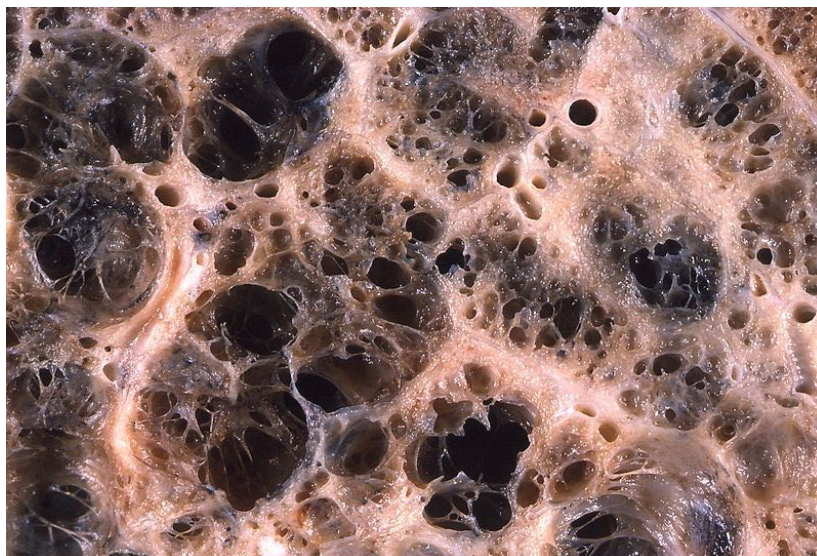
- **Aetiology:**
 - Smoking/Pollution
- **Pathogenesis:**
 - Smoking/Pollution → Acute & Chronic Inflammation of Bronchial Mucosa →
 - → Chronic, Excessive Mucous Production in Bronchial Tree → Excessive Sputum
 - Mucous Plugs Occlude Small Airways → ↑ Work of Breathing
- **Morphology:**
 - Acute & Chronic Inflammation of Bronchial Mucosa
 - ↑ Mucosal Thickness ("Mucus Gland Hyperplasia") (Note: not seen in terminal bronchioles)
 - Excess Mucous → Plugging
 - Lack of Cilia – Retention of Secretions → Recurrent Secondary Infections.
- **Clinical Features:**
 - "Blue Bloaters"
 - Productive Cough
 - Marked Cyanosis, Hypoxaemia & Hypercapnia
 - Mild Dyspnoea & Wheezing
 - Obese, Oedema
 - Infections Common – Fever
- **Investigations:**
 - Diagnostic Criteria = **"Persistent Productive Cough for >3mths/year for >2 Consecutive Years"**
 - **Spirometry** – (↓ FEV1 and FEV1/FVC (<80%); Minimal Change with Bronchodilators)
 - CXR – (Hyperinflation, Flattened Diaphragms, Bronchial Markings towards Periphery).
 - ABG – (↑ PCO₂, ↓ SpO₂, ↑ HCO₃)
- **Management:**
 - **Chronic (Symptomatic):**
 - **Bronchodilator:**
 - *Antimuscarinic [Ipratropium Bromide / Tiotropium]
 - Or Short Acting B₂-Agonist [Albuterol]
 - Or Long Acting B₂-Agonist [Formoterol]
 - **Inhaled Corticosteroids:**
 - Eg. Fluticasone
 - Eg. Budesonide
 - +/- Oxygen (BUT DO NOT KILL RESPIRATORY DRIVE)
 - + Quit Smoking – (Eg. Nicotine Replacement Therapy)
 - + Pneumovax / Fluvax
 - + Diuretics (If RV-Failure)
 - **Acute Exacerbation:**
 - As above...PLUS
 - 1. Theophylline
 - 2. Antibiotics – (Augmentin [Amoxicillin + Clavulanate] / Doxycycline / Trimethoprim / Tobramycin for Pseudomonas)
 - 3. +/- Mechanical Ventilation
- **Complications:**
 - Infective Exacerbations
 - Cor-Pulmonale & Heart Failure
 - Lung Cancer

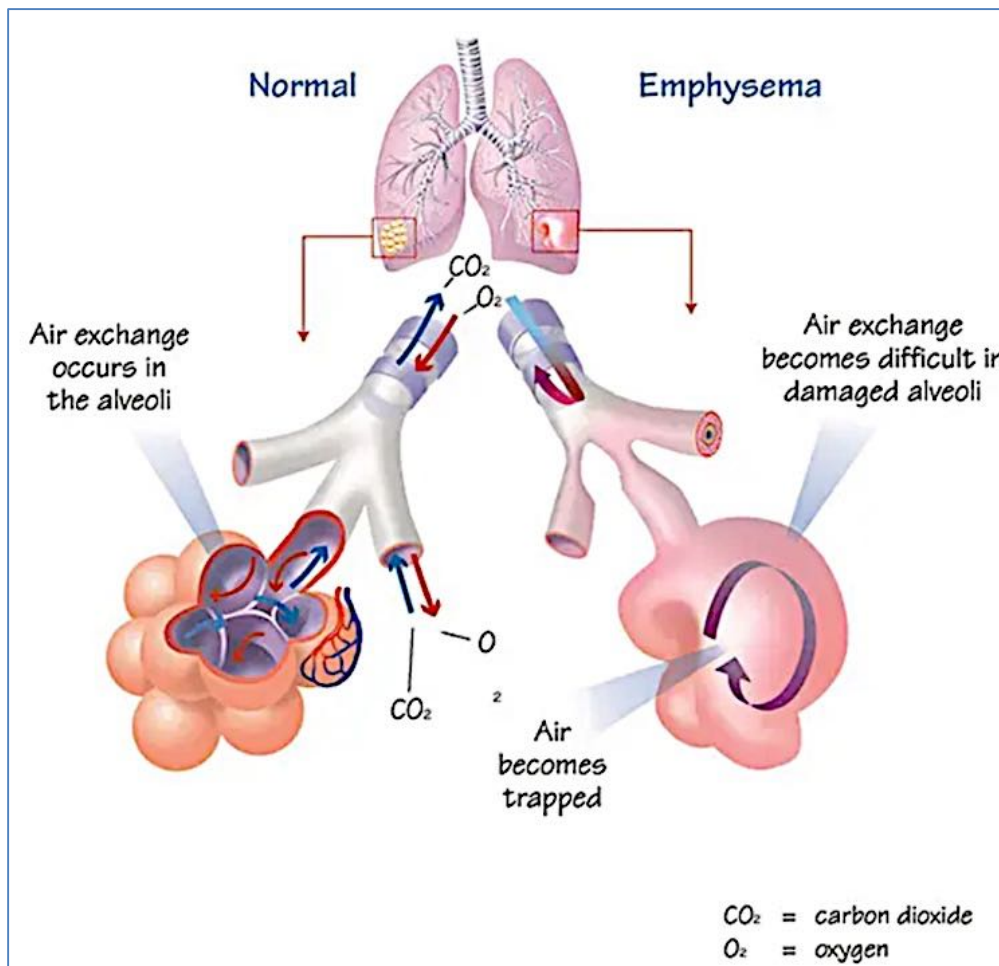


<https://chroniclungdiseases.com/en/copd/>

EMPHYSEMA:

- **Types & Aetiologies:**
 - **** 95% = Centrilobular – SMOKERS**
 - (Panacinar/Panlobular (Congenital - α 1-Antitrypsin Deficiency) → Early Age Emphysema)
- **Pathogenesis:**
 - **Smoking → O₂ Free Radicals → Inflammation (Elastase & Protease) → Direct Alveolar Damage → Loss of “Radial Traction” → Obstruction**
- **Clinical Features:**
 - **“Pink Puffers”**
 - Thin, No Oedema
 - Normal Blood Gasses
 - Little/No Cough
 - Severe Dyspnoea
 - Quiet Breath Sounds
 - No Peripheral Oedema
 - Hyperinflation → Barrel Chest
 - Forward Stooping
 - **Effects on Lung Capacities/Volumes:**
 - ↑TLC (Total Lung Capacity)
 - ↑RV (Residual Volume)
 - ↑FRC(Functional Residual Capacity)
 - ↓VC (Vital Capacity) - Because They Can't Expel All the Gas in their Lungs
 - ↓FEV₁ (Forced Expiratory Volume in 1 Sec) – Because of Dynamic Airway Compression
- **Diagnosis:**
 - **Clinical +**
 - **Spirometry – (Low FEV₁, ↑TLC, ↓Dco)**
 - **CXR – (Hyperinflated, Flattened Diaphragms, Upper-Zone Bullae, Narrow Mediastinum)**
- **Management:**
 - **Bronchodilator:**
 - ***Antimuscarinic [Ipratropium Bromide / Tiotropium]**
 - **Inhaled Corticosteroids:**
 - Eg. Fluticasone
 - Eg. Budesonide
 - **+/- Oxygen (BUT DO NOT KILL RESPIRATORY DRIVE)**
 - **+ Quit Smoking – (Eg. Nicotine Replacement Therapy)**
 - **+ Pneumovax / Fluvax**
 - **+ Diuretics (If RV-Failure)**

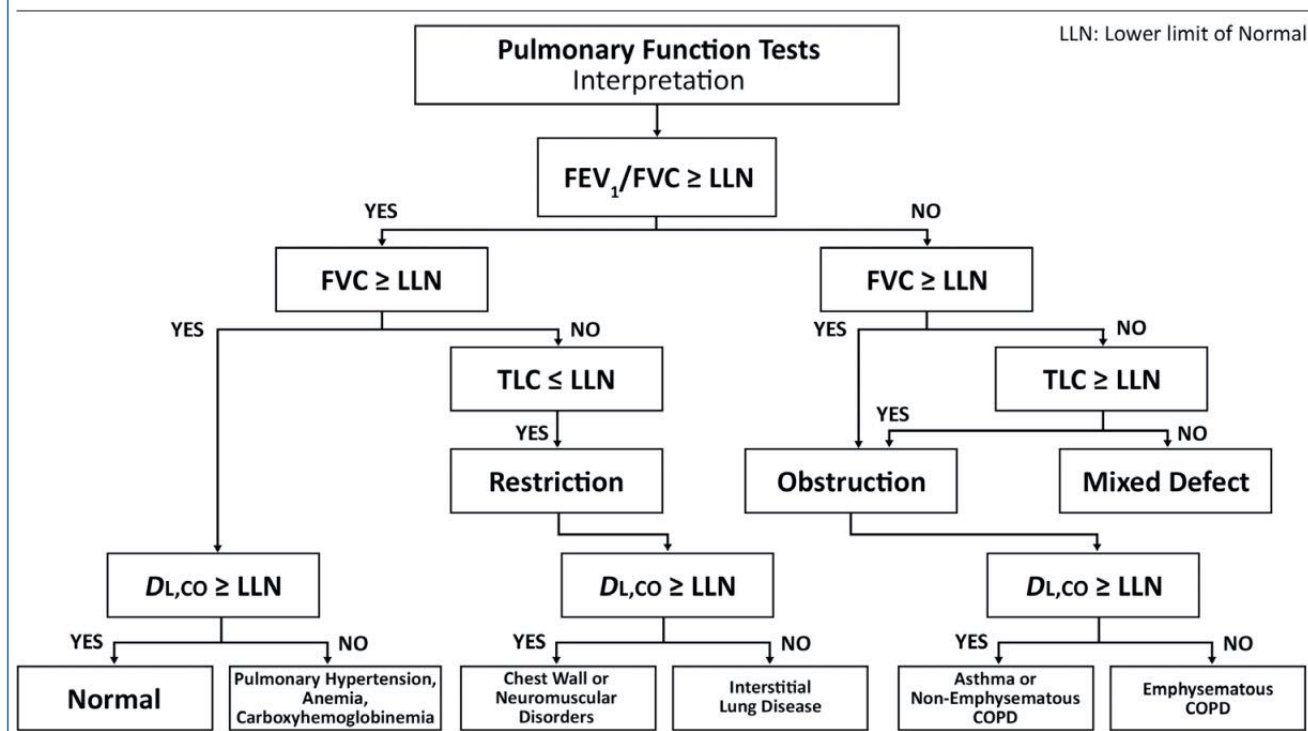




<https://chroniclungdiseases.com/en/copd/>

Pulmonary Function Tests

Interpretation

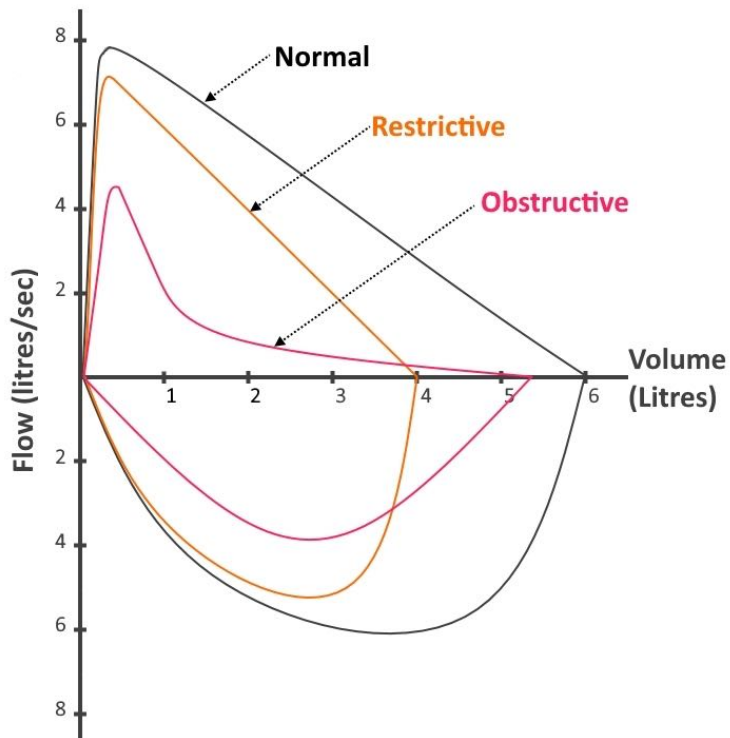


Adapted from: Pellegrino et al. Eur Respir j. 2005;26:948-968

RESTRICTIVE (INTERSTITIAL) LUNG DISEASES

- Loss of Lung/Chest-wall Compliance → Restricted Lung Expansion
 - **Pathogenesis:**
 - o Chronic Interstitial Inflammation → Fibrosis/Thickening/Stiffening of *Lung Parenchyma*
 - **Clinical Manifestations:**
 - o Normal PEF
 - o Normal FEV1/FVC (>80%)
 - o ↓TLC (Total Lung Capacity)
 - o ↓VC (Vital Capacity)
 - o ↓IC (Inspiratory Capacity)
- } Due to ↑Resistance to *Lung Expansion*

Flow Loop Patterns



Normal Pattern:

PEF Normal
Pattern Initially rapid, decreasing steadily
Volume Normal

Restrictive Pattern:

PEF Normal or slightly Reduced
Pattern Normal
Volume Reduced

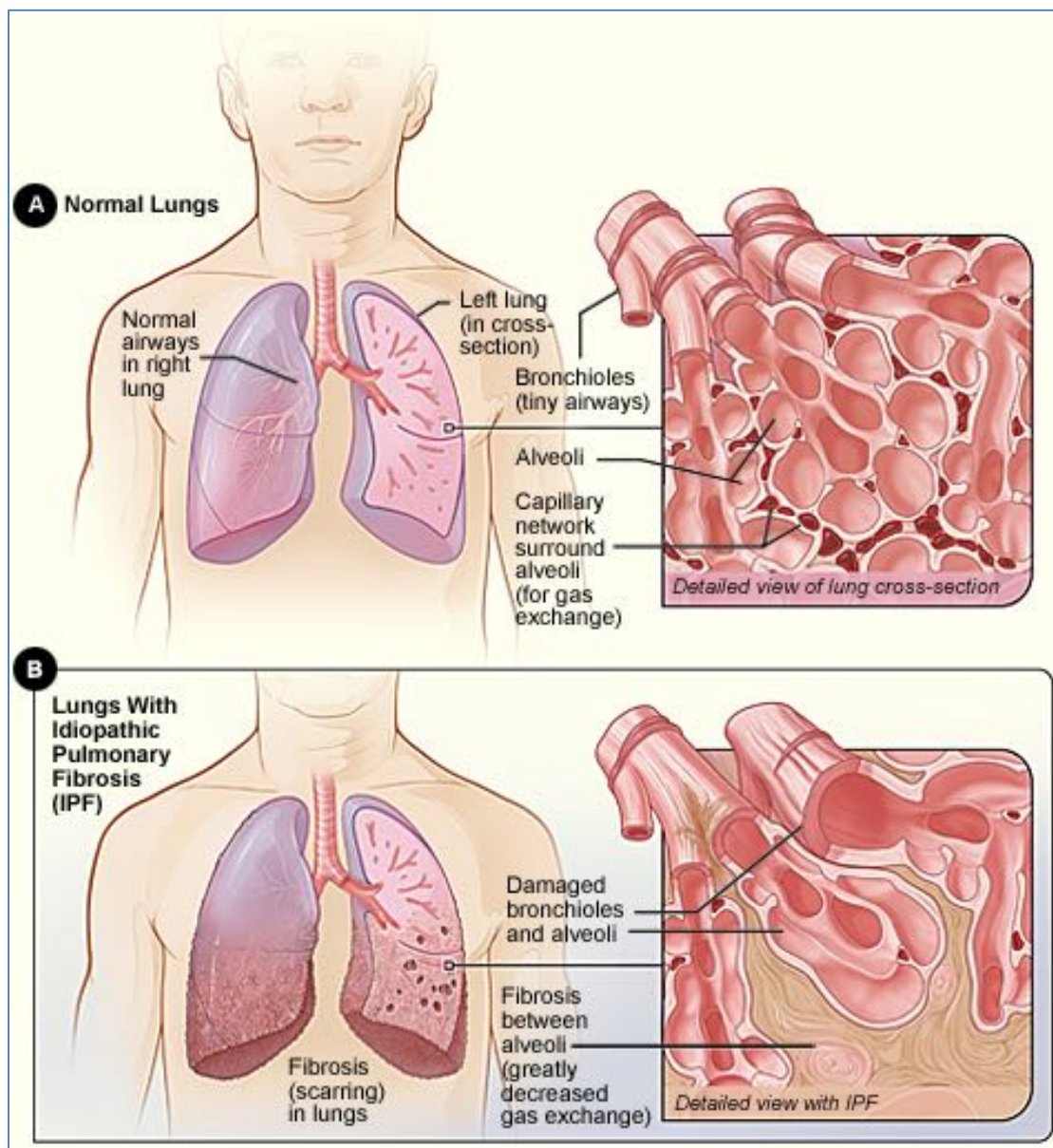
Obstructive Pattern:

PEF Reduced
Pattern Concave
Volume Normal or Reduced

Source: Unattributable

IDIOPATHIC PULMONARY FIBROSIS (IPF):

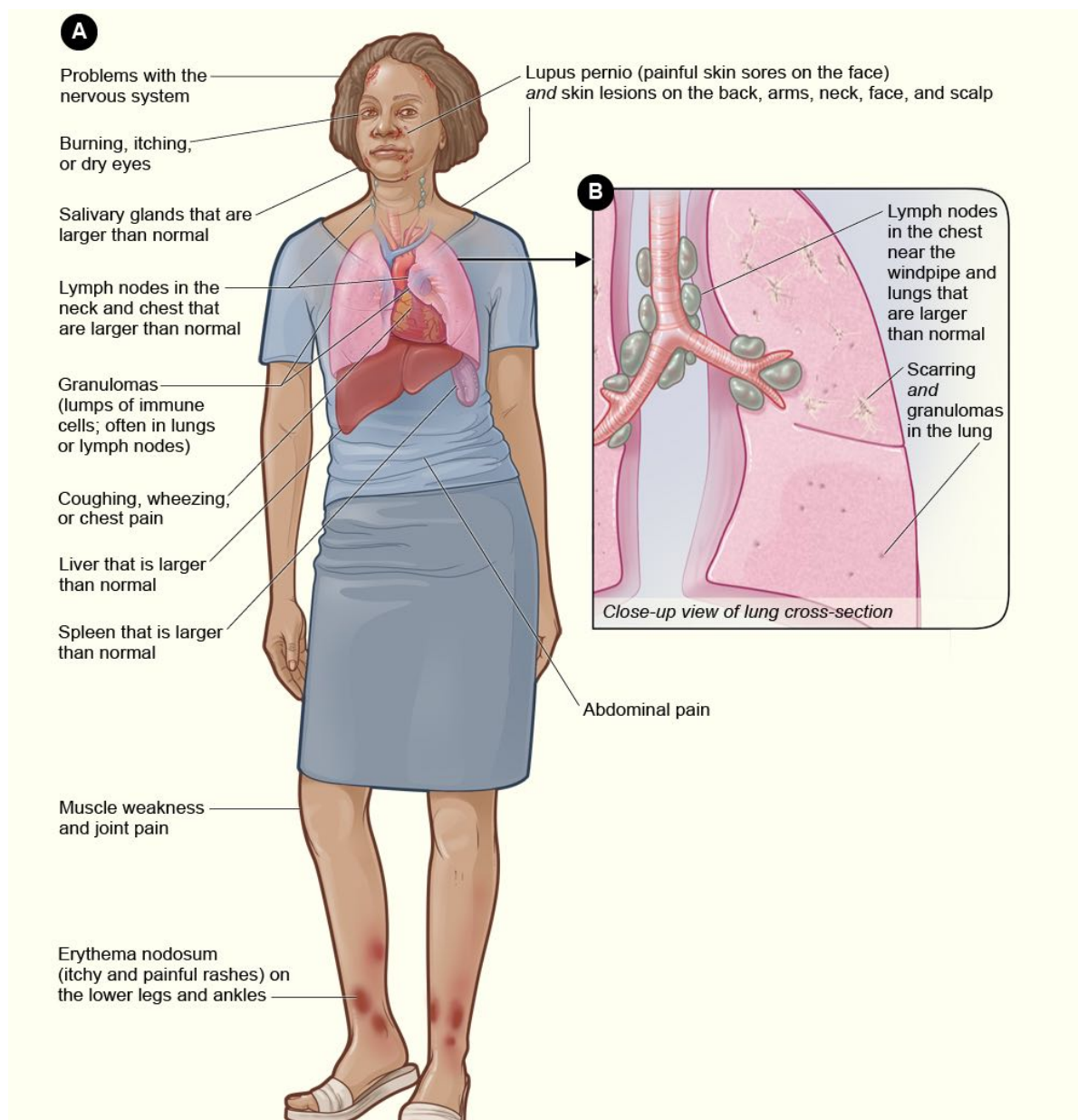
- **Aetiology:**
 - Unknown (*Idiopathic*)
- **Pathogenesis:**
 - Abnormal & Excessive Fibrosis of Pulmonary Interstitium (Mainly Alveolar Walls)
- **Morphology:**
 - Severe Interstitial Fibrosis & Scarring
- **Clinical Features:**
 - Typically in >50yo's
 - Gradual Onset of Symptoms
 - **Symptoms:** Progressive Dyspnoea, Dry Cough, Hypoxia/Cyanosis
 - **Signs:** Clubbing, Velcro-like Inspiratory Crackles
 - **Very poor prognosis (3 yrs)**
- **Investigation:**
 - **Spirometry** – (Restrictive Pattern).
- **Management:**
 - No Known Treatment



National Heart Lung and Blood Institute (NIH), Public domain, via Wikimedia Commons

SARCOIDOSIS:

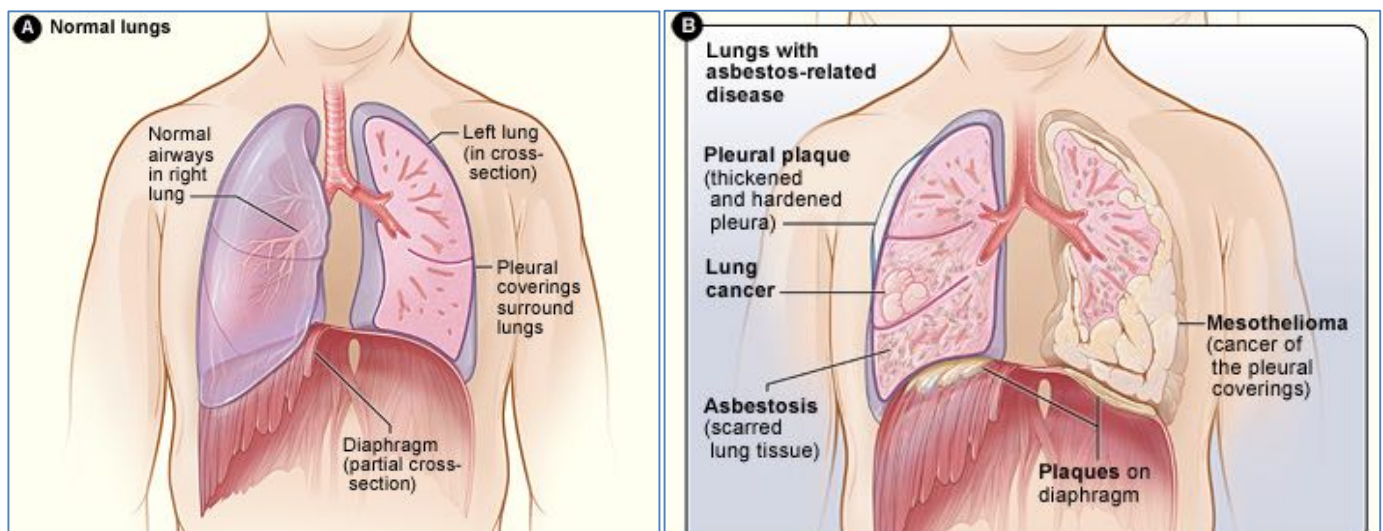
- **Aetiology:**
 - o Idiopathic Immune
- **Pathogenesis:**
 - o Infiltrating Non-Caseating Granulomas → Nodules in Multiple Organs
- **Morphology:**
 - o Multiple Fine Nodules in Multiple Organs (**Lungs, Heart, CNS, Lymph Nodes...)
 - o Pulmonary Fibrosis
- **Clinical Features:**
 - o **Lung Manifestations** – (Dyspnoea, Restrictive Lung Disease)
 - o **Other Organs** – (Lymphadenopathy, Erythema Nodosum, Kidney, Ocular, CNS Damage)
- **Investigations:**
 - o CT
 - o Guided Biopsy
- **Management:**
 - o Corticosteroids
- **Prognosis:**
 - o 50% Spontaneous Resolution within 1-3yrs
 - o Significantly Increased risk of Lung Cancer



PNEUMOCONIOSES:

- ASBESTOSIS:

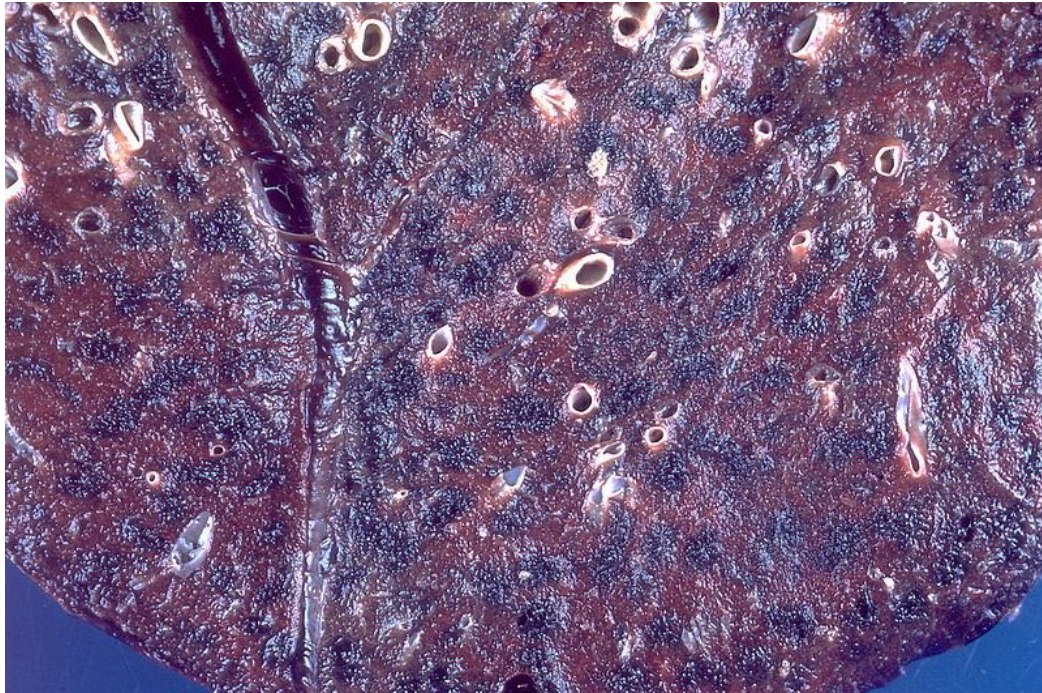
- **Aetiology:**
 - Inhalation of Asbestos Micro-Fibres
- **Pathogenesis:**
 - Asbestos Micro-Fibres in Alveoli → Macrophage Activation → Inflammation → Fibrosis
 - → ↓Elasticity & ↓Gas Diffusion
 - + → **Mesothelioma**
- **Morphology:**
 - Typically in the Lower Lobes
 - **Marked Interstitial (Parenchymal) Fibrosis/Scarring**
- **Clinical Features:**
 - **Symptoms:**
 - Long Latent Period (Several Decades after Exposure)
 - ****Severe Dyspnoea**
 - ****Productive Cough**
 - **Signs:**
 - Inspiratory Crackles
- **Investigations:**
 - **Spirometry** – (Restrictive Pattern, ↓VC, ↓TLC)
 - **CT** – (?Mesothelioma)
 - **Lung Biopsy** – (Ferruginous Bodies)
- **Treatment:**
 - None
 - Supportive Rx
 - Surgery (Pleurectomy if Mesothelioma)
- **Complications:**
 - **Mesothelioma**
 - **Pleural Effusions**
 - **Cor-Pulmonale**
 - **Respiratory Failure**



National Heart Lung and Blood Institute, Public domain, via Wikimedia Commons

- **ANTHRACOSIS ("Coal Miner's Lung"):**

- **Aetiology:**
 - Long Term Inhalation of Carbon Dust (Coal Dust)
- **Pathogenesis:**
 - Coal Dust → Macrophage Phagocytosis & Activation → Inflammation → Fibrosis
- **Clinical Features:**
 - Initially Benign
 - Dyspnoea, Chronic Cough
 - Lung function *reasonably* preserved



Yale Rosen from USA, CC BY-SA 2.0 <<https://creativecommons.org/licenses/by-sa/2.0/>>, via Wikimedia Commons

- **SILICOSIS:**

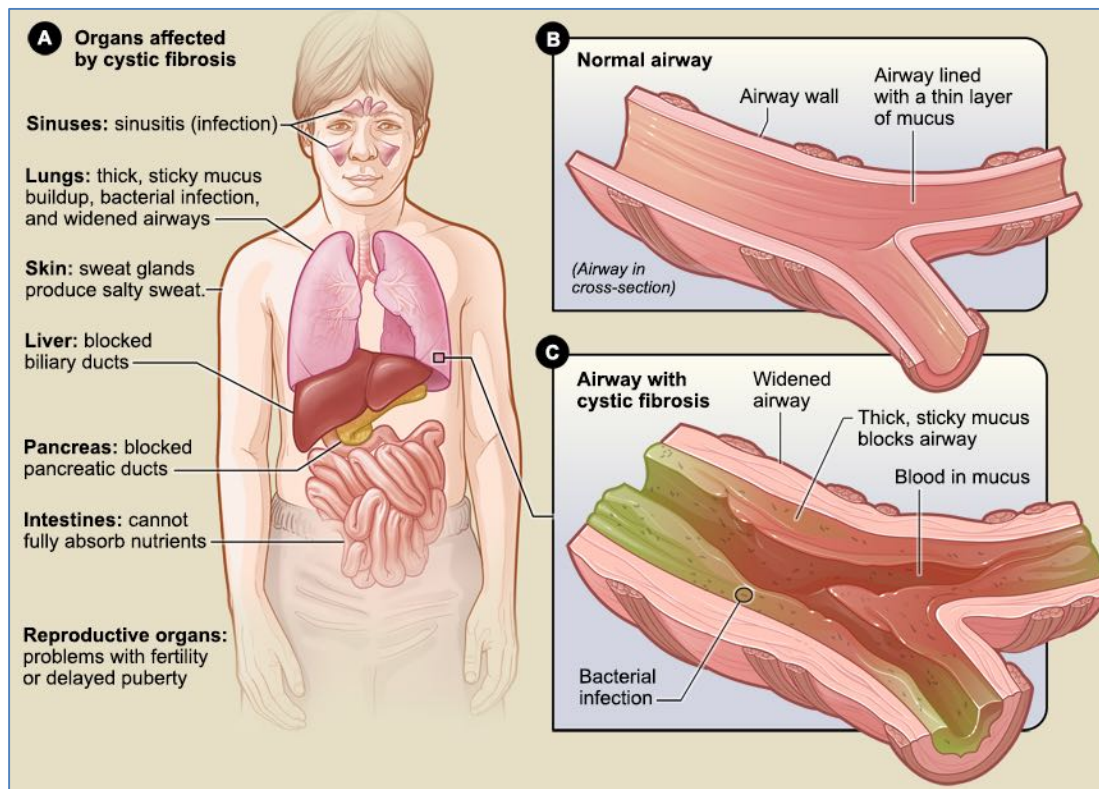
- **Aetiology:**
 - Inhalation of Sand & Stone Dust (Silicone)
- **Pathogenesis:**
 - Silicone Dust in Alveolar Walls → Macrophages Ingest Particles → Inflammation + Fibrosis
- **Clinical Features:**
 - Initially Asymptomatic
 - Dyspnoea, Cough, Cyanosis
- **Investigations:**
 - **Spirometry** – (Restrictive Pattern, ↓VC, ↓TLC)



<https://www.cdc.gov/mmwr/volumes/68/wr/mm6838a1.htm>

CYSTIC FIBROSIS (CF):

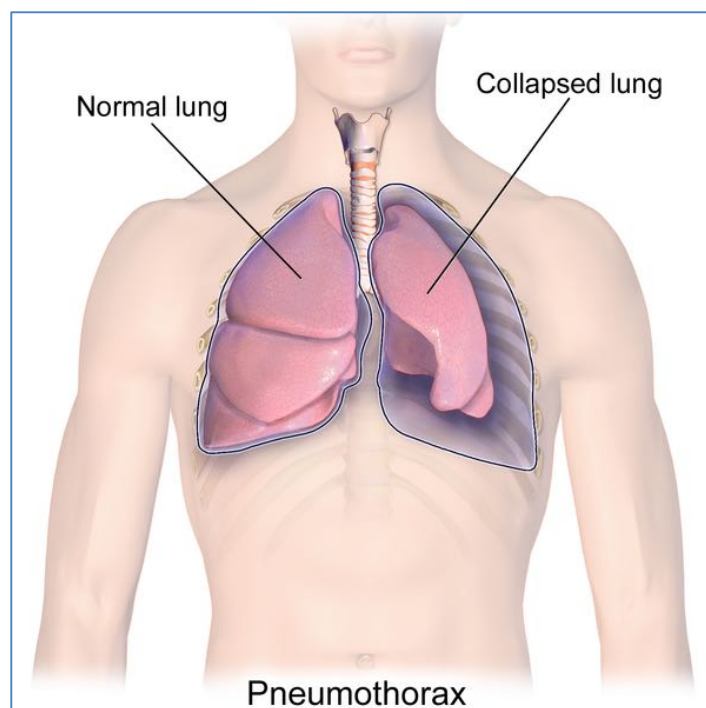
- **Aetiology:**
 - **Simple Autosomal Recessive CFTR Gene Mutation (Chromosome 7)**
 - $\approx 1/25$ people are carriers.
- **Pathogenesis:**
 - **CFTR** – Encodes for Active-Chloride Channels (Which Normally Regulates [Salt] in Secretions)
 - → Thick, Salty Exocrine Secretions → Mostly Affects **Lungs, Pancreas, Intestines & Skin.**
- **Clinical Features:**
 - **Lungs:**
 - Thicker Mucus & ↓ Clearance → **Frequent Lung Infections**
 - Crepitations (Crackling) & Rhonchi (Rattling/Whistling) – Heard through stethoscope
 - **Pancreas:**
 - Obstructed Pancreatic Duct → Chronic Subclinical Pancreatitis → **Pancreatic Failure**
 - **Intestines:**
 - Poor Digestion & Malabsorption → **Malnutrition**
 - **Reproductive Ducts:**
 - Obstructed Vas-Deferens → Infertility
 - **Sweat Glands:**
 - Salty Sweat → **Hyponatraemia** if not replaced.
- **Investigation:**
 - **Spirometry** – (Obstructive Pattern - ↓ FEV₁)
 - **CXR** – (Gas Trapping & Hyperinflated)
 - **Genetic Testing** – (Definitive)
- **Management:**
 - **Enzyme Replacement** – (“**Creon Forte**”)
 - **Salt Replacement** – (**Salt Tablets**)
 - **Fat-Sol. Vitamins** – (**ADEK**)
 - **Chest Physio** – (Percussion, Postural Drainage)
 - **Mucolytics** - (Eg. DNase → Destroys Extracellular DNA → ↓ Mucous Viscosity)
 - **Antibiotics for Recurrent Infections** – (**Tobramycin**)
- **Prognosis:**
 - **40yr Life Expectancy** (without lung transplant)



PENUMOTHORAX

PNEUMOTHORAX:

- **Aetiology:**
 - Penetrating Chest Trauma
 - Bullous Emphysema
 - Lung Cancer
- **Pathogenesis:**
 - Where Air/Fluid Enters The Pleural Space → Disrupts -ve Intrapleural Pressure → Lung Collapses
 - **Spontaneous Pneumothorax – (Eg. Bullous Emphysema):**
 - Rupture of small “Blebs” on Surface of Lung → Air enters Pleural Space From *Within*.
 - **Tension Pneumothorax – (Penetrating Injury):**
 - Penetrating Injury → Air Enters Pleural Space → Forms *Valve* (Air Enters But Can’t Escape).
 - Compresses Major Vessels
 - Impedes Venous Return
 - Causes Respiratory Distress
 - Causes Tachycardia
 - Causes Tracheal Deviation
- **Clinical Features – (Of Tension Pneumothorax):**
 - **Symptoms:**
 - Pleuritic Chest Pain
 - Dyspnoea
 - **Signs:**
 - Tracheal Deviation
 - Respiratory Distress
 - Tachycardia
- **Investigations:**
 - **CXR – (Air in Pleural Cavity, Displaced Mediastinum, Lung Markings Absent in Periphery)**
 - **CT – (If ?Rib# / ?Cancer / ?Haemothorax)**
- **Management:**
 - **Pleural Tap w. One-Way Valve**
 - **Correct Underlying Cause**
 - **O2 Supps.**
 - **Chest Physio** – (To Reinflate Lung & Prevent 2^o Pneumonia)



Blausen.com staff (2014). "Medical gallery of Blausen Medical 2014". WikiJournal of Medicine 1 (2). CC BY 3.0
<<https://creativecommons.org/licenses/by/3.0>>, via Wikimedia Commons



Carolyn M. Allen, Hamdan H. AL-Jahdali, Klaus L. Irion, Sarah Al Ghanem, Alaa Gouda, and Ali Nawaz Khan, CC BY 4.0
<<https://creativecommons.org/licenses/by/4.0>>, via Wikimedia Commons

HEAD AND NECK PATHOLOGIES

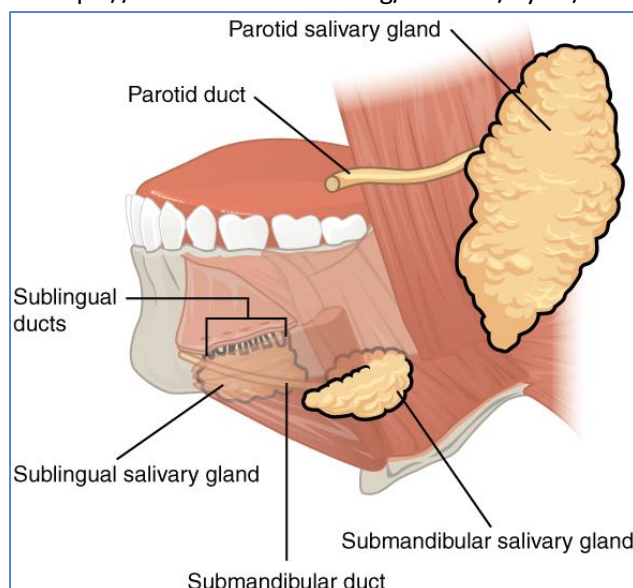
HEAD AND NECK PATHOLOGIES

SIALADENITIS (PAROTITIS):

- **Definition:**
 - Acute Inflammation of the salivary glands
 - (**Parotid** Most Common)
- **Aetiology:**
 - Dehydration/Dry Mouth (Xerostomia) - (Common in Post-Op Patients)
 - → Infection – Bacterial (Staph. Aureus), or Viral (**Mumps**)
- **Pathogenesis:**
 - Dry Mouth (Xerostomia) → Drying of Salivary Secretions in the glands → Infection (Bacterial/Viral)
 - (→ Duct obstruction → Recurrent Sialadenitis)
- **Morphology:**
 - Grossly Enlarged Parotid Gland.
- **Clinical Features:**
 - **Symptoms:**
 - Fever
 - Dry Mouth
 - Abnormal/Foul Tastes
 - ↓Ability to Open Mouth
 - Mouth or Facial Pain, (esp. when eating)
 - Redness over the side of the face or the upper neck
 - Swelling of the face
 - **Complications:**
 - Sialolithiasis (Salivary Gland Calculi) or Fibrosis → Duct Obstruction
- **Management:**
 - Antibiotics –(Penicillin / Metronidazole)



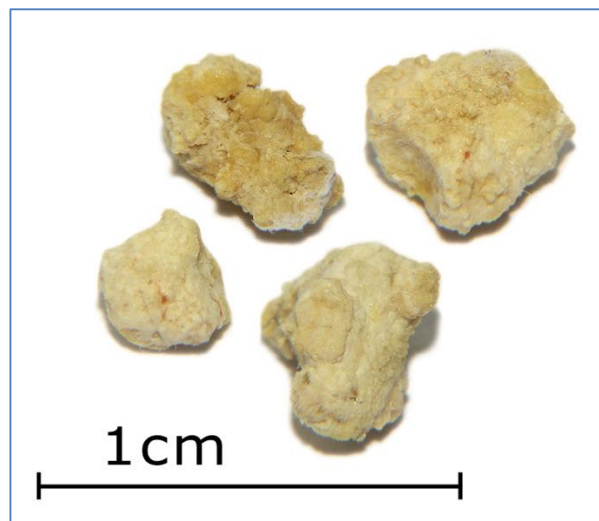
Afrodriguezg, CC BY-SA 3.0 <<https://creativecommons.org/licenses/by-sa/3.0/>>, via Wikimedia Commons



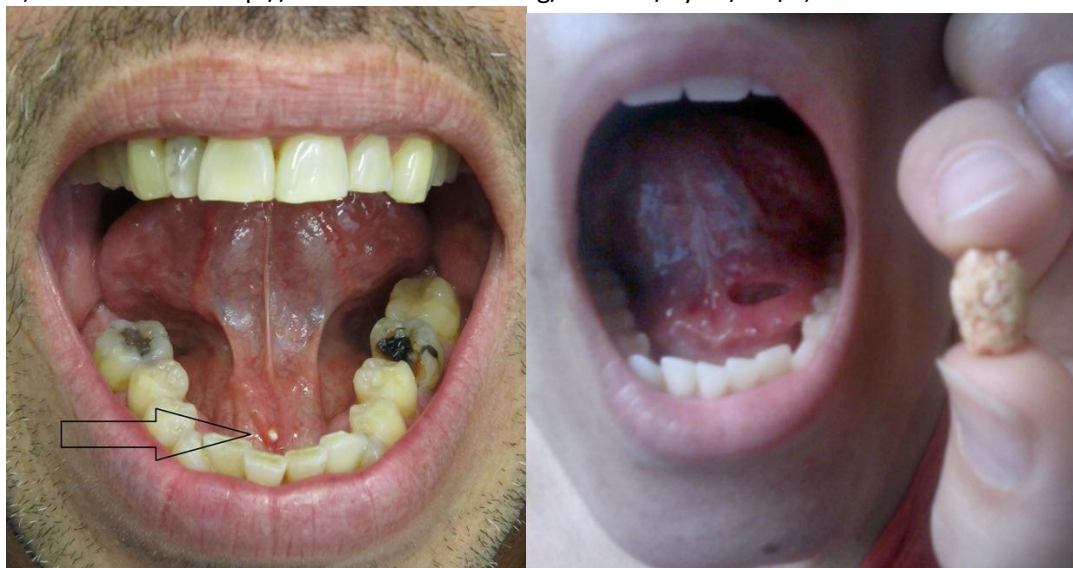
OpenStax College, CC BY 3.0 <<https://creativecommons.org/licenses/by/3.0/>>, via Wikimedia Commons

SIALOLITHIASIS (SALIVARY GLAND CALCULI):

- **Definition**
 - Ductal Stone (mainly hydroxyapatite) in Salivary Gland → Chronic Sialadenitis
 - **80% in submandibular gland**, <20% in parotid gland, -1% in sublingual gland
- **Risk Factors**
 - Anything causing Dry mouth (e.g. Dehydration, Diabetes, EtOH, Anticholinergics)
- **Clinical Features**
 - Painful, Tender Gland.
 - Swelling following Meals
 - Palpation of gland reveals Calculi
- **Investigations**
 - Sialogram
 - CT
- **Treatment**
 - May Resolve Spontaneously
 - Encourage Salivation To Clear Calculus
 - Dilation And Excision Through Floor Of Mouth
 - If Calculus Is in the Gland (not the duct) the Gland Must Be Excised



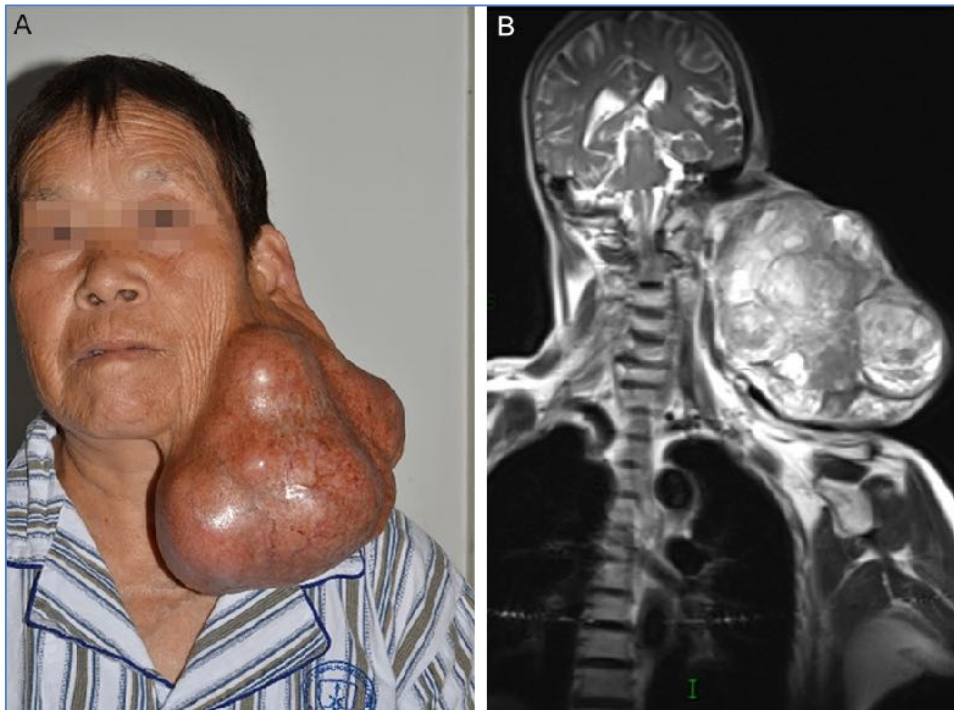
PGA, CC BY-SA 3.0 <<http://creativecommons.org/licenses/by-sa/3.0/>>, via Wikimedia Commons



1. James Heilman, MD, CC BY-SA 3.0 <<https://creativecommons.org/licenses/by-sa/3.0/>>, via Wikimedia Commons
2. Peternickson, CC0, via Wikimedia Commons

Salivary Gland Tumours

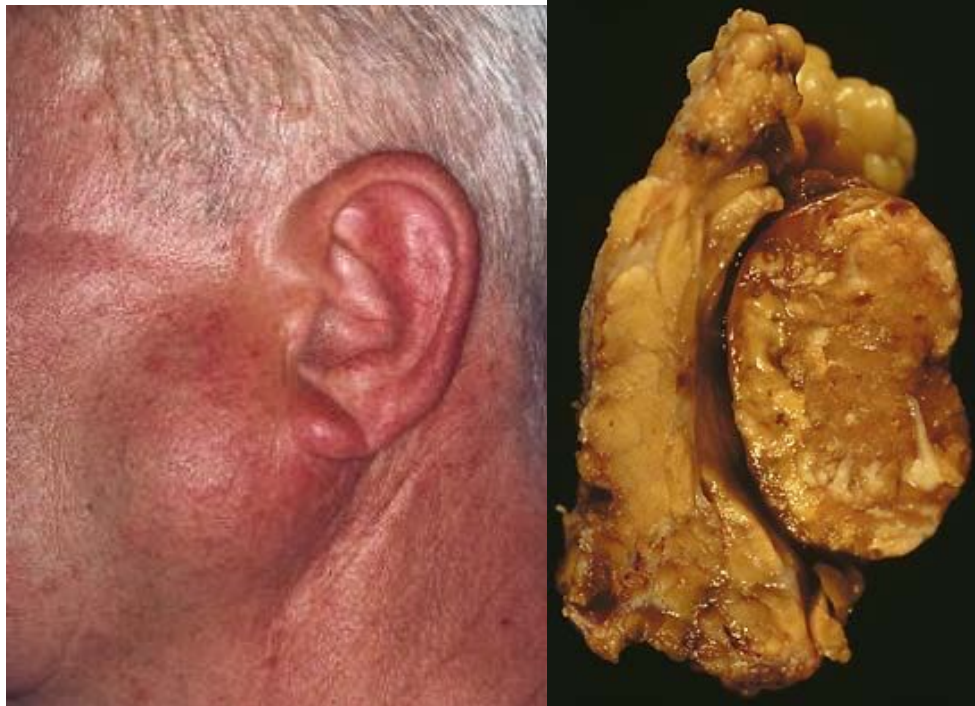
- **PLEOMORPHIC ADENOMA (80% in Parotid Gland)(80% of all Salivary Gland Tumours):**
 - **Definition:**
 - Benign Neoplastic Tumour Of The Salivary Glands
 - **Aetiology:**
 - Unknown – But Strong Association with Cigarette Smoking
 - **Pathogenesis:**
 - Slow-Growing
 - Benign
 - **Morphology:**
 - **Macro:**
 - Enlarged Parotid Gland
 - Firm, Mobile, Nodule/s.
 - **Micro:**
 - Architectural Pleomorphism (variable appearance) seen by light microscopy
 - Cysts lined by *Squamous Epithelium*
 - Anastomosing Trabeculae
 - Myxoid Areas (Mucoid/Mucous like)
 - Chondroid Areas (Cartilage)
 - Adenoma = Ductal Origin
 - Glands
 - Tumour is Not Enveloped, but is surrounded by a *Fibrous Pseudocapsule*.
 - **Clinical Features:**
 - Adults
 - Benign (But may transform to malignant “**Carcinoma Ex-Pleomorphic Adenoma**”)
 - Enlarged Parotid Gland
 - Painless & Slow-Growing, Firm Single Nodular Mass.
 - Asymptomatic
 - **Investigations:**
 - fine needle aspiration biopsy
 - CT or MRI to determine extent of tumour
 - **Treatment:**
 - Excision = Gold Standard for ALL Salivary Gland Tumours (Benign OR Malignant)



<https://www.semanticscholar.org/paper/Giant-pleomorphic-adenoma-of-the-parotid-gland-with-Weng-Guo/0f2396a0e8b176035b88dceb65463107c1aa9e73/figure/0>

- **WARTHIN'S TUMOUR (AKA: "PAPILLARY CYSTADENOMA LYMPHOMATOSUM") (10% of all Salivary Gland Tumours):**

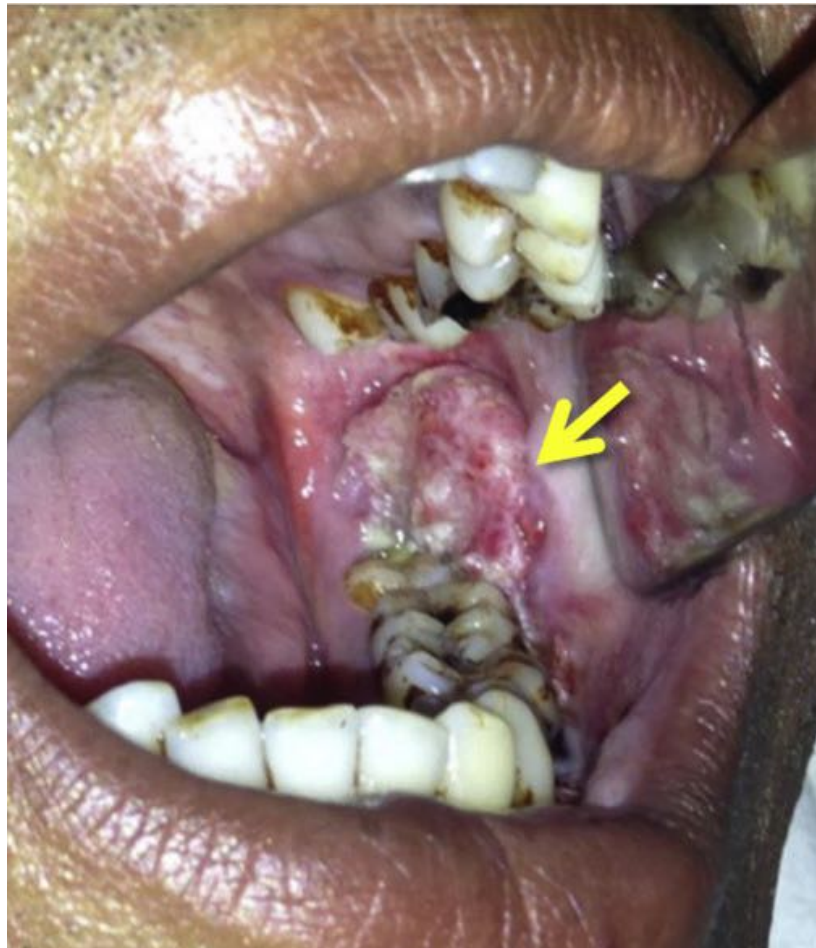
- **Aetiology:**
 - Unknown – But Strong Association with Cigarette Smoking
- **Pathogenesis:**
 - Benign
- **Morphology:**
 - Macro:
 - 80% in Parotid Gland
 - Parotid Swelling (Typically @ the tail near the angle of the Mandible)
 - Micro:
 - Epithelium-lined Lymphoid Tissue
 - Cystic Spaces surrounded by a 2-layered Epithelium with Central Pyknotic Nuclei.
 - Epithelium has Lymphoid Stroma with Germinal Centre Formation.
- **Clinical Features:**
 - Male; Old Age (60-70yrs).
 - Painless, Slow-Growing Parotid Gland (Typically @ the tail near the angle of the Mandible)
 - Benign (But risk of malignant transformation)
- **Investigations:**
 - fine needle aspiration biopsy
 - CT or MRI to determine extent of tumour
- **Treatment:**
 - Excision = Gold Standard for ALL Salivary Gland Tumours (Benign *OR* Malignant)



Ed Uthman, MD, Public domain, via Wikimedia Commons

ORAL SQUAMOUS CELL CARCINOMA:

- **Aetiology:**
 - Tobacco, Alcohol, HPV
- **Pathogenesis:**
 - Carcinogenesis of the Squamous Oral Mucosal Epithelium.
- **Clinical Features:**
 - Start as white-gray plaques → Nodular Masses or Necrotic Ulcers
 - **Common Sites:**
 - Border of Lower Lip
 - Floor of mouth
 - Lateral tongue
- **Investigations:**
 - Head CT
- **Management:**
 - Excisional Biopsy
 - +/- Sentinel Node Resection
- **Prognosis:**
 - (Note: 50% of oral SCC's have nodal involvement at diagnosis)
 - 30% 5yr survival rate.



[https://www.joms.org/article/S0278-2391\(17\)30365-8/fulltext#relatedArticles](https://www.joms.org/article/S0278-2391(17)30365-8/fulltext#relatedArticles)

Oral SCC Precursor Lesions = LEUKOPLAKIA/ERYTHROPLAKIA:

- **Aetiology:**
 - Associated with Tobacco
- **Pathogenesis:**
 - Hyperkeratosis & Parakeratosis
- **Morphology:**
 - Well-defined, white or red plaque (Due to Hyperkeratosis)
- **Clinical Features:**
 - 5-15% Transform to Cancer.



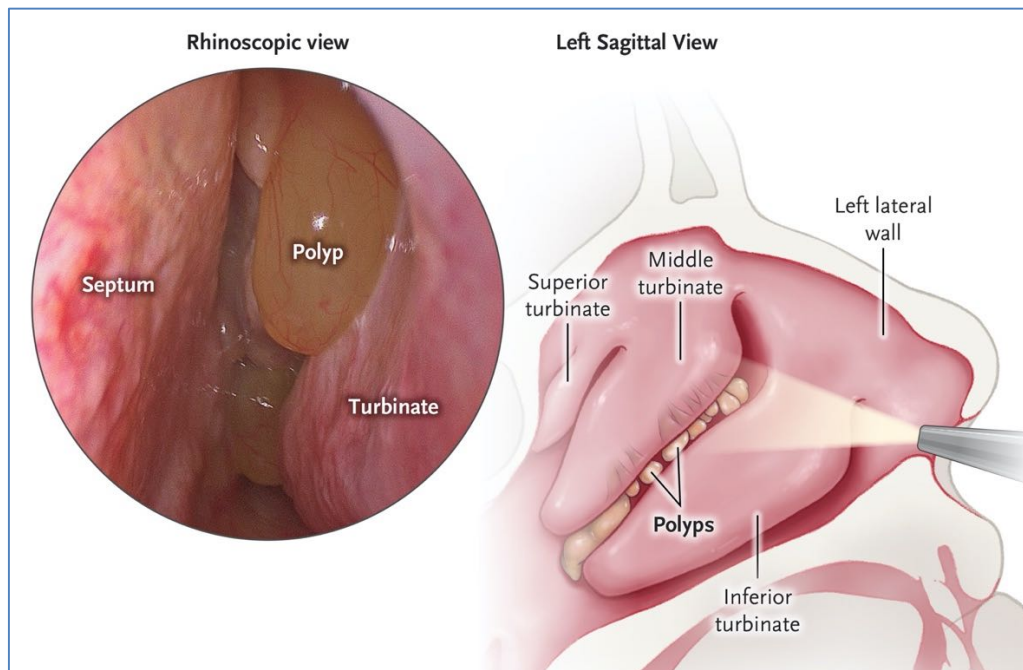
Photo uploaded by: dozenist., CC BY-SA 3.0 <<http://creativecommons.org/licenses/by-sa/3.0/>>, via Wikimedia Commons



<https://exodontia.info/erythroplakia-erythroplasia/>

NASAL POLYPS (Inflammatory):

- **Aetiology:**
 - Chronic/Recurrent URTI
 - Allergy, Hypersensitivity
- **Pathogenesis:**
 - Chronic Mucosal Inflammation + Oedema → Inflammatory Hyperplasia → Polyp
 - **Note: 100% Benign: NO Malignant Potential!**
- **Clinical Features:**
 - Mobile, Non-Tender Polypoid Masses
 - **Symptoms:**
 - Nasal Block
 - Sinusitis
 - Anosmia (Loss Of Smell)
 - Secondary Infection → Headache
- **Investigations:**
 - Rhinoscopy
- **Management:**
 - Nasal Corticosteroids (Nasonex)



<https://resident360.nejm.org/clinical-pearls/chronic-rhinosinusitis-with-nasal-polyps>



MathieuMD / Wikimedia Commons

NASOPHARYNGEAL CARCINOMA:

- (Most common cancer of the nasopharynx)
- **Aetiology:**
 - Risk Factors:
 - EBV
 - Chinese/African
 - Male
- **Pathogenesis:**
 - A Malignant Neoplasm, Arising From The Mucosal Epithelium Of The Nasopharynx
- **Morphology:**
 - Dedifferentiated Carcinoma (Often Keratinizing Squamous Cells)
 - Many Reactive Lymphocytes
 - Locally Invasive
- **Clinical Features:**
 - **Signs/Symptoms:**
 - Cervical lymphadenopathy
 - trismus (inability to normally open the mouth)
 - Pain
 - otitis media
 - nasal regurgitation due to paresis of the soft palate
 - hearing loss
 - cranial nerve palsies
 - nasal obstruction or bleeding
 - **Metastatic spread** → may result in bone pain or organ dysfunction

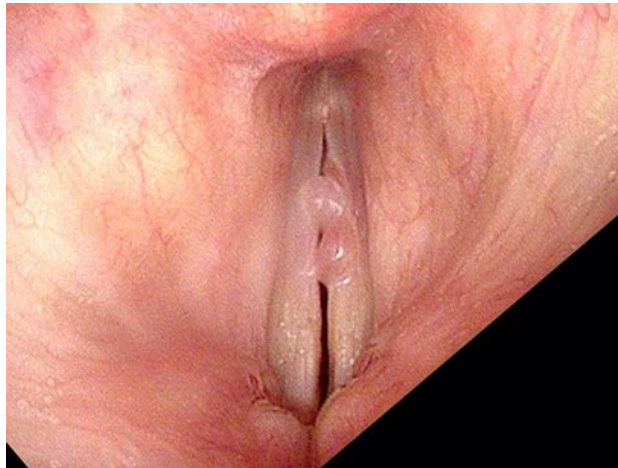


LARYNGEAL TUMOURS

Benign laryngeal Tumours:

- VOCAL CORD NODULES:

- **Aetiology:**
 - Chronic Irritation (Singers/Smokers)
- **Pathogenesis:**
 - Chronic Irritation → Fibrosis → Nodules
- **Morphology:**
 - Fibrous
- **Clinical Features:**
 - May Bleed
 - Non Malignant



<https://www.voicedoctor.net/media/behavioral-injury-overdoer-central/polyps/hemorrhagic-vocal-cord-polyp-1>

- RECURRENT PAPILLOMATOSIS:

- **Aetiology:**
 - HPV Infection in Children (From infected Mothers)
- **Pathogenesis:**
 - HPV Infection → Causes cell mutations → Dysregulated Cell Proliferation
- **Morphology:**
 - Warty Lesion on the Vocal Cords. (May extend down the trachea)
- **Clinical Features:**
 - Children



<https://www.voicedoctor.net/media/structural-injury-tumor-benign/papilloma/papilloma-tend-recur>

LARYNGEAL CARCINOMA (SCC) (MALIGNANT):

- **Aetiology:**
 - Smokers, Alcohol & Radiation
 - 7:1 - M:F
- **Pathogenesis:**
 - Squamous Cell Carcinoma
- **Morphology:**
 - Invasive SCC
- **Clinical Features:**
 - Persistent Hoarseness of Voice
 - Dysphagia
 - Sore Throat
- **Investigations:**
 - Head CT (Staging)
- **Management:**
 - Excisional Biopsy (Grading)
- **Prognosis:**
 - 30% Mortality (due to Metastasis)



https://upload.wikimedia.org/wikipedia/commons/a/a5/Tumor_Laryngis-01.jpg

CLINICAL APPROACH TO NECK MASSES

• Approach to a Neck Mass

- Ensure that the neck mass is not a normal neck structure (hyoid, transverse process of C1 vertebra)

Table 15. Acquired Causes of Neck Lumps According to Age

Age (yrs)	Possible Causes of Neck Lump
<20	• Congenital: lateral (branchial cleft cyst, laryngocele, cystic hygroma), midline (thyroglossal duct cyst) • Inflammatory neck nodes (tonsillitis, infectious mononucleosis, Kawasaki's) • Lymphoma
20-40	• HIV • Salivary gland (calculi, infection, tumour) • Thyroid (goitre, infection, tumour) • Granulomatous disease (TB, sarcoidosis)
>40	• Primary or secondary malignant disease

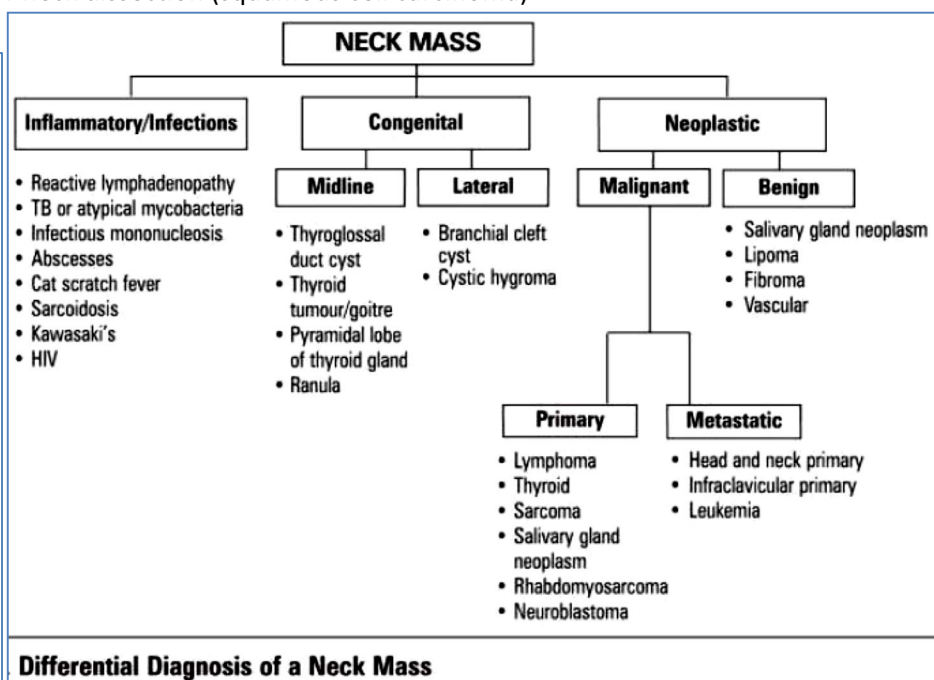
• Evaluation

○ Investigations

- **History And Physical (Including Nasopharynx And Larynx)**
- **Laboratory Investigations**
 - WBC - infection vs. Lymphoma?
 - Mantoux – TB?
 - TFTs and scan – Hypothyroid?
- **Imaging**
 - neck XR
 - CT scan
 - angiography - vascularity and blood supply to mass
 - radiologic exam of stomach, bowel and sinuses
- **Biopsy - For Histology**
 - fine needle aspiration (FNA) - least invasive
 - open biopsy - for lymphoma
- **Identification Of Primary Tumour**
 - Panendoscopy: Nasopharyngoscopy, Laryngoscopy, Esophagoscopy, Bronchoscopy With Washings, And Biopsy Of Suspicious Lesions
 - biopsy of normal tissue of nasopharynx, tonsils, base of tongue, and hypopharynx
 - **Primary identified 95% of time** - Stage and Treat
 - **Primary occult 5% of time** - Excisional Nodal Biopsy for Histology + Radiotherapy and/or neck dissection (squamous cell carcinoma)

Inflammatory vs. Neoplastic Neck Masses

	Inflammatory	Neoplastic
History		
Painful	Y	N
H&N infection	Y	N
Fever	Y	N
Weight loss	N	Y
CA risk factors	N	Y
Age	Younger	Older
Physical		
Tender	Y	N
Rubbery	Y	Occ.
Rock hard	N	Y
Mobile	Y	± fixed
Size	<2 cm	>2 cm



LUNG CANCERS

LUNG CANCERS

Lung Tumours:

- Classification of Malignant Lung Tumours:

- ****Bronchogenic Carcinomas (95% of Lung Cancers)** – From the Bronchi

	<u>Risk Factors?</u>	<u>Cent/Periph?</u>	<u>Aggressive?</u>	<u>Treatment</u>
Small Cell Ca. (SCC) 20%	Smoking Male	Central → Spreads	Highly Aggressive, Poorly Demarcated	Chemotherapy (NOT Surgery)
Non-Small Cell Ca (NSCC) 80%				
- Squamous Cell Carcinoma	Smoking Male	Central Local	Mildly Aggressive, Well Demarcated	Surgery (NOT Chemo)
- Adenocarcinoma	Female, Non-Smokers	Peripheral	Mildly Aggressive, Well Demarcated	Surgery (NOT Chemo)
- Large Cell Anaplastic Ca.	Male	Central Local	Mildly Aggressive, Well Demarcated	Surgery (NOT Chemo)

- (Metastasis from other organs) → Cannon-Ball appearance on X-Ray
- Mesothelioma – Asbestosis

BRONCHOGENIC LUNG CANCERS:

- Aetiologies:

- **STEPWISE ACCUMULATION OF GENETIC MUTATIONS DUE TO ENVIRONMENTAL INSULTS:**

- ***Smoking:** 90% Are Due to Smoking. (20x Risk if >40/day)
- **Occupational Exposure:** Asbestos / Coal Dust / Smoke
- **Radiation Exposure:** Nuclear scientists / Atomic Bomb Survivors

- Pathogenesis:

- Carcinogens → Mutations (Oncogene Activation → Promotion → Pleomorphism) → Cancer
- (Normal → Metaplasia → Dysplasia → Pleiomorphism → Neoplasia → Invasion)

- Clinical Features of Bronchogenic Carcinomas:

○ Symptoms:

- (**Note: Often Asymptomatic until Advanced Disease)
- Most Common Presenting Symptoms:
 - Dry Cough + Dyspnoea
 - Haemoptysis
 - Chest Pain
 - Weight Loss
- Other Symptoms:
 - Airway Obstruction →
 - → Pneumonia in the obstructed lobe only
 - → Atelectasis (Collapse of Lung) - (→ Tracheal + Mediastinal Deviation)
 - → Bronchiectasis (Overstretched Bronchial Tubes)
 - → Abscess
 - Tumour Invasion Of:
 - Pleura → Pleural Effusion
 - Pericardium → Pericarditis/Pericardial Effusion → Tamponade
 - Laryngeal Nerve → Hoarseness
 - Phrenic Nerve → Diaphragm Paralysis
 - Oesophagus → Dysphagia
 - Sympathetic Ganglia → Horner's Syndrome
 - SVC → SVC Syndrome (Pemberton's Sign)
 - Alveoli/Pulmonary Vessels → Haemoptysis
 - Bone → Bone Pain/Path-Fractures
 - Brain → Epilepsy/Focal Neurology

- **Complications:**

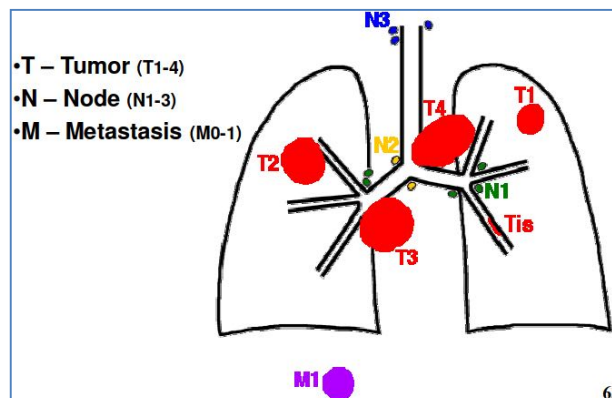
- **Paraneoplastic Syndromes** – Typically in **SMALL CELL CARCINOMAS**:
 - **Hypertrophic Pulmonary Osteo Arthropathy (HPOA)** → ↑PTH-like Hormone:
 - → Wrist Tenderness (Osteoarthropathy) + Finger Clubbing + Hypercalcaemia
 - **Carcinoid Syndrome** → Carcinoid Tumours Secrete ↑Serotonin:
 - → Hot Flushes + Diarrhoea + Abdo Cramps
 - **↑ADH (SIADH – Syndrome of Inappropriate Anti-Diuretic Hormone Secretion)**:
 - → Hyponatraemia
 - **Cushings Syndrome** → ↑ACTH (Adreno Cortico-Tropic Hormone) → ↑Cortisol:
 - → “Moon Facies” + Rapid Weight Gain + Hypertension + Insomnia + Impotence
- **Pancoast Tumours (Apical Lung Tumours)**:
 - **Horner’s Syndrome**:
 - Sympathetic Chain Compression → Symptoms mimic Loss of Sympathetic NS.
 - (Remember *Horny Pamela*):
 - **P** Ptosis (Unilateral) (Droopy Eyelid)
 - **A** Anhydrosis (Unilateral) (Loss of sweating)
 - **M** Miosis (Unilateral) (Pupillary Constriction)
 - **E** Enophthalmos (Unilateral) (Recession of Eyeball)
 - **L** Laryngeal Nerve Palsy → Hoarseness
 - **Pancoast Syndrome (Brachial Plexus Compression)**:
 - → Shoulder Pain
 - → Wasting of Intrinsic Hand Muscles
 - → Paraesthesia
 - → Motor Disturbances in Hands
 - **SVC Syndrome/Pemberton’s Sign (SVC Obstruction)**:
 - **Pemberton’s Sign**: Facial flushing & ↑JVP If arms are raised above Pt’s head

- **Investigations:**

- **Clinical:** History/Exam
- **Imaging:** CXR/CT/PET/Bone Scan
- **Cytology:** Sputum/Bronchial Lavage
- **Biopsy:** Needle Aspirate/Excision
- **Tumour Markers** (Monitoring only)

- **Grading Vs Staging:**

- **Grading** = Microscopic (Microscopic Features) (Higher = More Aggressive)
- **Staging** = Clinical (Progression – TNM)
 - **T** = Tumour Size
 - **N** = lymph node involvement – Primary/Secondary/Tertiary
 - **M** = Metastasis



- **Prognosis:**

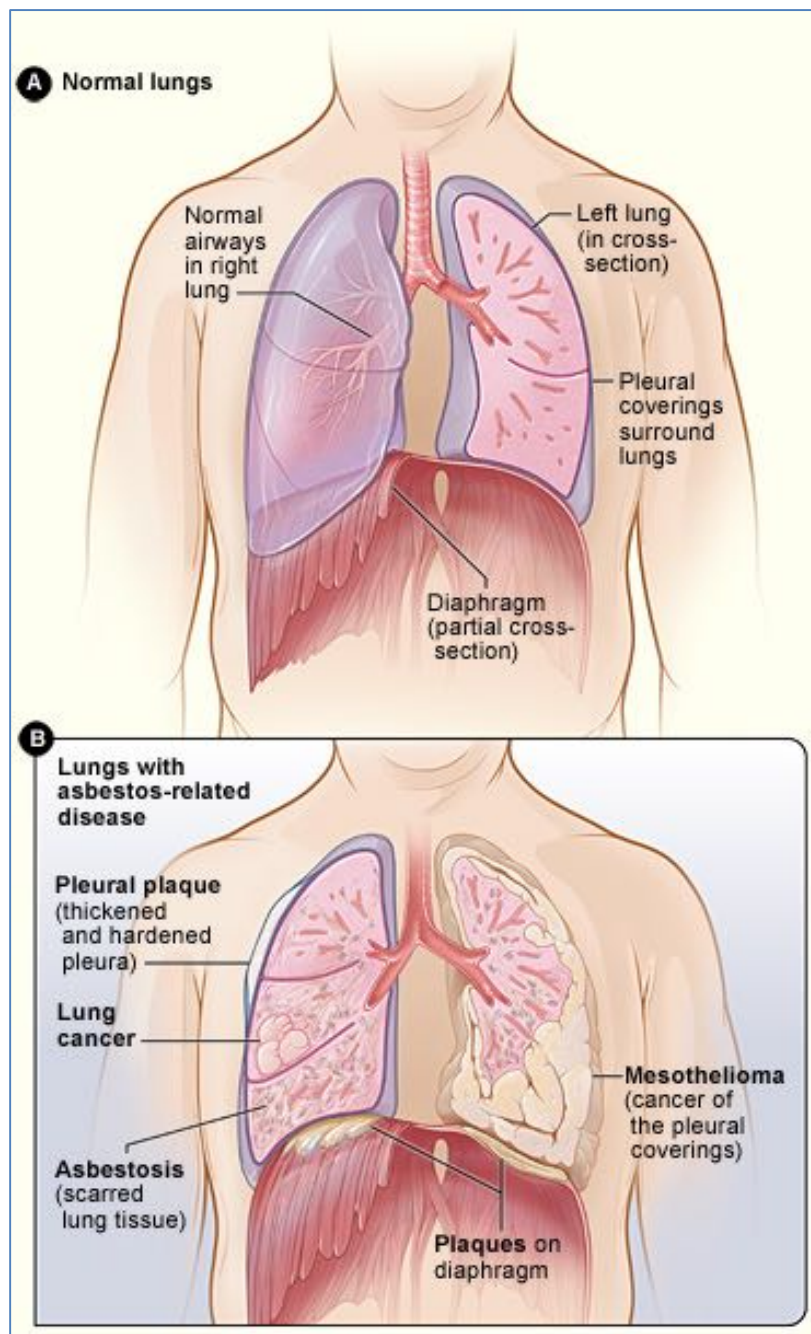
- **Poor** - 5yr average survival – (Depends on the Type, Grade & Stage of Cancer)

- **Treatment:**

- **Surgery** – (Usually Lobectomy, Unilateral Pneumonectomy).
- **Chemotherapy** – (Not curative *on its own*; but often a good Adjuvant to surgery or for Palliative).
- **Radiation** – (Can cure NSCLC, although not first line) – Also Effective in Palliative Care

MESOTHELIOMA:

- **Aetiology:**
 - *****Asbestosis – Risk Factor**
- **Pathogenesis:**
 - **Malignant Neoplasm of the Pleura** (No change in lung tissue)
 - See General Carcinogenesis
- **Morphology:**
 - **Poorly Demarcated**
 - **Spreads *Around the Pleura* → Encases the Lungs**
- **Clinical Features:**
 - Rare tumour
 - Looks like thickening of the pleural surface on CT
 - → **Recurring Pleural Effusions, Chest Pain, Dyspnoea
 - → Metastasis
 - **VERY Poor Prognosis (50% 1yr Mortality)**



National Heart Lung and Blood Institute, Public domain, via Wikimedia Commons

HYPOXIA AND HYPERCAPNIA

HYPOXIA AND HYPERCAPNIA

Common Outcomes of Respiratory Emergencies:

- HYPOXIA:

- **Types:**
 - **Hypoxic Hypoxia:**
 - Most common type.
 - Result of Insufficient oxygen available to the lungs (Eg. Obstruction/Drowning/Altitude)
 - **Stagnant Hypoxia:**
 - Not enough Cardiac Output → ↓Tissue Perfusion
 - **Anaemic Hypoxia:**
 - Not enough Haemoglobin → ↓O₂-Carrying Capacity of Blood.
 - **Histotoxic Hypoxia:**
 - Toxin which prevents Oxidative Metabolism @ the Cellular Level
 - Eg. Cyanide/Oligomycin
- **Effects:**
 - Reduced work Capacity of Muscles
 - Depressed Mental Capacity
- **Treatment:**
 - Supplemental O₂

- HYPERCAPNIA:

- **= Excess CO₂:**
 - Typically caused by Hypoventilation
 - (Normal pCO₂ Range = 35 - 45mmHg)
- **Effects:**
 - If pCO₂ > 60mmHg → Severe Dyspnoea
 - If pCO₂ > 80mmHg → Lethargy & Coma
 - If pCO₂ > 120mmHg → Anaesthesia, Respiratory Depression & Death
- **Note:**
 - CO₂ Diffuses 20x faster than O₂
 - CO₂ is a *More Potent* Respiratory Stimulus than O₂
 - Blood capacity for CO₂ is 3x More than O₂
- **Treatment:**
 - Encourage Hyperventilation
 - Assisted Breathing (if Unconscious)
 - Note: Supplemental O₂ can → Suppress Central Control of Breathing →Respiratory Arrest.

Some Common Respiratory Emergencies Cases:

- **Case 1:**

- Temperature
- Otherwise well
- 100% sats
- Cough & runny nose
- **Diagnosis** = Viral Infection
- **Treatment** = Reassurance, rest & fluids

- **Case 2:**

- Febrile
- Creps
- 90% Sats (Hypoxic)
- Tachypnoeic
- Mild indrawing & creps
- **Diagnosis** = Severe Pneumonia (Due to hypoxia)
- **Treatment** = IV Antibiotics, IV Fluids, Oxygen

- **Case 3:**

- Knife in man's chest on right hand side of sternum.
- Breathless
- Hypotensive
- JVP not raised
- **Diagnosis** = Massive Haemothorax (Because his JVP isn't distended → he probably has Hypovolaemia
∴ The knife has cut a blood vessel and is bleeding into his chest.)
 - Tension Pneumothorax is a possibility also
- **How would you confirm this?**
 - Clinical assessment
- **Treatment** =
 - If haemothorax → put a needle (drain) in his chest

- **Case 4:**

- Attempted hanging
- Unconscious
- Stridor
- **What is the 1st Priority?**
 - Airway Management – Probably use of Endotracheal Intubation (although may be very difficult)
 - Guedell airway is too short
 - If impossible to intubate → Emergency Tracheostomy

- **Case 5:**

- Car accident
- T-Boned at high speed
- Pale
- Sweaty
- Hypotensive (40mmHg)
- Whited out L Lung on Xray
- **Diagnosis** =
 - Haemothorax
 - Probably also has a pneumothorax but not visible on XRay
- **Treatment** =
 - Emergency chest drain with Intercostal Catheter (ICC)

- **Case 6:**

- Barking cough
- Inspiratory stridor
- Moderate indrawing
- Very dyspnoeic
- Agitated & crying
- **Diagnosis =**
 - Croup (Barking cough & stridor)
- **Assessment should include:**
 - Clinical Assessment should suffice (in ED situation)
 - Measure Sats.
- **Treatment =**
 - No antibiotics are needed (croup is viral)
 - Nebulised adrenaline & steroids to decrease swelling.

- **Case 7:**

- Ph 7.3 - Acidosis
- Co2 66 – Raised → Respiratory
- O3 150
- HCO3 28
- Base excess = -1 → No metabolic compensation
- **Diagnosis =**
 - Respiratory acidosis with no metabolic compensation



MEDSTUDENTNOTES.COM

Best Wishes From Our Team!

On behalf of the MedStudentNotes team, we sincerely hope you found these study notes valuable. Our mission is to help aspiring medicos like yourself to become the best clinicians that you can possibly be. We wish you all success in your ongoing studies and career ahead!!

If you have any comments, suggestions, or feedback, please feel free to email us at admin@medstudentnotes.com

If you've found these notes valuable and you haven't yet got our [Bundle Deal](#), (which includes ALL of our subjects at an 80% discount), then NOW is the time! To upgrade, simply visit the link below for instructions:

>>> <https://www.medstudentnotes.com/pages/bundle-upgrade-process> <<<

Just like you, we have big plans for the future, so be sure to keep an eye on your email inbox for updates.

